



BANSALCLASSES

TARGET IIT JEE 2007

ORGANIC CHEMISTRY

XII (ALL)

QUESTION BANK ON

**ACIDITY, BASICITY
H-BONDING
&
TAUTOMERISM**

ACIDITY, BASICITY, H-BONDING & TAUTOMERISM

Q.1 Write equations showing the Lewis acid-base reaction that takes place when

- (a) Methyl alcohol reacts with BF_3 .
- (b) Methyl chloride reacts with AlCl_3 .
- (c) Dimethyl ether reacts with BF_3 .

Q.2 Which of the following are Lewis acids & which are Lewis bases?

- (a) $\text{CH}_3\text{CH}_2-\overset{\cdot\cdot}{\underset{\text{CH}_3}{\text{N}}}-\text{CH}_3$ (b) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}^{\oplus}}$ (c) $(\text{C}_6\text{H}_5)_3\text{P}$
- (d) $:\ddot{\text{P}}:\text{---}$ (e) $(\text{CH}_3)_3\text{B}$ (f) H^+

Q.3 Which would you expect to be the stronger acid? Explain your reasoning in each instance.

- (a) $\text{CH}_2\text{ClCO}_2\text{H}$ or $\text{CHCl}_2\text{CO}_2\text{H}$
- (b) $\text{CCl}_3\text{CO}_2\text{H}$ or $\text{CHCl}_2\text{CO}_2\text{H}$
- (c) $\text{CH}_2\text{FCO}_2\text{H}$ or $\text{CH}_2\text{FCH}_2\text{CO}_2\text{H}$

Q.4 Write equations for the acid base reaction that would occur when each of the following compounds or solution are mixed. In each case label the stronger acid & stronger base, & the weaker acid & weaker base.

- (a) NaH is added to CH_3OH
- (b) NaNH_2 is added to $\text{CH}_3\text{CH}_2\text{OH}$
- (c) Gaseous NH_3 is added to ethyl lithium in hexane
- (d) NH_4Cl is added to NaNH_2 in liq. NH_3
- (e) $(\text{CH}_3)_3\text{CONa}$ is added to H_2O
- (f) NaOH is added to $(\text{CH}_3)_3\text{C-OH}$
- (g) $\text{C}_2\text{H}_5\text{OH}$ is added to a solution of $\text{HC}\equiv\text{C}^-\text{Na}^+$ in liquid NH_3 .

Q.5 $\text{CH}_3\text{CH}_2\text{MgBr} + \text{CH}_3\text{C}\equiv\text{CH} \longrightarrow \text{A} + \text{B}$

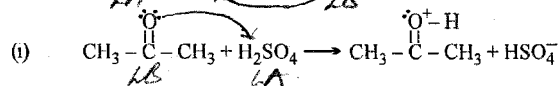
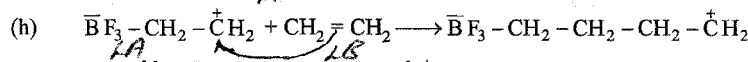
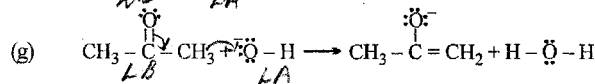
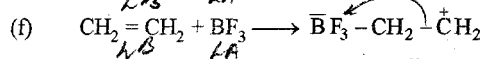
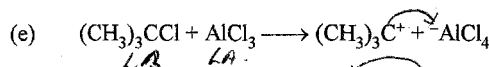
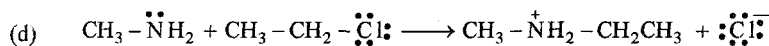
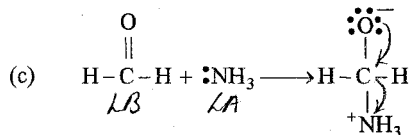
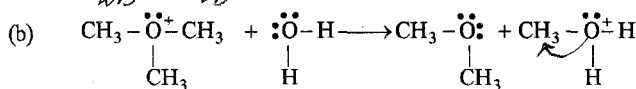
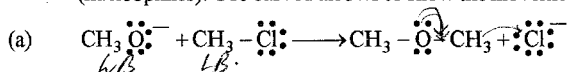
Q.6 Choose the member of each of the following pairs of compounds that is likely to be the stronger base.

- (a) NH_2^- or NH_3 (b) OH^- or H_2O (c) OH^- or SH^-
- (d) $\text{CH}_3\text{CH}_2\text{O}^-$ or  (e) $\text{CH}_2=\overset{\text{O}}{\text{C}}-\text{OH}$ or $\text{CH}_3\text{CH}_2\text{O}^-$ (f) NH_3 or NH_4^+

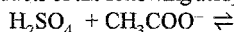
Q.7 Choose the member of each of the following pairs of compounds that is likely to be the weaker base.

- (a) H_2O or H_3O^+ (b) H_2S , HS^- , S^{2-} (c) Cl^- , SH^-
- (d) F^- , OH^- , NH_2^- , CH_3^- (e) HF , H_2O , NH_3 (f) OH^- , SH^- , SeH^-

Q.8 Label the reactants in these acid – base reactions as Lewis acids (electrophiles) or Lewis bases (nucleophiles). Use curved arrows to show the movement of electron pairs in the reactions.

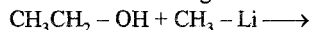


Q.9 Predict the products of the following acid – base reactions.



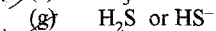
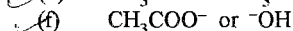
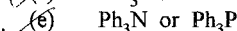
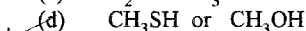
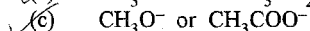
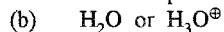
Q.10 Methyl lithium (CH_3Li) is often used as a base in organic reactions.

(a) Predict the products of the following acid – base reaction.

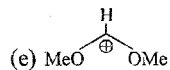
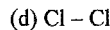
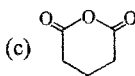
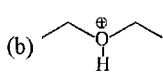
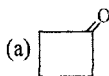


(b) What is the conjugate acid of CH_3Li ? Would you expect CH_3Li to be a strong base or a weak base?

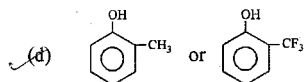
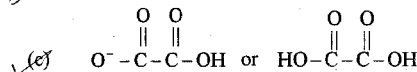
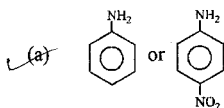
Q.11 Which reagent in each pair listed here would be the more reactive Nu in a protic solvent?



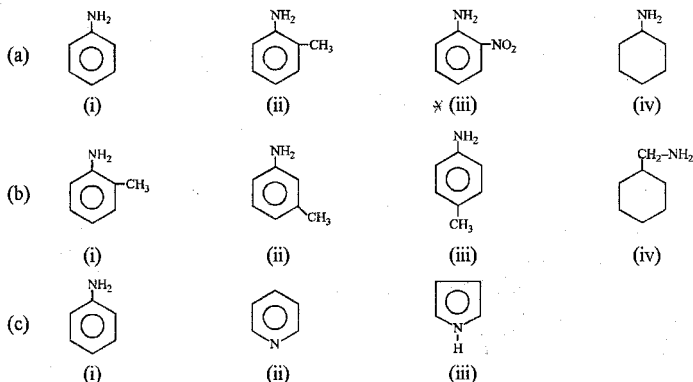
Q.12 Each of these molecules is electrophile. Identify the electrophilic atom & draw a mechanism for reaction with a generalised nucleophile Nu^- , giving the product in each case.



Q.19 Explain which compound is the weaker base.



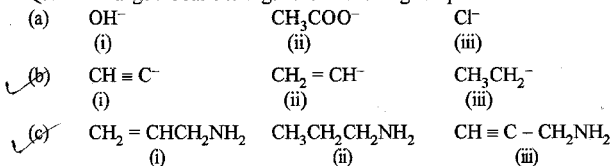
Q.20 Rank the following amines in increasing basic nature.



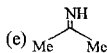
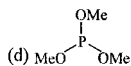
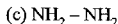
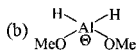
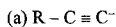
Q.21 Dimethyl furmamide (DMF) is an example of polar aprotic solvent, aprotic meaning it has no hydrogen atoms attached to highly electronegative atoms.

- (a) Draw what you predict to be its most important resonance forms.
 (b) DMF when used as the reaction solvent, greatly enhances the reactivity of nucleophiles. e.g.
 $\text{NaCN} + \text{CH}_3\text{CH}_2\text{Br} \longrightarrow \text{CH}_3\text{CH}_2\text{C} \equiv \text{N} + \text{NaBr}$
 Suggest an explanation for this effects.

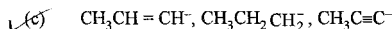
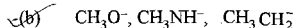
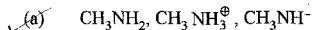
Q.22 Arrange the basic strength of the following compounds.



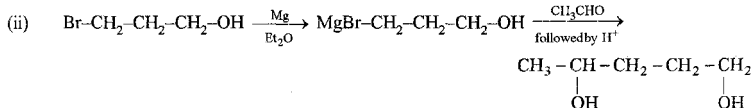
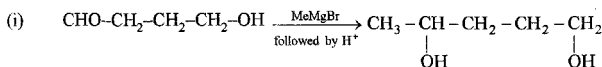
Q. 13 Each of these molecules is nucleophile. Identify the nucleophilic atom & draw a mechanism for reaction with a generalised electrophile E^+ , giving the product in each case.



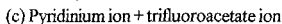
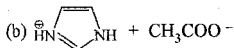
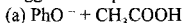
✓ Q. 14 Arrange the following compounds in order of increasing basicity.



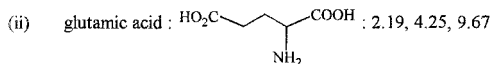
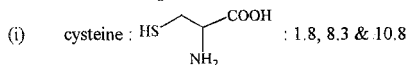
Q. 15 Neither of these methods of making pentan-1,4 diol will work. Explain why not – what will happen instead?



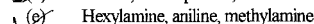
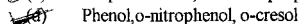
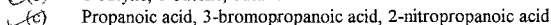
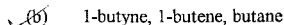
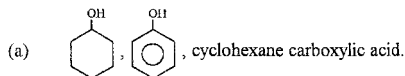
Q. 16 Suggest what species would be formed by each of the following combinations :

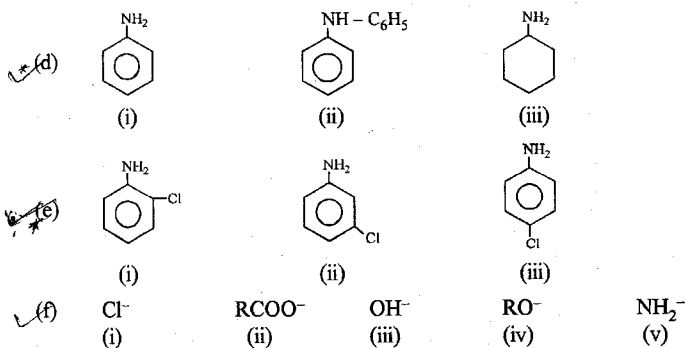


Q. 17 Say which pK_a belong to which functional group in case of following amino acids :

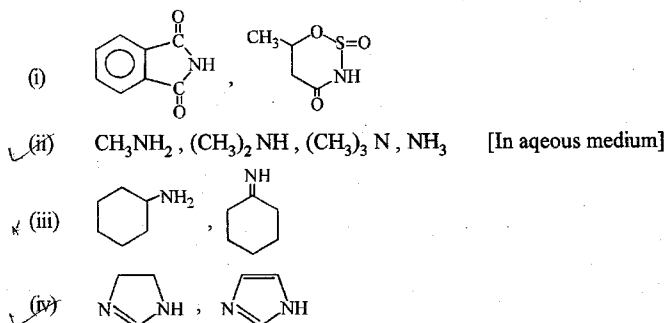


Q. 18 Record the following sets of compounds according to increasing pK_a ($= -\log K_a$)

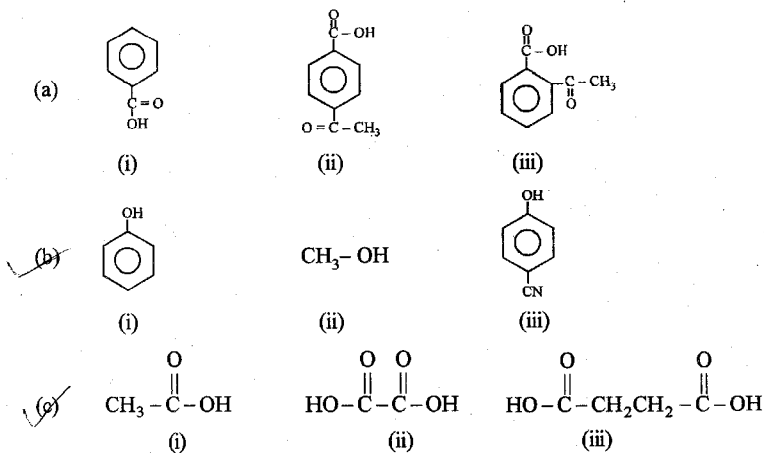




Q. 23 Set the following in increasing order of pK_b :



Q. 24 Arrange the following in increasing acid strength :

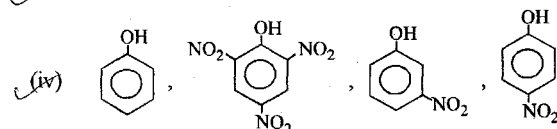


Q.25 Set the following in increasing order of pK_a :

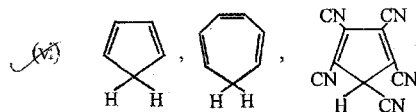
(i) Methane sulfonic acid, acetic acid & methanol.

(ii) $CH_3-CH_2-CH_3$, $CH_2=CH-CH_3$, CH_3-CHO , $CHO-CH_2-CHO$, CH_4 (vi) & (v)

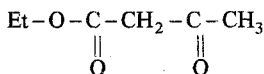
(iii) CH_3NO_2 , $(NO_2)_2-CH_2$, $(NO_2)_3-CH$



(v) CH_3COOH , NO_2-CH_2-COOH , $CH_3-C(=O)-CH_2-C(=O)-OH$



(vii) $CHO-CH_2-CHO$, $CH_3-C(=O)-CH_2-C(=O)-CH_3$, $Et-O-C(=O)-CH_2-C(=O)-O-Et$,

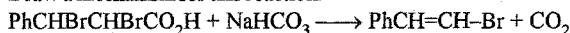


Q.26 Cyanic acid ($HO-C\equiv N$) & isocyanic acid ($H-N=C=O$) differ in the positions of their electrons but their structure do not represent resonance structures.

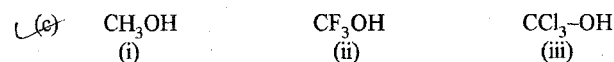
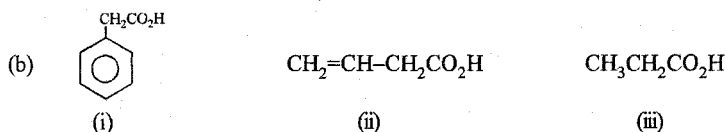
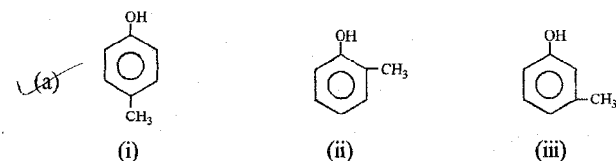
(a) Explain

(b) Loss of a proton from cyanic acid yields the same anion as that obtained by loss of a proton from isocyanic acid. Explain.

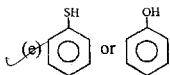
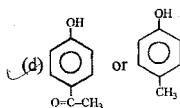
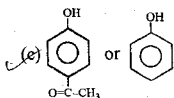
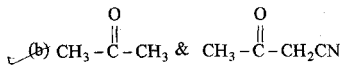
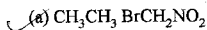
Q.27 Draw a mechanism for this reaction.



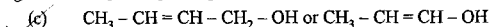
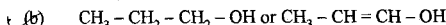
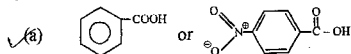
Q.28 Arrange the increasing order of acidic strength of the following compounds.



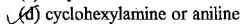
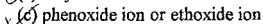
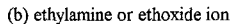
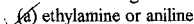
Q.29 Explain which is a stronger acid.



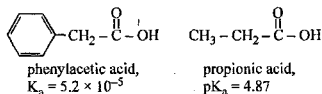
Q.30 Which of the following would you predict to be the stronger acid?



Q.31 Which is a stronger base? & Why.



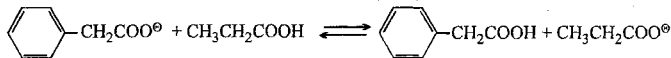
Q.32 The K_a of phenylacetic acid is 5.2×10^{-5} , and the pK_a of propionic acid is 4.87.



(a) Calculate the pK_a of phenylacetic acid and the K_a of propionic acid.

(b) Which of these is the stronger acid?

(c) Predict whether the following equilibrium will favor the reactants or the products.



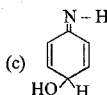
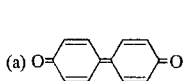
Q.33 Explain why the 2-methylpyrrolidine boils at a higher temperature than its isomers N-methylpyrrolidine.
H-bonding

Q.34 Although nitration usually causes elevation of B.P. yet the nitration of resorcinol to 2-nitroresorcinol decreases their B.P. then their parent compound.

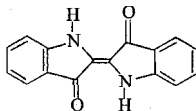
Q.35 o-nitrophenol is sparingly soluble in water & has lower B.P. but p-nitrophenol is completely soluble in water & has high boiling point.

Q.36 When 30 ml of ethanol and 5 ml of water are mixed the volume of the resulting solution is less than 35 ml. Explain.

- Q.37 Ammonium salts are much more soluble in water than are the corresponding sodium salts. Explain?
- Q.38 HF has a dipole moment of 1.82 D, its boiling point is 19.34°C. Ethyl fluoride has an almost identical dipole moment & has a larger molecular weight, yet its B.P. is -37.7°C. Explain.
- Q.39 HCO_2H & $\text{CH}_3\text{CO}_2\text{H}$ exists as dimer. Explain.
- Q.40 Cyclohexanol is more soluble in water than 1-hexanol.
- Q.41 Explain why CH_3OH and $\text{CH}_3\text{CH}_2\text{OH}$ are reasonably good solvents for many ionic compounds compare the solvent properties of ethanol and pentanol towards ionic compound.
- Q.42 1,5-Pentanediol is soluble & 1-Pentanol is slightly soluble in H_2O .
- Q.43 Compare the relative (i) Boiling points & (ii) Water solubilities of toluene & phenol.
- Q.44 Which isomer (o, m or p) of hydroxyacetophenone steam distills.
- Q.45 Give structure for ethanolamine showing two different intramolecular H-bonds & discuss their relative strength.
- Q.46 Which of the following system show H-bonding during tautomerism.

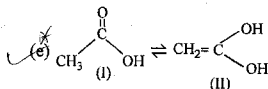
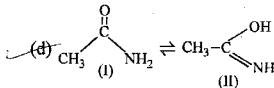
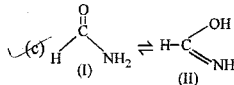
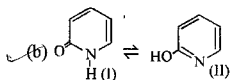
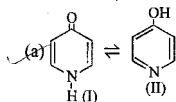


- Q.47 Trans isomer of indigotin is stable w.r.t cis isomer. Explain.



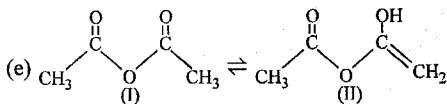
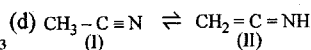
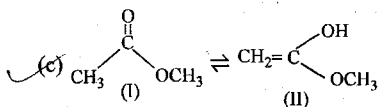
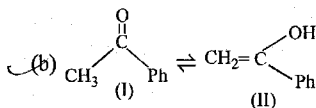
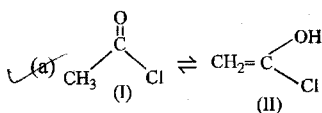
- Q.48 What is the attacking site of conjugate base of triketo form of phloroglucinol in protic & aprotic solvent.

- Q.49 In each of the following pairs which is more stable:



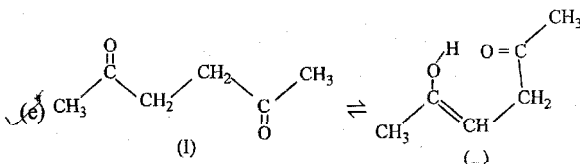
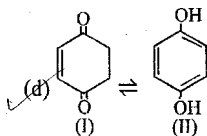
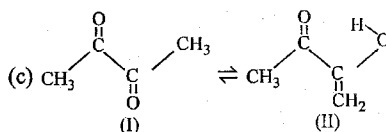
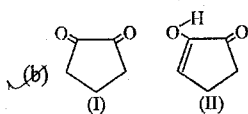
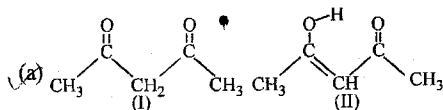
Q.50 In each of the following pairs which is less stable :

(l)

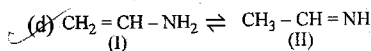
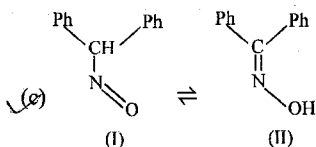
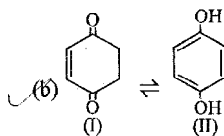
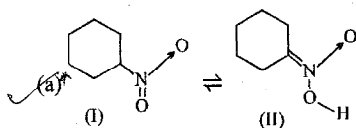


Q.51 In each of the following pairs which is more stable :

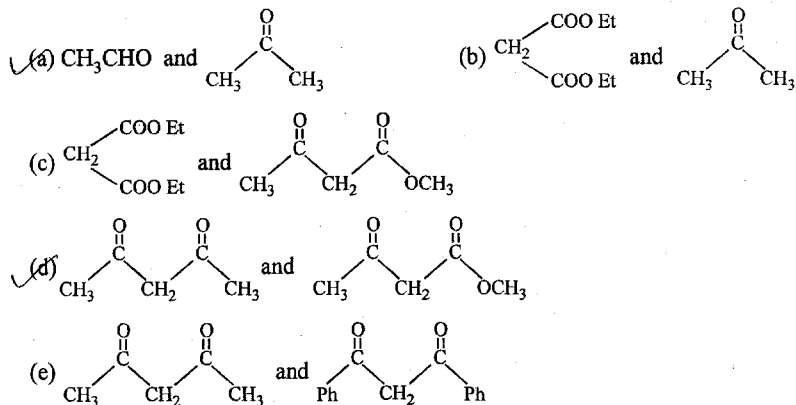
(l)



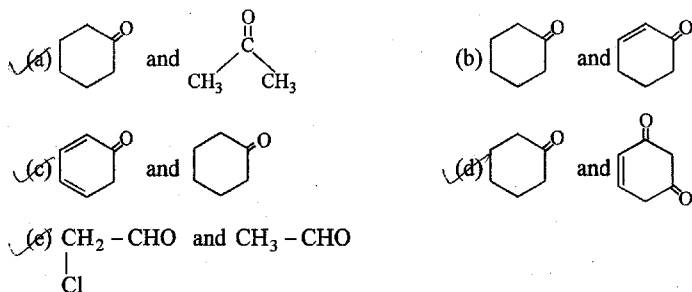
Q.52 In each of the following pairs which is less stable :



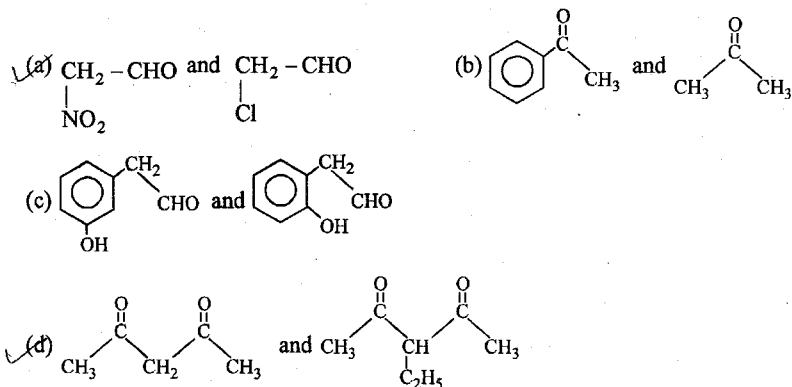
Q.53 In each of the following pairs which will have higher enol content :



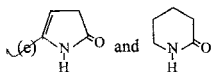
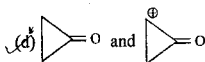
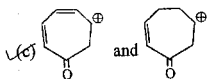
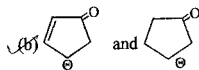
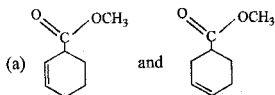
Q.54 In each of the following pairs which will have less enol content :



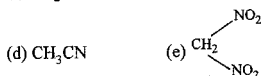
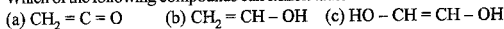
Q.55 In each of the following pairs which will have higher enol content :



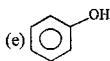
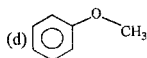
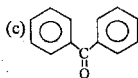
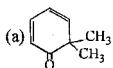
Q. 56 In each of the following pairs which will have less enol content :



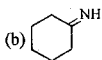
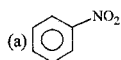
Q. 57 Which of the following compounds can exhibit tautomerism :



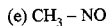
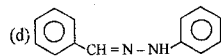
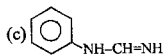
Q. 58 Which of the following compounds can not exhibit tautomerism :



Q. 59 Which of the following compounds can exhibit tautomerism :



Q. 60 Which of the following compounds can not exhibit tautomerism :

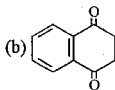
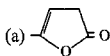


Q. 61 What is the relationship between these two molecules? Discuss the structure of the anion that would be formed by the deprotonation of each compound.

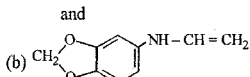
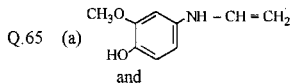
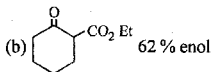
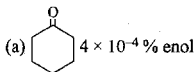


Q. 62 5,5 - dimethylcyclohexane-1,3 -dione (dimedone) shows tautomerism. Write down tautomeric structure.

Q. 63 Draw enol forms of these carbonyl compounds and comment on the stability of the enol forms.

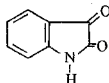


Q. 64 The proportion of enol in a sample of the two ketones below are shown. Why are they so different ?



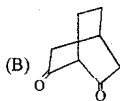
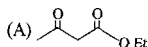
Shows tautomeric forms of (a) & (b).

Q. 66 Isatin was the first compound to show tautomerism.

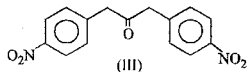
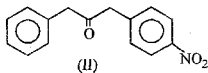
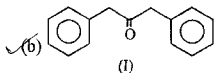
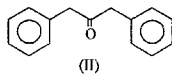
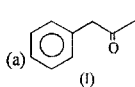


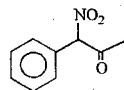
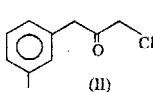
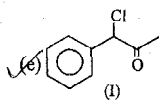
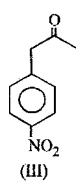
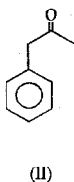
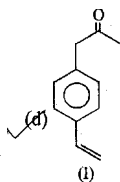
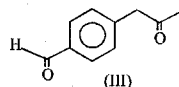
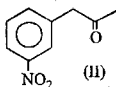
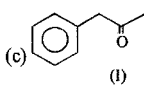
What is the tautomeric form of isatin.

Q. 67 1,3 - dicarbonyl compounds such as (A) are usually mostly enolized. Why is this ? Draw the enols available to compounds A & B comment on the different pattern of enolization.

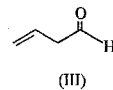
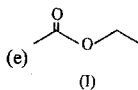
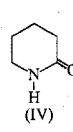
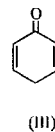
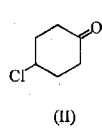
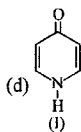
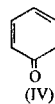
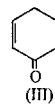
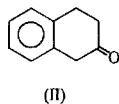
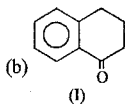
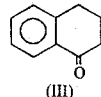
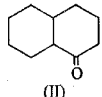


Q. 68 In each of the following sets of compounds write the increasing order of % enol content



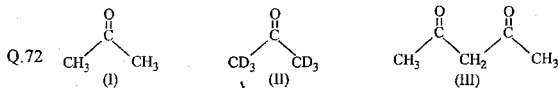


Q. 69 In each of the following sets of compounds write the decreasing order of % enol content.



Q.70 Enol form of cyclobutanone is very unstable vs enol form of triketocyclobutane, which is very stable.

Q.71  has lower boiling point than  even when former has -OH group.



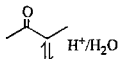
Among these give ease of enolization.

Q.73 % enol content of acetylacetone in following solvents is found as :

Solvent	% enol content
H ₂ O	15
Liquid state	76
hexane	92
gas phase	92

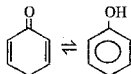
Explain the observation.

Q.74



Explain the observation.

Q.75



This tautomeric system exist almost exclusively in favour of phenol and it is insensitive to change in solvent.

ANSWER KEY

- Q.2 LA b,e LB acdf
Q.3 (a) 2; (b) 1; (c) 1
Q.5 $\text{CH}_3\text{CH}_3 + \text{CH}_3\text{C} \equiv \text{CMgBr}$
Q.6 (a) 1; (b) 1; (c) 1; (d) 1; (e) 2; (f) 1
Q.7 (a) 2; (b) 1; (c) 1; (d) 1; (e) 1; (f) 3
Q.11 (a) 1; (b) 1; (c) 1; (d) 1; (e) 2; (f) 2; (g) 2; (h) 1
Q.14 (a) $2 < 1 < 3$; (b) $1 < 2 < 3$; (c) $3 < 1 < 2$
Q.18 (a) $3 < 2 < 1$; (b) $1 < 2 < 3$; (c) $3 < 2 < 1$; (d) $2 < 1 < 3$; (e) $2 < 3 < 1$
Q.19 (a) 2; (b) 1; (c) 2; (d) 2
Q.20 (a) $3 < 2 < 1 < 4$; (b) $1 < 2 < 3 < 4$; (c) $3 < 1 < 2$
Q.22 (a) $1 > 2 > 3$; (b) $1 < 2 < 3$; (c) $3 < 1 < 2$; (d) $2 < 1 < 3$; (e) $1 < 2 < 3$; (f) $1 < 2 < 3 < 4 < 5$
Q.23 (i) $1 > 2$; (ii) $4 > 3 > 1 > 2$; (iii) $1 < 2$; (iv) $1 < 2$
Q.24 (a) $1 < 2 < 3$; (b) $3 > 1 > 2$; (c) $1 < 3 < 2$
Q.25 (i) $1 < 2 < 3$; (ii) $1 > 5 > 2 > 3 > 4$; (iii) $1 > 2 > 3$; (iv) $1 > 3 > 4 > 2$; (v) $1 > 3 > 2$; (vi) $2 > 1 > 3$; (vii) $3 > 4 > 2 > 1$
Q.28 (a) $3 > 1 > 2$; (b) $1 > 2 > 3$; (c) $2 > 3 > 1$
Q.29 (a) 2; (b) 2; (c) 1; (d) 1; (e) 1
Q.30 (a) 2; (b) 2; (c) 2
Q.31 (a) 1; (b) 2; (c) 2; (d) 1
Q.32 $4.25, 1.34 \times 10^{-5}$
Q.49 (a) 2; (b) 2; (c) 1; (d) 1; (e) 2
Q.50 (a) 2; (b) 2; (c) 2; (d) 2; (e) 1
Q.51 (a) 2; (b) 2; (c) 1; (d) 2; (e) 1
Q.52 (a) 2; (b) 1; (c) 1; (d) 1
Q.53 (a) 1; (b) 1; (c) 2; (d) 1; (e) 2
Q.54 (a) 1; (b) 1; (c) 2; (d) 1; (e) 2
Q.55 (a) 1; (b) 1; (c) 1; (d) 1
Q.56 (a) 2; (b) 2; (c) 2; (d) 1; (e) 2
Q.57 a, b, c, d, e
Q.58 a, c, d
Q.59 b
Q.60 b
Q.68 (a) $2 > 1 > 4 > 3$; (b) $3 > 2 > 1$; (c) $3 > 2 > 1$; (d) $3 > 1 > 2$; (e) $3 > 1 > 2$
Q.69 (a) $3 > 1 > 2$; (b) $4 > 2 > 1 > 3$; (c) $4 > 3 > 1 > 2$; (d) $1 > 3 > 4 > 2$; (e) $3 > 2 > 4 > 1$
Q.72 $3 > 1 > 2$



BANSAL CLASSES

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ORGANIC CHEMISTRY

XII (ALL)

ALCOHOL & ETHER

CONTENTS

EXERCISE - I(A)

EXERCISE I(B)

EXERCISE - II

EXERCISE - III

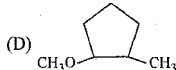
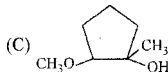
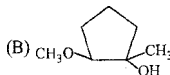
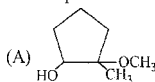
EXERCISE - IV(A)

EXERCISE - IV(B)

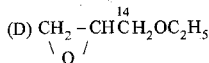
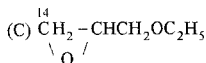
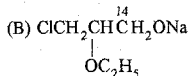
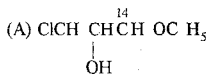
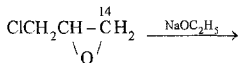
ANSWER KEY

EXERCISE-I(A)

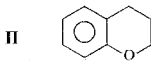
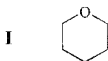
- Q.1 Which of the following reaction is called as 'Bouveault-Blanc reduction'
 (A) Reduction of acyl halide with $\text{H}_2\text{PdBaSO}_4$
 (B) Reduction of ester with $\text{Na/C}_2\text{H}_5\text{OH}$
 (C) Reduction of anhydride with LiAlH_4
 (D) Reduction of carbonyl compounds with Na/HgHCl
- Q.2 Glycol on treatment with PI_3 mainly gives—
 (A) Ethylene (B) Ethylene iodide (C) Ethyl iodide (D) Ethane
- Q.3 Acrolein is formed when glycerol is heated with—
 (A) Acidified KMnO_4 (B) Br_2 water (C) KHSO_4 (D) HNO_3
- Q.4 Glycerol on treatment with oxalic acid at 110°C forms—
 (A) Allyl alcohol (B) Formic acid (C) CO_2 and CO (D) Glyceric acid
- Q.5 If the starting material is 1-methyl-1,2-epoxy cyclopentane, of absolute configuration, decide which one compound correctly represent the product of its reaction with sodium methoxide in methanol.



- Q.6 When phenol is treated with PCl_5 , the yield of chlorobenzene is generally poor because of the formation of
 (A) Benzoyl chloride (B) p-chlorophenol (C) o-chlorophenol (D) Triphenyl phosphate
- Q.7 In the following reaction, final product is



Q. 8 Consider the reaction of HI with the following:



Which forms di-iodide on reaction with HI (excess)?

- (A) I and II both (B) II only (C) I only (D) none

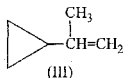
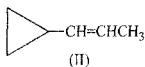
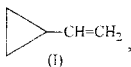
Q. 9 Ethanol on reaction with acetic anhydride gives

- (A) Acetic ester (B) Formic ester
(C) Ethanoic acid (D) Acetic ester and Ethanoic acid both

Q. 10 Ethanol cannot be dried by anhydrous CaCl_2 due to formation of the following solvated product

- (A) $\text{CaCl}_2 \cdot 2\text{C}_2\text{H}_5\text{OH}$ (B) $2\text{CaCl}_2 \cdot 3\text{C}_2\text{H}_5\text{OH}$ (C) $\text{CaCl}_2 \cdot 3\text{C}_2\text{H}_5\text{OH}$ (D) $\text{CaCl}_2 \cdot \text{C}_2\text{H}_5\text{OH}$

Q. 11 Rate of hydration of



will be in order:

- (A) I < II < III (B) I < III < II (C) II < I < III (D) III < II < I

Q. 12 The reaction of $\text{CH}_3\text{OC}_2\text{H}_5$ with HI gives

- (A) CH_3I (B) $\text{C}_2\text{H}_5\text{OH}$ (C) $\text{CH}_3\text{I} + \text{C}_2\text{H}_5\text{OH}$ (D) $\text{C}_2\text{H}_5\text{I} + \text{CH}_3\text{OH}$

Q. 13 The number of methoxy groups in a compound can be determined by treating it with

- (A) HI and AgNO_3 (B) Sodium carbonate (C) Sodium hydroxide (D) Acetic acid

Q. 14 Most acidic alcohol out of following compounds is

- (A) $(\text{CH}_3)_3\text{C-OH}$ (B) $\text{CH}_3\text{CH}_2\text{OH}$ (C) CH_3OH (D) PhOH

Q. 15 Action of HNO_2 on CH_3NH_2 gives following as major product

- (A) CH_3OH (B) $\text{CH}_3\text{-O-CH}_3$ (C) $\text{CH}_3\text{-O-N=O}$ (D) $\text{CH}_3\text{-NO}_2$

Q. 16 A compound 'X' with molecular formula $\text{C}_3\text{H}_8\text{O}$ can be oxidised to a compound 'Y' with the molecular formula $\text{C}_3\text{H}_6\text{O}_2$. 'X' is most likely to be

- (A) Primary alcohol (B) Secondary alcohol (C) Aldehyde (D) Ketone

Q. 17 Diethyl ether and air gives ether hydroperoxide. The mechanism of the reaction is

- (A) Nucleophilic substitution (B) Free radical addition
(C) Free radical substitution (D) None of the above

Q. 18 Ether on carbonylation gives

- (A) Alkanoic acid (B) Alkanone (C) Alkyl alkanoate (D) Alkanal

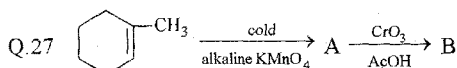
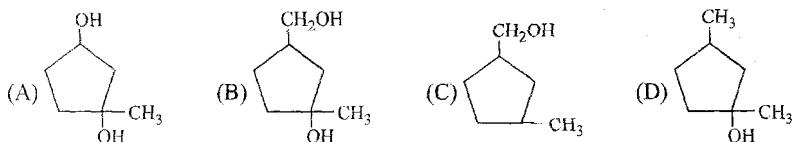
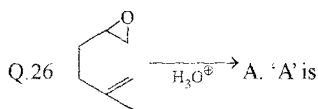
Q. 19 $\text{I}^\oplus\text{R} - \overset{\oplus}{\text{O}} - \text{R}' \longrightarrow \text{RI} + \text{R}'\text{OH}$ true about this mechanism

- (A) $\text{S}_{\text{N}}1$ in gas phase (B) $\text{S}_{\text{N}}2$ in aqueous phase
(C) both of the above (D) none

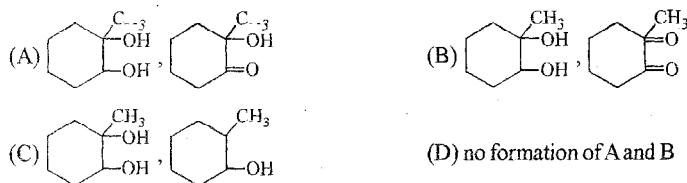
Q. 20 Phenol with Hinsberg's reagent gives

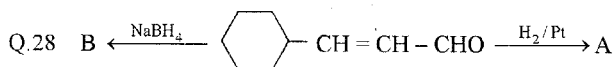
- (A) Sulphone (B) Sulphanilic acid (C) Sulphonic ester (D) Sulphonol

- Q. 21 Select the odd structure out
 (A) $\text{CH}_3\text{--CH(OH)--CH}_2\text{--CH}_3$ (B) $\text{CH}_3\text{--CH(OH)--CH}_2\text{--CH}_2\text{--CH}_3$
 (C) $\text{CH}_3\text{--CH(OH)--CH}_3$ (D) $\text{CH}_3\text{--CH}_2\text{--CH(OH)--C}_2\text{H}_5$
- Q. 22 Glycerol $\xrightarrow{\text{KHSO}_4}$ A $\xrightarrow{\text{LiAlH}_4}$ B
 A and B are:
 (A) Acrolein, allyl alcohol (B) Glyceryl sulphate, acrylic acid
 (C) Allyl alcohol, acrolein (D) Only acrolein (B is not formed)
- Q. 23 An ether is heated with phosphorous penta sulphide to give
 (A) Alkanthiol (B) Dialkyl sulphide (C) Hydrogen sulphide (D) Thioester
- Q. 24 Phenol $\xrightarrow[\text{(ii) CO}_2/140^\circ\text{C}]{\text{(i) NaOH}}$ A $\xrightarrow[\text{CH}_3\text{COOH}, \Delta]{\text{H}^+/\text{H}_2\text{O}}$ B $\xrightarrow[\text{CH}_3\text{COOH}, \Delta]{\text{Al}_2\text{O}_3}$ C
 In this reaction, the end product C is:
 (A) salicylaldehyde (B) salicylic acid (C) phenyl acetate (D) aspirin
- Q. 25 In the Liebermann's nitroso reaction changes in the colour of phenol occurs as:
 (A) Brown or red-greenish red-deep blue (B) red-deep blue-green
 (C) red-green-white (D) white-red-green

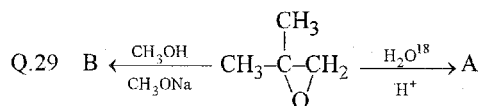
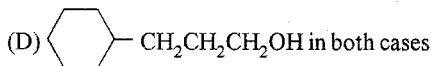
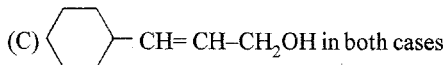
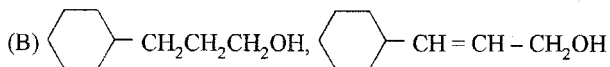
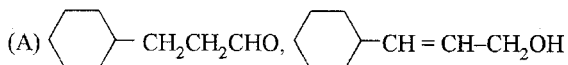


A and B are:

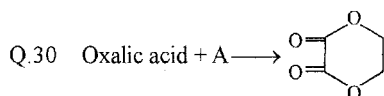
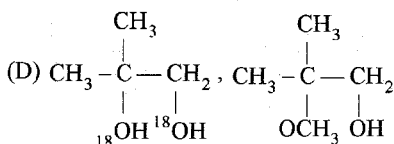
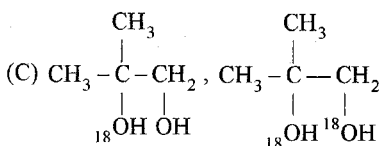
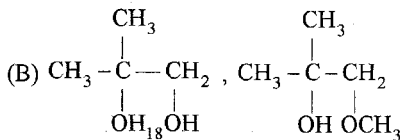
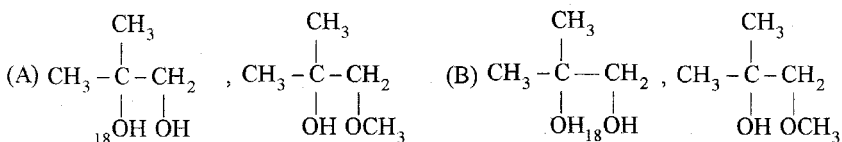




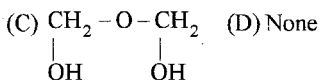
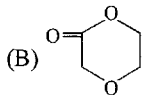
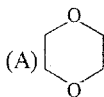
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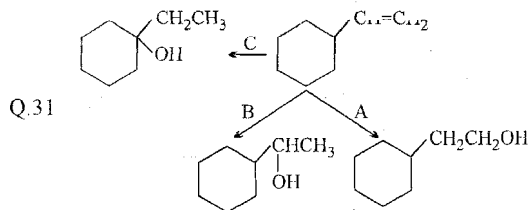


A and B are



hence A $\xrightarrow{\text{conc. H}_2\text{SO}_4}$ B, B is:





Select schemes A, B, C out of

I acid catalysed hydration

II

HBO

III

oxymercuration-demercuration

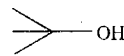
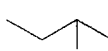
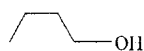
(A) I in all cases

(B) I, II, III

(C) II, III, I

(D) III, I, II

Q.32 Dehydration of the alcohols



(I)

(II)

(III)

(IV)

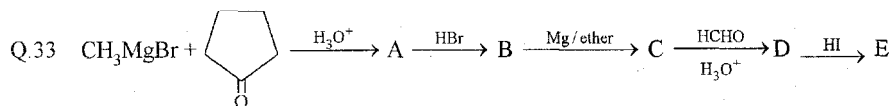
will be in order

(A) IV > III > II > I

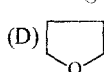
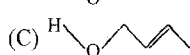
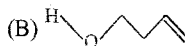
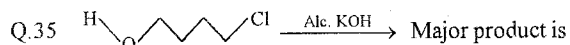
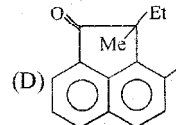
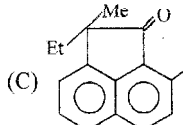
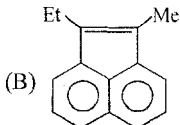
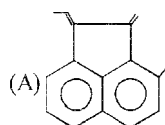
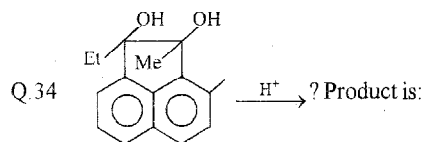
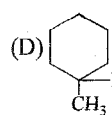
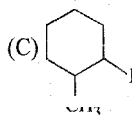
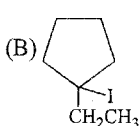
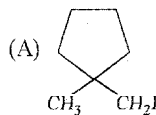
(B) I > II > III > IV

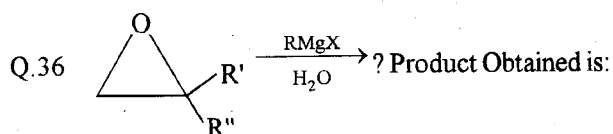
(C) IV > II > III > I

(D) II > IV > I > III

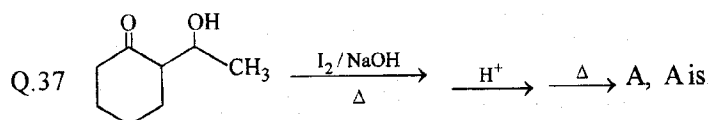


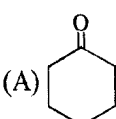
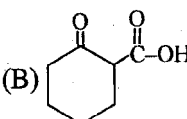
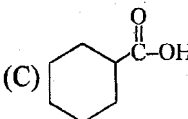
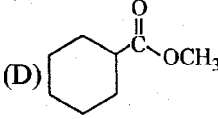
E is

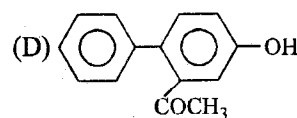
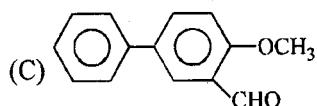
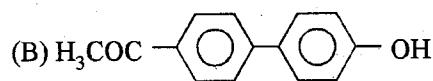
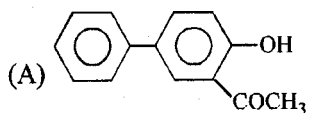
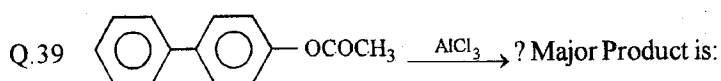
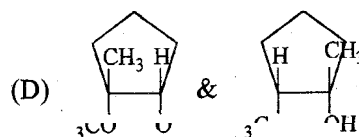
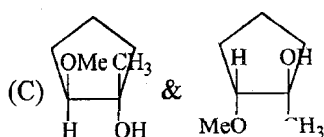
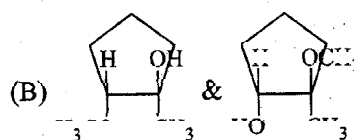
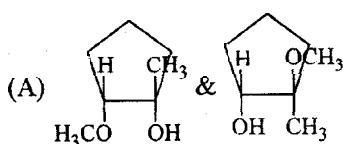
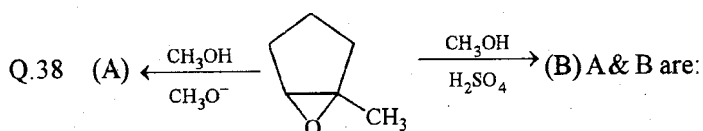


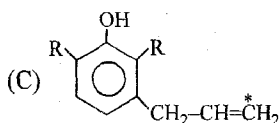
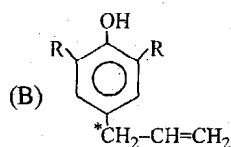
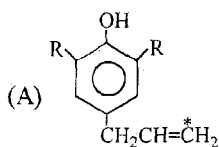
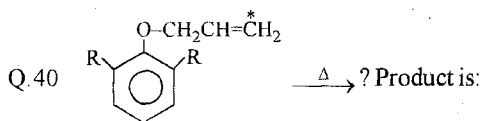


- (A) $\text{R}'\text{-}\overset{\text{R}}{\underset{\text{R}''}{\text{C}}}\text{-CH}_2\text{OH}$ (B) $\text{RCH}_2\text{-}\overset{\text{R}'}{\underset{\text{R}''}{\text{C}}}\text{-OH}$ (C) $\text{R}'\text{CH}_2\text{-}\overset{\text{R}}{\underset{\text{R}''}{\text{C}}}\text{-OH}$ (D) $\text{R}''\text{CH}_2\text{-}\overset{\text{R}}{\underset{\text{R}'}{\text{C}}}\text{-OH}$

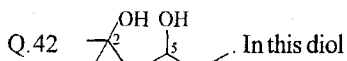
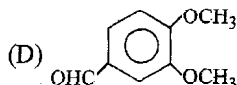
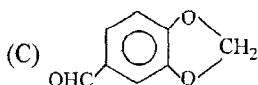
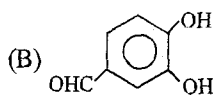
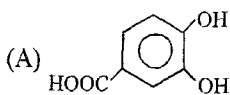
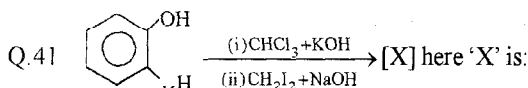


- (A)  (B)  (C)  (D) 



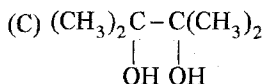
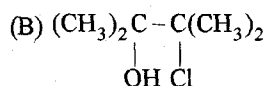
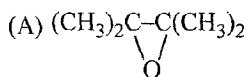
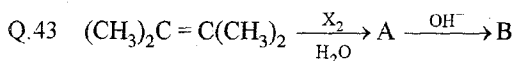


(D) no reaction

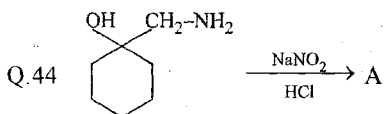


(A) OH at C₂ is more basic than that of at C₅
(C) both have same basicity

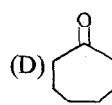
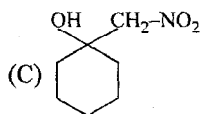
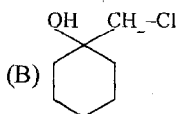
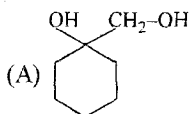
(B) OH at C₂ is more acidic than at C₅.
(D) both have same acidic strength



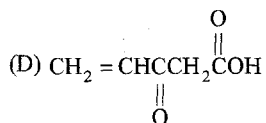
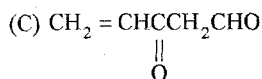
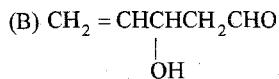
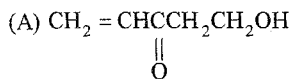
(D) None



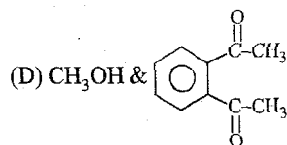
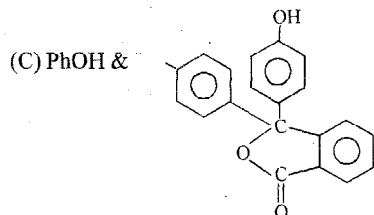
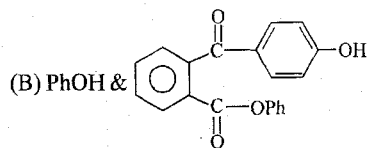
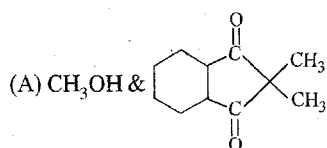
A is



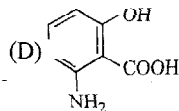
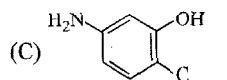
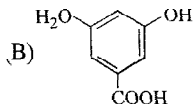
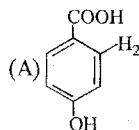
Q.45 $\text{CH}_2 = \underset{\text{OH}}{\text{CH}}\text{CHCH}_2\text{CH}_2\text{OH} \xrightarrow{\text{MnO}_2} \text{A. A is}$



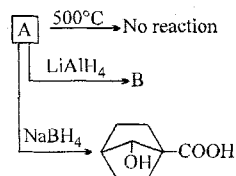
Q.46 Acetophenone $\xrightarrow{\text{HCO}_2\text{H}} \text{A} \xrightarrow{\text{H}_3\text{O}^+} \text{B} + \text{C} \xrightarrow[\text{H}^+]{\text{Phthalic Anhydride}} \text{Indicator (D)}$
C & D are



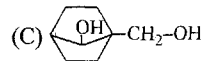
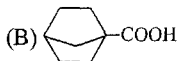
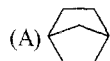
Q.47 m-Aminophenol on treatment with NaOH and CO_2 gives which of the following as major product?



Question No 48 to 50 (3 questions)



Q.48 Compound 'B' is?

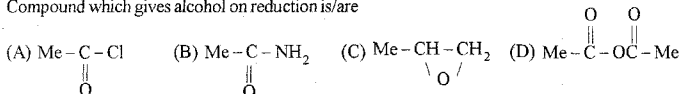


- Q.49 Organic compound A does not undergo decarboxylation reaction because?
 (A) Intermediate does not follow Saytzeff's rule
 (B) Intermediate does not follow Hofmann's rule
 (C) Intermediate does not follow Bredt's rule
 (D) Intermediate does not follow Markovnikov's rule

- Q.50 Total No. of stereoisomers of B are?
 (A) 2 (B) 4 (C) 3 (D) 6

EXERCISE-I(B)

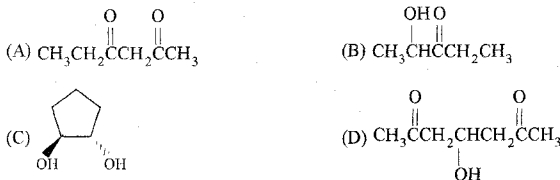
- Q.1 Compound which gives alcohol on reduction is/are



- Q.2 Phenol and ethanol are distinguished by the reaction with
 (A) Red litmus (B) NaHCO_3 (C) FeCl_3 (D) NaOH

- Q.3 In the reaction sequence, $\text{CaC}_2 \xrightarrow{\text{H}_2\text{O}} \text{A} \xrightarrow[\text{Hg}^{+2}]{\text{dil. H}_2\text{SO}_4} \text{B} \xrightarrow[\text{Ni}]{\text{H}_2} \text{C}$, true about the product C is
 (A) give yellow ppt. with NaOI (B) its final oxidation product is carbonyl compound
 (C) its final oxidation product is CO_2 and H_2O (D) its final oxidation product is CH_3COOH

- Q.4 Which can be cleaved by HIO_4 ?



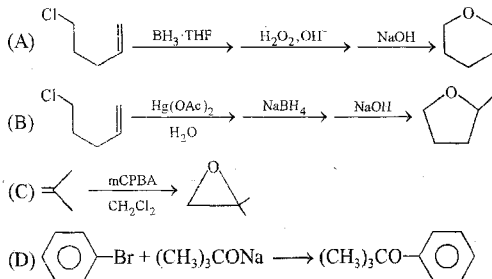
- Q.5 HBO , oxymercuration-demercuration and acid catalysed hydration will not give same product in



- Q.6 Diethyl ether reacts with PCl_5 to form



Q.7 Select the correct synthesis products



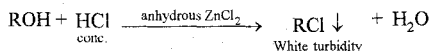
Q.8 The molecules of ether dehydrates in the presence of-

- (A) Al_2O_3 (B) H_3PO_4 (C) $\text{H}_2\text{S}_2\text{O}_7$ (D) liq. NH_3

Q.9 Anhydride of alcohol is

- (A) Ether (B) Aldehyde
(C) Alkyl hydrogen sulphure (D) Alkene

Q.10 Lucas test is used to make distinction between 1° , 2° and 3° alcohols



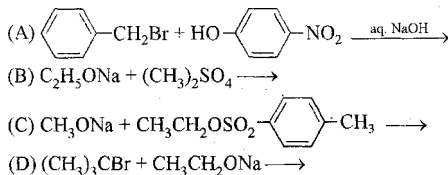
This shows that

- (A) ROH behaves as a base
(B) greater the value of pK_a (alcohol), greater the reactivity with conc. HCl and thus sooner the formation of white turbidity.
(C) alcohol which reacts fastest with Na metal, will give turbidity at fastest rate
(D) alcohol which gives red colour during Victor Mayor test, will give turbidity at slower rate than those giving blue or white colour during Victor Mayor test.

Q.11 If ethanol dissolves in water, then which of the following would be done

- (A) Absorption of heat (B) Emission of heat
(C) Increase in volume (D) Contraction in volume

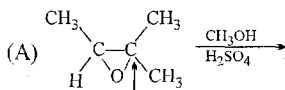
Q.12 Which method is useful for the synthesis of ether?



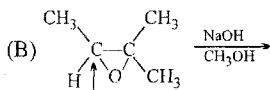
Q.13 Which of the following can react with TsCl

- (A) Glycerol (B) Oximinooacetone
(C) Oil of wintergreen (D) dimethyl amine

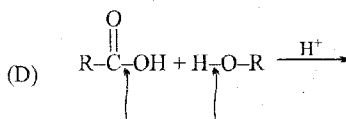
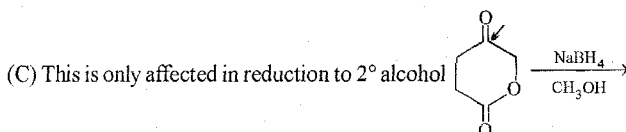
Q.14 Which is/are correct statements?



nucleophile attacks here when epoxy linkage is cleaved



nucleophile attacks here when epoxy linkage is cleaved

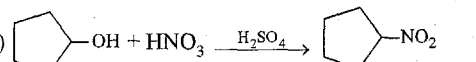
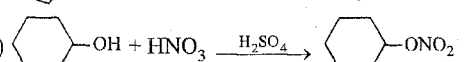
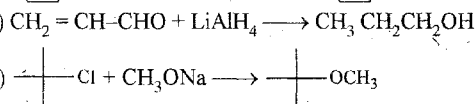


These bonds undergo cleavage in the reaction

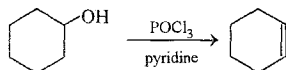
Q.15 3-methyl-3-hexanol can be prepared by

- (A) CH_3MgI and 3-hexanone, followed by hydrolysis
- (B) $\text{C}_2\text{H}_5\text{MgI}$ and 2-pentanone, followed by hydrolysis
- (C) $\text{C}_3\text{H}_7\text{MgI}$ and 2-butanone, followed by hydrolysis
- (D) $\text{C}_4\text{H}_9\text{MgI}$ and propanone, followed by hydrolysis

Q.16 In which cases product formed are not according to reaction?

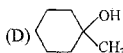
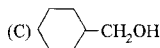
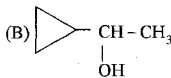
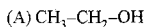
- (A) 
- (B) 
- (C) $\text{CH}_2 = \text{CH} - \text{CHO} + \text{LiAlH}_4 \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (D) 

Q.17 Dehydration of alcohols take place more rapidly with POCl_3 than with H_2SO_4 . Select the correct statement(s) about the following dehydration reaction.



- (A) It does not involve carbocation. *
- (B) It involves R-OPOCl_2 with $-\text{OPOCl}_2$ as a better leaving group.
- (C) It involves E2 mechanism as pyridine base abstracts proton from the adjacent carbon as the same time at which $-\text{OPOCl}_2$ is leaving.
- (D) It is E1 reaction without formation of carbocation.

Q. 18 Which of the following will get oxidised by Br_2/KOH into carboxylic acid?



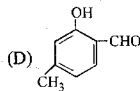
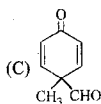
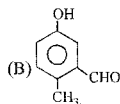
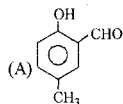
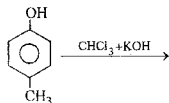
Q. 19 In Kolbe-Schmidt reaction, o-hydroxy benzoic acid is predominantly formed. This is because

- (A) salicylate anion is a stronger base than phenoxide ion
 (B) salicylate anion is a weaker base than phenoxide ion
 (C) p-hydroxy benzoate ion is a stronger base than phenoxide ion
 (D) p-hydroxy benzoate ion is a stronger base than salicylate ion

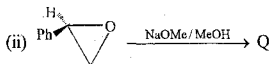
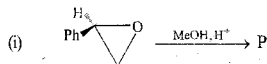
Q. 20 Methanol can be distinguished from ethanol by

- (A) Heating with I_2 and alkali
 (B) Treating with Schiff's reagent
 (C) Treating with CrO_3 solution in dil. H_2SO_4
 (D) Treating with Lucas reagent

Q. 21 Products form by following reactions are



Q. 22 For the reactions shown below, identify the correct statement(s) with regard to the products formed:

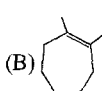
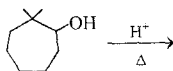


- (A) P and Q are identical
 (B) P is racemic and Q is optically active
 (C) P and Q are positional isomers
 (D) both are optically active

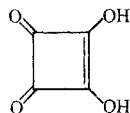
Q. 23 $\text{C}_2\text{N}_5\text{NH}_2 \xrightarrow{\text{Tilden reagent}} \text{(i)} \xrightarrow{\text{NH}_3} \text{(ii)} \xrightarrow[\text{HCl}]{\text{NaNO}_2} \text{(iii)}$. The product (iii) can be

- (A) Alcohol
 (B) Ether
 (C) Alkyl chloride
 (D) Alkyl nitrite

Q. 24 Products form during dehydration of following alcohols are



Q. 25 Consider the following compound A (below)



Select the correct statement(s)

(A) It is more acidic than CH_3OH

(C) It reacts very fast with Lucas reagent

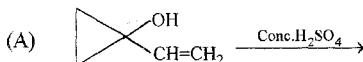
(B) It is more acidic than CH_3COOH

(D) It is a diacid base

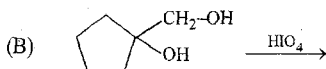
Q. 26 Match the following:

Reaction

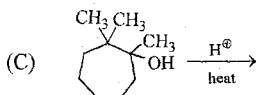
Mechanism



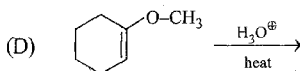
(P) change in number of carbon in cycle



(Q) Pinacol rearrangement



(R) Oxidative bond cleavage



(S) Ketone as product

Q. 27 Match the column:

Column I

Column II

- (A) Identification of 1° , 2° & 3° Alcohols
 (B) Identification of 1° , 2° & 3° Nitro alkanes
 (C) Formation of alcohol by anti Markovnikov's addition of H_2O
 (D) Formation of alcohol by Markovnikov's addition

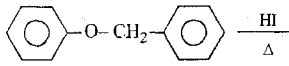
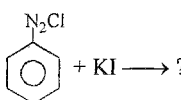
- (P) Oxymercuration demercuration
 (Q) $\text{Cu}/300^\circ$ heat
 (R) Victor Mayer's test
 (S) Hydroboration oxidation
 (T) Lucas test

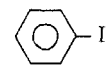
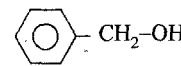
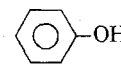
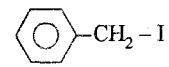
Q.28 Match the column:

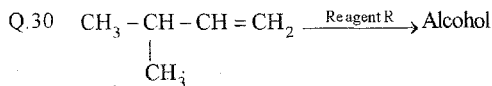
- Column I**
- (A) Oxidation of 1° alcohol in aldehyde
- (B) $\text{CrO}_3 \cdot \text{C}_6\text{H}_5\text{N} \cdot \text{HCl}$
- (C) $\text{CrO}_3 \cdot 2 \text{C}_6\text{H}_5\text{N}$
- (D) Oxidation of alkene into acid

- Column II**
- (P) KMnO_4/Δ
- (Q) Collin's reagent
- (R) Jones's reagent
- (S) P.C.C

Q.29 Match the products of following:

- Column I**
- (A) 
- (B) Violet color is obtain by the reaction of neutral FeCl_3 with
- (C) Reaction of benzaldehyde with $\text{LiAlH}_4/\text{H}_2\text{O}$ gives
- (D) 

- Column II**
- (P) 
- (Q) 
- (R) 
- (S) 



which is true about alcohol and R?

- Alcohol**
- (A) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{CH}_2\text{OH}$
- (B) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$
- (C) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}} - \text{CH}_2\text{CH}_3$
- (D) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}} - \text{CH}_2\text{CH}_3$

- Reagent**
- $\text{B}_2\text{H}_6, \text{H}_2\text{O}_2/\text{NaOH}$
- $\text{PdCl}_2, \text{H}_2\text{O}, \text{O}_2/\text{LAH}$
- $\text{Hg}(\text{OAc})_2, \text{H}_2\text{O}/\text{NaBH}_4$
- dil. H_2SO_4

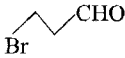
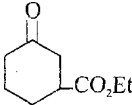
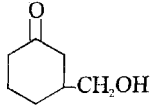
EXERCISE-II

- Q.1 What reagents could you use for the following conversions
- $\text{MeCO}(\text{CH}_2)_2\text{CO}_2\text{Et} \rightarrow \text{MeCHOH}(\text{CH}_2)_2\text{CO}_2\text{Et}$
 - $\text{HO}_2\text{C}(\text{CH}_2)_4\text{COCl} \rightarrow \text{HO}_2\text{C}-(\text{CH}_2)_4\text{CH}_2\text{OH}$
 - $\text{O}_2\text{N}(\text{CH}_2)_2\text{CN} \rightarrow \text{O}_2\text{N}(\text{CH}_2)_2\text{CH}_2\text{NH}_2$
 - $\text{O}_2\text{N}(\text{CH}_2)_2\text{CH}=\text{CH}_2 \rightarrow \text{H}_2\text{N}(\text{CH}_2)_2\text{CH}=\text{CH}_2$
 - $\text{Me}_2\text{CHCOCl} \rightarrow \text{Me}_2\text{CHCHO}$
 - $\text{O}_2\text{N}(\text{CH}_2)_3\text{CHO} \rightarrow \text{O}_2\text{N}(\text{CH}_2)_3\text{CH}_2\text{OH}$
 - $\text{O}_2\text{N}(\text{CH}_2)_2\text{CH}=\text{CH}_2 \rightarrow \text{O}_2\text{N}(\text{CH}_2)_3\text{CH}_3$

Q.2 Following compounds undergoes reaction with MnO_2 . What are the products in each case?

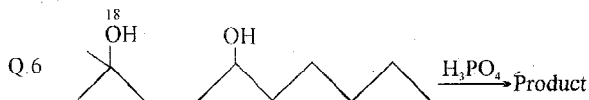
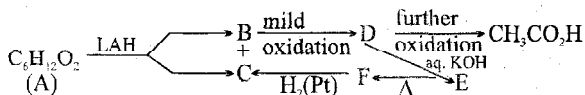
- $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{OH}$
- PhCH_2OH
- $\text{PhCH}(\text{OH})\text{CH}_2\text{CH}_2\text{OH}$
- $\text{MeCH}=\text{CH}(\text{CH}_2)_3\text{OH}$
- 

Q.3 Carry out the following conversion

-  $\rightarrow$ 
- Acraldehyde $\rightarrow$ Glyceraldehyde
- $\text{Br}-\text{CH}_2\text{CHO}$ $\rightarrow$ $\text{PhCHOHCH}_2\text{CHO}$
-  $\rightarrow$ 
- $\text{CH}_2(\text{Cl})-\text{CH}_2\text{CHO}$ $\rightarrow$ Glyceraldehyde

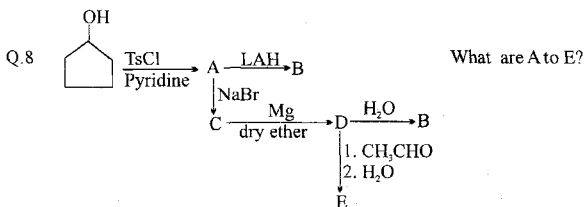
Q.4 Outline a mechanism to account for different isomer formed when $\text{Me}_2\text{C}(\text{O})\text{CH}_2$ reacts with CH_3OH in acidic and in basic medium.

Q.5 Find the structures of A to F.

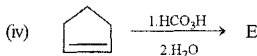
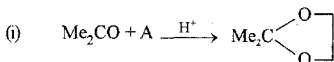


Explain product formation. How two OH groups behave differently.

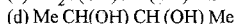
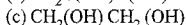
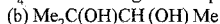
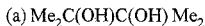
Q.7 $\text{R}_2\text{C}=\text{O} + \text{Et}_3\text{Al} \rightarrow \text{C}_2\text{H}_4 + \text{R}_2\text{CHO AlEt}_2$
Suggest a mechanism for the reaction.



Q. 9 Identify A to E in the following reactions.

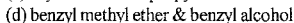
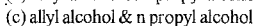
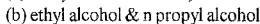
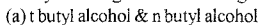


Q. 10 What are the order of rates of oxidation with HIO_4 of the following diols. Explain with reasons.

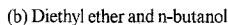
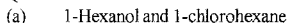


Q. 11 t-Butanol in presence of fenton's reagent forms 2,5 dimethyl hexane 2,5 diol as one of the product. Explain with the mechanism.

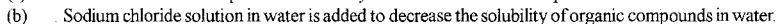
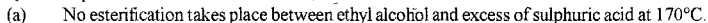
Q. 12 How you distinguish the following compounds?



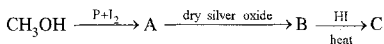
Q. 13 Differentiate:



Q. 14 Explain:



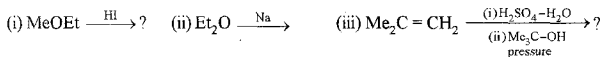
Q. 15 Predict A, B & C.



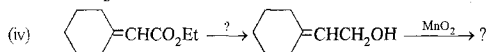
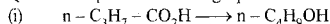
Q. 16 Carry out conversion of methyl alcohol into ethyl alcohol and vice versa.

Q.17 Arrange the compounds (i) Ethanol, propane and 1-pentanol, (ii) Butane 1,2,3-pentanetriol and butyl alcohol, (iii) Pentane, 1-pentanol and 1-hexanol, on the basis of (a) in order of increasing boiling points and (b) in order of increasing solubility in water.

Q.18 Complete the following equations & comment:



Q.19 Complete the following equations:



Q.20 t-butyl alcohol reacts less rapidly with metallic sodium than the primary alcohol. Explain why?

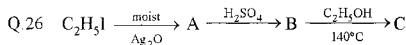
Q.21 Diethyl ether behaves as base. Why?

Q.22 What is the significance of proof spirit?

Q.23 Sodium metal can be used for drying diethyl ether but not for ethanol. Why?

Q.24 Sometimes explosion occurs during distillation of ether sample. Give the reason.

Q.25 Ethyl alcohol is denatured with methyl alcohol. Why?



Q.27 Ethyl alcohol reacts with HI but not with HCN. Explain why?

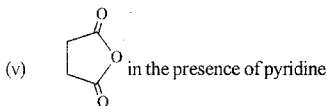
Q.28 Write the structure of the principal organic product formed in the reaction of 1-propanol with each of the following reagents:

(i) Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) in aqueous sulfuric acid, heat

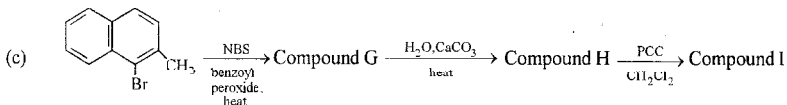
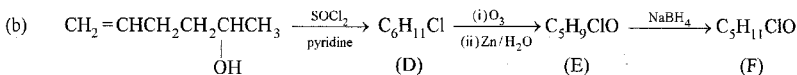
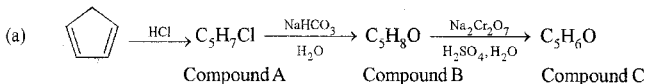
(ii) Acetic acid CH_3COOH in the presence of dissolved hydrogen chloride.

(iii) $\text{CH}_3\text{-C}_6\text{H}_4\text{-SO}_2\text{Cl}$ in the presence of pyridine

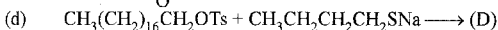
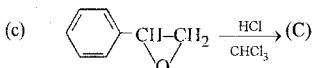
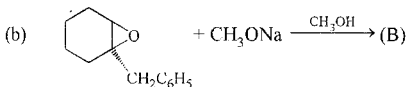
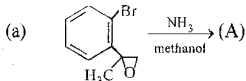
(iv) $\text{C}_6\text{H}_5\text{COCC}_6\text{H}_5$ in the presence of pyridine



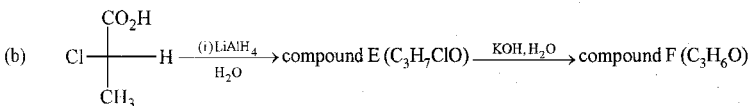
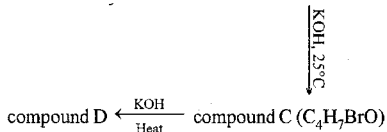
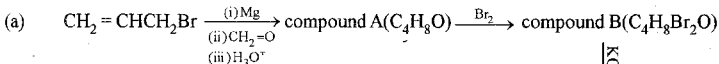
Q. 29 Complete the following series of equations by writing structural formula for compounds A through I:



Q. 30 Predict the principal organic product of each of the following reactions. Specify stereochemistry where appropriate.



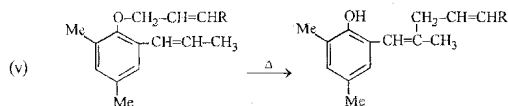
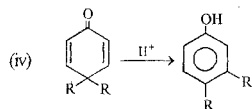
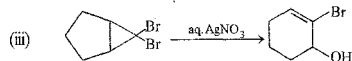
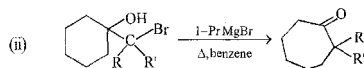
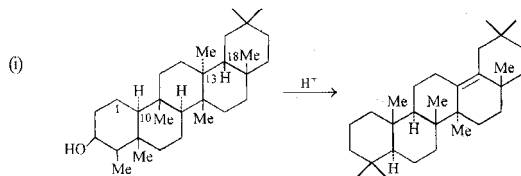
Q. 31 Deduce the identity of the missing compounds in the following reaction sequences. Show stereochemistry in parts (b) through (d).



- Q.32 Choose the reaction in each of the following pairs that proceeds at the faster rate. Explain your reasoning.
- Base-promoted hydrolysis of phenyl acetate or m-nitrophenyl acetate
 - Base-promoted hydrolysis of m-nitrophenyl acetate or p-nitrophenyl acetate
 - Reaction of ethyl bromide with phenol or with the sodium salt of phenol.
 - Reaction of ethylene oxide with the sodium salt of phenol or with the sodium salt of p-nitrophenol
 - Bromination of phenol or phenyl acetate.

Q.33 Explain why ArOR ethers are cleaved to give RI and ArOH rather than ArI and ROH.

Q.34 Explain the mechanism of following :



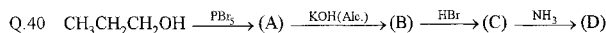
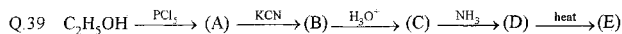
Q.35 Outline a synthesis of each alcohol from the indicated starting materials:

- Isopropyl alcohol from a hydrocarbon
- n-butyl alcohol from acetylene
- allyl alcohol from propane, and
- t-butyl alcohol from t-butyl chloride

Q.36 What product is expected from the reaction of  with (a) LiAlH_4 and (b) H_2/Pt

Q.37 Give the product of the reaction of $\text{Ph}_2\text{CHCH}_2\text{OH}$ with HBr and explain its formation.

Q.38 Give the product and write a mechanism for the acid dehydration of cyclobutylcarbinol.



EXERCISE-III

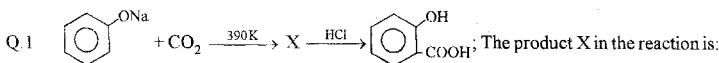
- Q. 1 Compound (A) gives Lucas test within 5 minutes. 6g (A) when reacts with Na, 1120 mL of H_2 is generated at STP. (A) having one oxygen per molecule. What is the structural formula of (A)? Compound (A) when treated with PBr_3 gives (B) which when treated with benzene in presence of anhydrous $AlCl_3$ gives (C). What are (B) and (C)?
- Q. 2 An open chain compound (A) C_3H_8O is optically active. When (A) is hydrogenated in presence of Pd as catalyst, it absorbs two moles of H_2 per mole of (A) to produce compound (B) $C_3H_{12}O$ which is optically inactive. However when (A) is warmed with dilute H_2SO_4 in presence of $HgSO_4$ it gives compound C ($C_3H_{10}O$) which is still optically active. C responds to iodoform test. What are the structures of A to C?
- Q. 3 An organic compound (A) $C_{10}H_{12}O$ gives red colour when treated with ceric ammonium nitrate & also decolourises Br_2 in CCl_4 to give B ($C_{10}H_{12}OBr_2$). A can show both geometrical & optical isomerism. A on treatment with I_2 & NaOH gives iodoform & an acid (C) after acidification. Give the structures of A to C & also the stereoisomers of A.
- Q. 4 $A(C_7H_{14})$ decolorises Br_2 in CCl_4 reacts with $Hg(OAc)_2$, THF- $NaBH_4$ to form B (resolvable compound)
 $A \xrightarrow[2. Zn/H_2O]{1. O_3} C \xleftarrow{KMnO_4} 3 \text{ hexanol}$
 $D(\text{isomer of } A) \xrightarrow[2. H_2O_2/OH^-]{1. B_2H_6, THF} E(\text{chiral}) \xrightarrow{KMnO_4} F(\text{chiral carboxylic acid})$
 $D(\text{isomer of } A) \xrightarrow[2. Zn/H_2O]{1. O_3} G \xleftarrow{KMnO_4/alkaline} 2 \text{ methyl 3 pentanol}$
 Find out A to G
- Q. 5 Suggest short, efficient reaction sequence suitable for preparing each of the following compounds from the given starting materials and any necessary organic or inorganic reagents.
- (i)  CH_2OCH_3 from  $COCH_3$ (ii)  from bromobenzene and cyclohexanol
- (iii) $C_6H_5CH_2CHCH_3$ from bromobenzene and isopropyl alcohol
 $\quad \quad \quad |$
 $\quad \quad \quad OH$
- (iv) $C_6H_5CH_2CH_2CH_2OCH_2CH_3$ from benzyl alcohol and ethanol.
- (v)  from 1,3-cyclohexadiene and ethanol
- (vi) $C_6H_5CHCH_2SCH_2CH_3$ from styrene and ethanol
 $\quad \quad \quad |$
 $\quad \quad \quad OH$
- Q. 6 A compound (X) reacts with thionyl chloride to give a compound (Y). (Y) reacts with Mg to form a Grignard reagent, which is treated with acetone and the product is hydrolysed to give 2-methyl-2-butanol. What are structural formulae of (X) and (Y)?

- Q.7 The compound (D) reacts with BH_3 –Tetrahydrofuran and then alkaline H_2O_2 to give chiral (E). Oxidation of (E) with KMnO_4 (acidified) gives a chiral carboxylic acid (F). Ozonolysis of 'D' gives after reduction with Zn the same compound (G), which is obtained by oxidation of 2-methyl-3-pentanol with acidified KMnO_4 . Identify (D), (E), (F) and (G).
- Q.8 An organic compound (A) contains 60.12% Carbon, 13.13% Hydrogen and has vapour density 30. On oxidation it gives (B) $\text{C}_3\text{H}_6\text{O}$ which on further oxidation by $\text{NaOH} + \text{I}_2$ produces a salt of the acid (C) $\text{C}_2\text{H}_4\text{O}_2$. When (B) is treated with hydroxyl amine, a compound of molecular formula $\text{C}_3\text{H}_7\text{NO}$ (D) is formed. What are (A), (B), (C) and (D)?
- Q.9 Compound (A) $\text{C}_6\text{H}_{10}\text{O}$ reacts rapidly with metallic sodium, but undergoes almost no reaction with Lucas reagent. When (A) is treated with hot concentrated sulphuric acid, a new compound (B) C_4H_8 is formed. If C_4H_8 is hydrated with sulphuric acid a new compound (C) $\text{C}_4\text{H}_9\text{OH}$ is formed, which is almost inert to metallic sodium but reacts rapidly with Lucas reagent. What are (A), (B) and (C)?
- Q.10 A compound (X) containing C, H and O is unreactive towards sodium. It does not add bromine. It does also not react with Schiff's reagent. On refluxing with an excess of HI , (X) gives only (Y). (Y) on hydrolysis gives (Z) which can be converted to (Y) by the action of $\text{P} + \text{I}_2$. Compound (Z) on oxidation gives an acid of equivalent weight 60. What are (X), (Y) and (Z)?
- Q.11 A compound (A) $\text{C}_4\text{H}_{10}\text{O}_2$ yields on acetylation (B) of formula $\text{C}_{12}\text{H}_{18}\text{O}_8$. How many hydroxyl groups are present in compound? Also write structures of (A) and (B).
- Q.12 An optically active alcohol (A) ($\text{C}_6\text{H}_{10}\text{O}$) absorbs two moles of hydrogen per mole of (A) upon catalytic hydrogenation and gives a product (B). The compound (B) is resistant to oxidation by CrO_3 and does not show any optical activity. Deduce the structures of (A) and (B).
- Q.13 An unsaturated hydrocarbon (A) on hydration gave alcohol (B). Oxidation of (B) gave a ketone (C). When the hydrocarbon (A) was submitted to ozonolysis it gave formaldehyde and acetaldehyde. What one formula would you assign to A, B and C? Give the reactions involved.
- Q.14 Compound (A) reacts with sodium to liberate H_2 forming a compound (B). When (A) is treated with $\text{HCl} + \text{ZnCl}_2$, an immediate reaction takes place liberating (C). When (A) is heated with H_2SO_4 , (D) is formed. Ozonolysis of (D) gives a three carbon ketone and another isomeric compound. What are (A), (B), (C) and (D)?
- Q.15 An organic compound (A) ($\text{C}_6\text{H}_{10}\text{O}$) reacts with HI giving a compound (B) ($\text{C}_6\text{H}_5\text{I}$) which on reduction gives a normal hydrocarbon having four carbon atoms. On oxidation A gives compound (C) ($\text{C}_4\text{H}_8\text{O}$) and then an acid (D) ($\text{C}_4\text{H}_8\text{O}_2$). Deduce the structures of, B, C and D giving reasons.
- Q.16 An organic compound (A) gives hydrogen on reacting with sodium. It also gives iodoform test and forms an aldehyde of molecular formula (B) ($\text{C}_2\text{H}_4\text{O}$) on oxidation. Name the compound (A) and (B).
- Q.17 An organic compound (X) on analysis gives 24.24% C, 4.04% H. Further sodium extract of 1.0 g of (X) gives 2.90 g of AgCl with acidified AgNO_3 solution. The compound (X) may be represented by two isomeric structures (Y) and (Z). (Y) on treatment with aqueous KOH solution gives a dihydroxy compound, while (Z) on similar treatment gives ethanal. Find out (X), (Y) and (Z).

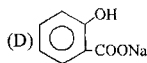
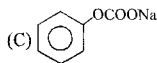
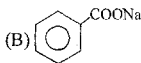
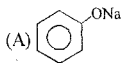
- Q. 18 Compound (X) (C_5H_8O) does not react appreciably with Lucas reagent at room temperature but gives a precipitate with ammonical silver nitrate. With excess of $MeMgBr$, 0.42 g of (X) gives 224 ml CH_4 at STP. Treatment of (X) with H_2 in presence of Pt catalyst followed by boiling with excess HI gives n-pentane. Suggest structure for (X) and write the equations involved.
- Q. 19 An alcohol (A) when heated with concentrated H_2SO_4 gives an alkene (B). When (B) is bubbled through bromine water and the product obtained is dehydrohalogenated with excess of sodamide, a new compound (C) is obtained. The compound (C) give (D) when treated with dilute H_2SO_4 in presence of $HgSO_4$. (D) can also be obtained either by oxidising (A) with $KMnO_4$ or from acetic acid through its calcium salt. Identify (A), (B), (C) and (D).
- Q. 20 Compound (A) contains only carbon and hydrogen. It decolorizes bromine in CCl_4 solution and reacts slowly with concentrated H_2SO_4 . Compound (A) reacts with HBr to form (B). (B) reacts with $NaOH$ to form (C). (C) on oxidation gave hexanone-3. Write structures of (A), (B) and (C).
- Q. 21 A hydrocarbon (A) (molecular formula C_5H_{10}) yield 2-methylbutane on catalytic hydrogenation. (A) adds HBr in accordance with Markownikoff's rule to form a compound (B), which on reaction with silver hydroxide forms an alcohol (C) $C_5H_{12}O$. Alcohol (C) on oxidation gives a ketone (D). Deduce (A) to (D) and give the reactions involved.
- Q. 22 (A) $\xrightarrow[250^\circ C]{Al_2O_3}$ (B) $\xrightarrow[(ii) AgOH]{(i) HI}$ (C) $\xrightarrow[150^\circ C]{Al_2O_3}$ (B) $\xrightarrow[(ii) H_2O_2, OH^-]{(i) B_2H_6}$ (A)

In the above reactions scheme (A) and (C) are isomers. (B) has a formula of C_5H_{10} . (C) Can also be obtained from the product of the reactions of CH_3CH_2MgBr and $(CH_3)_2CO$. Give structures of (A), (B) and (C).

EXERCISE-IV (A)



[REE 1990]



Q.2 The products of combustion of an aliphatic thiol (RSH) at 298 K are:

[JEE 1992]

(A) CO₂(l), H₂O (g) and SO₂(g)

(B) CO₂(g), H₂O (g) and SO₂(g)

(C) CO₂(l), H₂O (l) and SO₂(g)

(D) CO₂(g), H₂O (l) and SO₂(l)

Q.3 An organic compound C₃H₆O does not give a precipitate with 2,4-dinitrophenyl hydrazine reagent and does not react with sodium metal. It could be:

[JEE 1993]

(A) CH₃-CH₂-CHO (B) CH₃-CO-CH₃ (C) CH₂=CH-CH₂OH (D) CH₂=CH-OCH₃

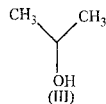
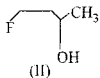
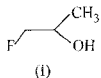
Q.4 The reaction products of C₆H₅OCH₃ + HI $\xrightarrow{\Delta}$ are:

[JEE 1995]

(A) C₆H₅OH + CH₃I (B) C₆H₅I + CH₃OH (C) C₆H₅CH₃ + HOI (D) C₆H₆ + CH₃OI

Q.5 The order of reactivity of the following alcohols:

[JEE 1997]



towards conc. HCl is:

(A) I > II > III > IV (B) I > III > II > IV (C) IV > III > II > I (D) IV > II > III > I

Q.6 Among the following compounds, the strongest acid is:

[JEE 1998]

(A) HC≡CH

(B) C₆H₆

(C) C₂H₆

(D) CH₃OH

Q.7 Benzenediazonium chloride on reaction with phenol in weakly basic medium gives:

[JEE 1998]

(A) Diphenyl ether

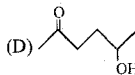
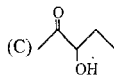
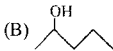
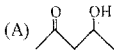
(B) p-hydroxyazobenzene

(C) Chlorobenzene

(D) Benzene

Q.8 Which one of the following will most readily be dehydrated in acidic condition:

[JEE 2000]



Q.9 1-propanol & 2-propanol can be best distinguished by:

[JEE 2001]

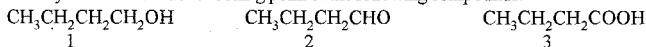
(A) Oxidation with alkaline KMnO₄ followed by reaction with Fehling solution

(B) Oxidation with acetic dichromate followed by reaction with Fehling solution

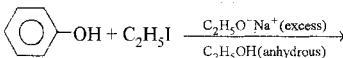
(C) Oxidation by heating with copper followed by reaction with Fehling solution

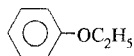
(D) Oxidation with concentrated H₂SO₄ followed by reaction with Fehling

Q. 10 Identify the correct order of boiling point of the following compounds: [JEE 2002]



- (A) $1 > 2 > 3$ (B) $3 > 1 > 2$ (C) $1 > 3 > 2$ (D) $3 > 2 > 1$

Q. 11  [JEE 2003]

- (A)  (B)  (C) $\text{C}_6\text{H}_5\text{OC}_6\text{H}_5$ (D) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$

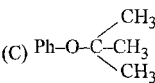
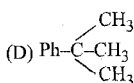
Q. 12 Reaction of enantiomerically pure acid with 1 chiral carbon and racemic alcohol with 1 chiral carbon gives an ester which is: [JEE 2003]

- (A) Meso (B) Optically active mixture
(C) Racemic mixture (D) Enantiomerically pure

Q. 13 On acid catalysed hydration, 2-phenyl propene gives: [JEE 2004]

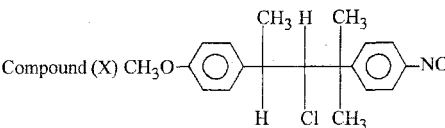
- (A) 3-phenyl-2-propanol (B) 2-phenyl-1-propanol
(C) 1-phenyl-3-propanol (D) 2-phenyl-2-propanol

Q. 14 Phenyl magnesium bromide reacting with t-Butyl alcohol gives [JEE 2005]

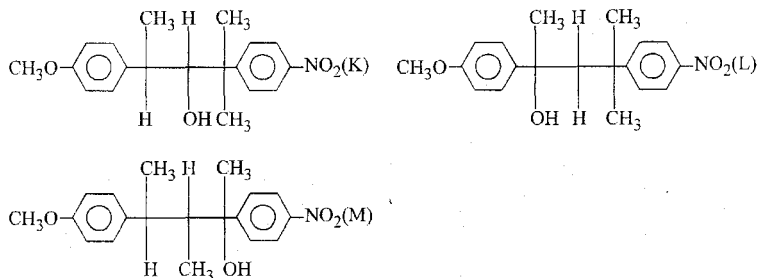
- (A) $\text{Ph} - \text{OH}$ (B) $\text{Ph} - \text{H}$ (C)  (D) 

Q. 15 Which is the best reagent to convert cyclohexanol into cyclohexene. [JEE 2005]

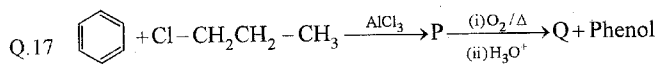
- (A) conc. HCl (B) conc. HBr (C) conc. H_3PO_4 (D) $\text{HCl} + \text{ZnCl}_2$

Q. 16 Compound (X)  is reacted with aqueous acetone it gives

following products:

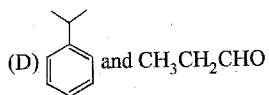
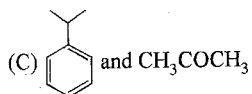
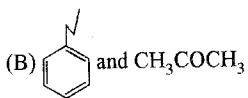
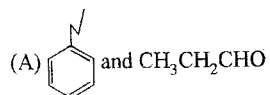


- (A) K, L (B) K, M (C) L only (D) M only [JEE 2005]



[JEE 2006]

The major products P and Q are



EXERCISE-IV (B)

Q.1 When t-butanol and n-butanol are separately treated with a few drops of dilute KMnO_4 in one case only, purple colour disappears and a brown precipitate is formed. Which of the two alcohols gives the above reaction and what is the brown precipitate? [JEE 1994]

Q.2 3,3-Dimethylbutan-2-ol loses a molecule of water in the presence of a concentrated sulphuric acid to give tetramethyl ethylene as a major product. Suggest a suitable mechanism. [JEE 1996]

Q.3 A compound D ($\text{C}_8\text{H}_{10}\text{O}$) upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid (E) ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of (D) and (E) and explain the formation of (E). [JEE 1996]

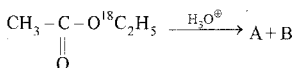
Q.4 Which of the following is the correct method for synthesising methyl-t-butyl ether and why?

(i) $(\text{CH}_3)_3\text{CBr} + \text{NaOMe} \longrightarrow$ (ii) $\text{CH}_3\text{Br} + \text{tert-BuONa} \longrightarrow$ [JEE 1997]

Q.5
$$\begin{array}{c} \text{O} = \text{C} - \text{OH} \\ | \\ \text{O} = \text{C} - \text{OH} \end{array} + \begin{array}{c} \text{HOCH}_2 \\ | \\ \text{HOCH}_2 \end{array} \xrightarrow{\text{Conc. H}_2\text{SO}_4} (\text{A})$$
 [JEE 1997]

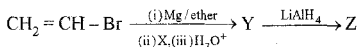
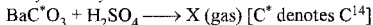
Q.6 Discuss why o-hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid? [JEE 1999]

Q.7 Write the structures of the product A & B [JEE 2000]



Q.8 Cyclobutyl bromide on treatment with magnesium in dry ether forms an organometallic A. The organometallic reacts with ethanal to give an alcohol B after mild acidification. Prolonged treatment of alcohol B with an equivalent amount of HBr gives 1-bromo-1-methylcyclopentane (C). Write the structures of A, B and explain how C is obtained from B. [JEE 2001]

Q.9 Identify X, Y and Z in the following synthetic scheme and write their structure. Explain the formation of labelled formaldehyde ($\text{H}_2\text{C}^*\text{O}$) as one of the products when compound Z is treated with HBr and subsequently ozonolysed. Mark the C^* carbon in the entire scheme. [JEE 2001]



Q.10 Mention two esters produced when a racemic mixture of 2-phenyl propanoic acid is treated with (+) 2-butanol. What is the stereochemical relationship between these esters? [JEE 2003]

Q.11 An organic compound P ($\text{C}_5\text{H}_{10}\text{O}$) Reacts 10^{15} times faster than ethylene with dil. H_2SO_4 to give two products Q and R. Both Q and R give positive iodoform test. Identify P, Q and R and also give reason for very high reactivity of P. [JEE 2004]

Q.12 (X) $\text{C}_5\text{H}_{13}\text{N} \xrightarrow[\text{-N}_2]{\text{NaNO}_2, \text{HCl}} (\text{Y})$ (Tertiary alcohol + other products)

(Optically active)

Find X and Y. Is Y optically active? Write the intermediate steps.

[JEE 2005]

ANSWER KEY

EXERCISE-I(A)

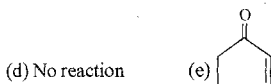
Q.1 B	Q.2 A	Q.3 C	Q.4 B	Q.5 B	Q.6 D	Q.7 D
Q.8 C	Q.9 D	Q.10 C	Q.11 A	Q.12 C	Q.13 A	Q.14 C
Q.15 B	Q.16 A	Q.17 B	Q.18 C	Q.19 D	Q.20 C	Q.21 D
Q.22 A	Q.23 B	Q.24 D	Q.25 A	Q.26 B	Q.27 A	Q.28 B
Q.29 A	Q.30 A	Q.31 C	Q.32 C	Q.33 D	Q.34 D	Q.35 D
Q.36 B	Q.37 A	Q.38 B	Q.39 A	Q.40 A	Q.41 C	Q.42 A
Q.43 A	Q.44 D	Q.45 A	Q.46 C	Q.47 C	Q.48 C	Q.49 C
Q.50 A						

EXERCISE-I(B)

Q.1 A,C,D	Q.2 C	Q.3 A,C	Q.4 B
Q.5 A,B,D	Q.6 A,B	Q.7 A,B,C	Q.8 A,B,C
Q.9 A,D	Q.10 A,B,D	Q.11 A,C	Q.12 A,B,C
Q.13 A,B,C,D	Q.14 A,B,C,D	Q.15 A,B,C	Q.16 A,C,D
Q.17 A,B,C	Q.18 A,B	Q.19	Q.20 A
Q.21 A,C	Q.22 C,D	Q.23 A,B,C,D	Q.24 A,B,C,D
Q.25 A,B	Q.26 (A) P, Q, S; (B) R, S; (C) P; (D) S		
Q.27 (A) Q, R, T, (B) R, (C) S, (D) P	Q.28 (A) Q, R, S (B) S (C) Q (D) P		
Q.29 (A) R, S; (B) R; (C) Q; (D) P	Q.30 A,B,D		

EXERCISE-II

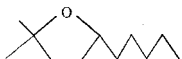
- Q.2 (a) $\text{Et}-\text{CH}=\text{CH}-\text{CHO}$ (b) $\text{Ph}-\text{CHO}$ (c) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{CH}$



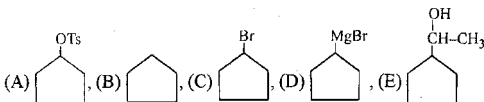
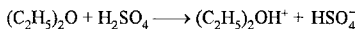
- Q.4 In acidic medium $\text{Me}_2-\overset{\text{OCH}_3}{\underset{\text{OH}}{\text{C}}}-\text{CH}_2$ and In basic medium $\text{Me}_2-\overset{\text{OCH}_3}{\underset{\text{O}^\ominus}{\text{C}}}-\text{CH}_2$

- Q.5 (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{C}-\text{C}-\text{C}-\text{C}$, (B) $\text{CH}_3-\text{CH}_2\text{OH}$, (C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$,
 (D) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$, (E) $\text{CH}_3-\overset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CHO}$ (F) $\text{CH}_3-\text{CH}=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$

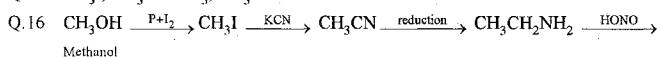
Q.6



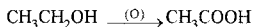
Q.8

Q.9 (A) $\text{CH}_2 = \text{CH} - \text{OH}$ Q.10 $a > b > d > c$ Q.12 (a) Lucas reagent (b) Iodoform (c) PBr_3 & AgNO_3 (d) NaQ.13 (a) 1-hexanol reduces orange red CrO_3 to green Cr^{3+} ; while 1-chlorohexane gives white ppt. of AgCl on warming with ethanolic AgNO_3 .(b) n-Butanol gives a positive test with CrO_3 in acid and evolves H_2 with sodium; while dry ethyl ether is negative to both tests(c) Unlike n-pentane, diethyl ether is basic and dissolves in conc. H_2SO_4 Q.14 (a) Ethyl alcohol first reacts with sulphuric acid to form ethyl hydrogen sulphate which readily loses a molecule of sulphuric acid when heated with concentrated sulphuric acid at 170°C to form ethylene.

(b) When sodium chloride is dissolved in water it reduces the polarity of water the space between water molecules are occupied by sodium and chloride ions with the result the solubility of the organic compound is reduced.

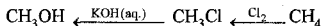
Q.15 CH_3I , $\text{CH}_3\text{O}-\text{CH}_3$, CH_3I 

Methanol



Ethylamine

Ethanol



Q.17 (a) In order of increasing boiling points:

(i) Propane < Ethanol < Pentanol-1

(ii) Butane < Butyl alcohol < 1, 2, 3-Pentanetriol

(iii) Pentane < 1-Pentanol < 1-Hexanol

The explanation is based on molecular weight and hydrogen bonding

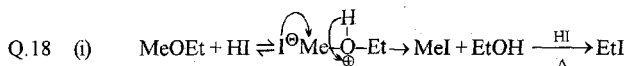
(b) In order of increasing solubility in water:

(i) Propane < Pentanol-1 < Ethanol

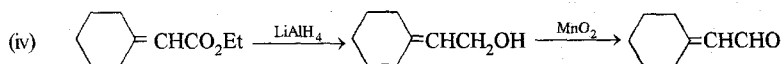
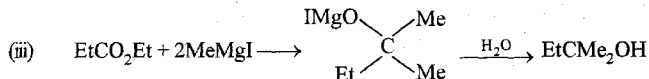
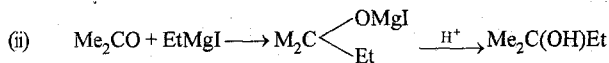
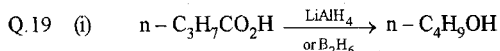
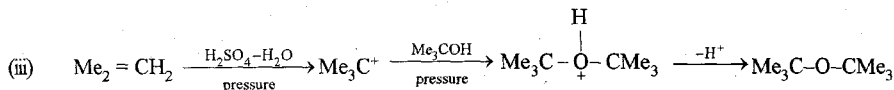
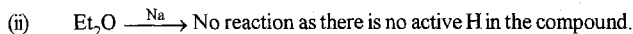
(ii) Butane < butyl alcohol < 1, 2, 3-Pentanetriol

(iii) Pentane < 1-Hexanol < 1-Pentanol

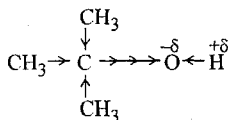
The explanation is based on hydrogen bonding



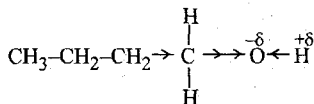
The reaction is $\text{S}_{\text{N}}2$ the smaller group forms the iodide (steric effect) & stops at the first stage in the cold. When heated, the reaction proceeds to the second stage.



Q. 20 The +I.E. of three methyl groups on central C-atom of tert-butyl alcohol makes it partially negative with the result that it pushes the electron pair of $-\text{OH}$ bond towards H-atom and thus H-atom is not replaced easily.

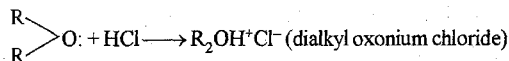


(less partial +ve charge)



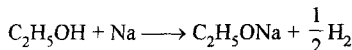
(more partial +ve charge)

Q. 21 Oxygen atom in diethyl ether molecule has two lone pair of electrons available for co-ordination. Therefore ethers behave as Lewis base forming oxonium salts.



Q. 22 Proof spirit is a technical term representing 57.1% by volume ethanol-water mixture for the purpose of tax levied on its production. The percentage represents the weakest possible percentage which allows the gun powder to catch fire.

Q. 23 Sodium metal reacts with ethanol to form sodium ethoxide ion because of active H atom in ethanol. Ether do not have such active H and thus can be dried by sodium



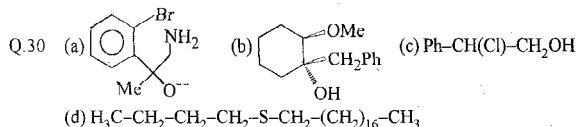
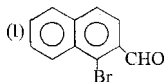
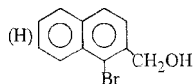
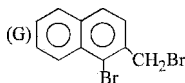
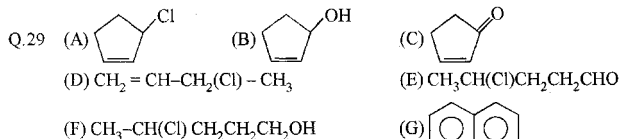
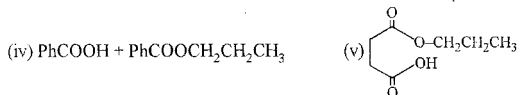
Q. 24 Due to the formation of explosive peroxide by oxidation.

Q. 25 Addition of methyl alcohol or naphtha or pyridine is made into ethyl alcohol to make sure that it will not be used for beverage purposes. This process is known as denaturation of alcohol.

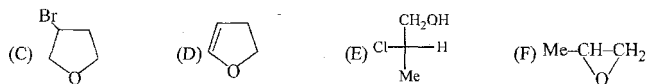
Q.26 C_2H_5OH , $C_2H_5HSO_4$, $C_2H_5-O-C_2H_5$

Q.27 Ethanol being very weak Lewis base and thus reacts with stronger acid HI

Q.28 (i) CH_3CH_2COOH (ii) $CH_3COOCH_2CH_2CH_3$ (iii) $MePhSO_2OCH_2CH_2CH_3$

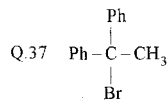
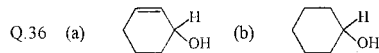


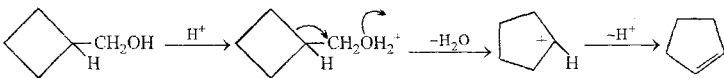
Q.31 (A) $CH_2=CH-CH_2-CH_2OH$ (B) $Br-CH_2CH(Br)CH_2CH_2OH$



Q.32 (a) m-nitrophenyl acetate (better leaving group) (b) p-nitrophenyl acetate (better leaving group)
 (c) sodium salt of phenol (better nucleophile) (d) sodium salt of phenol (better nucleophile)
 (e) phenol (more activated aromatic ring)

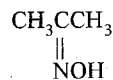
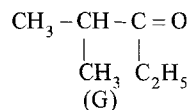
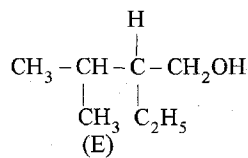
Q.33 S_N2 attack on a carbon of a benzene ring does not occur nor does the high energy $C_6H_5^+$ form by an S_N1 reaction. Hence ArI cannot be a product even in an excess of conc. HI



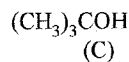
- Q.38 
- Q.39 (A) $\text{C}_2\text{H}_5\text{Cl}$; (B) $\text{C}_2\text{H}_5\text{CN}$; (C) $\text{C}_2\text{H}_5\text{COOH}$; (D) $\text{C}_2\text{H}_5\text{COONH}_4$
 (E) $\text{C}_2\text{H}_5\text{CONH}_2$
- Q.40 (A) $\text{C}_3\text{H}_7\text{Br}$; (B) $\text{CH}_3\text{CH}=\text{CH}_2$ (C) $\text{CH}_3\text{CHBrCH}_3$ (D) $\text{CH}_3\text{CHNH}_2\text{CH}_3$

EXERCISE-III

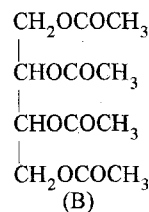
- Q.1 A: $\text{CH}_3 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$
- Q.2 (A) $\text{C} - \text{C} - \overset{\text{H}}{\underset{\text{OH}}{|}} - \text{C} \equiv \text{C}$ (B) $\text{C} - \text{C} - \overset{\text{H}}{\underset{\text{OH}}{|}} - \text{C} - \text{C}$ (C) $\text{C} - \text{C} - \overset{\text{H}}{\underset{\text{OH}}{|}} - \overset{\text{O}}{\parallel} \text{C} - \text{C}$
- Q.3 (A) $\text{Ph} - \text{CH} = \text{CH} - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$, (B) $\text{Ph} - \overset{\text{Br}}{\underset{|}{\text{CH}}} - \overset{\text{Br}}{\underset{|}{\text{CH}}} - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$, (C) $\text{Ph} - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{O} - \text{H}$
- Q.4 (A) $\text{C} - \text{C} - \overset{\text{CH}_2}{\parallel} \text{C} - \text{C} - \text{C} - \text{C}$, (B) $\text{CH}_3 - \text{CH}_2 - \overset{\text{CH}_3}{\underset{\text{OH}}{|}} \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$, (C) $\text{C} - \text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{C} - \text{C} - \text{C}$
- (D) $\text{Et} - \overset{\text{CH}_2}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CH} = \text{CH}_2$, (E) $\text{Et} - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CH}_2 - \text{OH}$, (F) $\text{Et} - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CO}_2\text{H}$
- (D) $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \overset{\text{CH}_2}{\parallel} \text{CH} - \text{CH}_2 - \text{CH}_3$, (G) $\text{C} - \overset{\text{C}}{\parallel} \text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{C} - \text{C}$, $\text{C} - \overset{\text{C}}{\parallel} \text{C} - \overset{\text{OH}}{\underset{|}{\text{C}}} - \text{C} - \text{C}$
- Q.6 $\text{C}_2\text{H}_5\text{OH}$ Ethyl alcohol (X) $\text{C}_2\text{H}_5\text{Cl}$ Ethyl chloride (Y)



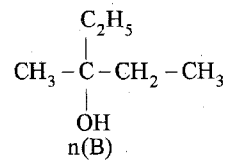
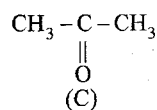
(D)


$$(\bar{Z})$$

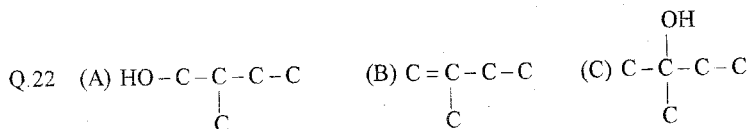
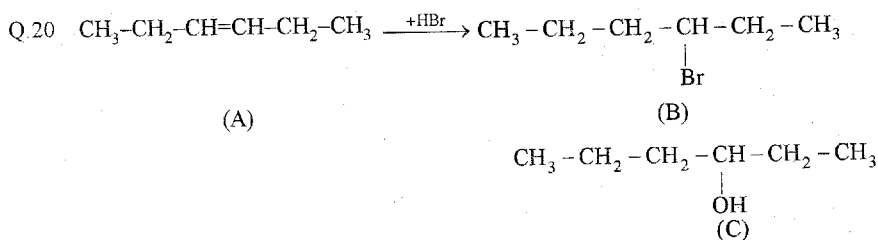
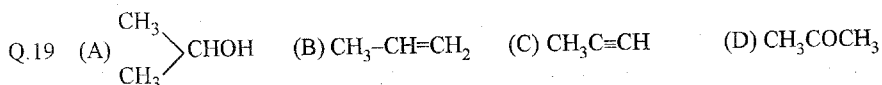
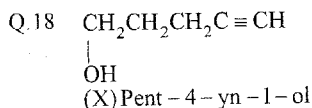
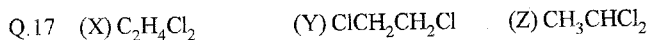
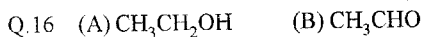
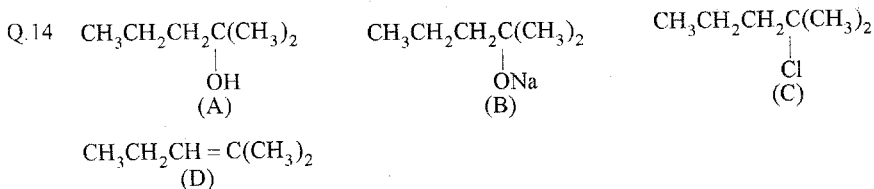
Q.11 Four OH groups are present in compound A.



(B)


$$n(B)$$


(C)

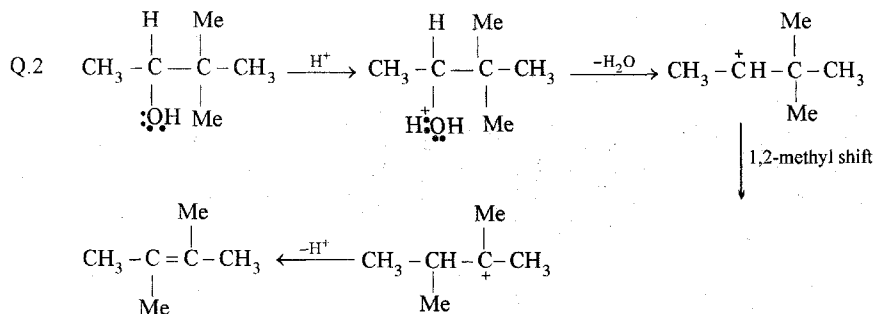
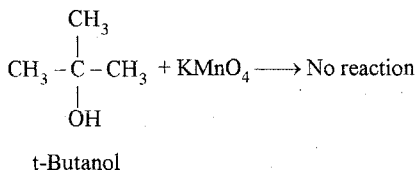
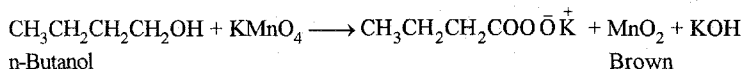


EXERCISE-IV (A)

Q.1	D	Q.2	B	Q.3	D	Q.4	A	Q.5	C
Q.6	D	Q.7	B	Q.8	A	Q.9	C	Q.10	B
Q.11	D	Q.12	B	Q.13	D	Q.14	B	Q.15	C
Q.16	A								

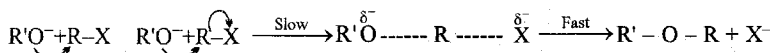
EXERCISE-IV (B)

- Q.1 n-butanol is oxidised by KMnO_4 and not t-butanol as the latter does not contain H atom attached to carbinol carbon atom.

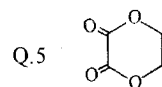


- Q.3 (D) $\text{Ph} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$ (E) $\text{C}_6\text{H}_5\text{COOH}$

- Q.4 The ether formation involves nucleophilic substitution of alkoxide ion for halide ion.

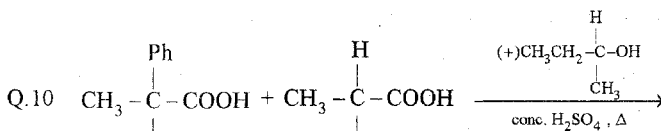
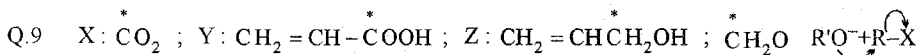
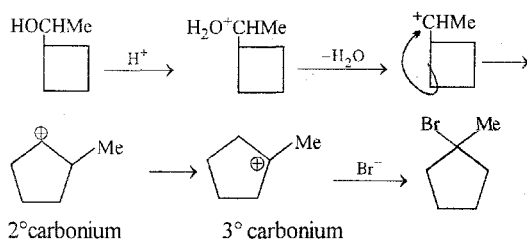
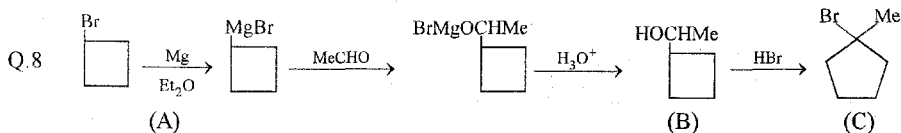


3° alkyl halide can also involve elimination of HX to give alkene in the presence of a base. So, it is better to start with 3° alkoxide and 1° alkyl halide, i.e., equation (b)

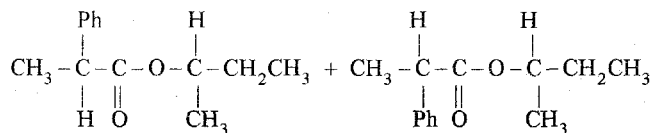


- Q.6 Due to intermolecular H-Bonding

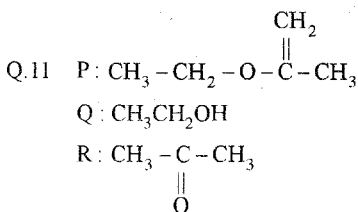
- Q.7 A = $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$, B = $\text{C}_2\text{H}_5\text{O}^{18}\text{H}$



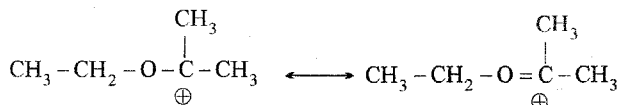
(racemic mixture)



during esterification reaction only $-COOH$ and $-OH$ participates. There is no effect on structure of configuration of carbon adjacent of these groups. So when $(\pm)$ acid reacts with pure $(+)$ alcohol two esters are produced which are **diastereoisomers** of each other.



when ethylene reacts with dil. H_2SO_4 , $CH_3CH_2^+$ is produced during rate determining step, whereas P gives resonance stabilized intermediate.



due to extra stability of intermediate the rate of reaction is very fast



BANSALCLASSES

TARGET IIT JEE 2007

ORGANIC CHEMISTRY

XIII (XYZ)

***AROMATIC
COMPOUNDS***

CONTENTS

EXERCISE - I

EXERCISE - II

EXERCISE - III

EXERCISE - IV(A)

EXERCISE - IV(B)

ANSWER KEY

EXERCISE - I

Q.1 How many π electron are there in the following species :



- (A) 2 (B) 4 (C) 6 (D) 8

Q.2 Number of π electrons present in naphthalene is:

- (A) 2 (B) 4 (C) 10 (D) 14

Q.3 Which of the following is not the property of benzene:

- (A) Characteristic smell (B) Inflammable
(C) Unsaturation (D) Colourless

Q.4 Benzene undergoes substitution reaction more easily than addition because:

- (A) It has a cyclic structure (B) It has three double bonds
(C) It has six hydrogen atoms (D) Of resonance

Q.5 The order of reactivity of following compounds:

- (I) $\phi - \text{ClI}_3$ (II) $\phi - \text{CH}_2 - \text{CH}_3$ (III) $\phi - \text{CH}(\text{CH}_3)_2$ (IV) $\phi - \text{C}(\text{CH}_3)_3$
towards electrophilic substitution will be-[where $\phi = \text{C}_6\text{H}_5$]
(A) I > II > III > IV (B) IV > III > II > I (C) II > I > III > IV (D) III > II > I > IV

Q.6 Which one of the following is the most basic compound :

- (A) $\text{C}_6\text{H}_5 - \text{NH}_2$ (B) $\text{C}_6\text{H}_5 - \text{NHCH}_3$ (C) $\text{C}_6\text{H}_5 - \text{N}(\text{CH}_3)_2$ (D) $\text{C}_6\text{H}_5\text{N}(\text{C}_2\text{H}_5)_2$

Q.7 Find odd compound out :

- (A) Lindane (B) Gammexane (C) 666 (D) HCB

Q.8 Which of the following reagents and conditions convert benzene to chloro-benzene:

- (A) Cl_2 , sunlight, heat (B) HCl , heat (C) HCl , sunlight, heat (D) Cl_2 , AlCl_3 , cold

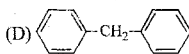
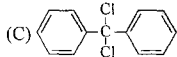
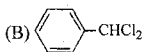
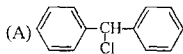
Q.9 Which order is correct for the decreasing reactivity to ring monobromination of the following compounds:

- (I) $\text{C}_6\text{H}_5\text{CH}_3$ (II) $\text{C}_6\text{H}_5\text{COOH}$ (III) C_6H_6 (IV) $\text{C}_6\text{H}_5\text{NO}_2$
(A) I > II > III > IV (B) I > III > II > IV (C) II > III > IV > I (D) III > I > II > IV

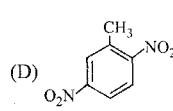
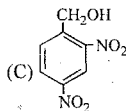
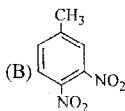
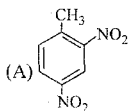
Q.10 Benzene on treatment with a mixture of conc. HNO_3 and conc. H_2SO_4 at 90°C gives:

- (A) Nitrobenzene (B) m-Dinitrobenzene (C) p-Dinitrobenzene (D) o-Dinitrobenzene

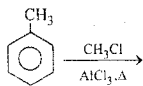
Q.11 Which of the following structures correspond to the product expected, when excess of C_6H_6 reacts with CH_2Cl_2 in presence of anhydrous AlCl_3 :



- Q.12 For preparing monoalkyl benzene, acylation process is preferred than direct alkylation because
 (A) In alkylation, a poisonous gas is evolved
 (B) In alkylation, large amount of heat is evolved
 (C) In alkylation, polyalkylated product is formed
 (D) Alkylation is very costly
- Q.13 Benzene reacts with n-propyl chloride in the presence of anhydrous AlCl_3 to give predominantly:
 (A) n-Propylbenzene (B) Isopropylbenzene
 (C) 3-Propyl-1-chlorobenzene (D) No reaction
- Q.14 In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be:
 (A) SO_3^+ , CH_3^+ , $\text{C} \equiv \text{O}^+$, HCO^+ (B) SO_3 , $\text{CH}_3 - \text{C} \equiv \text{O}^+$, HCO^+
 (C) SO_3 , CH_3CHO , $\text{CO} + \text{HCl}$ (D) HSO_3 , CH_3CO , HCO
- Q.15 Number of benzene derivatives of $\text{C}_7\text{H}_7\text{Cl}$ is:
 (A) 2 (B) 4 (C) 5 (D) 6
- Q.16 The number of benzene derivatives of the formula $\text{C}_7\text{H}_8\text{O}$ is:
 (A) 2 (B) 3 (C) 4 (D) 5
- Q.17 Ethylbenzene + $\text{Cl}_2 \xrightarrow{\text{Light}}$ (main) compound is:
 (A) o- & p-chloroethylbenzene (B) 1-Chloroethylbenzene
 (C) 2-Chloroethylbenzene (D) m-Chloroethylbenzene
- Q.18 Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives:
 (A) o-Cresol (B) p-Cresol
 (C) 2,4-Dihydroxytoluene (D) Benzyl alcohol
- Q.19 p-Nitrotoluene on further nitration gives:



- Q.20 In the reaction of p-chlorotoluene with KNH_2 in liquid NH_3 , the major product is:
 (A) o-Toluidine (B) m-Toluidine (C) p-Toluidine (D) p-Chloroaniline
- Q.21 Major product of this reaction will be :

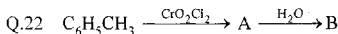


(A) o-xylene

(B) p-xylene

(C) both

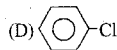
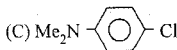
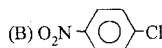
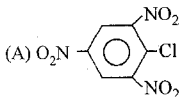
(D) m-xylene



The functional group present in B and name of the reaction would be

- (A) $-\text{CHO}$, Gattermann aldehyde synthesis (B) $-\text{CHO}$, Etard reaction
(C) $-\text{COCH}_3$, Friedel Crafts reaction (D) $-\text{CHO}$, Oxo reaction

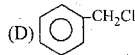
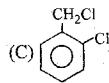
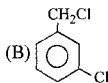
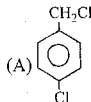
- Q.23 Which chloroderivative of benzene among the following would undergo-hydrolysis most readily with aq.NaOH to furnish the corresponding hydroxy derivative.



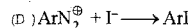
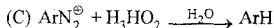
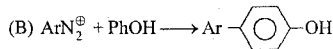
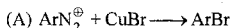
- Q.24 The order of relative reactivity of the given halides towards $\text{S}_{\text{N}}2$ reaction is:

- (A) $\text{PhCH}_2\text{Cl} > \text{PhCHClCH}_3 > \text{PhCCl}(\text{CH}_3)_2$
(B) $\text{PhCH}_2\text{Cl} < \text{PhCHCl}(\text{CH}_3) < \text{PhCCl}(\text{CH}_3)_2$
(C) $\text{PhCHCl}(\text{CH}_3) > \text{PhCH}_2\text{Cl} > \text{PhCCl}(\text{CH}_3)_2$
(D) $\text{PhCHCl}(\text{CH}_3) > \text{PhCCl}(\text{CH}_3)_2 > \text{PhCH}_2\text{Cl}$

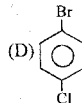
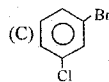
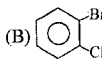
- Q.25 An aromatic compound 'A' $\text{C}_7\text{H}_6\text{Cl}_2$, gives AgCl on bonding with alcoholic AgNO_3 solution, and yields $\text{C}_7\text{H}_7\text{OCl}$ on treatment with sodium hydroxide. 'A' on oxidation gives a mono chlorobenzoic acid which affords only one mononitro derivative. The compound A is:



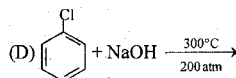
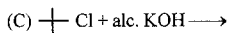
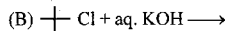
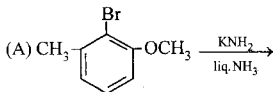
- Q.26 Which of the following reactions will not take place:



- Q.27 Which of the following substituted benzene derivatives would furnish three isomers in significant amount when one more substituent is introduced:



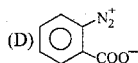
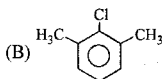
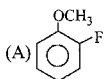
Q.28 Which of the following reaction does not take place?



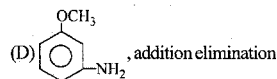
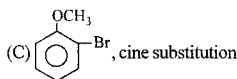
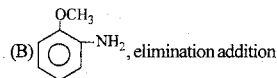
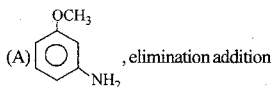
Q.29 m-Bromotoluene is prepared by:

- (A) Bromination of toluene
 (B) Friedel Craft's reaction of bromobenzene with CH3Cl
 (C) Bromination of nitrobenzene and subsequent replacement of -NO2 group with methyl group
 (D) Bromination of aceto-p-toluidine followed by hydrolysis and deamination

Q.30 Benzyne intermediate is not observed in:



Q.31 COc1cc(Br)ccc1 $\xrightarrow{\text{NaNH}_2}$ A, major product A and reaction R are:



Q.32 Chloral + Clc1ccccc1 $\xrightarrow{\text{Conc. H}_2\text{SO}_4}$ product. The product is:

- (A) Lindane (B) DDT (C) Teflon (D) Ethaneperchlorate

Q.33 Which of the following group is divalent:

- (A) Benzoyl (B) Benzyl (C) Benzal (D) p-Tolyl

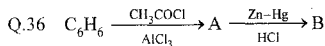
Q.34 C6H6 + A $\xrightarrow{\text{AlCl}_3}$ C6H5CONH2

A in the above reaction is:

- (A) NH2CONH2 (B) ClCONH2 (C) CH3CONH2 (D) CH2(Cl)CONH2

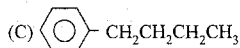
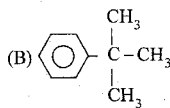
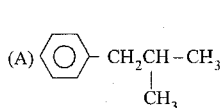
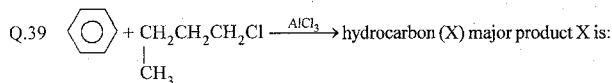
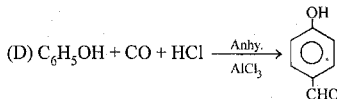
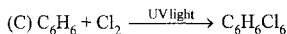
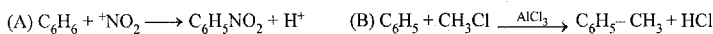
Q.35 When benzene is heated with acetic anhydride in the presence of anhydrous aluminium chloride at 80°C , the product formed is:

- (A) Benzoic acid (B) Benzophenone (C) Acetophenone (D) Ethyl phenyl ketone



The end product in the above sequence is:

- (A) Toluene (B) Ethyl benzene (C) Both the above (D) None
- Q.37 The species responsible for nitration and sulphonation by nitric acid conc. H_2SO_4 and fuming H_2SO_4 are:
- (A) NO_2 and SO_3 (B) NO_2^+ and SO_3 (C) NO^+ and SO_2 (D) NO_2 and SO_2
- Q.38 Which of the following reactions is not an example of electrophilic substitution:

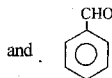


(D) None is correct

Q.40 Reaction of SO_3 is easier in:

- (A) Benzene (B) Toluene (C) Nitrobenzene (D) chlorobenzene

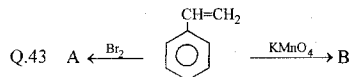
Q.41 If the mixture of the following four aromatic compounds on oxidation by strong oxidising agent gives:



- (A) Mixture of $\text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{C}_6\text{H}_5\text{COOH}$ (B) Mixture of $\text{C}_6\text{H}_5\text{CHO} + \text{C}_6\text{H}_5\text{COOH}$
(C) Only $\text{C}_6\text{H}_5\text{COOH}$ (D) None of the above

Q.42 Methyl group attached to benzene can be oxidised to carboxyl group by reacting with:

- (A) Fe_2O_3 (B) AgNO_3 (C) KMnO_4 (D) CrO_3



Compound A and B respectively are:

- (A) o-Bromostyrene, benzoic acid (B) p-Bromostyrene, benzaldehyde
(C) m-Bromostyrene, benzaldehyde (D) Styrene dibromide, benzoic acid
- Q.44 Which of the following is the strongest o,p-directing group?
- (A) OH (B) Cl (C) Br (D) C_6H_5

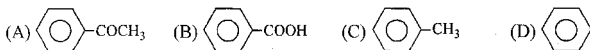
Q.45 The reaction of benzene with CO and HCl in the presence of anhydrous AlCl_3 gives:

- (A) Chlorobenzene (B) Toluene (C) Benzyl chloride (D) Benzaldehyde

Q.46 The number of benzylic hydrogen atoms in ethylbenzene is:
(A) 3 (B) 5 (C) 2 (D) 7

Q.47 $X \xrightarrow{\text{Cl}_2} \text{Benzotrichloride} \xrightarrow{\text{Hydrolysis}} Y$
X and Y respectively are:
(A) Benzene, Benzaldehyde (B) Toluene, Benzaldehyde
(C) Toluene, Benzoic acid (D) Benzene, Benzoic acid

Q.48 Product obtained when benzoyl acetic acid is heated with soda-lime is:



Q.49 Etard reaction in the following is:

- (A) Oxidation of toluene to benzaldehyde by chromylchloride
(B) Oxidation of toluene to benzaldehyde by alkaline KMnO_4
(C) Dry distillation of calcium benzoate
(D) Reaction of benzene with Cl_2 in the presence of UV light

Q.50 The highest yield of m-product is possible by the electrophilic substitution of the following:

- (A) $\text{C}_6\text{H}_5\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}_2\text{COOC}_2\text{H}_5$
(C) $\text{C}_6\text{H}_5\text{CH}(\text{COOC}_2\text{H}_5)_2$ (D) $\text{C}_6\text{H}_5\text{C}(\text{COOC}_2\text{H}_5)_3$

Q.51  + $\text{H}_2 \xrightarrow[\text{high pressure}]{\text{Ni, high temp.}}$ (A). Which of the following can be isolated as the product of this reaction.



Q.52 Which of the following is/are produced when a mixture of benzene vapour and oxygen is passed over V_2O_5 catalyst at 775 K?

- (A) Oxalic acid (B) Glyoxal (C) Fumaric acid (D) Maleic anhydride

Q.53 Which of the following is the least reactive in the case of bromination:

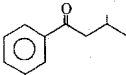
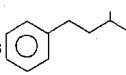
- (A) Phenol (B) Aniline (C) Nitrobenzene (D) Anisole

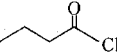
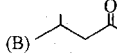
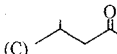
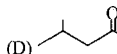
Q.54 No. of π electrons in  is:

- (A) 2 (B) 4 (C) 5 (D) 6

Q.55 Which of the following is not an aromatic compound:



Q.56 Benzene on reaction with 'A' forms  which on reaction with 'B' forms 
'A' and 'B' are:

- (A) $\text{Zn(Hg)} + \text{conc. HCl}$,  (B) , LiAlH_4
(C) , NaBH_4 (D) , $\text{Zn(Hg)} + \text{conc. HCl}$

Q.57 In a reaction of C_6H_5Y , the major product (>60%) is m-isomer, so the group Y is:

- (A) $-COOH$ (B) $-Cl$ (C) $-OH$ (D) $-NH_2$

Q.58 Which of the following will undergo sulphonation at fastest rate?



Q.59 Aniline under acidic medium, when chlorinated, produces:

- (A) o-chloro aniline (B) m-chloro aniline
(C) p-chloro aniline (D) Mixture of ortho and para-chloro aniline

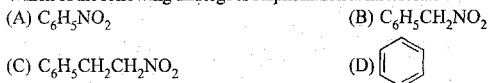
Q.60 Which of the following undergoes chlorination at fastest rate?



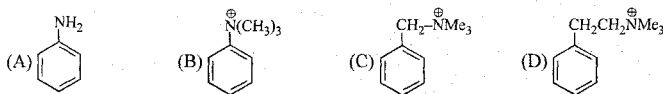
Q.61 Which of the following is most reactive towards sulphonation?

- (A) m-Xylene (B) o-Xylene (C) Toluene (D) p-Xylene

Q.62 Which of the following undergoes sulphonation at fastest rate?



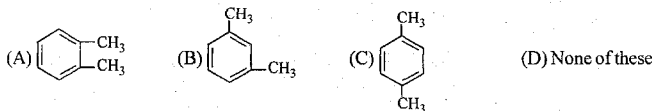
Q.63 Which of the following undergoes nitration fast?



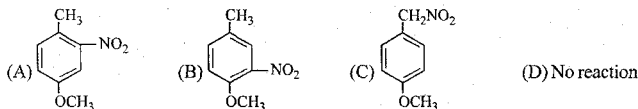
Q.64 When sulphonilic acid ($p-H_2NC_6H_4SO_3H$) is treated with excess of bromine, the product is:

- (A) tribromo product (B) dibromo product
(C) monobromo product (D) tetrabromo product

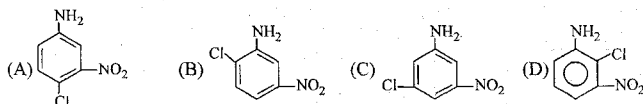
Q.65 Ring nitration of dimethyl benzene results in the formation of only one nitro dimethyl benzene. The dimethyl benzene is:



Q.66 If p-methoxy toluene is nitrated, the major product is:



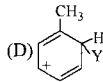
Q.67 If meta-nitroaniline is chlorinated, the major product is:



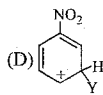
Q.68 An aromatic compound of molecular formula $C_6H_4Br_2$ was nitrated when three isomers of formula $C_6H_3Br_2NO_2$ were obtained. The original compound is:

- (A) o-dibromobenzene (B) m-dibromobenzene
(C) p-dibromobenzene (D) None of these

Q.69 Which of the following carbocations is expected to be most stable?



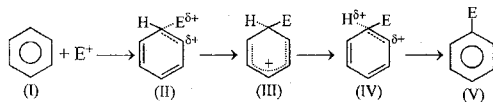
Q.70 Which of the following carbocations is expected to be most stable?



Q.71 The number of disubstituted products of benzene is

- (A) 2 (B) 3 (C) 4 (D) 5

Q.72 Which of the following species is expected to have maximum enthalpy in an electrophilic aromatic substitution reaction?



- (A) Species (II) (B) Species (III) (C) Species (IV) (D) Species (V)

Q.73 For the electrophilic substitution reaction involving nitration, which of the following sequence regarding the rate of reaction is true?

- (A) $k_{C_6H_6} > k_{C_6D_6} > k_{C_6T_6}$ (B) $k_{C_6H_6} < k_{C_6D_6} < k_{C_6T_6}$
(C) $k_{C_6H_6} = k_{C_6D_6} = k_{C_6T_6}$ (D) $k_{C_6H_6} > k_{C_6D_6} < k_{C_6T_6}$

Q.74 For the electrophilic substitution reaction involving sulphonation, which of the following sequence regarding the rate of reaction is true?

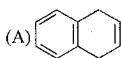
- (A) $k_{C_6H_6} > k_{C_6D_6} > k_{C_6T_6}$ (B) $k_{C_6H_6} < k_{C_6D_6} < k_{C_6T_6}$
(C) $k_{C_6H_6} = k_{C_6D_6} = k_{C_6T_6}$ (D) $k_{C_6H_6} > k_{C_6D_6} < k_{C_6T_6}$

Q.75 The order of relative reactivity of the given halides towards S_N1 reaction is

- (A) benzylchloride > p-methoxybenzylchloride > p-nitrobenzylchloride
(B) p-methoxybenzylchloride > benzylchloride > p-nitrobenzylchloride
(C) p-methoxybenzylchloride > p-nitrobenzylchloride > benzylchloride
(D) benzylchloride > p-nitrobenzylchloride > p-methoxybenzylchloride

EXERCISE - II

Q.1 Which of the following will show aromatic character:



Q.2 Which aromatic compound is obtained when n-octane undergoes catalytic hydroforming:

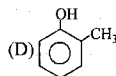
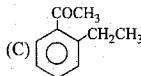
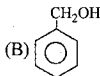
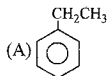
(A) ethyl benzene

(B) m-Xylene

(C) o-Xylene

(D) p-Xylene

Q.3 Benzoic acid may be prepared by the oxidation of:



Q.4 In which of the following reaction t-butylbenzene is formed:

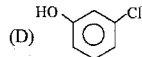
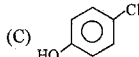
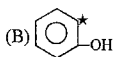
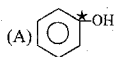
(A) Benzene + iso-butyl chloride, AlCl_3

(B) Benzene + $(\text{CH}_3)_2\text{C}=\text{CH}_2 \xrightarrow{\text{BF}_3, \text{HF}}$

(C) Benzene + t-butyl alcohol $\xrightarrow{\text{H}_2\text{SO}_4}$

(D) Benzene + $(\text{CH}_3)_2\text{C}=\text{CH}_2 \xrightarrow{\text{AlCl}_3}$

Q.5 Product is:



Q.6 Which of the following reactions of benzene proves the presence of three carbon-carbon double bonds in it:

(A) Formation of a triozonide

(B) Hydrogenation of benzene to cyclohexane

(C) Formation of $\text{C}_6\text{H}_6\text{Cl}_6$ by addition of chlorine

(D) Formation of nitrobenzene on heating benzene with a mixture of concentrated nitric acid and sulphuric acid

Q.7 Electrophile NO_2^+ attacks the following in which cases NO_2^+ will be at meta position:



Q.8 Benzene is a resonance hybrid mainly of two Kekule structures. Hence:

(A) Half of the molecules correspond to one structure, and half of the second structure

(B) At low temperatures benzene can be separated into two structures

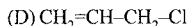
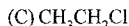
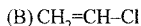
(C) Two structures make equal contribution to resonance hybrid

(D) An individual benzene molecule changes back and forth between two structures

- Q.9 The reaction of replacement of a hydrogen atom in benzene by alkyl group can be brought about with the following reagents:
 (A) Alkyl chloride and AlCl_3 (B) Alkene and AlCl_3
 (C) Alkanol and alkali (D) Alkanol and acid
- Q.10 Which of the following is/are fitting reaction:
 (A) $\text{C}_6\text{H}_5\text{I} + 2\text{Na} + \text{ICH}_3 \longrightarrow \text{C}_6\text{H}_5 - \text{CH}_3$
 (B) $\text{C}_6\text{H}_5\text{I} + 2\text{Na} + \text{IC}_6\text{H}_5 \longrightarrow \text{C}_6\text{H}_5 - \text{C}_6\text{H}_5$
 (C) $\text{CH}_3\text{I} + 2\text{Na} + \text{ICH}_3 \longrightarrow \text{CH}_3 - \text{CH}_3$
 (D) $\text{C}_6\text{H}_5\text{CH}_2\text{-Cl} + 2\text{Na} + \text{ClCH}_2\text{C}_6\text{H}_5 \longrightarrow \text{C}_6\text{H}_5\text{CH}_2 - \text{CH}_2\text{C}_6\text{H}_5$
- Q.11 Which of the following statements is correct:
 (A) Bromination of toluene occurs faster than that of benzene
 (B) Nitration of toluene is faster than that of ethylbenzene
 (C) The bromonium ion is a good nucleophile
 (D) Effective nitrating agent is nitrate ion
- Q.12 Electrophilic substitution in cinnamic acid will take place at:
 (A) Ortho-position (B) Para-position (C) Meta-position (D) On side chain
- Q.13 Which of the following gives Friedel Crafts reaction?

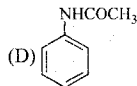
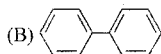


- Q.14 Which of the following can be used in Friedel Crafts reaction?

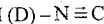


- Q.15 The good method for converting benzene into propyl benzene is:
 (A) $\text{C}_6\text{H}_6 + \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{Anhyd. AlCl}_3$
 (B) $\text{C}_6\text{H}_6 + \text{CH}_3\text{CH}_2\text{COCl} + \text{Anhyd. AlCl}_3$ and then treatment with Zn/Hg/HCl
 (C) $\text{C}_6\text{H}_6 + \text{CH}_3\text{CH}_2\text{COCl} + \text{Anhyd. AlCl}_3$ and then treatment with H_2Ni
 (D) $\text{C}_6\text{H}_6 + \text{Anhyd. AlCl}_3 + \text{cyclopropane}$

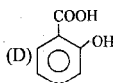
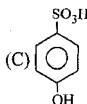
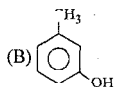
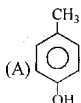
- Q.16 Which of the following will undergo nitration slow than benzene?



- Q.17 Which of the following is ortho-para directing group?

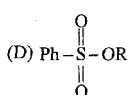


- Q.18 Which of the following statements is true?
 (A) Ortho-and para-directing groups increase electron density at ortho-and para-positions
 (B) Meta-directing groups increase electron density at meta-position
 (C) Meta-directing groups decrease electron density at meta-position
 (D) Ortho-and para-directing groups decrease electron density at meta-position
- Q.19 Which of the following statements is/are not true?
 (A) All ortho-para directing groups activates the ring
 (B) All ortho-para directing groups except halogens activate the ring
 (C) All meta-directing groups have π -bond on the atom directly attached to the ring
 (D) All meta directing groups are deactivating.
- Q.20 Which of the following compounds can be classified as aromatic?
 (A) 1, 2, 3-Triphenylcyclopropenium cation (B) Cyclooctatetraenyl dianion
 (C) Azulene (D) 10-Azulene
- Q.21 The structure of the compound that gives a tribromo derivative on treatment with bromine water is:



- Q.22 Amongst the following, the moderately activating group is
 (A) —NHR (B) —NHCOCH_3 (C) —NR_2 (D) —CH_3
- Q.23 False statement is / are :
 (A) Although benzene contains three double bonds, normally it does not undergo addition reaction.
 (B) m-chlorobromobenzene is an isomer of m-bromochlorobenzene.
 (C) In benzene, carbon uses all the three p orbitals for hybridization.
 (D) An electron donating substituent in benzene orients the incoming electrophilic group to the meta position.

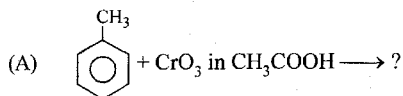
- Q.24 Of the species PhSH , PhSR , PhSO_2R and PhSO_2OR the meta-substituted product is obtained from



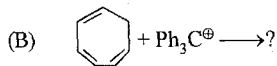
Q.25 Match the column:

Column A

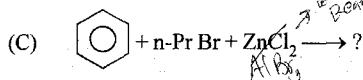
Column B



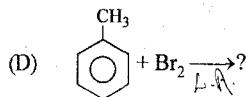
(P) Tropylium ion



(Q) Benzaldehyde



(R) Two types of products



(S) Cumene

Q.26 Match the following:

Column I

Column II

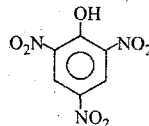
(A) $\text{CO}_2 \uparrow$ is evolved from
NaHCO₃ by the reaction of

(P)



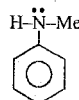
(B) Libermann nitroso test is given by

(Q)



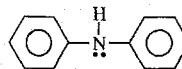
(C) Compounds gives yellow oily
liquid on reaction with $\text{NaNO}_2 + \text{HCl}$

(R)



(D) Compounds evolve a colourless gas
on reaction with active metals

(S)

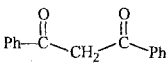
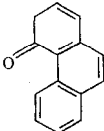


Q.27 Match the following:

Column I

- (A) Compounds which on reaction with neutral FeCl_3 gives violet / Blue colour are
- (B) Compounds which on reaction with $\text{NaO}_3\text{S}-\text{C}_6\text{H}_4-\text{N}_2\text{Cl}$ gives para red dye.
[CN coupling rxn with end of S]
- (C) Compounds which do not give yellow ppt. On reaction with NaOI
- (D) Compound which have maximum % of enol form

Column II

- (P) 
- (Q) 
- (R) $\text{Ph}-\text{CH}_2-\text{C}(=\text{O})-\text{CH}_2\text{CH}_3$
- (S) 

Q.28 Match the following:

Column I

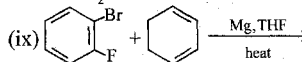
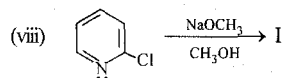
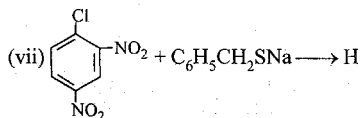
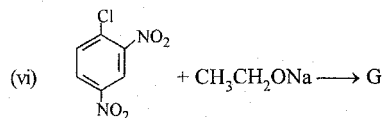
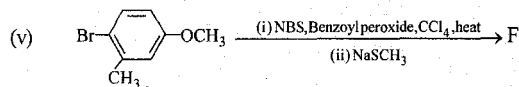
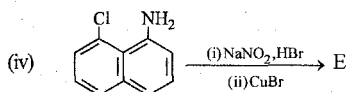
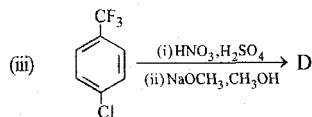
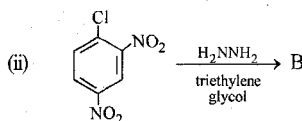
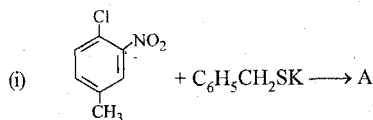
- (A) Compounds which gives Diels Alder's reaction in greater extent with dienophile
- (B) Compounds which undergoes ArSN reaction are
- (C) Compound which has highest aromatic characters
- (D) Strongest Base

Column II

- (P) 
- (Q) 
- (R) 
- (S) 

EXERCISE - III

Q.1 Write the principal organic product in each of the following reactions:



Q.2 Write the most stable resonance structure for the cyclohexadienyl anion formed by reaction of methoxide ion with o-fluoronitrobenzene.

Q.3 Reaction of 1,2,3-tribrom-5-nitrobenzene with sodium ethoxide in ethanol gave a single product, $\text{C}_6\text{H}_7\text{Br}_2\text{NO}_3$, in quantitative yield. Suggest a reasonable structure for this compound.

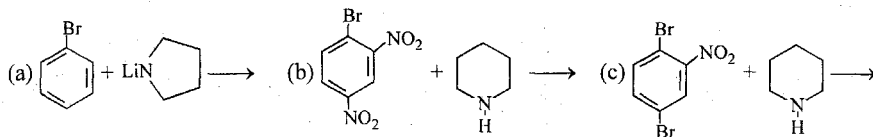
Q.4 Offer an explanation for the observation that 4-chloropyridine is more reaction toward nucleophiles than 3-chloropyridine.

Q.5 2-Bromo-1,3-dimethylbenzene is inert to nucleophilic aromatic substitution on treatment with sodium amide in liquid ammonia. It is recovered unchanged even after extended contact with the reagent. Suggest an explanation for this lack of reactivity.

Q.6 Two isomeric phenols are obtained in comparable amounts on hydrolysis of p-iodotoluene with 1 M sodium hydroxide at 300°C. Suggest reasonable structures for these two products.

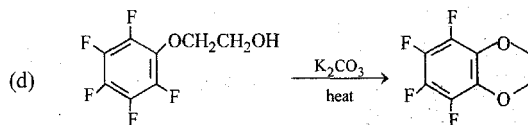
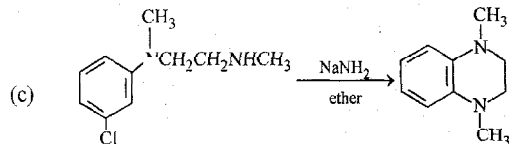
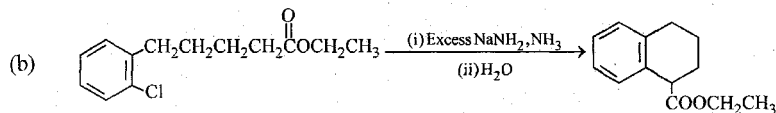
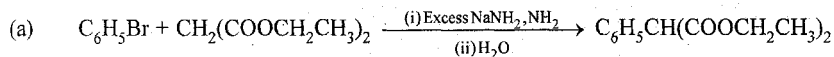
Q.7 Given the structure of the cycloaddition product formed when benzyne is generated in the presence of furan.

Q.8 In each of the following reactions, an amine or a lithium amide derivative reacts with an aryl halide. Give the structure of the expected product and specify the mechanism by which it is formed.



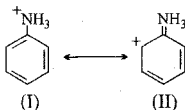
Q.9 1,2,3,4,5-Pentafluoro-6-nitrobenzene reacts readily with sodium methoxide in methanol at room temperature to yield two major products, each having the molecular formula $C_7H_3F_4NO_3$. Suggest reasonable structures for these two compounds.

Q.10 Suggest a reasonable mechanism for each of the following reactions:



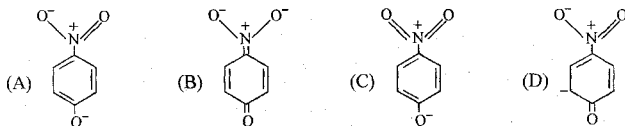
EXERCISE - IV(A)

- Q.1 The chlorination of toluene in presence of ferric chloride gives predominately: [JEE 1986]
(A) Benzyl chloride (B) m-chlorotoluene (C) Benzal chloride (D) o-and p-chlorotoluene
- Q.2 Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halide due to
(A) The formation of less stable carbonium ion (B) Resonance stabilization [JEE 1990]
(C) Longer carbon-halogen bond (D) The inductive effect
- Q.3 The most basic compound among the following is: [JEE 1990]
(A) Benzylamine (B) Aniline (C) Acetaniline (D) p-nitro aniline
- Q.4 Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives:
(A) o-cresol (B) p-cresol
(C) 2,4-dihydroxytoluene (D) Benzoic acid [JEE 1990]
- Q.5 When nitrobenzene is treated with Br_2 in presence of FeBr_3 the major product formed is m-bromonitrobenzene. Statements which are related obtain the m-isomer are: [JEE 1992]
(A) The electron density on meta carbon is more than on ortho and para position.
(B) The intermediate carbonium ion formed after initial attack of Br^+ attack the meta position is least destabilized.
(C) Loss of aromaticity when Br^+ attacks at the ortho and para positions and not at meta position
(D) Easier loss of H^+ to regain aromaticity from the meta position than from ortho and para position.
- Q.6 Choose the correct statement from the ones given below for two aniline: [JEE 1993]



- (A) II is not an acceptable canonical structure because carbonium ions are less stable than ammonium ions
(B) II is not an acceptable canonical structure because it is non aromatic
(C) II is not an acceptable canonical structure because the nitrogen has 10 valence electrons
(D) II is an acceptable canonical structure
- Q.7 Most stable carbonium ion is: [JEE 1995]
(A) $\text{p-NO}_2-\text{C}_6\text{H}_4-\text{CH}_2^+$ (B) $\text{C}_6\text{H}_5\text{CH}_2^+$
(C) $\text{p-Cl-C}_6\text{H}_4-\text{CH}_2^+$ (D) $\text{p-CH}_3\text{O-C}_6\text{H}_4-\text{CH}_2^+$
- Q.8 Arrange in order of decreasing trend towards S_E reactions: [JEE 1995]
(I) Chlorobenzene (II) Benzene (III) Anilinium chloride (IV) Toluene
(A) $\text{II} > \text{I} > \text{III} > \text{IV}$ (B) $\text{III} > \text{I} > \text{II} > \text{IV}$ (C) $\text{IV} > \text{II} > \text{I} > \text{III}$ (D) $\text{I} > \text{II} > \text{III} > \text{IV}$
- Q.9 Among the following statements on the nitration of aromatic compounds, the false one is: [JEE 1997]
(A) The rate of benzene is almost the same as that of hexadeuterobenzene
(B) The rate of nitration of toluene is greater than that of benzene.
(C) The rate of nitration of benzene is greater than that of hexadeuterobenzene
(D) Nitration is an electrophilic substitution reaction
- Q.10 Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 . In the nitrating mixture HNO_3 acts as a: [JEE 1997]
(A) Base (B) Acid (C) Reducing agent (D) Catalyst
- Q.11 Benzyl chloride ($\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$) can be prepared from toluene by chlorination with: [JEE 1998]
(A) $\text{SO}_2\text{Cl}_2\text{h}\nu$ (B) SOCl_2 (C) $\text{Cl}_2\text{h}\nu$ (D) NaOCl

Q.12 The most unlikely representation of resonance structure of p-nitrophenoxide ion is: [JEE 1998]



Q.13 A solution of (+)-2-chloro-2-phenylethane in toluene racemises slowly in the presence of small amount of SbCl_5 , due to the formation of: [JEE 1999]

- (A) Carbanion (B) Carbene (C) Free-radical (D) Carbocation

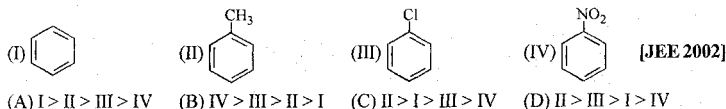
Q.14 Toluene, when treated with Br_2/Fe , gives p-bromotoluene as the major product, because the CH_3 group: [JEE 1999]

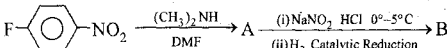
- (A) is para directing (B) is meta directing
(C) activates the ring by hyperconjugation (D) deactivates the ring

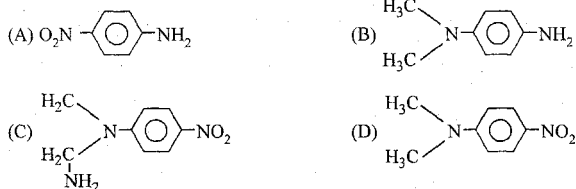
Q.15 Amongst the following the strongest base is: [JEE 2000]

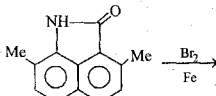
- (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $\text{p-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (C) $\text{m-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

Q.16 Identify the correct order of reactivity in electrophilic substitution reactions of the following compounds:

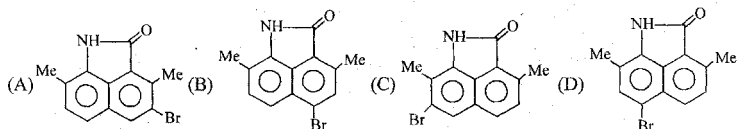


Q.17  [JEE 2003]

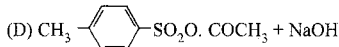
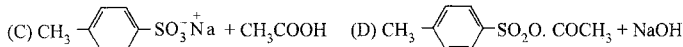
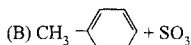
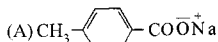


Q.18  [JEE 2004]

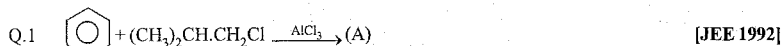
Major product of above reaction is:



- Q.19 Which of the following is obtained when 4-Methylbenzenesulphonic acid is hydrolysed with excess of sodium acetate? [JEE 2005]



EXERCISE - IV (B)

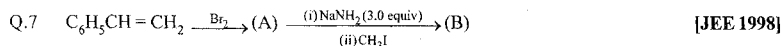
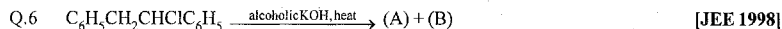


- Q.3 An organic compound (A), C_8H_6 on treatment with dilute sulphuric acid containing mercuric sulphate gives a compound (B), which can also be obtained from a reaction of benzene with an acid chloride in the presence of anhydrous aluminium chloride. The compound (B), when treated with iodine in aqueous KOH, yields (C) and a yellow compound (D). Identify (A), (B), (C) and (D) with justification. Show how (B) is formed from (A)? [JEE 1994]

- Q.4 Toluene reacts with bromine in the presence of light to give benzyl bromine while in presence of FeBr_3 it gives p-bromotoluene. Give explanation for the above observations. [JEE 1996]

- Q.5 Show the steps to carry out the following transformations: [JEE 1998]

- (a) Ethylbenzene $\longrightarrow$ benzene
 (b) Ethylbenzene $\longrightarrow$ 2-phenylpropionic acid



- Q.8 Normally, benzene gives electrophilic substitution reaction rather than thus electrophilic addition reaction although it has double bonds. [JEE 2000]

- Q.9 How would you synthesis 4 methoxyphenol from bromobenzene in NOT more than five steps? State clearly the reagents used in each step and show the structures of the intermediate compounds in your synthetic scheme. [JEE 2001]

- Q.10 Carry out following conversions in 3 or less steps.



[JEE 2003]

Q.11 A compound $C_9H_7O_2Cl$ exists in keto form A and enolic form B. Enolic form B predominates at equilibrium. On oxidation with $KMnO_4$ it gives m-chlorobenzoic acid gives structures of A and B.

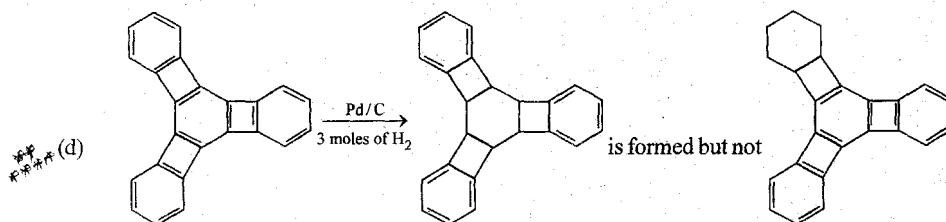
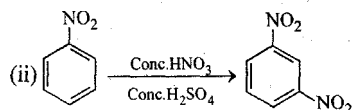
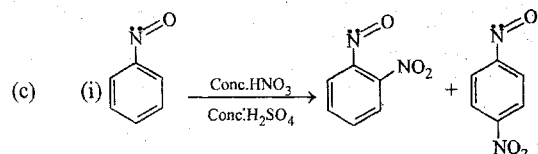
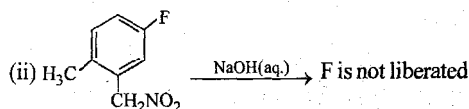
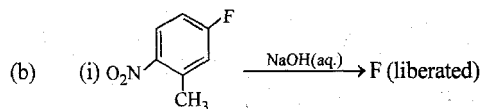
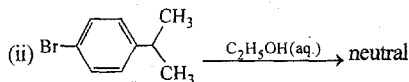
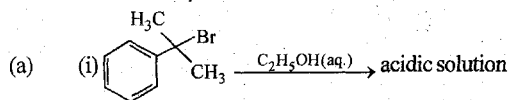
[JEE 2003]

Q.12 7-bromo-1, 3, 5-cycloheptatriene is ionic compound, whereas 5-bromo-1,3-cyclopentadiene can't ionise even in the presence of Ag^+ , Explain why?

[JEE 2004]

Q.13 Give reasons:

[JEE 2005]



Q.14 (Brown fumes and pungent smell) $B \xleftarrow{NaBr+MnO_2} A \xrightarrow{\text{Conc. } HNO_3} C(\text{intermediate}) \xrightarrow{\text{C}_6\text{H}_5\text{CH}_3} D(\text{Explosive product})$
Find A, B, C and D. Also write equations A to B and A to C.

[JEE 2005]

ANSWER KEY

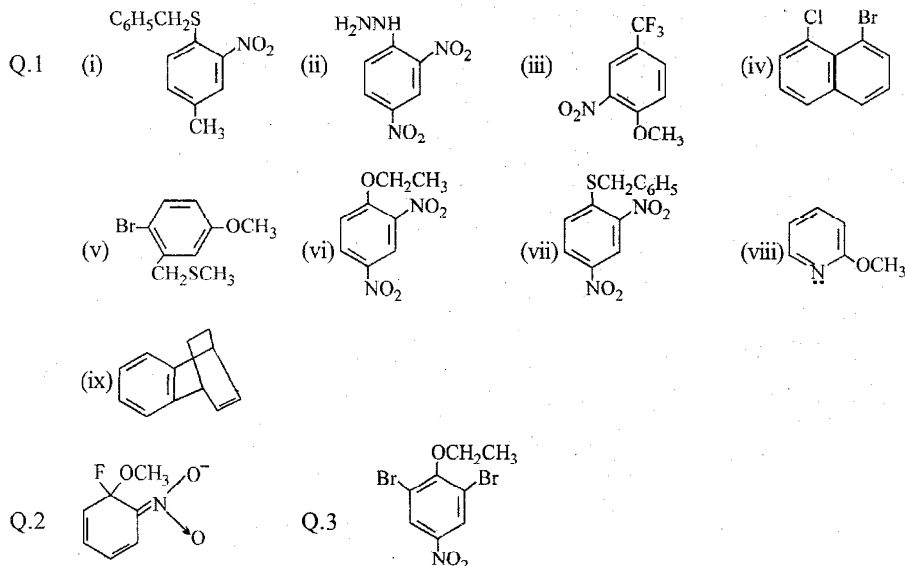
EXERCISE - I

Q.1	C	Q.2	C	Q.3	C	Q.4	D	Q.5	A	Q.6	D	Q.7	D
Q.8	D	Q.9	B	Q.10	B	Q.11	D	Q.12	C	Q.13	B	Q.14	B
Q.15	B	Q.16	D	Q.17	B	Q.18	D	Q.19	A	Q.20	B	Q.21	D
Q.22	B	Q.23	A	Q.24	A	Q.25	A	Q.26	B	Q.27	C	Q.28	A
Q.29	D	Q.30	B	Q.31	A	Q.32	B	Q.33	C	Q.34	B	Q.35	C
Q.36	B	Q.37	B	Q.38	C	Q.39	D	Q.40	B	Q.41	C	Q.42	C
Q.43	D	Q.44	A	Q.45	D	Q.46	C	Q.47	C	Q.48	A	Q.49	A
Q.50	D	Q.51	A	Q.52	D	Q.53	C	Q.54	D	Q.55	B	Q.56	D
Q.57	A	Q.58	B	Q.59	B	Q.60	D	Q.61	A	Q.62	D	Q.63	A
Q.64	A	Q.65	C	Q.66	B	Q.67	B	Q.68	B	Q.69	B	Q.70	D
Q.71	B	Q.72	A	Q.73	C	Q.74	A	Q.75	B				

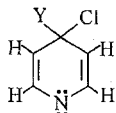
EXERCISE - II

Q.1	A,B,D	Q.2	A,C	Q.3	A,B	Q.4	A,B,C,D
Q.5	A,B	Q.6	A,B,C	Q.7	A,B,C	Q.8	C
Q.9	A,B,D	Q.10	B	Q.11	A,B	Q.12	A,B
Q.13	A	Q.14	C,D	Q.15	B,D	Q.16	C
Q.17	C	Q.18	A	Q.19	A,B,C	Q.20	A,B,C
Q.21	B,C,D	Q.22	B	Q.23	B,C,D	Q.24	C,D
Q.25	(A) Q, (B) P, (C) R, S, (D) R			Q.26	(A) Q, (B) P,R,S, (C) R,S, (D) P,Q,R,S		
Q.27	(A) P,Q,R,S, (B) Q, (C) P,Q,R,S, (D) P			Q.28	(A) R, (B) S, R, (C) P, (D) S		

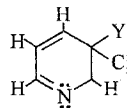
EXERCISE - III



Q.4 Nitrogen bears a portion of the negative charge in the anionic intermediate formed in the nucleophilic addition step in 4-chloropyridine, but not in 3-chloropyridine.

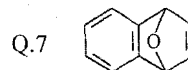


is more stable and formed faster than



Q.5 A benzene intermediate is impossible because neither on the carbons ortho to the intended leaving group bears a proton.

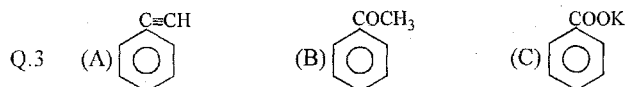
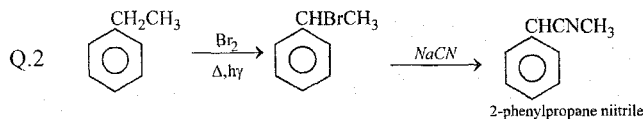
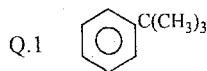
Q.6 3-Methylphenol and 4-methylphenol (m-cresol and p-cresol)



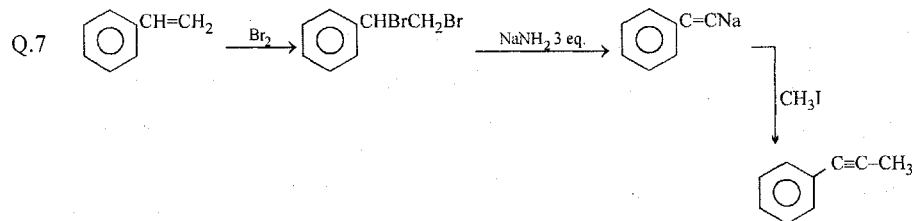
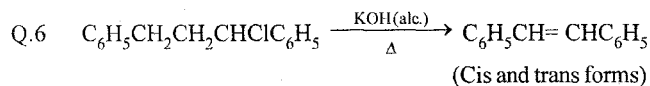
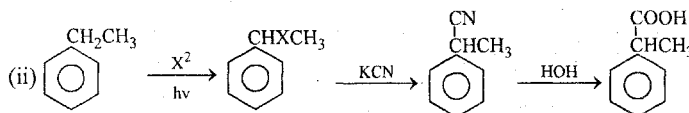
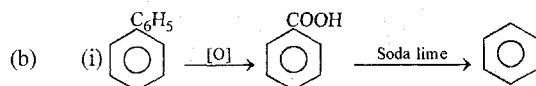
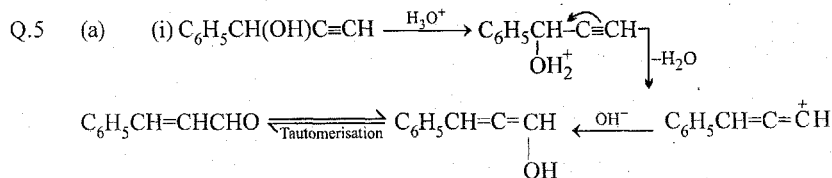
EXERCISE - IV(A)

- | | | | | | | |
|--------|--------|--------|----------|---------|--------|----------|
| Q.1 D | Q.2 B | Q.3 A | Q.4 D | Q.5 A,B | Q.6 C | Q.7 D |
| Q.8 C | Q.9 C | Q.10 A | Q.11 A,C | Q.12 C | Q.13 D | Q.14 A,C |
| Q.15 D | Q.16 C | Q.17 B | Q.18 D | Q.19 C | | |

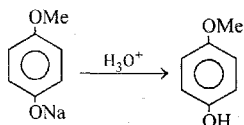
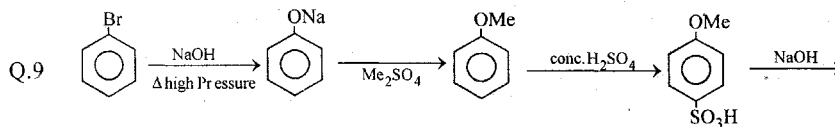
EXERCISE - IV (B)



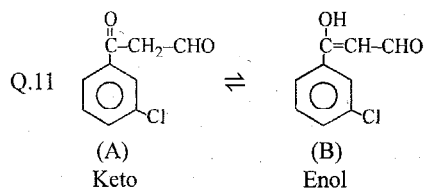
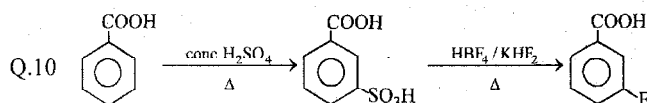
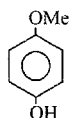
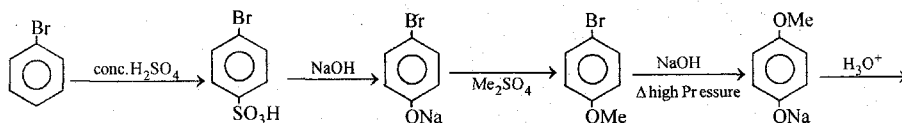
Q.4 In presence of FeBr_3 , Br_2 produces Br^+ (an electrophile) which attacks the benzene ring at o-, p-position to give p-bromotoluene. In presence of light, side chain is attacked to produce benzyl bromide.



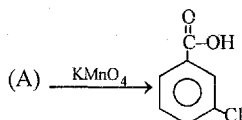
- Q.8 Benzene has resonance stabilization due to delocalisation of π -electrons. Also during electrophilic addition reactions, it loses its aromaticity. In electrophilic substitution reaction aromaticity is retained.



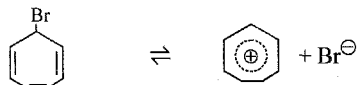
Alternative rout



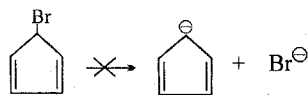
Enol form is more stable due to extended conjugation



- Q.12 On ionization 7-bromo-1,3,4-cycloheptatriene gives tropylium ion which is aromatic with 6π electrons.



5-bromo-1,3-cyclopentadiene can't ionise as it will in that case give highly unstable antiaromatic cation with 4π electrons.



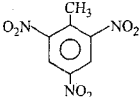
Q.13 (a) (i) $\text{H}_5\text{C}_6 - \overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}} - \text{OC}_2\text{H}_5 + \text{HBr}(\text{acid})$; (ii) no reaction due to partial double bond character

(b) (i)  + F^- is liberated; (ii) Bimolecular mechanism is not possible in (ii) case

(c) (i) due to presence of lone pair of nitrogen atom NO group is electron donating and ortho, para directing

(ii) NO_2 group is electron withdrawing and meta directing

(d) Due to reduction of central ring, three four membered antiaromatic rings become stable while on reduction of terminal ring only one antiaromatic ring can be stabilized.

Q.14. (A) H_2SO_4 , (B) Br_2 , (C) NO_2^+ , (D) 

ORGANIC CHEMISTRY

XII (ALL)

CARBONYL COMPOUND

CONTENTS

EXERCISE I(A)

EXERCISE I(B)

EXERCISE II

EXERCISE III

EXERCISE IV (A)

EXERCISE IV (B)

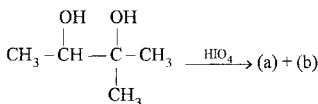
ANSWER KEY

EXERCISE-I(A)

Q.1 Gem dihalide on hydrolysis gives:

- (A) Vic diol (B) Gem diol
(C) Carbonyl compound (D) Carboxylic acid

Q.2 In the given reaction:



(a) and (b) respectively be:

- (A) CH_3CHO and CH_3CHO (B) CH_3COCH_3 and CH_3CHO
(C) CH_3COCH_3 and CH_3COCH_3 (D) CH_3COOH and CH_3COCH_3

Q.3 Acetophenone can be obtained by the distillation of:

- (A) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ (B) $(\text{CH}_3\text{COO})_2\text{Ca}$
(C) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{COO})_2\text{Ca}$ (D) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$

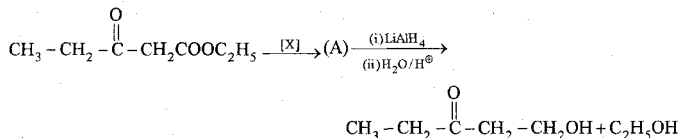
Q.4 Arrange these compounds in decreasing order of reactivity for the nucleophilic addition reaction:

- (I) Acid chloride (II) Aldehyde (III) Ketone (IV) Ester
Select the correct answer from the codes given below:
(A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{III} > \text{II} > \text{I} > \text{IV}$ (D) $\text{I} > \text{IV} > \text{II} > \text{III}$

Q.5 Acetal or ketal is:

- (A) Vic dialkoxo compound (B) α, ω -dialkoxo compound
(C) α -alkoxo alcohol (D) Gem dialkoxo compound

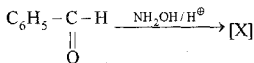
Q.6 In the given reaction



[X] will be:

- (A) HCHO (B) $\begin{array}{c} \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array} + \text{H}^+$
(C) $\begin{array}{c} \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array} + \text{OH}^-$ (D) HCN

Q.7 In the given reaction:



[X] will be:

- (A) Only syn oxime (B) Only anti oxime
(C) mixture of syn and anti oxime (D) secondary amide

Q.8 Schiff's base is prepared from:

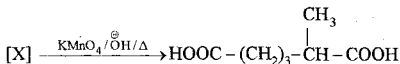
- (A) Carbonyl compound and primary amine (B) Carbonyl compound and secondary amine
(C) Carbonyl compound and tertiary amine (D) All of these

Q.9 Schiff's reagent is used for the differentiation between:

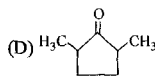
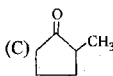
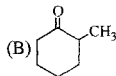
- (A) HCHO and CH_3CHO
(B) CH_3COCH_3 and CH_3CHO

- (C) $\text{C}_6\text{H}_5 - \text{CH}_2 - \overset{\text{O}}{\overset{\parallel}{\text{C}}} - \text{CH}_3$ and $\text{C}_6\text{H}_5 - \overset{\text{O}}{\overset{\parallel}{\text{C}}} - \text{CH}_2 - \text{CH}_3$
(D) HCHO and $\text{C}_6\text{H}_5\text{CHO}$

Q.10 In the reaction sequence, [X] is ketone:



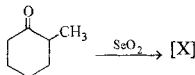
[X] will be:



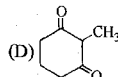
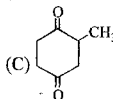
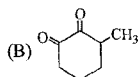
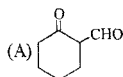
Q.11 Which one of the following compounds will give dimethyl glyoxal with SeO_2 :

- (A) Acetone (B) Acetophenone (C) Ethyl methyl ketone (D) Propanaldehyde

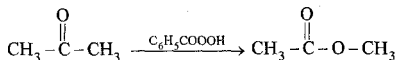
Q.12 In the given reaction



[X] will be:



Q.13 Consider the given reaction:

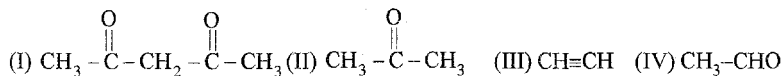


The above reaction is known as:

- (A) Baeyer-villiger oxidation (B) Oppenauer oxidation
(C) Periodate oxidation (D) Peroxide oxidation

- Q. 14 Acetone can be converted into pinacol by :
 (A) $\text{Mg/Hg/H}_2\text{O}$ (B) Zn/Hg/HCl (C) $\text{Na/Hg/H}_2\text{SO}_4$ (D) All of these

- Q. 15 Arrange acidity of given four compounds in decreasing order:

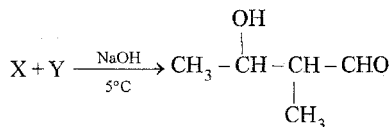


Select correct answer from the codes given below:

Codes:

- (A) $\text{I} > \text{IV} > \text{III} > \text{II}$ (B) $\text{I} > \text{IV} > \text{II} > \text{III}$ (C) $\text{III} > \text{I} > \text{IV} > \text{II}$ (D) $\text{II} > \text{IV} > \text{I} > \text{III}$

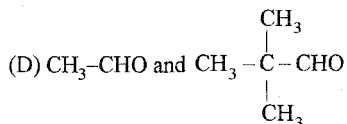
- Q. 16 In the given reaction



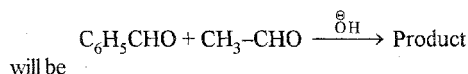
(X) and (Y) will respectively be:

- (A) $\text{CH}_3-\text{CH}_2-\text{CHO}$ and $\text{CH}_3-\text{CH}_2-\text{CHO}$ (B) CH_3-CHO and $\text{CH}_3-\text{CH}_2-\text{CHO}$

- (C) CH_3-CHO and CH_3-CHO

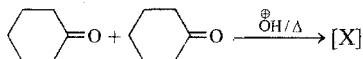


- Q. 17 Number of products in the given reaction :

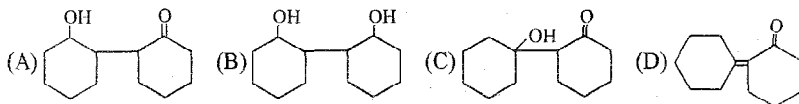


- (A) One (B) Three (C) Two (D) Four

- Q. 18 In the reaction:



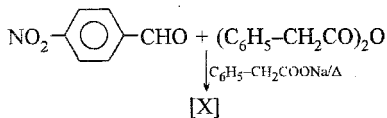
[X] will be :



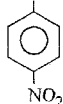
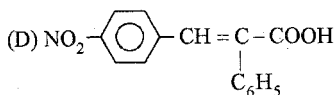
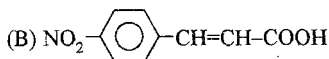
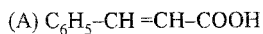
- Q. 19 Product of Perkin reaction is:

- (A) α, β -unsaturated aldehyde (B) β -cyclohexyl α, β -unsaturated aldehyde
 (C) β -Aryl- α, β -unsaturated acid (D) All of these

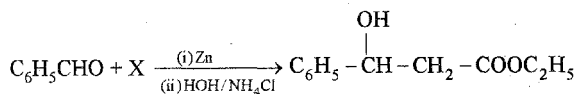
Q. 20 The product of the reaction:



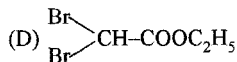
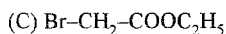
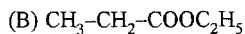
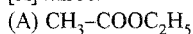
will be :



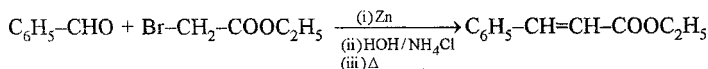
Q. 21 In the given reaction



[X] will be:



Q. 22 The given reaction



is known as :

(A) Perkin reaction

(B) Knoevenagel reaction

(C) Reformatsky reaction

(D) Claisen-Schmidt reaction

Q. 23 Cannizzaro reaction is example of:

(A) Redox reaction (B) Disproportionation (C) Both (A) and (B) (D) Only oxidation

Q. 24 Acetaldehyde can be converted into $\text{HO}-\text{CH}_2-\overset{\text{CH}_2\text{OH}}{\underset{\text{CH}_2\text{OH}}{\text{C}}}-\text{CH}_2\text{OH}$ by which reagent?

(A) KOH

(B) KOH followed by LAH

(C) excess of HCHO and KOH

(D) KCN followed by SBH

Q. 25 Metaformaldehyde is:

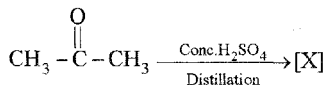
(A) Dimer of HCHO

(B) Trimer of formaldehyde

(C) Tetramer of formaldehyde

(D) Polymer in which number of HCHO unit is more than 100

Q.26 In the given reaction :



[X] will be :

- (A) Methyl oxide (B) Phorone
(C) 1, 3, 5-Trimethylbenzene (D) 2-Butyne

Q.27 Which will give silver mirror test with Tollens reagent :

- (A) $\text{C}_6\text{H}_5\text{CHO}$ (B) $\text{CH}_3\text{-CHO}$ (C) HCOOH (D) All of these

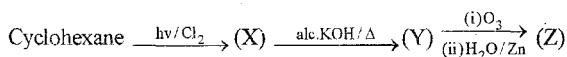
Q.28 Which one of the combinations will give propanaldehyde on dry distillation?

- (A) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$ (B) $(\text{CH}_3\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{CH}_2\text{-COO})_2\text{Ca}$
(C) $(\text{CH}_3\text{-CH}_2\text{-COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$ (D) $(\text{CH}_3\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{COO})_2\text{Ca}$

Q.29 Grignard reagents do not give carbonyl compounds with :

- (A) CO_2 (B) RCOCl (C) RCN (D) RCOOR

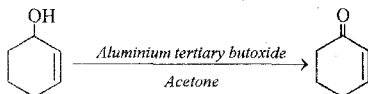
Q.30 In the reaction sequence:



(Z) will be :

- (A) Hexanal (B) 2-Hexanone (C) 3-Hexanone (D) Hexanedial

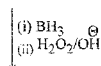
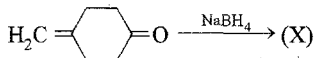
Q.31 The given reaction



is known as :

- (A) Kolbe reaction (B) Tischenko reaction (C) MPV reaction (D) Oppenauer oxidation

Q.32 In the given reaction:



(Y)

(X) and (Y) are :

- (A) $\text{CH}_2=\text{C}_6\text{H}_9\text{-OH}$ and $\text{HOCH}_2\text{-C}_6\text{H}_9=\text{O}$
(B) $\text{CH}_3\text{-C}_6\text{H}_9=\text{O}$ and $\text{HOCH}_2\text{-C}_6\text{H}_9=\text{O}$
(C) $\text{CH}_2=\text{C}_6\text{H}_9\text{-OH}$ and $\text{HOCH}_2\text{-C}_6\text{H}_9\text{-OH}$
(D) $\text{CH}_2\text{-C}_6\text{H}_9\text{-OH}$ and $\text{CH}_3\text{-C}_6\text{H}_9\text{-OH}$

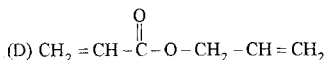
- Q.33 Cyanohydrin of which compound on hydrolysis will give lactic acid?
 (A) C_6H_5CHO (B) $HCHO$ (C) CH_3CHO (D) CH_3-CH_2-CHO

- Q.34 In the reaction sequence:

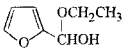


(Y) will be:

- (A) $CH_2=CH-CHO$
 (B) $CH_2=CH-CH_2OH$
 (C) Mixture of $CH_2=CH-COOH$ and $CH_2=CH-CH_2OH$



- Q.35 Acetaldehyde cannot give:
 (A) Iodoform test (B) Lucas test (C) Benedict test (D) Tollens test
- Q.36 The reaction in which $NaCN/C_2H_5OH/HOH$ is used is:
 (A) Perkin reaction (B) Benzoin condensation
 (C) Reimer-Tiemann reaction (D) Rosenmunds reduction

- Q.37 Compound  formed by the reaction of furfural () with ethanol is
 (A) an aldol (B) an acetal (C) a ketal (D) a hemiacetal

Question No. 38 to 40 (3 questions)

An alkene (A) $C_{16}H_{16}$ on ozonolysis gives only product (B) C_8H_8 . (B) also can be obtained by hydrolysis of the product obtained by reaction between cyano benzene and CH_3MgBr . (A) can show geometrical isomerism and it can decolourise Br_2 water. (B) on treatment with SeO_2 produces (C).

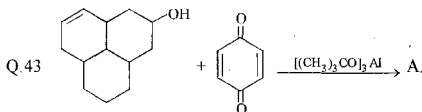
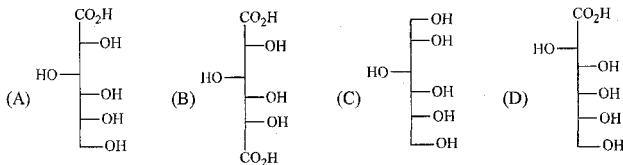
- Q.38 Which is not correct about (A)?
 (A) A is optically inactive
 (B) On catalytic hydrogenation 'trans' form of A produces racemic mixture
 (C) A can be prepared by Wittig reaction on acetophenone with $Ph_3P=C(CH_3)Ph$.
 (D) On treatment with per acid followed by hydrolysis 'trans' form of A produces racemic mixture
- Q.39 Which is not correct about B?
 (A) It gives iodoform test
 (B) On treatment with $LiAlH_4$, H_2O it produces a compound which also responds to iodoform test.
 (C) It gives Tollen's test
 (D) On treatment with NH_2NH_2 , EtO^- at high temperature it produces ethyl benzene
- Q.40 Which is not correct about C?
 (A) On treatment with $NaBH_4$ it will produce a diol.
 (B) On treatment with OH^- (conc.) followed by acidification it produces an optically active acid.
 (C) It gives Tollen's test
 (D) It can take part in aldol condensation

- Q. 41 Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:
 (A) if both (A) and (R) are true and (R) is the correct explanation of (A)
 (B) if both (A) and (R) are true but (R) is not correct explanation of (A)
 (C) if (A) is true but (R) is false
 (D) if (A) is false and (R) is true

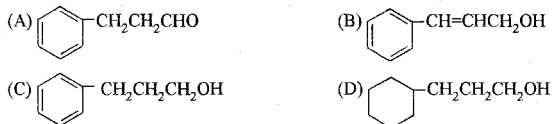
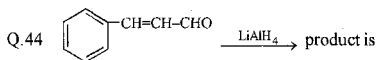
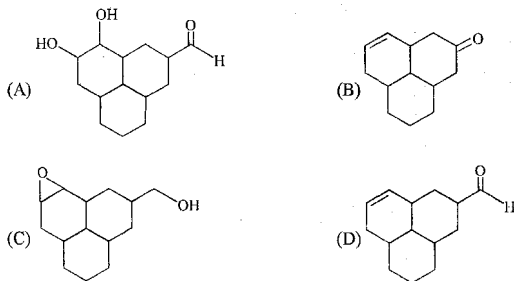
Assertion : Benzaldehyde with HCN gives two isomeric compounds

Reason : Both nitrile and isonitrile compounds are possible when HCN reacts with carbonyl group.

- Q. 42 When D-glucose is treated with Br_2 water then the product is:



A is:

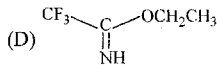
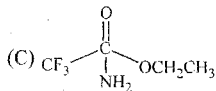
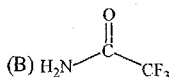
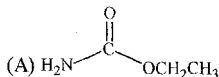


Question No.45 to 49 (5 questions)

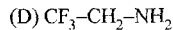
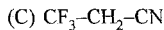
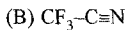
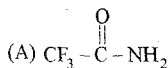
Compound 'C' $\left[\text{CF}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{C}(\text{CH}_3)_3 \right]$ was prepared in a three step sequence from ethyl trifluoroacetate.

The first step in a sequence involved treating ethyl trifluoroacetate with NH_3 to give a compound A. A on treatment with reagent 'X' (alongwith gentle heating) produces a compound B (which on hydrolysis can produce an acid). B on treatment with an orango metallic, 'Y', followed by hyydrolysis produces C. Based on above passing attempt the following questions:

Q.45 Structure of 'A' would be



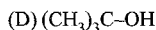
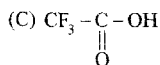
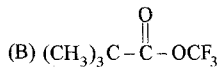
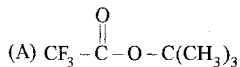
Q.46 Structure of B would be



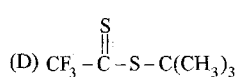
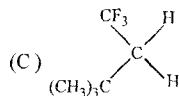
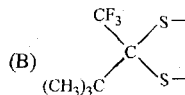
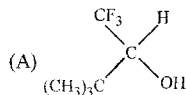
Q.47 'X' should be



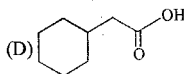
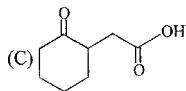
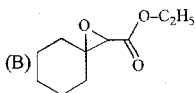
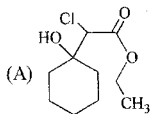
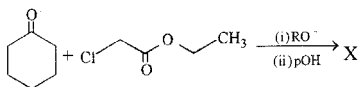
Q.48 When 'C' is treated with perbenzoic acid it will produce



Q.49 $\text{C} \xrightarrow{\begin{smallmatrix} \text{SH} \\ | \\ \text{SH} \end{smallmatrix}} \text{D} \xrightarrow[\text{Raney Ni}]{\text{H}_2} \text{E}$: 'E' is :

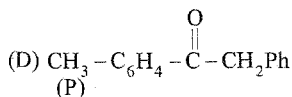
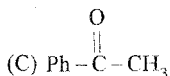
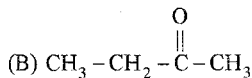
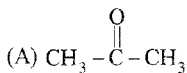


Q. 50 What is X in the following reaction?

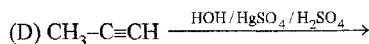
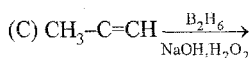
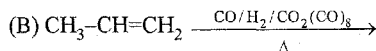
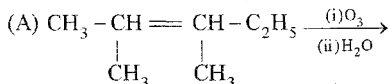


EXERCISE-I(B)

Q.1 Which one of the following is mixed ketone:



Q.2 In which of the following reactions product will be aldehyde?



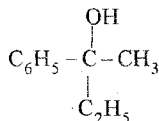
Q.3 Which one of the following alcohols can be oxidised by K_2CrO_4 ?

- (A) Ethanol (B) Tert butyl alcohol (C) Isopropyl alcohol (D) Allyl alcohol

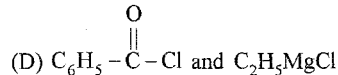
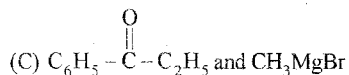
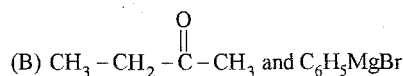
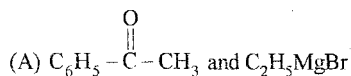
Q.4 Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by:

- (A) I_2 / NaOH (B) NaSO_3H (C) NaCN / HCl (D) 2,4-DNP

Q.5 Consider the structure of given alcohol:



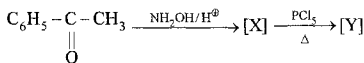
This alcohol can be prepared from:



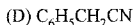
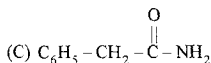
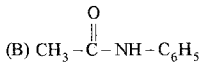
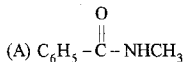
Q.6 Stability of hydrates of carbonyl compounds depends on:

- (A) Steric hindrance (B) Presence of -I group on gemdiol carbon
(C) Intramolecular hydrogen bonding (D) angle strain in carbonyl compound

Q.7 In the reaction sequence:



[Y] will be:



Q.8 Fehling solution gives red precipitate with:

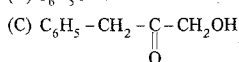
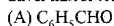
(A) Aromatic aldehyde

(B) Aliphatic aldehyde

(C) Ketones

(D) α -hydroxy ketones

Q.9 Silver mirror test with Tollens reagent is given by:



Q.10 Which one of the following compounds will not give aldol:

(A) Acetaldehyde

(B) Formaldehyde

(C) Pivaldehyde

(D) Crotonaldehyde

Q.11 Knoevenagel reaction is catalysed by:

(A) NaOH

(B) NH_3

(C) EtOk

(D) Pyridine

Q.12 Cross Cannizzaro reaction is an example of:

(A) Redox reaction

(B) Disproportionation

(C) Isomerization

(D) Dimerization

Q.13 Which one of the following reagent(s) is/are used for the conversion of ketone into hydrocarbons?

(A) LAH

(B) $\text{N}_2\text{H}_4/\text{H}_2\text{O}_2$

(C) $\text{Mg Hg}/\text{H}_2\text{O}$

(D) $\text{Na Hg}/\text{HCl}$

Q.14 Schiff's reagent gives pink colour with:

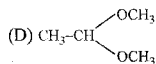
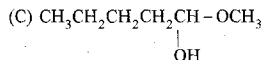
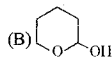
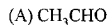
(A) Acetaldehyde

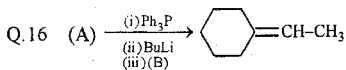
(B) Formic acid

(C) Acetic acid

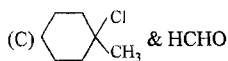
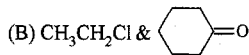
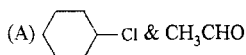
(D) Methyl acetate

Q.15 Which of the following compound will give positive Tollens test

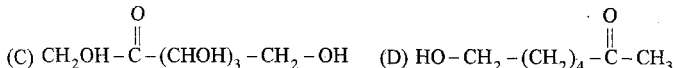
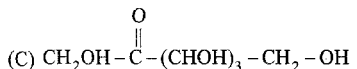
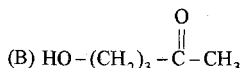
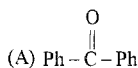




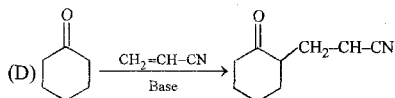
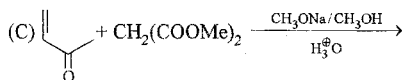
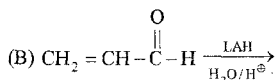
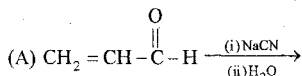
In above reaction (A) and (B) will respectively be



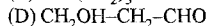
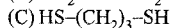
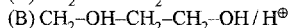
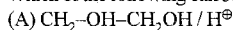
Q. 17 Which of the following form stable hemiketal



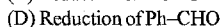
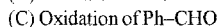
Q. 18 Which of the following is example of 1,4-addition reaction



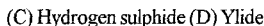
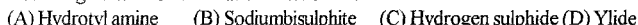
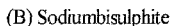
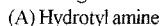
Q. 19 Which of the following can be used for protection of carbonyl group



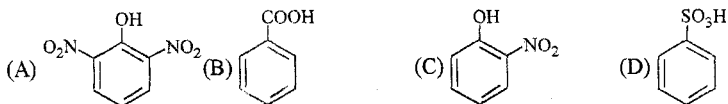
Q. 20 Mixture of $\text{Ph}-\text{CHO}$ & HCHO is treated with NaOH then Cannizzaro reaction involves:



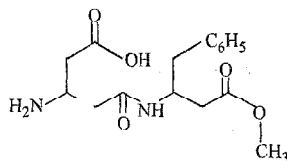
Q. 21 Acetone give addition elimination reaction with



Q. 22 Which of the following will be soluble in sodium bicarbonate solution?



Q. 23 Aspartame, an artificial sweetener is a peptide and has the following structure. Which of the following is correct about the molecule?



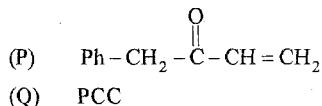
- (A) It has four functional groups
 (B) It has three functional groups
 (C) on hydrolysis it produces only one amino acid
 (D) on hydrolysis it produces a mixture of amino acids

Q. 24 Match List-I with List-II and select the correct answer using the codes given below the lists.

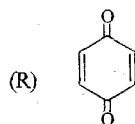
List-I (Compound)

List-II (Used as)

- (A) 1,4 addition
 (B) Tautomerism



- (C) Allylic bromination



- (D) Preparation of Ketone from 2° alcohol

(S) NBS

Q. 25 Match list-I with list-II and select the correct answer using the codes give below the lists:

List-I

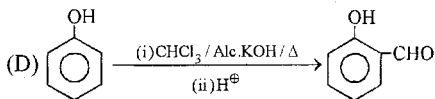
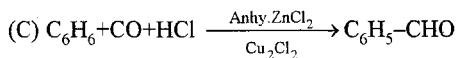
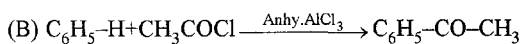
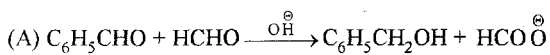
List-II

- (A) $\text{CH}_2=\text{CH}-\text{CHO} \xrightarrow{\text{NaBH}_4}$
 (B) $\text{C}_6\text{H}_5\text{CHO} + \text{Ph}-\text{NH}_2 \xrightarrow{\text{H}^+}$
 (C) $\text{C}_6\text{H}_5\text{COCH}_3 + \text{CH}_3-\text{CH}_2-\text{NH}_2 \xrightarrow{\text{H}^+}$
 (D) $\text{RCHO} + 2\text{RCH}_2\text{OH} \xrightarrow{\text{H}^+}$

- (P) Acetal
 (Q) Schiff's base
 (R) Unsaturated alcohol
 (S) Enamine

Q.26 Match list-I with list-II and select the correct answer using the codes give below the lists:

List-I



List-II

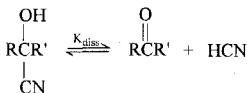
(P) Cannizzaro reaction

(Q) Friedel Crafts reaction

(R) Reimer-Tiemann reaction

(S) Gattermann-koch aldehyde synthesis

Q.3(b) Equilibrium constants for the dissociation (K_{diss}) of cyanohydrins according to the equation

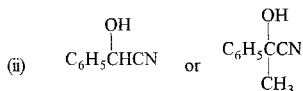
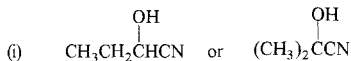


Cyanohydrin

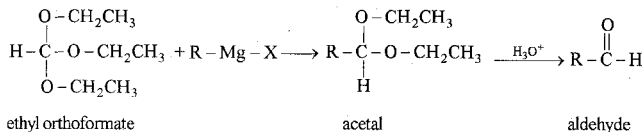
Aldehyde
or ketone

Hydrogen cyanide

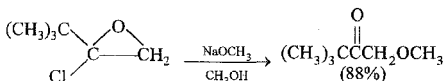
have been measured for a number of cyanohydrins. Which cyanohydrin in each of the following pairs has the greater dissociation constant?



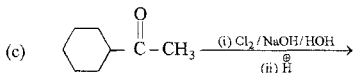
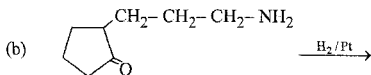
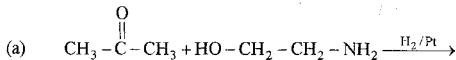
Q 4 Some Grignard reagents react with ethyl orthoformate, followed by acidic hydrolysis, to give aldehydes. Propose mechanisms for the two steps in this synthesis.



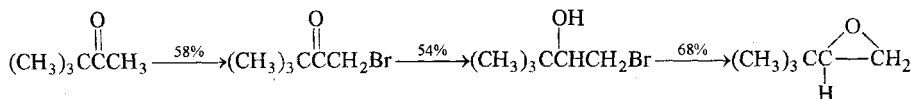
Q.5 Suggest a reasonable mechanism for each of the following reactions :



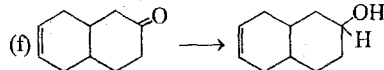
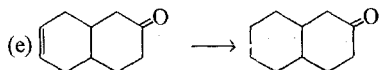
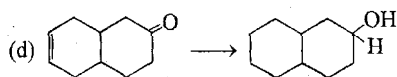
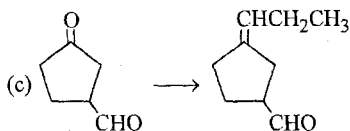
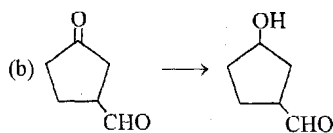
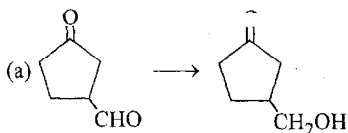
Q.6 Predict the organic products :



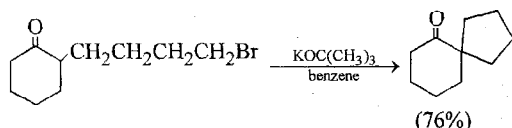
- Q.7 A synthesis that begins with 3,3-dimethyl-2-butanone gives the epoxide shown. Suggest reagents appropriate for each step in the synthesis.



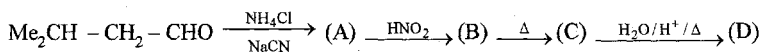
- Q.8 Show how you would accomplish the following syntheses efficiently and in good yield. You may use any necessary reagents.



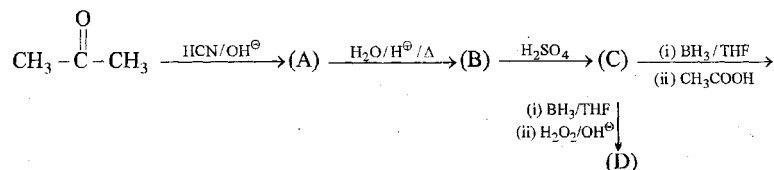
- Q.9 Outline reasonable mechanisms for each of the following reactions:



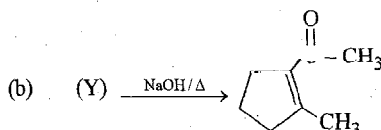
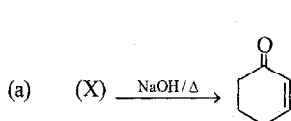
- Q.10 Identify A to D:

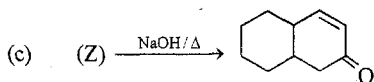


- Q.11 Give the structures of A to E in the given set of reaction:

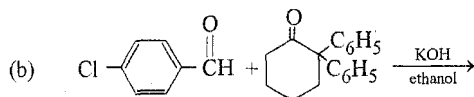
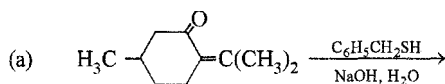


- Q.12 Give the structure of the substance for each of the following reaction:

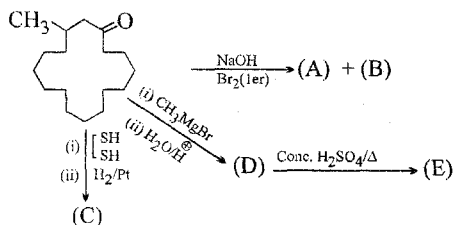




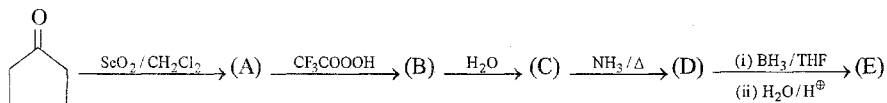
Q. 13 Each of the following reactions has been reported in the chemical literature. Write the structure of the product (s) formed in each case.



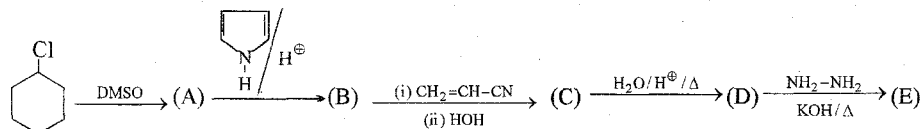
Q. 14 Identify A to E :



Q. 15 Identify A to E in the given sequence of reaction :

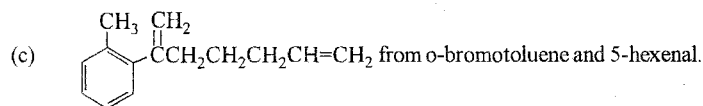
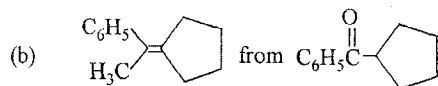


Q. 16 Identify (A) to (E) in the given sequence of reaction :



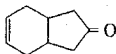
Q. 17 Syntheses of each of the following compounds have been reported in the chemical literature. Using the indicated starting material and any necessary organic or inorganic reagents, describe short sequence of reactions that would be appropriate for each transformation.

(a) 1,1,5-Trimethylcyclononane from 5,5-dimethylcyclononanone

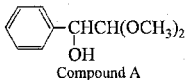


EXERCISE-III

- Q.1 Compound (A) $C_9H_{10}O$ is inert to Br_2 / CCl_4 and on vigorous oxidation with hot alkaline $KMnO_4$ yields benzoic acid. (A) gives red precipitate with 2, 4 DNP / HCl / Δ .
 (a) Write all possible structures of [A] and
 (b) How can these isomers be distinguished by chemical tests?
- Q.2 Compound (A) C_8H_8O is water insoluble and gives yellow precipitate with $I_2 / NaOH$. It reacts with NH_2OH in the presence of NH_4Cl / NH_4OH to give two geometrical isomers (B) and (C) having MF C_8H_9ON . These two compounds do not undergo dehydration. What will be structures of (A), (B) and (C) ?
- Q.3 A dithiane synthesis can convert an aldehyde to a ketone. The aldehyde is first converted to its dithiane derivative, which is deprotonated and alkylated. A mercuric chloride - assisted hydrolysis gives the ketone. Show how this technique might be used to convert benzaldehyde to benzyl phenyl ketone
- Q.4 Wolff-Kishner reduction (hydrazine, KOH , ethylene glycol, $130^\circ C$) of the compound shown gave compound A. Treatment of compound A with *m*-chloroperoxybenzoic acid gave compound B, which on reduction with lithium aluminium hydride gave compound C. Oxidation of compound C with chromic acid gave compound D ($C_9H_{14}O$). Identify compounds A through D in this sequence.



- Q.5 Reaction of benzaldehyde with 1, 2-octanediol in benzene containing a small amount of *p*-toluenesulfonic acid yields almost equal quantities of two products in a combined yield of 94%. Both products have the molecular formula $C_{15}H_{22}O_2$. Suggest reasonable structure for these products.
- Q.6 The sex attractant of the female winter moth has been identified as the tetraene $CH_3(CH_2)_8CH=CHCH_2CH=CHCH_2CH=CHCH=CH_2$. Devise a synthesis of this material from 3, 6-hexadecadien-1-ol and allyl alcohol.
- Q.7 Hydrolysis of a compound A in dilute aqueous hydrochloric acid gave (along with methanal) a compound B, mp $164-165^\circ C$. Compound B had the molecular formula $C_{16}H_{16}O_4$. What is the reasonable structure for compound B?

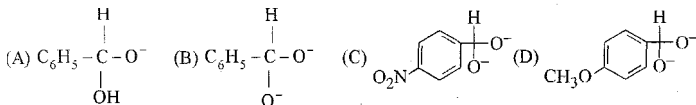


- Q.8 When $HO-CH_2-CH_2-CH_2-CH_2-CHO$ this compound is dissolved in water it exist in the equilibrium of isomeric forms. Both forms are ring-chain isomers. Write the structure of isomeric cyclic form and give mechanism for the formation of cyclic form. When (I) is treated with CH_3OH in dry HCl a new product (iii) is obtained. Give structure of this compound.
- Q.9 Compound (A) C_5H_8 , decolourises bromine water and on reduction with H_2 / Pt gives compound (B) having MF C_5H_{12} . A gives compound (C) with $H_2 / Lindlars$ catalyst and compound (D) with Na / NH_3 . Compound (C) and (D) are isomers having MF C_5H_{10} . (A) on hydration gives two isomeric compounds (E) and (F) having MF $C_5H_{10}O$. (E) gives yellow precipitate with $I_2 / NaOH$ but (F) does not. What will be the structures of (A) to (F) ?
- Q.10 An organic compound (A) contains 40% carbon and 6.7% hydrogen. Its vapour density is 15. On reacting with a concentrated solution of potassium hydroxide, it gives two compounds (B) and (C). When (B) is oxidised, original compound (A) is obtained. When (C) is treated with concentrated HCl , it gives a compound (D) which reduces Fehling and ammonical $AgNO_3$ solution and also gives effervescences with sodium bicarbonate solution. Write structures of (A), (B), (C) and (D) and explain reaction.

EXERCISE-IV (A)

- Q.1 The formation of cyanohydrin from a ketone is an example of: [IIT 1990]
(A) Electrophilic addition (B) Nucleophilic addition
(C) Nucleophilic substitution (D) Electrophilic substitution
- Q.2 The enolic form of acetone contains: [IIT 1990]
(A) 9 sigma bonds, 1 pi bond and 2 lone pairs (B) 8 sigma bonds, 2 pi bonds and 2 lone pairs
(C) 10 sigma bonds, 1 pi bond and 1 lone pair (D) 9 sigma bonds, 2 pi bonds and 1 lone pair
- Q.3 m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives: [IIT 1991]
(A) Potassium m-chlorobenzoate and m-hydroxybenzaldehyde
(B) m-hydroxybenzaldehyde and m-chlorobenzyl alcohol
(C) m-chlorobenzyl and m-hydroxybenzyl alcohol
(D) Potassium m-chlorobenzoate and m-chlorobenzyl alcohol
- Q.4 Hydrogenation of benzoyl chloride in the presence of Pd and BaSO₄ gives: [IIT 1992]
(A) Benzyl alcohol (B) Benzaldehyde (C) Benzoic acid (D) Phenol
- Q.5 An organic compound C₃H₆O does not give a precipitate with 2,4-Dinitrophenyl hydrazine reagent and does not react with metallic sodium. It could be: [IIT 1993]
(A) CH₃CH₂CHO (B) CH₃COCH₃
(C) CH₂=CH-CH₂OH (D) CH₂=CH-O-CH₃
- Q.6 Under Wolff Kishner reduction conditions, the conversions which may be brought about is?
(A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane
(C) Cyclohexanone into Cyclohexanol (D) Benzophenone into Diphenylmethane [IIT 1995]
- Q.7 In the reaction, P is [IIT 1995]
$$\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CO} \\ \diagdown \\ \text{CH}_3 \end{array} \xrightarrow{\text{SeO}_2} \text{P} + \text{Se} + \text{H}_2\text{O}$$

(A) CH₃COCHO (B) CH₃COOCH₃ (C) CH₃COCH₂OH (D) None
- Q.8 In the Cannizzaro reaction given below, 2Ph-CHO $\xrightarrow{\text{OH}^-}$ Ph-CH₂OH + PhCO₂⁻ the slowest step is:
(A) the attack of OH⁻ at the carbonyl group
(B) the transfer of hydride to the carbonyl group
(C) the abstraction of proton from the carboxylic acid
(D) the deprotonation of Ph-CH₂OH [IIT 1996]
- Q.9 Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is
(A) MeCOCl (B) MeCHO (C) MeCOOMe (D) MeCOOCOMe [IIT 1997]
- Q.10 In a Cannizzaro reaction the intermediate which is the best hydride donor is: [IIT 1997]



Q. 11 $\text{CH}_3\text{CHO} + \text{H}_2\text{NOH} \longrightarrow \text{CH}_3 - \text{CH} = \text{N} - \text{OH}$. The above reaction occurs at: [IIT 1997]
 (A) pH = 1 (B) pH = 4.5 (C) Any value of pH (D) pH = 12

Q. 12 Among the following compounds, which will react acetone to give a product containing $>\text{C}=\text{N}-$
 (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $(\text{CH}_3)_3\text{N}$ (C) $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$ (D) $\text{C}_6\text{H}_5\text{NHNH}_2$ [IIT 1998]

Q. 13 The product obtained via oxymercuration ($\text{HgSO}_4-\text{H}_2\text{SO}_4$) of 1-butyne would be

- (A) $\text{CH}_3\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
 (C) $\text{CH}_3\text{CH}_2\text{CHO} + \text{HCHO}$ (D) $\text{CH}_3\text{CH}_2\text{COOH} + \text{HCOOH}$ [IIT 1998]

Q. 14 Which of the following will undergo aldol condensation: [IIT 1998]
 (A) Acetaldehyde (B) Propanaldehyde (C) Benzaldehyde (D) Trideutero acetaldehyde

Q. 15 Which of the following will react with water: [IIT 1998]
 (A) CHCl_3 (B) Cl_3CCHO (C) CCl_4 (D) $\text{ClCH}_2\text{CH}_2\text{Cl}$

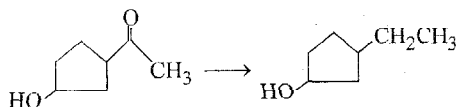
Q. 16 A new carbon-carbon bond formation is possible in [IIT 1998]
 (A) Cannizzaro reaction (B) Friedel-Crafts alkylation
 (C) Clemmensen reduction (D) Reimer-Tiemann reaction

Q. 17 The enol form of acetone, after treatment with D_2O gives: [IIT 1999]

- (A) $\text{CH}_3-\overset{\text{OD}}{\underset{|}{\text{C}}}=\text{CH}_2$ (B) $\text{CD}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CD}_3$ (C) $\text{CH}_2=\overset{\text{OH}}{\underset{|}{\text{C}}}-\text{CH}_2\text{D}$ (D) $\text{CD}_2=\overset{\text{OD}}{\underset{|}{\text{C}}}-\text{CD}_3$

Q. 18 Which of the following has the most acidic hydrogen: [IIT 2000]
 (A) 3-hexanone (B) 2,4-hexanedione (C) 2,5-hexanedione (D) 2,3-hexanedione

Q. 19 The appropriate reagent for the following transformation: [IIT 2000]

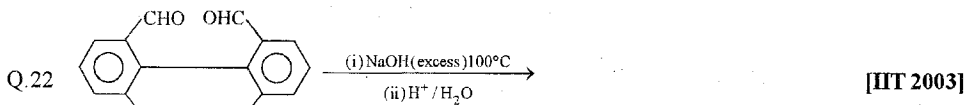


- (A) $\text{Zn}(\text{Hg}), \text{HCl}$ (B) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (C) H_2/Ni (D) NaBH_4

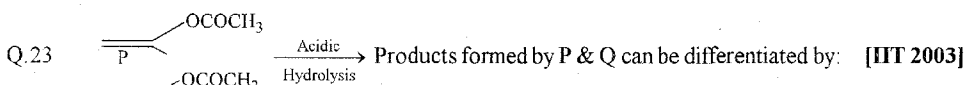
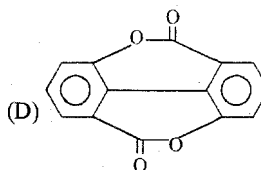
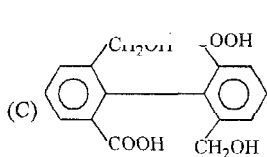
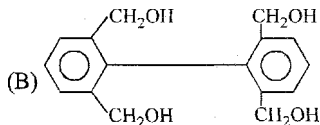
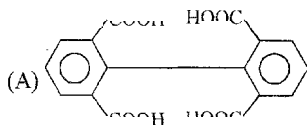
Q. 20 A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives:
 (A) benzyl alcohol and sodium formate (B) sodium benzoate and methyl alcohol
 (C) sodium benzoate and sodium formate (D) benzyl alcohol and methyl alcohol [IIT 2001]

Q. 21 Compound A (molecular formula $\text{C}_3\text{H}_8\text{O}$) is treated with acidified potassium dichromate to form a product B (molecular formula $\text{C}_3\text{H}_6\text{O}$). B forms a shining silver mirror on warming with ammonical silver nitrate. B when treated with an aqueous solution of $\text{H}_2\text{NCONHNH}_2$, HCl and sodium acetate gives a product C. Identify the structure of C.

- (A) $\text{CH}_3\text{CH}_2\text{CH}=\text{NNHCONH}_2$ (B) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{C}}}=\text{NNHCONH}_2$
 (C) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{C}}}=\text{NCONHNH}_2$ (D) $\text{CH}_3\text{CH}_2\text{CH}=\text{NCONHNH}_2$ [IIT 2002]



any one of the products formed is :



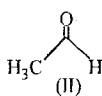
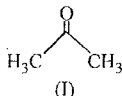
(A) 2, 4 DNP

(B) Lucas reagent (ZnCl_2) conc. HCl

(C) NaHSO_3

(D) Fehlings solution

Q.24 The order of reactivity of phenyl Magnesium Bromide with the following compounds is [IIT 2004]

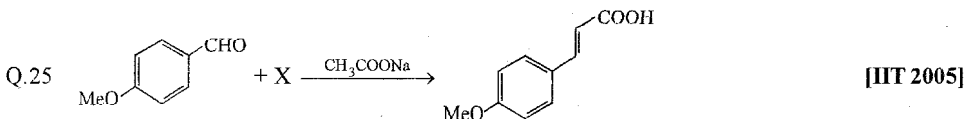


(A) $\text{II} > \text{III} > \text{I}$

(B) $\text{I} > \text{III} > \text{II}$

(C) $\text{II} > \text{I} > \text{III}$

(D) All react with the same rate



What is X?

(A) CH_3COOH

(B) BrCH_2COOH

(C) $(\text{CH}_3\text{CO})_2\text{O}$

(D) $\text{CHO}-\text{COOH}$

Q.26 The smallest ketone and its next homologue are reacted with NH_2OH to form oxime.

(A) Two different oximes are formed

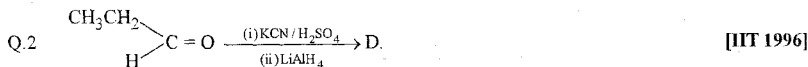
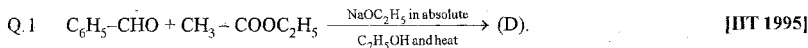
(B) Three different oximes are formed

(C) Two oximes are optically active

(D) All oximes are optically active

[JEE 2006]

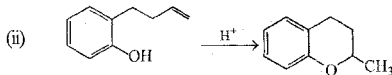
EXERCISE-IV (B)



Q.3 Acetophenone on reaction with hydroxylamine-hydrochloride can produce two isomeric oximes. Write structures of the oximes. [IIT 1997]

Q.4 An aldehyde (A) ($\text{C}_{11}\text{H}_{18}\text{O}$), which does not undergo self aldol condensation, gives benzaldehyde and two mole of (B) on ozonolysis. Compound (B), on oxidation with silver ion, gives oxalic acid. Identify the compounds (A) and (B). [IIT 1998]

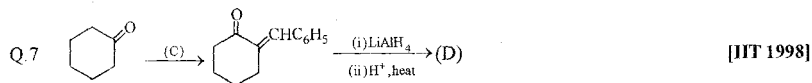
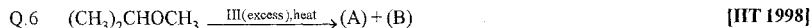
Q.5 (a) Write the intermediate steps for each of the following reaction. [IIT 1998]



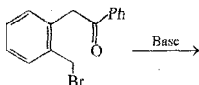
(b) Show the steps to carry out the following transformations.

(i) Ethylbenzene $\rightarrow$ benzene

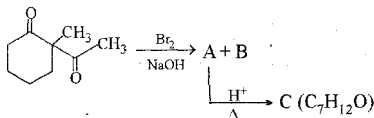
(ii) Ethylbenzene $\rightarrow$ 2-phenylpropionic acid



Q.8 What would be the major product in each of the following reaction? [IIT 2000]



Q.9 Identify (A), (B) and (C), and give their structures. [IIT 2000]



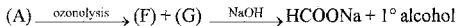
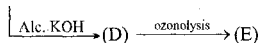
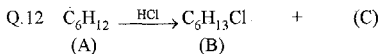
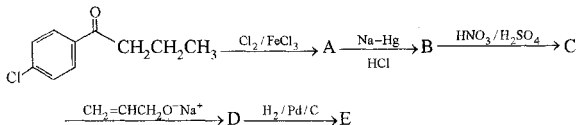
Q.10 Five isomeric para-disubstituted aromatic compounds A to E with molecular formula $\text{C}_8\text{H}_8\text{O}_2$ were given for identification. Based on the following observations, give structure of the compounds.

- Both A and B form a silver mirror with Tollen's reagent; also, B gives a positive test with FeCl_3 solution
- C gives positive iodoform test
- D is readily extracted in aqueous NaHCO_3 solution.
- E on acid hydrolysis gives 1,4-dihydroxybenzene.

[IIT 2002]

Q. 11 Write structures of the products A, B, C, D and E in the following scheme.

[IIT 2002]



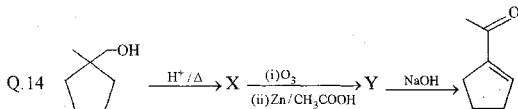
(D) is isomer A. E gives negative test with Fehling solution but gives iodoform test. F and G gives Tollen's test but do not give iodoform test. Identify A to G

[IIT 2003]

Q. 13 An organic compound (P), $\text{C}_5\text{H}_{10}\text{O}$ reacts with dil. H_2SO_4 to give (Q) and (R). Both (Q) and (R) give positive iodoform test. The reactivity of organic compound (P) is 10^{15} times more than ethylene with respect to dil. H_2SO_4 .

[IIT 2004]

- Identify the compounds (P), (Q) and (R).
- Give reason for the extraordinary reactivity of compound (P).

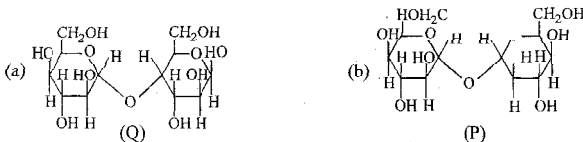


[IIT 2005]

Identify X and Y.

Q. 15 Which of the following disaccharide will not reduce Tollen's reagent?

[IIT 2005]



ANSWER KEY

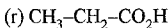
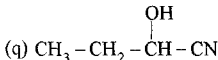
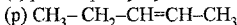
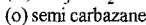
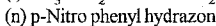
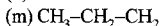
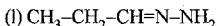
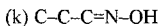
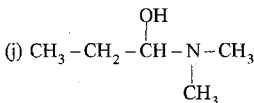
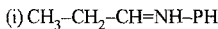
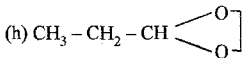
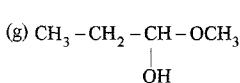
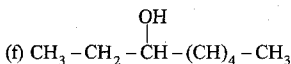
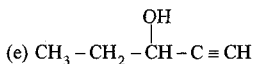
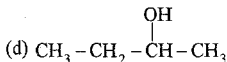
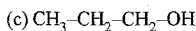
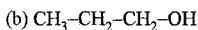
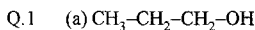
EXERCISE-II(A)

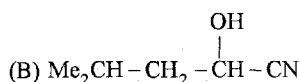
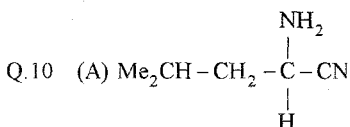
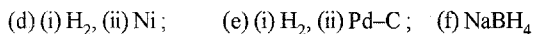
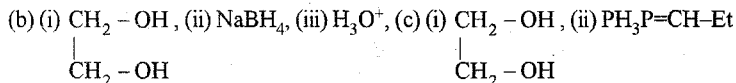
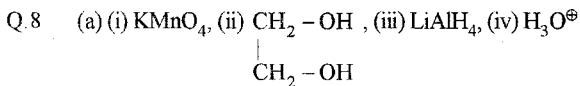
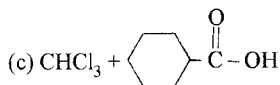
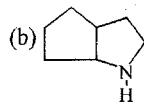
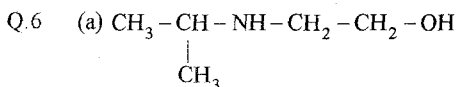
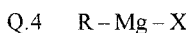
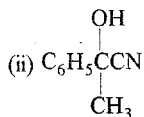
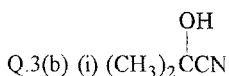
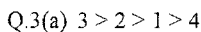
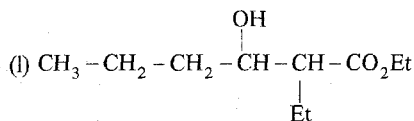
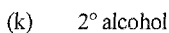
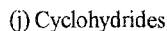
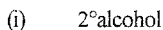
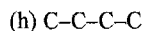
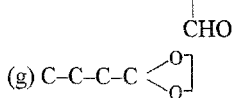
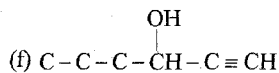
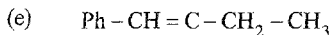
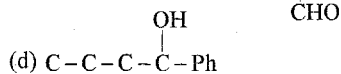
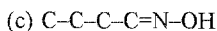
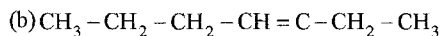
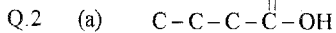
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Q.8 A	Q.9 B	Q.10 B	Q.11 C	Q.12 B	Q.13 A	Q.14 A
Q.15 B	Q.16 B	Q.17 C	Q.18 D	Q.19 C	Q.20 D	Q.21 C
Q.22 C	Q.23 A	Q.24 C	Q.25 B	Q.26 C	Q.27 D	Q.28 C
Q.29 A	Q.30 D	Q.31 D	Q.32 A	Q.33 C	Q.34 D	Q.35 B
Q.36 B	Q.37 D	Q.38 D	Q.39 C	Q.40 D	Q.41 C	Q.42 A
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Q.50 B						

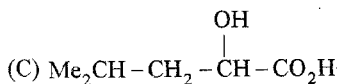
EXERCISE-II(B)

Q.1 C,D	Q.2 B,C	Q.3 A,C,D	Q.4 A,B
Q.5 A,B,C	Q.6 A,B,C,D	Q.7 A,B	Q.8 B,D
Q.9 A,B,C,D	Q.10 B,C	Q.11 B,D	Q.12 A
Q.13 D	Q.14 A	Q.15 A,B,C	Q.16 A,B
Q.17 B,C,D	Q.18 A,C,D	Q.19 A,B,C	Q.20 A,D
Q.21 A,C,D	Q.22 A,B,D	Q.23 A,D	
Q.24 (A) P, R, (B) P, (C) S, (D) Q, S	Q.25 (A) P, (B) Q, S, (C) S, (D) P		
Q.26 (A) P, (B) Q, (C) S, (D) R			

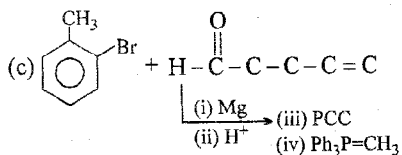
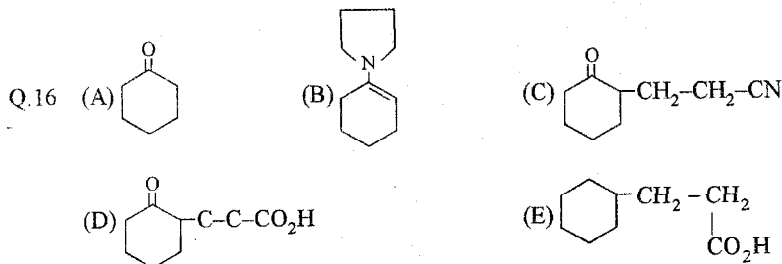
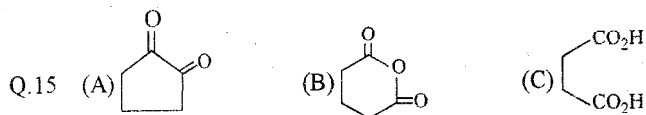
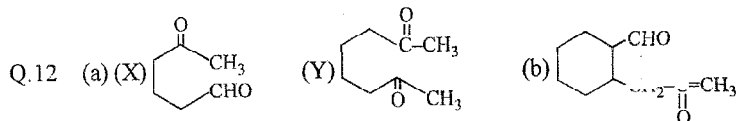
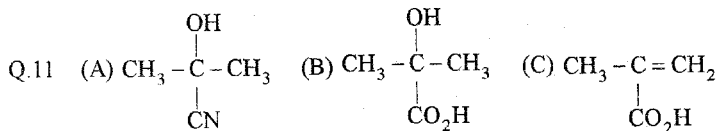
EXERCISE - II





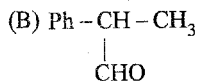
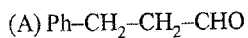


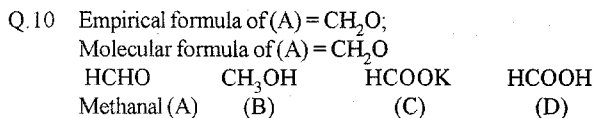
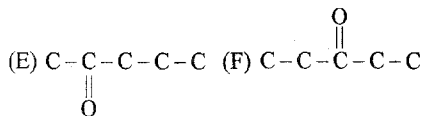
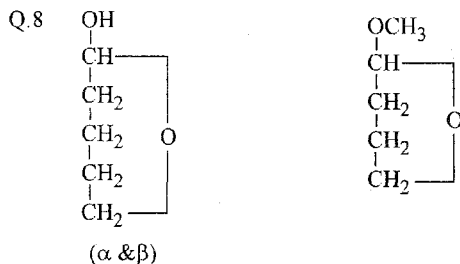
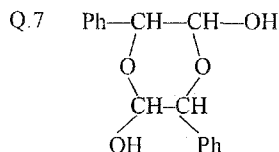
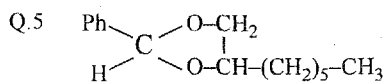
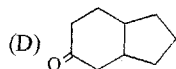
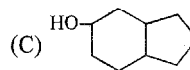
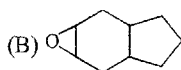
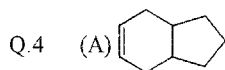
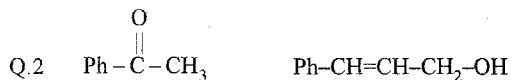
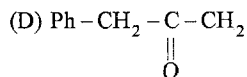
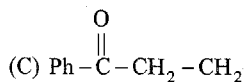
(D) Lactide



EXERCISE-III

Q.1 Possible structure an

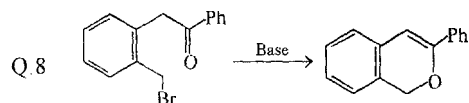
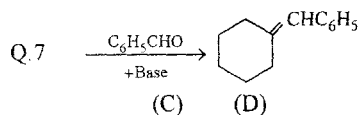
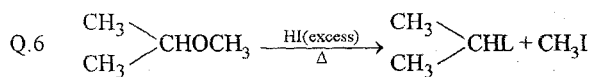
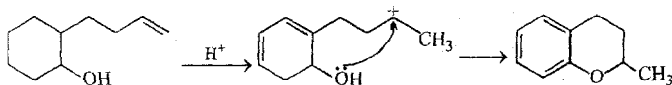
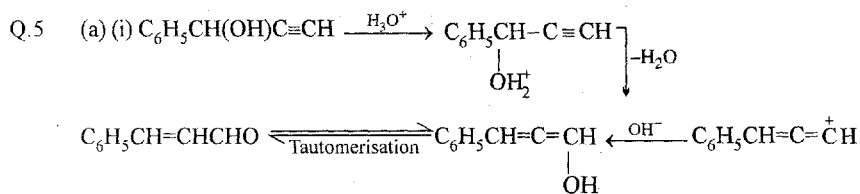
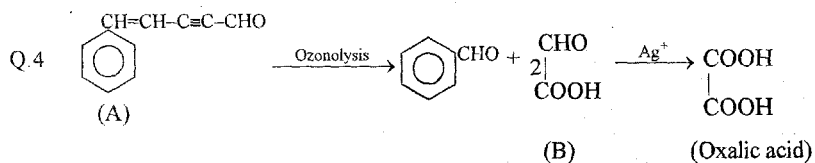
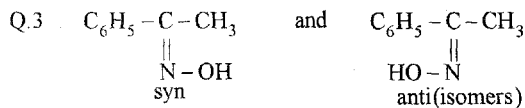
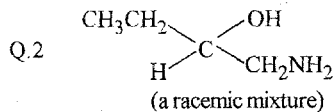
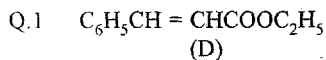


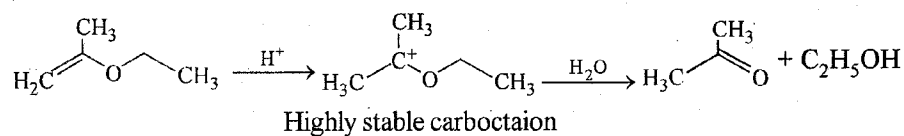
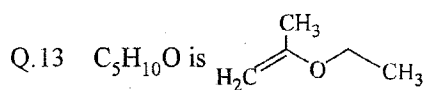
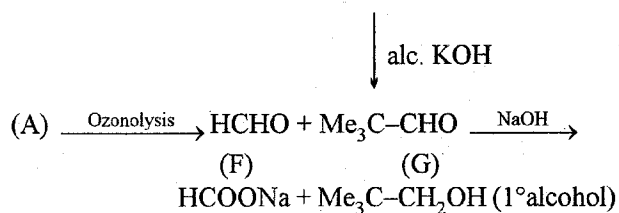
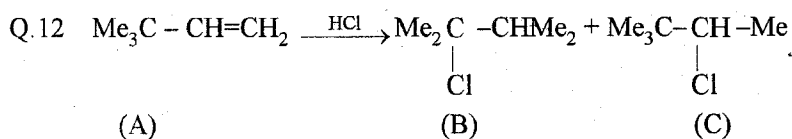
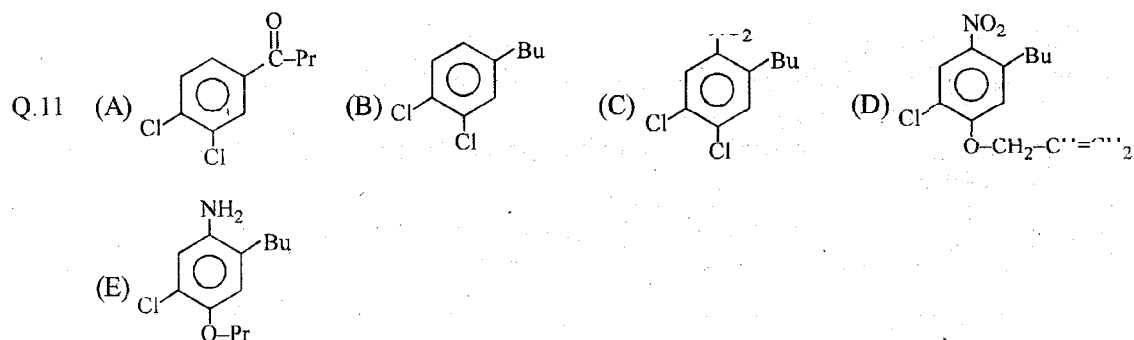
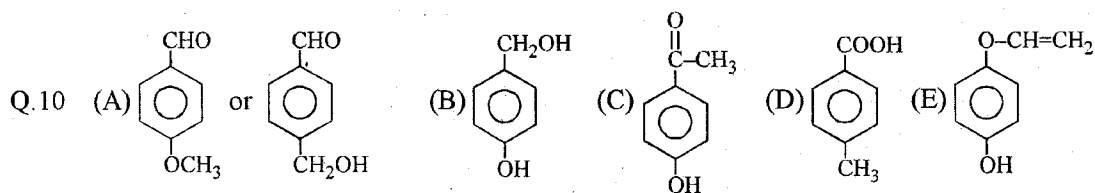
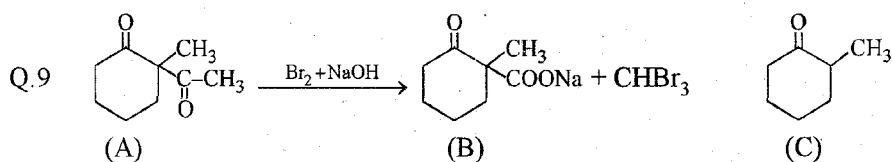


EXERCISE-IV (A)

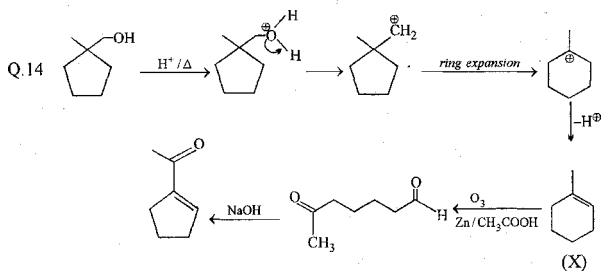
Q.1 B	Q.2 A	Q.3 D	Q.4 B	Q.5 D	Q.6 D	Q.7 A
Q.8 B	Q.9 A	Q.10 D	Q.11 B	Q.12 A,D	Q.13 A	Q.14 A,B,D
Q.15 B	Q.16 B,D	Q.17 B	Q.18 B	Q.19 B	Q.20 A	Q.21 A
Q.22 C	Q.23 D	Q.24 C	Q.25 C	Q.26 B		

EXERCISE-IV (B)





P is stabilized by resonance



- Q. 15 In structure (P) both the rings are present in acetyl form therefore it will not hydrolyse in solution that why Fehling solution cannot react with this.
In structure (Q) one ring present in the form of hemiacetal. This will hydrolysed in solution it can reduce Fehling solution.



BANSALCLASSES

TARGET IIT JEE 2007

ORGANIC CHEMISTRY

XII (ALL)

***CARBOXYLIC ACIDS AND
IT'S DERIVATIVE,
ALIPHATIC AMINES***

CONTENTS

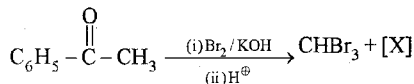
*EXERCISE - I
EXERCISE - II
EXERCISE - III
EXERCISE - IV(A)
EXERCISE - IV(B)
ANSWER KEY*

EXERCISE -I(A)

Q.1 Which one of the following is an acyl group:

- (A) $R-O-$ (B) $R-\overset{\overset{O}{\parallel}}{C}-O-$ (C) $R-\overset{\overset{O}{\parallel}}{C}-$ (D) $R-\overset{\overset{O}{\parallel}}{C}-NH-$

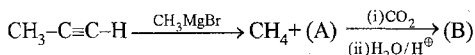
Q.2 In the given reaction :



[X] will be:

- (A) C_6H_5-CHO (B) C_6H_5COOH (C) $C_6H_5-CH_2OH$ (D) CH_3COOH

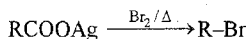
Q.3 In the reaction sequence :



(B) will be:

- (A) $CH_3-C\equiv C-CH_3$ (B) $CH_3-C\equiv C-MgBr$ (C) $CH_3-C\equiv C-COOH$ (D) $CH_3-CH=CH-COOH$

Q.4 Consider the given reaction



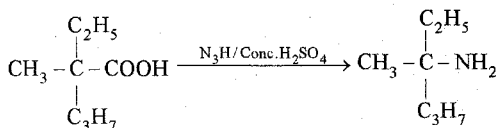
which one of the following acid will give maximum yield of R-Br in the above reaction?

- (A) $CH_3-\underset{\underset{CH_3}{|}}{CH}-COOH$ (B) $CH_3-CH_2-CH_2-COOH$
 (C) $CH_3-\underset{\underset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-COOH$ (D) All will give same yield

Q.5 Benzoic acid on treatment with hydrazoic acid in the presence of concentrated sulphuric acid gives:

- (A) Benzamide (B) Sodium benzoate (C) Aniline (D) $C_6H_5CON_3$

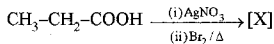
Q.6 The given reaction



is called :

- (A) Schmidt reaction (B) Curtius reaction
 (C) Hofmann rearrangement (D) Lossen rearrangement

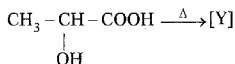
Q.7 In the given reaction :



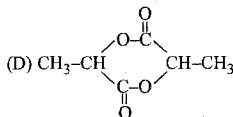
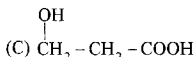
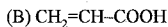
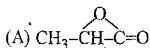
[X] will be:

- (A) Ethyl bromide (B) Propyl bromide (C) Propyl propanoate (D) All of these

Q.8 In the reaction sequence:



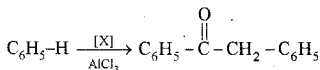
[Y] will be:



Q.9 Acetylation reaction with acetyl chloride is carried out in the presence of which solvent?

- (A) HOH (B) CH_3OH (C) Pyridine (D) NaOH

Q.10 In the reaction:

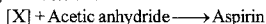


[X] will be:

- (A) $\text{C}_6\text{H}_5\text{CH}_2\text{COCl}$
(C) Both (A) and (B)

- (B) $(\text{C}_6\text{H}_5\text{-CH}_2\text{CO})_2\text{O}$
(D) $\text{C}_6\text{H}_5\text{-CHCl}_2$

Q.11 In the given reaction :

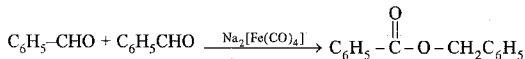


[X] will be:

- (A) Benzoic acid
(C) *o*-Hydroxybenzoic acid

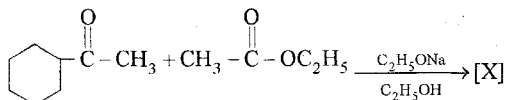
- (B) *o*-methoxybenzoic acid
(D) *p*-Hydroxybenzoic acid

Q.12 The given reaction is :

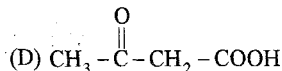
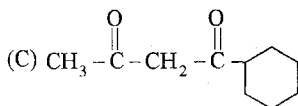
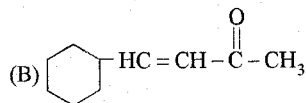
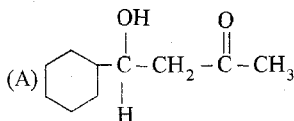


- (A) Claisen reaction (B) Tischenko reaction (C) Perkin reaction (D) Cannizzaro reaction

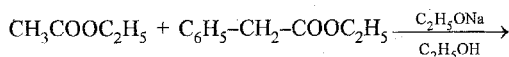
Q. 13 In the given reaction:



[X] will be:



Q. 14 Number of cross products in the given reaction:



(A) One

(B) Three

(C) Two

(D) four

Q. 15 Hofmann degradation is given by:

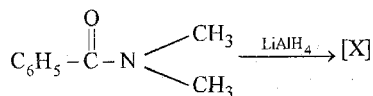
(A) Imide

(B) Acid chloride

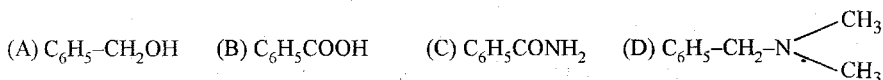
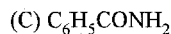
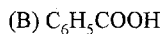
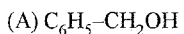
(C) Acid anhydride

(D) None of these

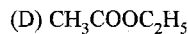
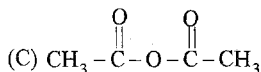
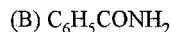
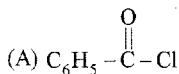
Q. 16 In the given reaction:



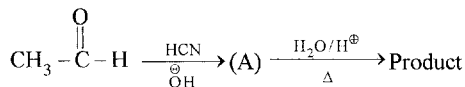
[X] will be:



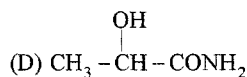
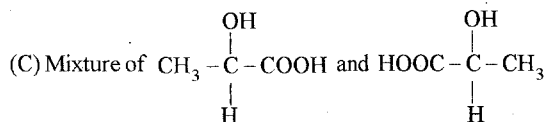
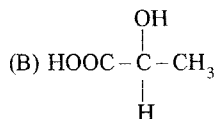
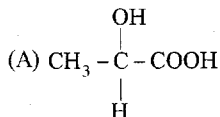
Q. 17 Which one of the following compounds gives carboxylic acid with HNO_2 ?



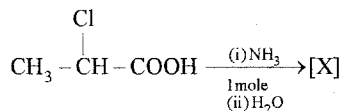
Q. 18 In the reaction sequence:



Product will be:



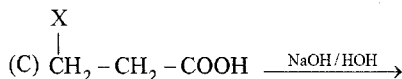
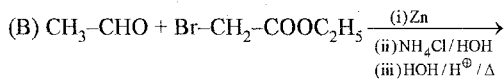
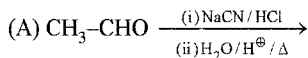
Q. 19 In the given reaction:



Product [X] will be:

- (A) α -Amino acid (B) α -Aminoamide (C) β -Amino acid (D) β -Aminoamide

Q. 20 Which of the following reactions will give α -hydroxy acid as a product:

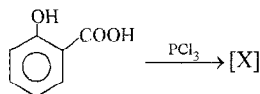


(D) All of these

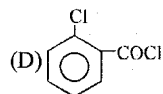
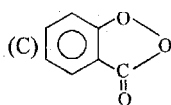
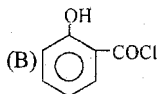
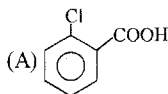
Q. 21 Rochelle salt is dimetal salt of:

- (A) Tartaric acid (B) Citric acid (C) Oxalic acid (D) Salicylic acid

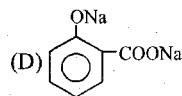
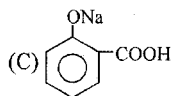
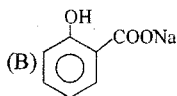
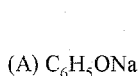
Q.22 In the given reaction :



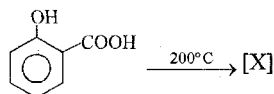
[X] will be :



Q.23 Sodium bicarbonate reacts with salicylic acid to form:



Q.24 In the given reaction :



[X] will be:

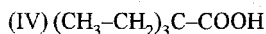
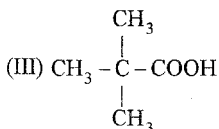
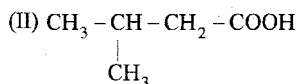
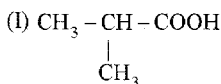
(A) Phenyl salicylate

(B) Aspirin

(C) Phenol

(D) Benzoic acid

Q.25 Arrange these esters in decreasing order of ease of esterification with $\text{CH}_3\text{OH}/\text{H}^+$:



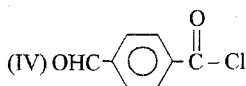
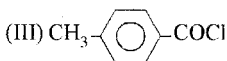
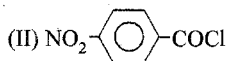
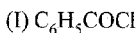
(A) II > I > III > IV

(B) I > II > III > IV

(C) III > IV > II > I

(D) IV > III > II > I

Q.26 Arrange following compounds in decreasing order of reactivity for hydrolysis reaction :



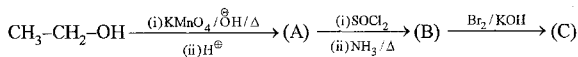
(A) II > IV > I > III

(B) II > IV > III > I

(C) I > II > III > IV

(D) IV > III > II > I

Q. 27 In the given reaction sequence:



(C) will be :

- (A) Methylamine (B) Ethylamine (C) Propylamine (D) Acetamide

Q. 28 Which one of the following on heating gives unsaturated acid:

- (A) α -Hydroxy acid (B) β -Hydroxy acid (C) γ -Hydroxy acid (D) δ -Hydroxy acid

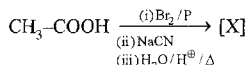
Q. 29 Which will form lactone on treatment with NaOH ?

- (A) α -Bromo acid (B) β -Bromo acid (C) β -Hydroxy acid (D) δ -Bromo acid

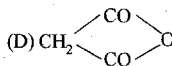
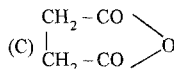
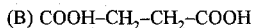
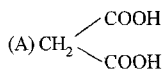
Q. 30 Which one of the following will go decarboxylation on heating?

- (A) Succinic acid (B) Phthalic acid (C) Malonic acid (D) Adipic acid

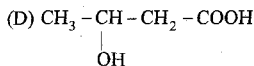
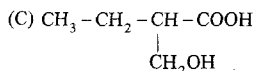
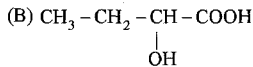
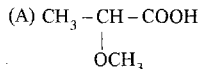
Q. 31 In the given reaction :



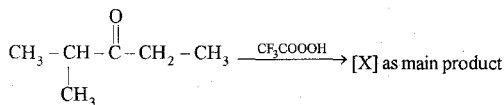
[X] will be:



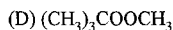
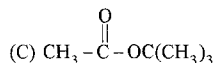
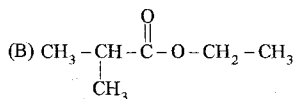
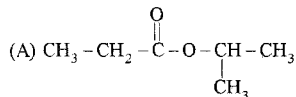
Q. 32 Which optically active compound on reduction with LiAlH_4 will give optically inactive compound?



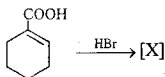
Q. 33 In the given reaction:



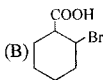
[X] will be:



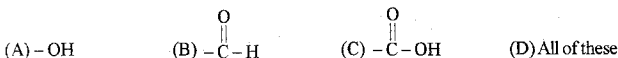
- Q.34 Which one of the following reactions can be used for the preparation of β -hydroxy acid:
 (A) Perkin reaction (B) Reformatsky reaction
 (C) Aldol condensation (D) Claisen condensation
- Q.35 Which acid can be oxidised by Fehling solution:
 (A) Malonic acid (B) Acetic acid (C) Oxalic acid (D) formic acid
- Q.36 In the given reaction:



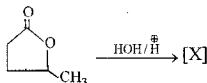
[X] will be:



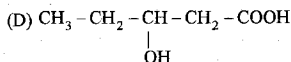
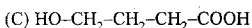
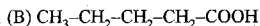
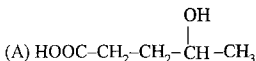
- Q.37 Reducing property of formic acid is due to the presence of:



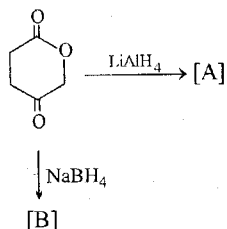
- Q.38 In the given reaction:



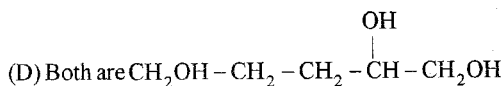
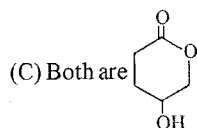
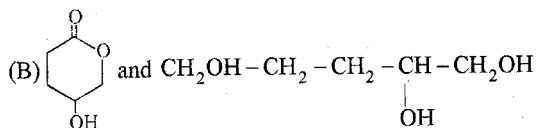
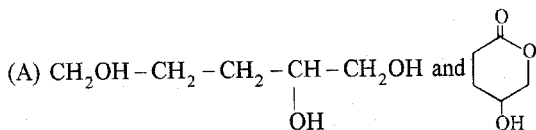
[X] will be:



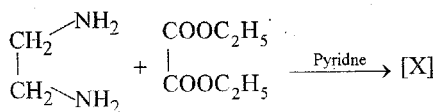
Q.39 In the given reaction:



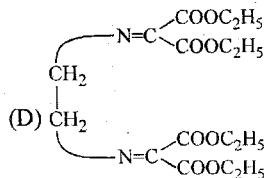
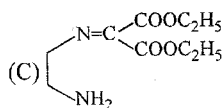
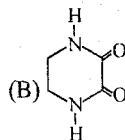
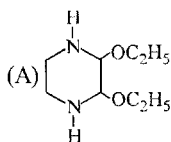
[A] and [B] respectively be:



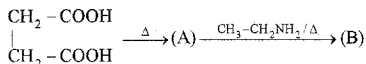
Q.40 In the given reaction:



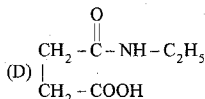
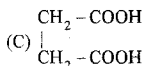
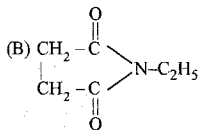
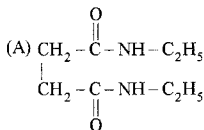
[X] will be :



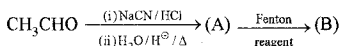
Q. 41 In the given reaction sequence:



(B) will be:



Q. 42 In the given reaction :



(B) will be :

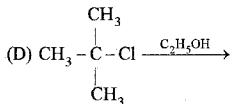
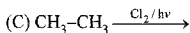
(A) Acetic acid

(B) Oxalic acid

(C) Pyruvic acid

(D) Citric acid

Q. 43 In which reaction product is hydrocarbon?



Q. 44 Sodium benzoate on heating with sodalime gives:

(A) Benzene

(B) Benzophenone

(C) Methane

(D) Calcium benzoate

Q. 45 Which of the following compounds gives carbondioxide with NaHCO_3 ?

(A) Acetic acid

(B) Hexanol

(C) Phenol

(D) Acetylene

Q. 46 When propanoic acid is treated with aqueous sodium bicarbonate, carbon dioxide is liberated. The carbon of the CO_2 comes from :

(A) Methyl group

(B) Carboxylic group

(C) Methylene group

(D) Bicarbonate

Q. 47 The reduction of benzoyl chloride with Pd and $\text{BaSO}_4 / \text{CaCO}_3$ produces:

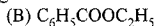
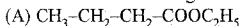
(A) Benzyl chloride

(B) Benzoic acid

(C) Benzaldehyde

(D) All of these

Q. 48 Which one of the following esters cannot undergo self Claisen condensation?



Q. 49 Amides may be converted into amines by a reaction named after :

(A) Kekule

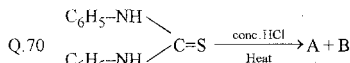
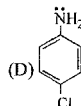
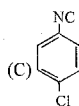
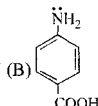
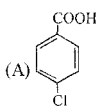
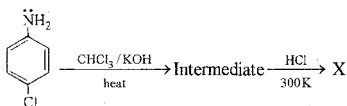
(B) Perkin

(C) Hofmann

(D) Claisen

- Q.50 The treatment of an ester with LiAlH_4 followed by acid hydrolysis produces :
 (A) Two aldehyde (B) One carboxylic acid and one alcohol
 (C) Two alcohols (D) Two acids
- Q.51 Which of the following is hydrolysed to give secondary amine:
 (A) Alkyl cyanide (B) $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{N} \begin{matrix} \nearrow \text{CH}_3 \\ \searrow \text{CH}_3 \end{matrix}$
 (C) Nitro paraffins (D) Acid amide
- Q.52 N-Ethyl phthalimide on hydrolysis gives:
 (A) Methyl alcohol (B) Ethyl amine (C) Dimethyl amine (D) Diethyl amine
- Q.53 Which amine will not react with nitrous acid:
 (A) Methyl amine (B) Ethyl amine
 (C) Dimethyl amine (D) N, N Dimethyl ethane amine
- Q.54 Which gas will be evolved out when $[\text{CH}_3\text{CH}_2\text{NH}_2 + (\text{CH}_3)_2\text{CHNH}_2]$ is treated with sodium nitrite and HCl:
 (A) Chlorine (B) Ammonia (C) Nitrogen (D) NO_2
- Q.55 Boiling Ethylamine on reaction with acidified KMnO_4 gives:
 (A) Acetaldehyde (B) Ethanoic Acid (C) Ethanol (D) Acetamide
- Q.56 Methyl amine on reaction with chlorine in the presence of NaOH gives:
 (A) Chloroform (B) Methyl chloride
 (C) N-Methyl chloramine (D) Chloramine
- Q.57 Highest value of pK_b will be of:
 (A) NH_3 (B) $(\text{CH}_3)_2\text{NH}$ (C) $(\text{CH}_3)_3\text{N}$ (D) CH_3NH_2
- Q.58 Lowest boiling point will be of the compound:
 (A) Ethylamine (B) Ethylmethylamine
 (C) 1-Propaneamine (D) N,N-Dimethylmethanamine
- Q.59 Amines are basic in nature because:
 (A) They produce OH^- ions when treated with water
 (B) They have replaceable H atoms on N atoms
 (C) They have lone pair of electron on N atom
 (D) None of these
- Q.60 Which of the following diazonium salt is relatively stable of $0-5^\circ\text{C}$:
 (A) $\text{CH}_3-\text{N} \equiv \text{N} \}^+ \text{Cl}^-$ (B) $\text{CH}_3\text{C}(\text{CH}_3)-\text{N} \equiv \text{N} \}^+ \text{Cl}^-$
 (C) $\text{C}_6\text{H}_5-\text{N} \equiv \text{N} \}^+ \text{Cl}^-$ (D) $(\text{CH}_3)_3\text{C}-\text{N} \equiv \text{N} \}^+ \text{Cl}^-$
- Q.61 Alkylamine dissolve in hydrochloric acid to form alkylammonium chloride. The nitrogen in the latter salt is:
 (A) Quadricovalent only (B) Tricovalent only
 (C) Unielectrovalent only (D) Quadricovalent, Unielectrovalent
- Q.62 Which of the following can be detected by carbylamine reaction:
 (A) Urea (B) CH_3CONH_2 (C) $\text{C}_2\text{H}_5\text{NH}_2$ (D) All of above

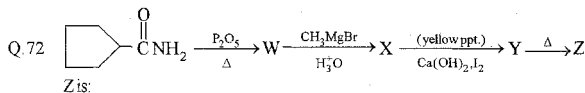
- Q. 63 Cyanides exists in:
 (A) Tautomeric form (B) Geometrical form (C) In both form (D) None
- Q. 64 Hydrolysis of alkyl isocyanide yields:
 (A) Primary amine (B) Tert. amine (C) Alcohol (D) Aldehyde
- Q. 65 Which of the following compound gives the smell of mustard oil:
 (A) Alkyl isocyanate (B) Alkyl isothiocyanate
 (C) Alkyl isocyanide (D) Alkyl isonitrile
- Q. 66 The compound obtained by the reaction between primary amine and aldehyde is:
 (A) An amide (B) Imine (C) Nitrite (D) Nitro
- Q. 67 When propionamide reacts with Br_2 in the presence of alkali the product is:
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ (B) $\text{CH}_3\text{CH}_2\text{NH}_2$ (C) $\text{C}_3\text{H}_7\text{CN}$ (D) $\text{C}_2\text{H}_5\text{CN}$
- Q. 68 A reaction of ethyl amine & acetic anhydride leads to the formation of:
 (A) $\text{CH}_3\text{NHCOCCH}_3$ (B) $\text{C}_2\text{H}_5\text{CONHCH}_3$ (C) $\text{CH}_3\text{CONHC}_2\text{H}_5$ (D) $\text{CH}_3-\text{CH}=\text{NOC}_2\text{H}_5$
- Q. 69 Identify X in the reaction



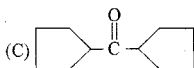
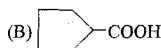
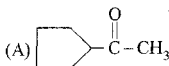
The unknown compounds A and B in the above reaction are:

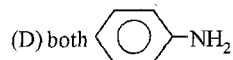
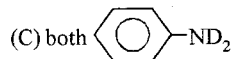
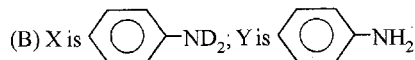
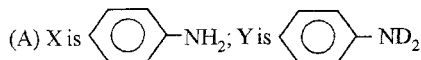
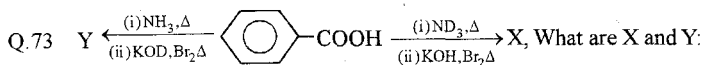
- (A) Phenyl mercaptan and aniline (B) Thiophenol and aniline hydrochloride
 (C) $\text{C}_6\text{H}_5-\text{N}=\text{C}=\text{S}$ and $\text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{SH}$ and $\text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$

- Q. 71 Aniline with solid KOH and CS_2 gives:
 (A) Thiourea (B) Phenyl thio urea (C) Phenol (D) Diphenyl thio urea

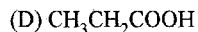
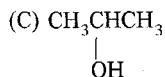
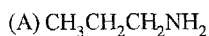
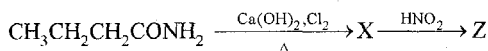


Z is:

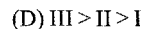
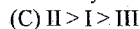
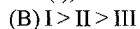
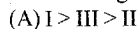




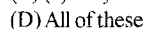
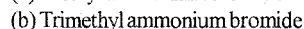
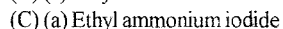
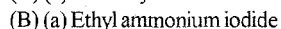
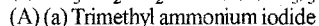
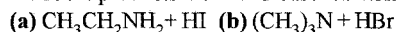
Q.74 Major end product of the following sequence of reaction is:



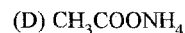
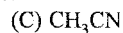
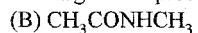
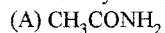
Q.75 The bond angles in methane (I), ammonia (II) and trimethyl amine (III) increase in the order:



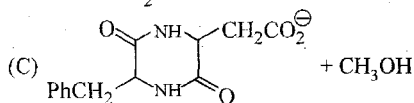
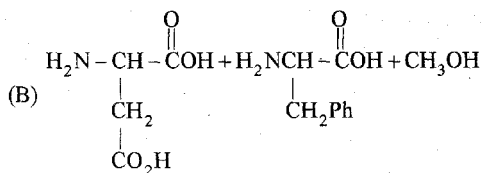
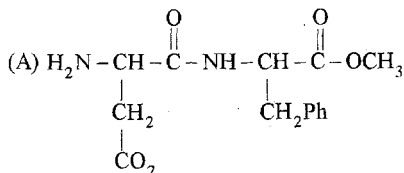
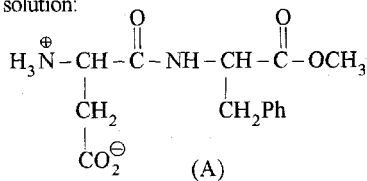
Q.76 Name the products in the acid-base reaction:



Q.77 Acetic anhydride and ammonia gives the product:

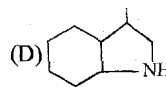
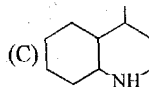
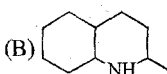
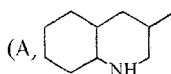
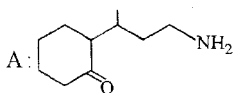


Q. 78 The artificial sweetener aspartame (A) is converted to _____ on storage for extended periods of time in aqueous solution:

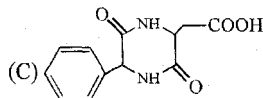
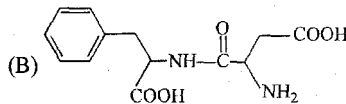
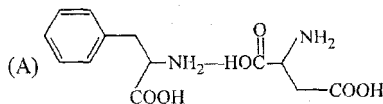
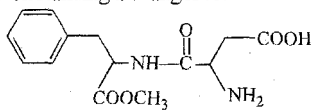


(D) no change, remains as (A)

Q. 79 Reductive amination of A forms:



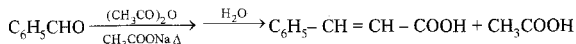
Q. 80 The synthetic sweetener aspartame, known commercially as Nutra sweet, is carboxylic acid. In aqueous solution on standing it changes to:



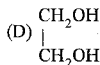
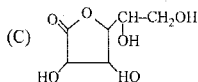
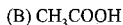
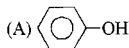
(D) no hydrolysis is possible

EXERCISE - I(B)

- Q.1 Consider the following statements regarding the following reaction:

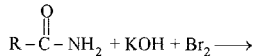


- (A) Acetate ion acts as base in this reaction
(B) The anhydride having atleast one α -hydrogen, provides an enolate for the reaction
(C) The acetate deprotonates the anhydride to form the nucleophile needed for the reaction
(D) EtONa can also be used as base
- Q.2 Which of the following will liberate CO_2 on reaction with NaHCO_3



- Q.3 RCOOR' can be prepared by:
- (A) esterification of RCOOH
(B) esterification of $(\text{RCO})_2\text{O}$
(C) Baeyer-Villiger oxidation of RCOR' with peroxy acid
(D) reaction of RCOCl with $\text{R}'\text{OH}$

- Q.4 In the Hofmann bromamide degradation reaction

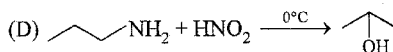
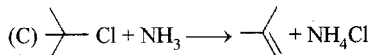
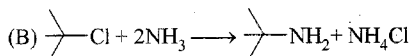
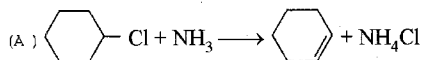


Intermediates are:

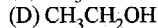
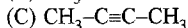
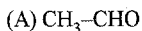
- (A) RCONHBr (B) RNCO (C) RNH_2 (D) None
- Q.5 $\text{C}_4\text{H}_{11}\text{N} + \text{HNO}_2 \longrightarrow \text{C}_4\text{H}_{10}\text{O}$ (3° alcohol)
(X)
hence X will give:
- (A) carbyl amine reaction
(B) Hofmann mustard oil reaction
(C) diazonium salt (as the intermediate) with HNO_2
(D) base insoluble product with Hinsburg reagent

- Q.6 Mixture of 1°, 2° and 3° amines can be separated by:
- (A) Hinsberg's method (B) Hofmann's isocyanide test
(C) fractional distillation (D) NaNO_2HCl

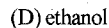
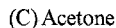
Q.7 Which is/are correct reaction(s):



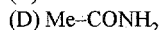
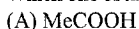
Q.8 Which of the following compounds will give acetic acid with $\text{KMnO}_4/\text{H}^+/\Delta$:



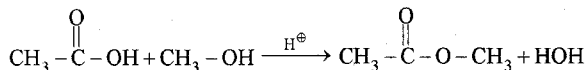
Q.9 Acetic acid can be used for the preparation of:



Q.10 Which one of the following acids will form acetyl chloride with PCl_5 ?



Q.11 Consider the following reaction:



True about the above reaction is:

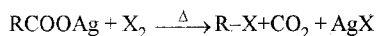
(A) Product is having smell like fruits

(B) Nucleophilic addition followed by elimination reaction

(C) follows A_{AC} mechanism

(D) it is irreversible reaction

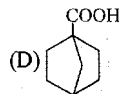
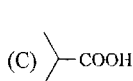
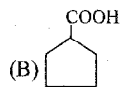
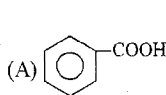
Q.12 In the given reaction:



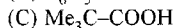
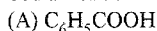
X_2 can be:



Q.13 Which one of the following compounds will give HVZ reaction?

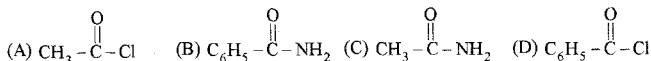


Q.14 Sodium salt of which one of the monobasic acids on electrolysis does not give hydrocarbon:



Q. 15 Which one of the following acids undergoes decarboxylation on strong heating:
 (A) Pyruvic acid (B) 3-butenic acid (C) Formic acid (D) Salicylic acid

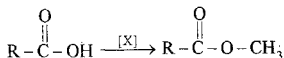
Q. 16 Which one of the following compounds is least reactive with water?



Q. 17 Acetic anhydride is used as:

(A) Solvent (B) Dehydrating agent (C) Acetylating agent (D) Antiseptic

Q. 18 In the given reaction:



[X] will be:

(A) CH_2N_2 (B) $\text{CH}_3\text{OH}/\text{H}^+$ (C) MeCOOH (D) Me_2SO_4

Q. 19 Which of the following gives silver mirror test?

(A) HCOOH (B) $\text{CH}_3\text{COCHOHCH}_3$
 (C) Tartaric acid (D) Glucose

Q. 20 An example of a primary amine is:

(A) n-Propylamine (B) Isopropylamine (C) t-Butyl amine (D) Sec. butyl amine

Q. 21 Which compound is soluble in water:

(A) $[(\text{CH}_3)_2\text{NH}_2]^+ \text{Cl}^-$ (B) $[(\text{CH}_3)_3\text{NH}_3]^+ \text{Cl}^-$
 (C) $[(\text{CH}_3)_3\text{NH}]^+ \text{Cl}^-$ (D) PhNO_2

Q. 22 Which compound will liberate CO_2 from NaHCO_3 solution:

(A) $\text{CH}_3\text{CO NH}_2$ (B) CH_3NH_2 (C) $(\text{CH}_3)_4\text{N}^+ \text{OH}^-$ (D) $\text{CH}_3\text{N}^+ \text{H}_3 \text{Cl}^-$

Q. 23 Which of the following compound can be produced if 1-propane amine is treated with NaNO_2 and HCl

(A) Propane-1-ol (B) Propane-2-ol (C) 2-Chloropropane (D) 2-Propaneamine

Q. 24 Which of the following amine does not react with Hinsberg reagent to give base soluble product

(A) Neopentyl amine (B) secpropyl amine (C) diethyl amine (D) Ethyl methyl amine

Q. 25 Ethylamine can be prepared by

(A) Curtius reaction (B) Hofmann reaction (C) Mendius reaction (D) Reduction of formaldoxime

Q. 26 The presence of primary amines can be confirmed in laboratory by:

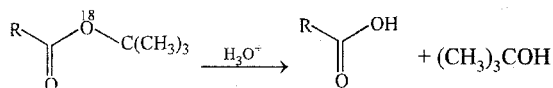
(A) Reaction with HNO_2 (B) reaction with CHCl_3 and alc. KOH
 (C) Reaction with Grignard reagent (D) Reaction with acetyl chloride

Q. 27 Select correct statement (s):

- (A) Cyclohexanone is oxidised to caprolactone which is hydrolysed under the reaction conditions
 (B) Baeyer-Villiger oxidation involves migration of H/alkyl/aryl groups which is in order $H > \text{tert-alkyl} > \text{sec-alkyl} > \text{phenyl} > \text{primary alkyl} > \text{methyl}$
 (C) Pinacolone $(\text{CH}_3)_3\text{CCCH}_3$ is oxidised to t-butyl acetate by $\text{C}_6\text{H}_5\text{CO}_3\text{H}$

(D) All of the above are correct statements

Q. 28 Following equilibrium is favoured in the forward side because of:



- (A) $(\text{CH}_3)_3\text{CO}^-$ (alkoxide) – a strong nucleophile is formed
 (B) $(\text{CH}_3)_3\text{C}^+$ (carbocation) is formed

(C) $\text{R}-\text{C}(=\text{O})-\text{O}^-$ formed is stabilised by resonance

(D) $\text{R}-\text{C}(=\text{O})-\text{OH}$ (a weak acid) is formed

Q. 29

Column I

(organic compounds oxidised by HIO_4)

- (A) CH_3COCHO
 (B) 1,2-cyclohexane dione
 (C) $\text{PhCH}(\text{OH})\text{CHO}$
 (D) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{COCH}_3$

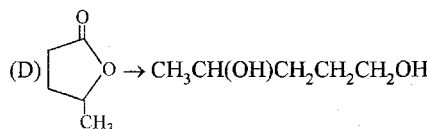
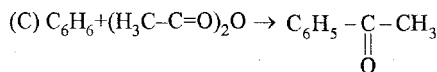
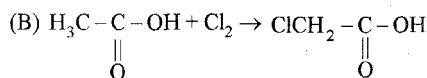
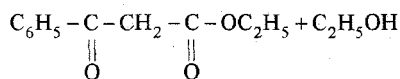
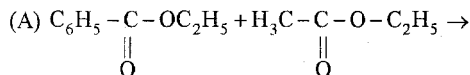
Column II

(products of HIO_4 oxidation)

- (P) $\text{PhCH}=\text{O} + \text{HCOOH}$
 (Q) $\text{CH}_3\text{CH}_2\text{CHO} + \text{HOOCCH}_3$
 (R) $\text{HOOC}(\text{CH}_2)_4\text{COOH}$
 (S) $\text{CH}_3\text{COOH} + \text{HCOOH}$

Q. 30

Column I



Column II

(P) Anhydrous ZnCl_2

(Q) $\text{C}_2\text{H}_5\text{O}^-$

(R) Red P

(S) LiAlH_4

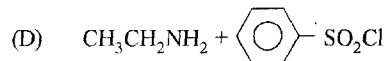
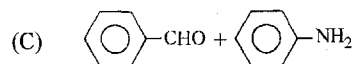
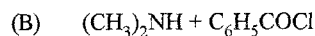
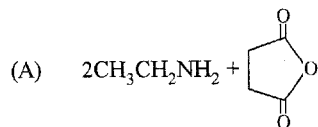
Q.31

Column I
(Reagents reacting with PhCH_2COOH)

- (A) CH_3MgBr
(B) PCl_5
(C) NH_3 , followed by heating
(D) CH_3OH in the presence of conc. H_2SO_4

Q.32

Column I (Reactions)



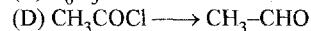
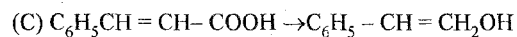
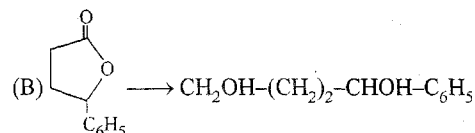
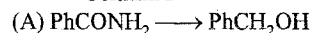
Q.33

Column I

- (A) $\text{R}'\text{NH}_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}}$
(B) $\text{PhNH}_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}}$
(C) $\text{PhN}(\text{CH}_3)_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}}$
(D) $\text{RR}'\text{NH}_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}}$

Q.34

Column I



Q.35

Column I

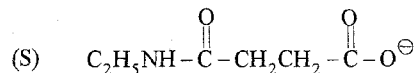
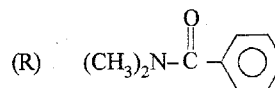
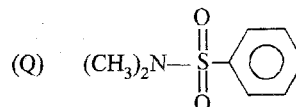
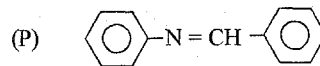
- (A) $\text{RCN} \xrightarrow{\text{reduction}}$
(B) $\text{RCN} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) CH}_3\text{MgBr}}$
(C) $\text{RNC} \xrightarrow{\text{hydrolysis}}$
(D) $\text{RNH}_2 \xrightarrow{\text{HNO}_2}$

Column II

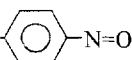
(Product formed)

- (P) PhCH_2COCl
(Q) $\text{PhCH}_2\text{COOCH}_3$
(R) CH_4
(S) $\text{PhCH}_2\text{CONH}_2$

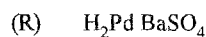
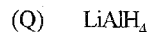
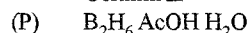
Column II (Products)



Column II

- (P) $\text{RR}'\text{N}-\text{N}=\text{O}$
(Q) $\text{Ph}-\text{N}^+ \equiv \text{NCl}^-$
(R) $(\text{CH}_3)_2\text{N}-$  $=\text{O}$
(S) $\text{R}'\text{OH} + \text{N}_2$

Column II

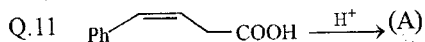
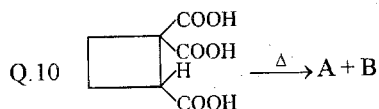
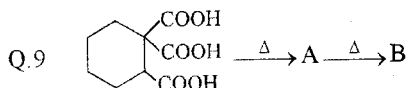
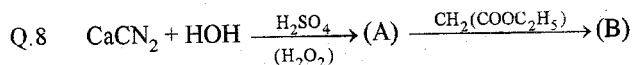
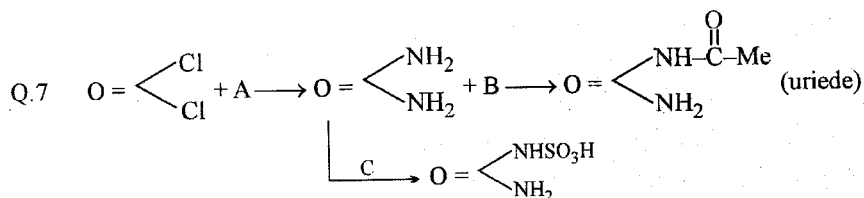
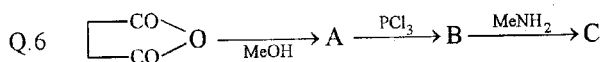
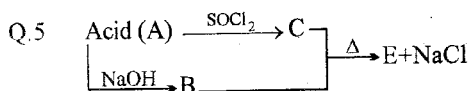
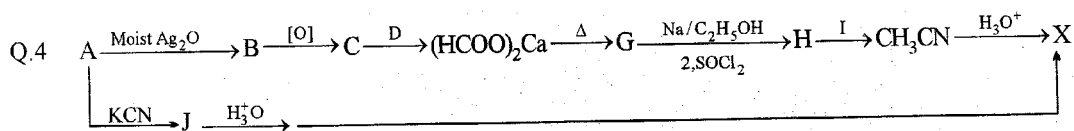
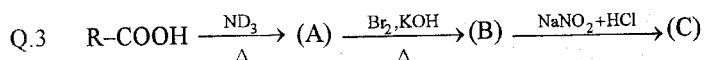
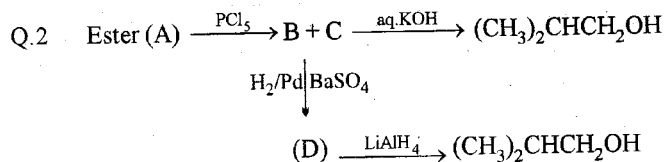
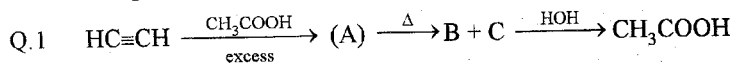


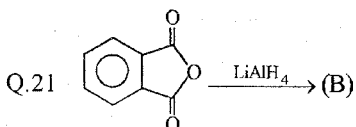
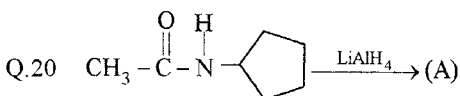
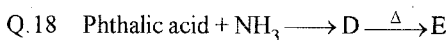
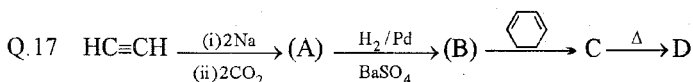
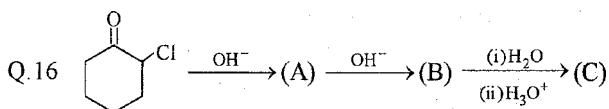
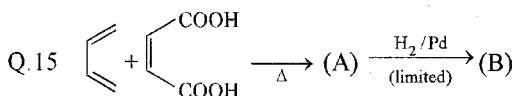
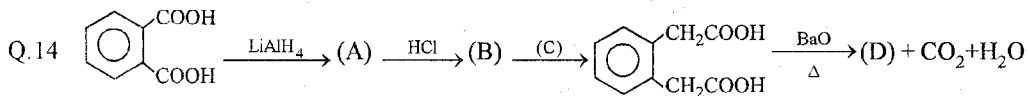
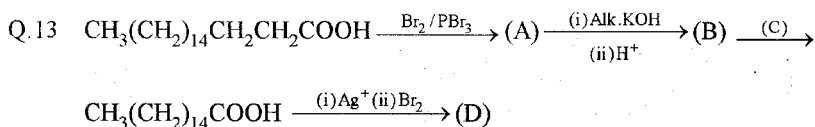
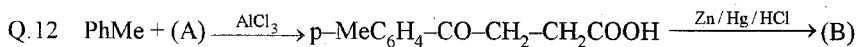
Column II

- (P) 1° Amine
(Q) Alcohol
(R) Ketone
(S) Acid

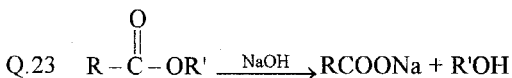
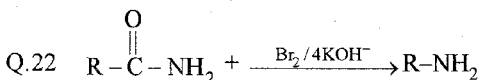
EXERCISE - II

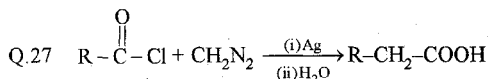
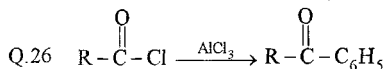
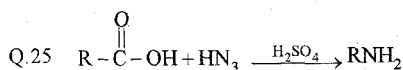
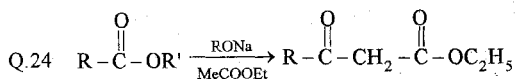
Complete the following equation by writing the missing A, B, C, D.....etc.





Give the mechanism for the reaction with suitable examples:





Account for following:

- Q.28 Fluoro acetic acid is stronger than chloro acetic acid.
- Q.29 C-O bond length in formic acid are 1.23 Å and 1.36 Å but in sodium formate both carbon and oxygen bonds have same value i.e. 1.27 Å.
- Q.30 Acetic acid can be halogenated in presence of phosphorous and chlorine but formic acid cannot be halogenated in the same way.
- Q.31 The C-G bond in acid derivative is expected to be shorter and stronger than alkyl derivative.
- Q.32 An acyl chloride undergoes nucleophilic attack more rapidly than does an alkyl halide.
- Q.33 Primary and secondary amide exist as dimer in solid and pure liquid state.

How will you bring about the following transformation:

- Q.34 Propanoic acid into lactic acid.
- Q.35 Nitroethane to acetic acid.
- Q.36 Benzoic acid to benzamide.
- Q.37 Ethyl benzene to 2-phenyl propionic acid.
- Q.38 Acetamide from acetone.
- Q.39 Tartaric acid from ethylene.
- Q.40 Lactic acid from acetic acid.
- Q.41 Pyruvic acid from ethyl bromide.
- Q.42 β-hydroxy propionic acid from ethylene oxide.

EXERCISE - III

Q. 1 Show by a series of equations how you could synthesize each of the following compounds from the indicated starting material and any necessary organic or inorganic reagents.

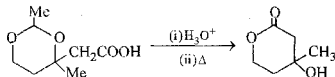
- (a) 2-Methylpropanoic acid from tert-butyl alcohol
- (b) 3-Methylbutanoic acid from tert-butyl alcohol
- (c) 3,3-Dimethylbutanoic acid from tert-butyl alcohol
- (d) $\text{HO}_2\text{C}(\text{CH}_2)_5\text{CO}_2\text{H}$ from $\text{HO}_2\text{C}(\text{CH}_2)_3\text{CO}_2\text{H}$
- (e) 3-Phenyl-1-butanol from $\text{CH}_3\text{CHCH}_2\text{CN}$



- (f)  from cyclopentyl bromide

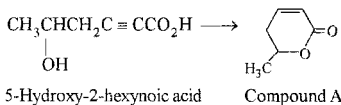
- (g)  from (E)- $\text{ClCH}=\text{CHCO}_2\text{H}$

Q. 2 On standing in dilute aqueous acid, compound A is smoothly converted to mevaionolactone.

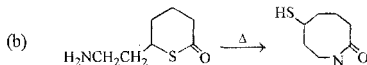
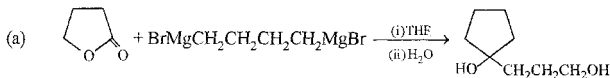


Suggest a reasonable mechanism for this reaction. What other organic product is also formed?

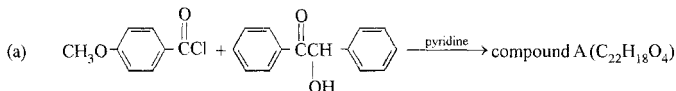
Q. 3 Suggest reaction conditions suitable for the preparation of compound A from 5-hydroxy-2-hexynoic acid.

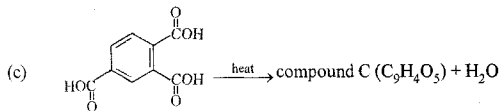
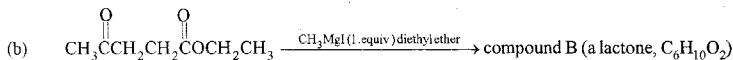


Q. 4 Outline reasonable mechanism for each of the following reactions



Q. 5 Identify compounds A through C in the following equations:



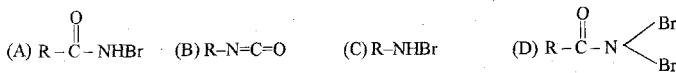


- Q. 6 An organic compound $\text{C}_4\text{H}_6\text{O}_4$ (A) when heated with acetic anhydride give another compound $\text{C}_4\text{H}_4\text{O}_3$ (B) which in turn react with ammonia to give a third compound $\text{C}_4\text{H}_5\text{O}_2\text{N}$ (C). Both (B) and (C) may be hydrolysed to A. With chlorine A give monochloro-compound (D) which reacts with caustic potash to give either $\text{C}_4\text{H}_5\text{O}_3\text{K}_2$ (E) or $\text{C}_4\text{H}_2\text{O}_4\text{K}_2$ (F) depending upon the condition of reaction. Identify the compound A to F.
- Q. 7 An organic compound A ($\text{C}_4\text{H}_9\text{NO}$) on treatment with bromine and alkali form another compound B ($\text{C}_3\text{H}_5\text{N}$) on treatment with sodium nitrite and dilute hydrochloric acid. B yield C ($\text{C}_3\text{H}_8\text{O}$). C can be oxidized to D ($\text{C}_3\text{H}_6\text{O}$). Which can also synthesised from methyl acetylene. Identify the compound A to D.
- Q. 8 With bromine and caustic potash the straight chain compound $\text{C}_4\text{H}_9\text{ON}$ (A) give $\text{C}_3\text{H}_9\text{N}$ (B), with caustic soda $\text{C}_4\text{H}_7\text{O}_2\text{Na}$ (C) and with phosphorous pentoxide gives $\text{C}_4\text{H}_7\text{N}$ (D). With hydrochloric acid D give $\text{C}_4\text{H}_8\text{O}_2$ (E) of which C is the sodium salt. B may be converted to an alcohol (F) and then to a halide the grignard reagent from which react with carbon dioxide to give E. On oxidation F give. Identify the compound A to H.
- Q. 9 Compound (A) with an empirical formula, $\text{C}_7\text{H}_9\text{N}$ on diazotization give a product which undergoes reaction with Cu_2Cl_2 & HCl to give a compound (B). B on oxidation give a compound (C). Compound (A) on treatment with $\text{Br}_2/\text{H}_2\text{O}$ form $\text{C}_7\text{H}_6\text{NBr}_3$ (D). Give the structural formula of A, B, C & D.
- Q. 10 An organic compound A ($\text{C}_3\text{H}_5\text{N}$) on boiling with alkali give ammonia and sodium salt of an acid B ($\text{C}_3\text{H}_6\text{O}_2$). A on reduction give a compound C ($\text{C}_3\text{H}_7\text{N}$) which with nitrous acid give D ($\text{C}_3\text{H}_8\text{O}$). Give the structural formula of A, B, C and D.
- Q. 11 An organic compound (A) $\text{C}_3\text{H}_7\text{ON}$ on heating with POCl_3 give a compound B which on hydrolysis give a compound (C) $\text{C}_3\text{H}_6\text{O}_2$ with the liberation of ammonia. Compound A on treatment with LiAlH_4 form compound $\text{C}_3\text{H}_9\text{N}$ (D). Compound D treating with HNO_2 liberate N_2 gas on refluxing with chloroform and alcoholic KOH give compound E which have foul smell. Identify compound A, B, C, D and E.
- Q. 12 An organic compound A ($\text{C}_8\text{H}_8\text{O}$) on treatment with NH_2OH . HCl give B and C. Compound B and C can be converted into D and E respectively by treatment of H_2SO_4 . Compound B, C, D and E are all isomer of molecular formula $\text{C}_8\text{H}_9\text{NO}$. When D is boiled with KOH, and oil F ($\text{C}_6\text{H}_7\text{N}$) separate out. F react rapidly with CH_3COCl to give back (D). On other hand. E on boiling with alkali followed by acidification give a white solid G ($\text{C}_7\text{H}_6\text{O}_2$). Identify A to G.
- Q. 13 An organic compound A, ($\text{C}_8\text{H}_9\text{NO}$) on heating with conc. H_2SO_4 give another isomer of (A) i.e. (B) $\text{C}_8\text{H}_9\text{NO}$. Compound B on hydrolysis with dil HCl give an aliphatic acid C, $\text{C}_2\text{H}_3\text{O}_2$ and (D) on treatment with NaNO_2/HCl at 0°C give a compound (E) which give a white ppt. of AgCl on addition of AgNO_3 . Compound (E) react with alkaline β -naphthol to produce orange dye what are A to E.
- Q. 14 An organic compound A of the molecular formula $\text{C}_4\text{H}_8\text{O}_3$ give effervescence with sodium bicarbonate solution and can be acetylated. It can be separated into dextro and laevo isomers on heating it forms an acid B with formula $\text{C}_4\text{H}_6\text{O}_2$. Compound B give positive Bayer's test. Identify A and B.

- Q.15 An aromatic compound A, $C_8H_7NO_2$ can be condensed with an acid derivative (B) $C_7H_{11}O_4$ Br in presence of alkali to form compound (C), $C_{15}H_{15}NO_6$. Compound (B) can be prepared by action of sodium followed by bromine gas on diethyl malonate. Compound (C) condensed again with ethyl chloro acetate in presence of sodium to give compound (D) $C_{19}H_{21}NO_8$. D on heating with a base can be converted into E and a salt of acid F ($C_8H_6O_4$). Compound A can be obtained by heating F with ammonia. E on boiling with mineral acid produce amino acid G ($C_4H_7NO_2$). Identify A to C.
- Q.16 An organic acid A ($C_3H_4O_3$) is catalytically reduced in presence of ammonia to give (B) $C_3H_7NO_2$. B react with acetyl chloride hydrochloric acid and alcohol. It can also react with nitrous acid to give another compound (C) $C_3H_6O_3$ along with evolution of N_2 . What are (A), (B) and (C)?
- Q.17 A dibasic acid (A), $C_4H_4O_6$, on heating with $KHSO_4$ give a monobasic acid (B), $C_3H_4O_3$, with the elimination of CO_2 and H_2O . Acid (B) on reduction give another monobasic acid (C), $C_3H_6O_3$. Both (B) and (C) on heating with I_2 and NaOH solution give yellow precipitate of iodoform and sodium oxalate. Acid (A) react with two moles of acetyl chloride, while (C) reacts with one mole of acetyl chloride. Acid (A) reacts with four moles of HI to form succinic acid, while (C) react with two moles of HI to give propionic acid. Identify A, B, C and give equation for reaction.
- Q.18 An organic compound of the formula C_3H_6O (A) gave iodoform test but not Tollen's and Fehling tests. Two molecules of (A) condenses in presence of dry HCl gas to yield (B) of formula $C_6H_{10}O$. Compound (B) decolourised Br_2/CCl_4 and cold dilute neutral $KMnO_4$, but no ppt. with ammonical $AgNO_3$ or $CuCl$ solution. It gave iodoform test and potassium salt of a compound on heating with I_2 and KOH. The potassium salt on treatment with H_2SO_4 gave compound (C) of formula $C_5H_8O_2$. Compound (C) on heating with sodalime gave (D) of formula C_4H_8 . Identify compound A to D & write the chemical equation.
- Q.19 An organic compound (A) of formula C_3H_6O from a monoxime with hydroxylamine and form iodoform on heating with I_2 and NaOH and sodium acetate. Compound A on reaction with NaCN and dil. H_2SO_4 gave product B. Which on hydrolysis produced (C). Compound (C) on heating gave (D), which on decarboxylation gave (E) of formula C_3H_6 . Compound (E) on ozonolysis gave one molecule of acetaldehyde and one of methanal. What are A to E?
- Q.20 Two moles of an ester (A) are condensed in the presence of sodium ethoxide to give a β -keto ester (B) and ethanol. On heating in an acidic solution (B) gives ethanol and β -keto acid (C). On decarboxylation (C) gives 3-pentanone. Identify (A), (B) and (C) with proper reasoning. Name the reaction involved in the conversion of (A) to (B).
- Q.21 An acidic compound (A), $C_4H_8O_3$ loses its optical activity on strong heating yielding (B), $C_4H_6O_2$ which reacts readily with $KMnO_4$. (B) forms a derivatives (C) with $SOCl_2$, which on reaction with $(CH_3)_2NH$ gives (D). The compound (A) on oxidation with dilute chromic acid gives an unstable compound (E) which decarboxylates readily to give (F), C_3H_6O . The compound (F) gives a hydrocarbon (G) on treatment with amalgamated Zn and HCl. Give structures of (A) to (G) with proper reasoning.
- Q.22 An organic compound (A), $C_5H_8O_3$ on heating with soda-lime gives (B), which reacts with HCN to give (C). The compound (C) reacts with thionylchloride to produce (D), which on reaction with KCN gives compound (E). Alkaline hydrolysis of (E) gives a salt (F), which on heating with soda-lime produces butane. Careful oxidation of (A) with dichromate gives acetic and malonic acid. Give structures of (A) to (F) with proper reasoning.
- Q.23 An acid (A), $C_8H_7O_2Br$ on bromination in the presence of FeBr gives two isomers, (B) and (C) of formula $C_8H_6O_2Br_2$. Vigorous oxidation of (A), (B) and (C) gives acids (D), (E) and (F) respectively. (D), $C_7H_5O_2Br$ is the strongest acid among all of its isomers, whereas (E) and (F) each has a molecular formula of $C_7H_4O_2Br_2$. Give structures of (A) to (F) with justification.

EXERCISE - IV (A)

- Q.1 The reaction of $R-\overset{\overset{O}{\parallel}}{C}-NH_2$ with a mixture of Br_2 and KOH gives $R-NH_2$ as a product. The intermediates involved in this reaction are: [IIT 1992]



- Q.2 Which of the following carboxylic acids undergo decarboxylation easily: [IIT 1995]



- Q.3 The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to: [IIT 1996]



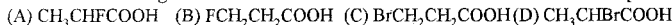
- Q.4 When propionic acid is treated with aqueous $NaHCO_3$, CO_2 is liberated. The 'C' of CO_2 comes from [IIT 1999]

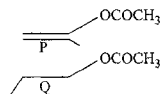


- Q.5 Benzoyl chloride is prepared from benzoic acid by: [IIT 2000]



- Q.6 Which of the following acids has the smallest dissociation constant? [IIT 2002]



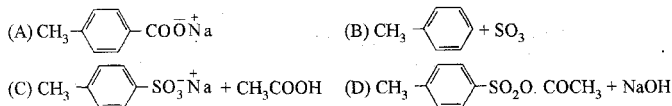
- Q.7  Products formed by P & Q can be differentiated by [IIT 2003]



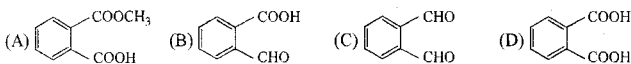
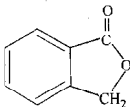
- Q.8 When benzamide is treated with $POCl_3$, the product is: [IIT 2004]



- Q.9 Which of the following is obtained when 4-Methylbenzenesulphonic acid is hydrolysed with excess of sodium acetate? [IIT 2005]

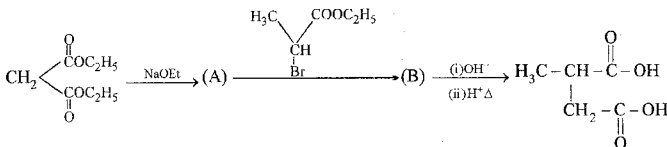


- Q. 10 The correct IUPAC name of C_6H_5COCl is
 (A) Benzoyl chloride (B) Benzene chloro ketone
 (C) Benzene carbonyl chloride (D) Chloro phenyl ketone [IIT 2006]
- Q. 11 Which of the following reactants on reaction with conc. NaOH followed by acidification gives the following lactone as the only product? [IIT 2006]



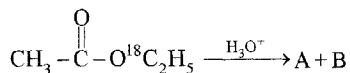
EXERCISE - IV (B)

- Q. 1 Compound (A) ($C_6H_{12}O_2$) on reduction with $LiAlH_4$ yielded two compounds (B) and (C). The compound (B) on oxidation gave (D) which on treatment with aqueous alkali and subsequent heating furnished (E). The latter on catalytic hydrogenation gave (C). The compound (D) was oxidised further to give (F) which was found to be monobasic acid (molecular mass = 60.0). Deduce the structure of (A), (B), (C), (D) and (E). [IIT 1990]
- Q. 2 In the following reactions, identify the compounds (A), (B), (C) and (D) [IIT 1994]
- $$PCl_5 + SO_2 \longrightarrow (A) + (B)$$
- $$(A) + CH_3COOH \longrightarrow (C) + SO_2 + HCl$$
- $$2(C) + b(CH_3)_2Cd \longrightarrow 2(D) + CdCl_2$$
- Q. 3 Complete the following equations by writing the missing A and B [IIT 1995]
- (i) $CH_3CH_2COOH \xrightarrow{P/Bz} (A) \xrightarrow[(ii) H^+]{(i) Alc. KOH(excess)} (B)$
- (ii) $C_4H_8O_3(A) \xrightarrow{CrO_3} (B) \xrightarrow{Warm} CH_3-C(=O)-CH_3 + CO_2$
- Q. 4 A mixture of an acid anhydride (A) and a monobasic acid (B) on heating produces another monobasic acid (C) of equivalent weight 74 and an anhydride (D). The acids and anhydrides remain in equilibrium. The anhydride (D) contains two identical fluoro-alkyl groups. The acid (B) contains a trifluoromethyl group and has an equivalent weight of 128. Give structures of (A) to (D) with proper reasoning. (Atomic weight of fluorine = 19). [IIT 1998]
- Q. 5 Identify A and B in the following equations: [IIT 1999]



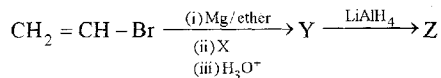
Q.6 Write the structure of product A & B:

[IIT 2000]



Q.7 Identify X, Y and Z in the following synthetic scheme and write their structure. Explain the formation of labelled formaldehyde ($\text{H}_2\text{C}^*\text{O}$) as one of the products when compound Z is treated with HBr and $\text{BaC}^*\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{C (gas)}$ [C^* denotes C^{14}]

[IIT 2001]

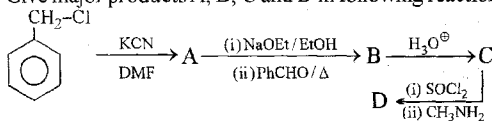


Q.8 Mention two esters produced when a racemic mixture of 2-phenyl propanoic acid is treated with (+) 2-butanol. What is the stereochemical relationship between these esters?

[IIT 2003]

Q.9 Give major products A, B, C and D in following reaction sequence.

[IIT 2004]



ANSWER KEY
EXERCISE - I(A)

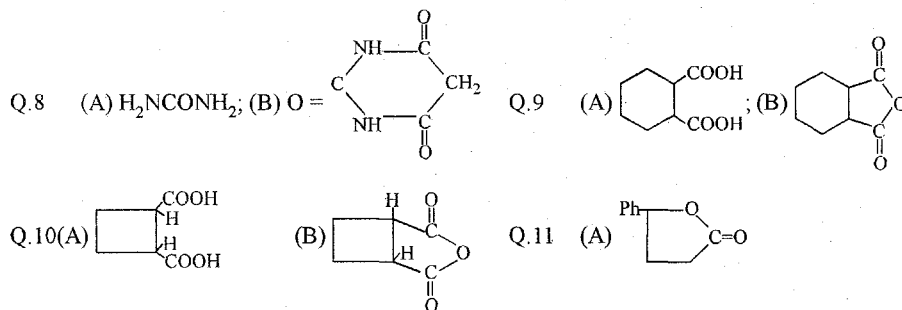
Q.1	C	Q.2	B	Q.3	C	Q.4	B	Q.5	C	Q.6	A	Q.7	A
Q.8	D	Q.9	C	Q.10	C	Q.11	C	Q.12	B	Q.13	C	Q.14	C
Q.15	A	Q.16	D	Q.17	B	Q.18	C	Q.19	A	Q.20	A	Q.21	A
Q.22	B	Q.23	B	Q.24	A	Q.25	A	Q.26	A	Q.27	A	Q.28	B
Q.29	D	Q.30	C	Q.31	A	Q.32	C	Q.33	A	Q.34	B	Q.35	D
Q.36	B	Q.37	B	Q.38	A	Q.39	A	Q.40	B	Q.41	B	Q.42	C
Q.43	A	Q.44	A	Q.45	A	Q.46	D	Q.47	C	Q.48	B	Q.49	C
Q.50	C	Q.51	B	Q.52	B	Q.53	D	Q.54	C	Q.55	A	Q.56	C
Q.57	A	Q.58	D	Q.59	C	Q.60	C	Q.61	D	Q.62	C	Q.63	A
Q.64	A	Q.65	B	Q.66	B	Q.67	B	Q.68	C	Q.69	D	Q.70	C
Q.71	D	Q.72	C	Q.73	A	Q.74	C	Q.75	A	Q.76	C	Q.77	A
Q.78	C	Q.79	C	Q.80	C								

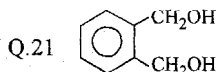
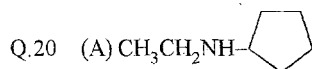
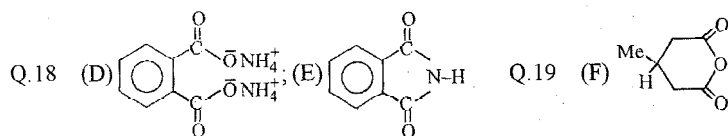
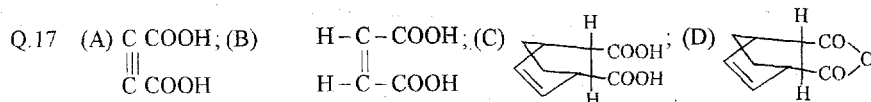
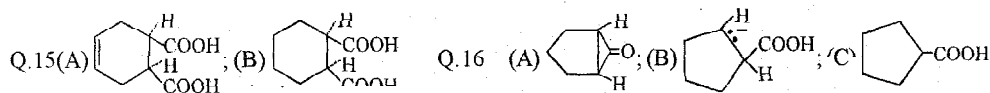
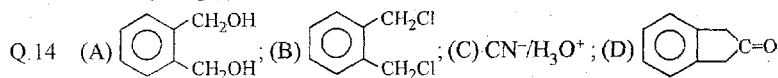
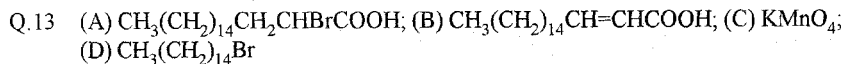
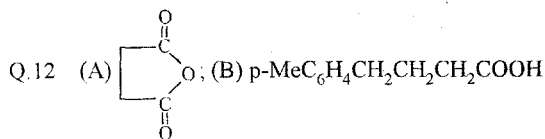
EXERCISE - I(B)

Q.1	A,B,C	Q.2	B,C	Q.3	A,B,C,D	Q.4	A,B
Q.5	A,B,C	Q.6	A,C	Q.7	A,C,D	Q.8	A,B,C,D
Q.9	A,B,C,D	Q.10	A,B,C	Q.11	A,B	Q.12	A,B
Q.13	B,C	Q.14	B,C	Q.15	A,B,C,D	Q.16	B
Q.17	A,B,C	Q.18	A,B,D	Q.19	A,B,C,D	Q.20	A,B,C,D
Q.21	A,B,C	Q.22	D	Q.23	A,B,C	Q.24	A,B
Q.25	A,B,C	Q.26	B	Q.27	D	Q.28	B
Q.29	(A) S, (B) R, (C) P, (D) Q			Q.31	(A) R, (B) P, (C) S, (D) Q		
Q.30	(A) Q, (B) R, (C) P, (D) S			Q.33	(A) S, (B) Q, (C) P, (D) R		
Q.32	(A) Q, (B) S, (C) P, (D) Q			Q.35	(A) P, (B) R, (C) P,S, (D) Q		
Q.34	(A) S, (B) Q, (C) S, (D) R						

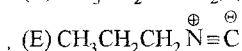
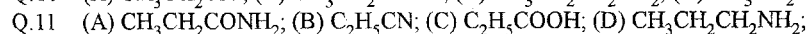
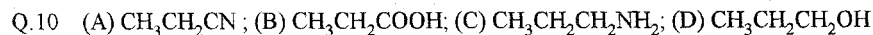
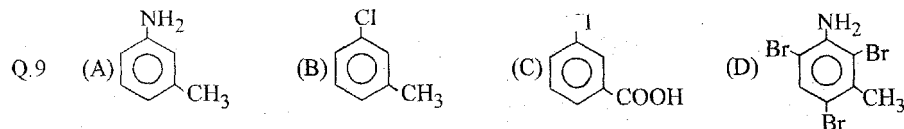
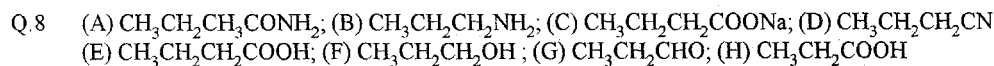
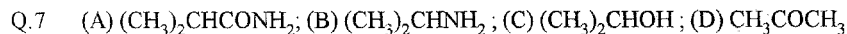
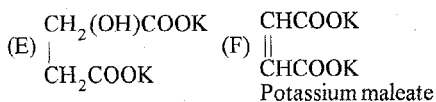
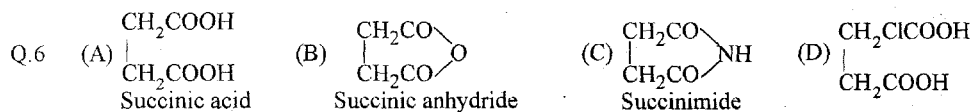
EXERCISE - II

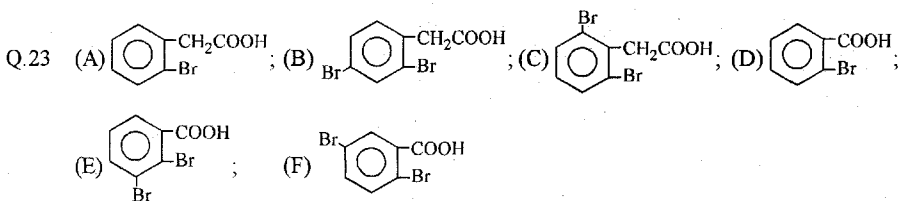
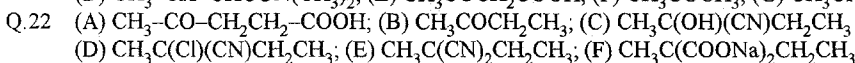
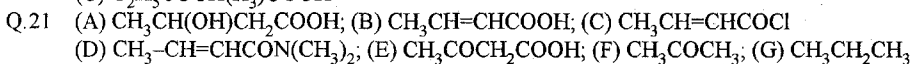
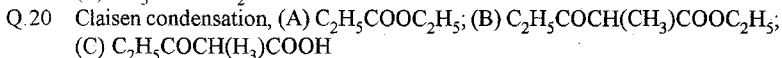
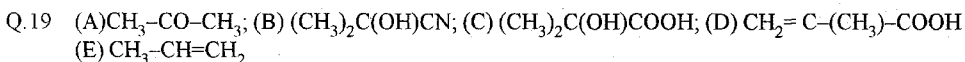
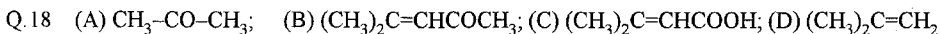
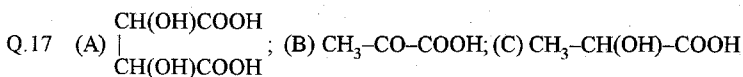
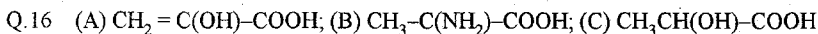
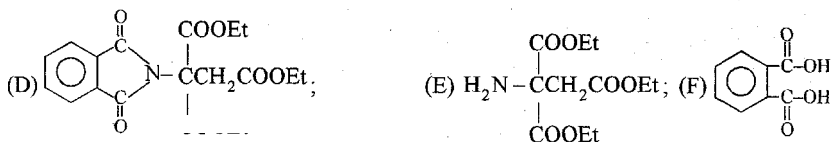
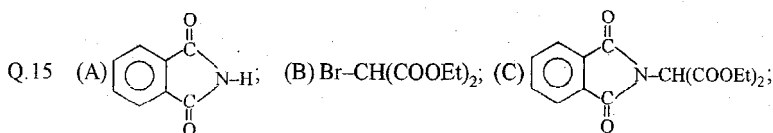
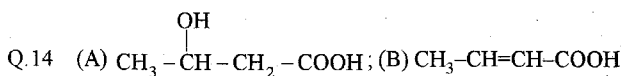
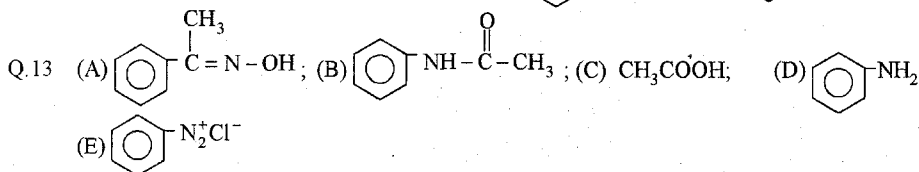
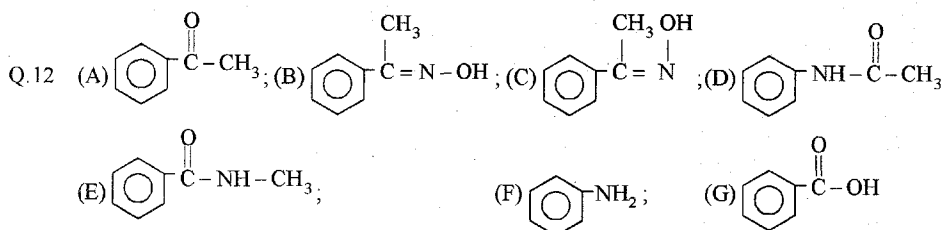
- Q.1 (A) $\text{CH}_3\text{CH}(\text{OOCCH}_3)_2$; (B) CH_3CHO ; (C) $(\text{CH}_3\text{CO})_2\text{O}$
 Q.2 (A) $(\text{CH}_3)_2\text{CHCOOCH}_2\text{CH}(\text{CH}_3)_2$; (B) $(\text{CH}_3)_2\text{CHCOCl}$; (C) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$
 Q.3 (A) RCONH_2 ; (B) RNH_2 ; (C) ROH
 Q.4 (A) CH_3Cl ; (B) CH_3OH ; (C) HCOOH ; (D) $\text{Ca}(\text{OH})_2$; (G) HCHO ; (H) CH_3Cl ; (I) KCN ; (J) CH_3CN ; (X) CH_3COOH
 Q.5 (A) CH_3COOH ; (B) CH_3COONa ; (C) CH_3COCl ; (E) $(\text{CH}_3\text{CO})_2\text{O}$
 Q.6 (A) $\text{HOOCCH}_2\text{CH}_2\text{COOMe}$; (B) $\text{ClOOCCH}_2\text{CH}_2\text{COOMe}$; (C) $\text{MeNHOCCH}_2\text{CH}_2\text{COOMe}$
 Q.7 (A) 2NH_3 ; (B) CH_3COCl ; (C) $\text{H}_2\text{SO}_4 + \text{SO}_3$





EXERCISE - III





EXERCISE - IV (A)

- Q.1 A, B Q.2 A Q.3 B Q.4 D Q.5 C Q.6 C Q.7 D
Q.8 A Q.9 C

EXERCISE - IV (B)

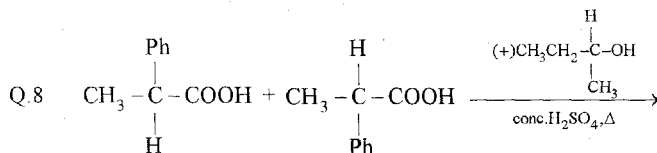
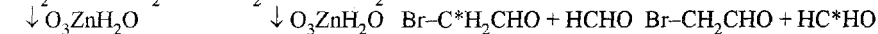
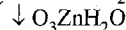
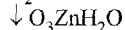
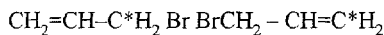
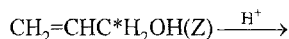
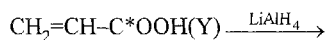
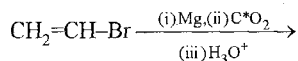
- Q.1 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$; (B) $\text{CH}_3\text{CH}_2\text{OH}$; (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$; (D) CH_3CHO ; (E) $\text{CH}_3\text{CH}=\text{CHCHO}$
Q.2 (A) SOCl_2 ; (B) POCl_3 ; (C) CH_3COCl ; (D) CH_3COCH_3
Q.3 (i) (A) $\text{CH}_3\text{CHBrCOOH}$; (B) $\text{CH}_2=\text{CHCOOH}$;
(ii) (A) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$; (B) $\text{CH}_3\text{COCH}_2\text{COOH}$
Q.4 (A) $(\text{CH}_3\text{CH}_2\text{CO})_2\text{O}$; (B) $\text{CF}_3\text{CH}_2\text{COOH}$; (C) $\text{CH}_3\text{CH}_2\text{COOH}$; (D) $(\text{CF}_3\text{CH}_2\text{CO})_2\text{O}$



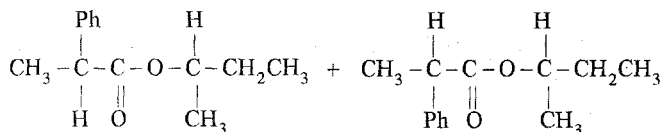
- Q.5 (A) $\text{NaCH}(\text{COC}_2\text{H}_5)_2$; (B) $\text{CH}_3-\text{HC}-\text{COOC}_2\text{H}_5$

- Q.6 (A) CH_3COOH ; (B) $\text{C}_2\text{H}_5\text{O}^{18}\text{H}$

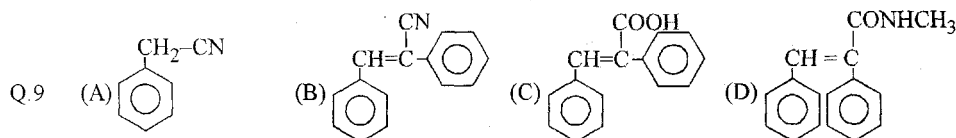
- Q.7 $\text{BaC}^*\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{C}^*\text{O}_2(\text{X}) + \text{BaSO}_4$



(racemic mixture)



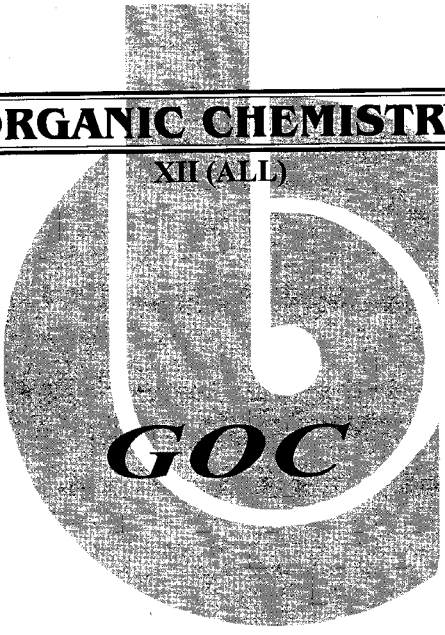
during esterification reaction only $-\text{COOH}$ and $-\text{OH}$ participates. There is no effect on structure or configuration of carbon adjacent to these groups. So when $(\pm)$ acid reacts with pure $(+)$ alcohol two esters are produced which are diastereoisomers of each other.





BANSALCLASSES

TARGET IIT JEE 2007



ORGANIC CHEMISTRY

XII (ALL)

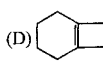
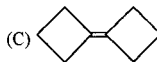
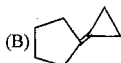
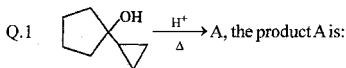
GOC

CONTENTS

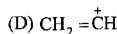
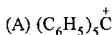
EXERCISE - I
EXERCISE - II
EXERCISE-III
EXERCISE IV
ANSWER KEY

EXERCISE-I

(Choose the correct option. Only one is correct)



Q.2 Which carbocation is least likely to form as intermediate?



Q.3 2-chloropentane on halogenation with chlorine gives 2,3-dichloropentane. What will be the structure of free radical species form in the reaction?

(A) Planar

(B) Trigonal planar

(C) Square planar

(D) Pyramidal

Q.4 What will be major product, when 2-methyl butane undergoes bromination in presence of light?

(A) 1-bromo-2-methyl butane

(B) 2-bromo-2-methyl butane

(C) 2-bromo-3-methyl butane

(D) 1-bromo-3-methyl butane

Q.5 The smallest compound, which on photochlorination produces diastereomers is

(A) 2-methyl, 3-chloro hexane

(B) 2-chloro butane

(C) 1-bromo propane

(D) 1-bromo-3-methyl butane

Q.6 Which of the following will give two isomeric monochloro derivatives?

(A) 2,3-dimethyl butane

(B) n-hexane

(C) Isopentane

(D) Butane

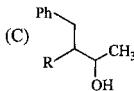
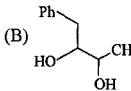
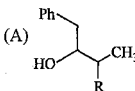
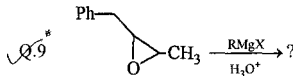
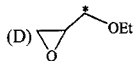
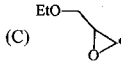
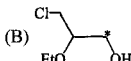
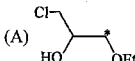
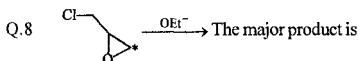
Q.7 How many isomers can be produced from the molecular formula $C_4H_{10}O$?

(A) 8

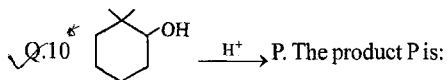
(B) 7

(C) 6

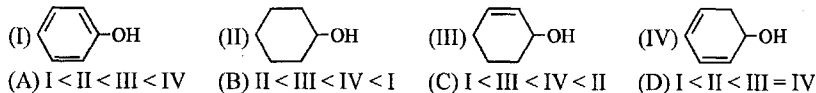
(D) 5



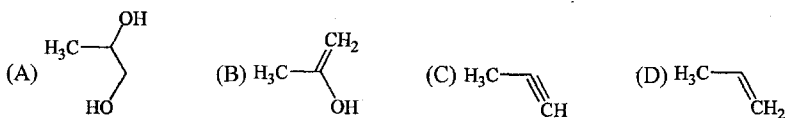
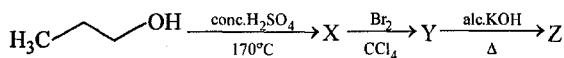
(D) None of these



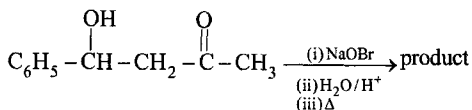
Q.11 Among the given compounds, the correct dehydration order is:



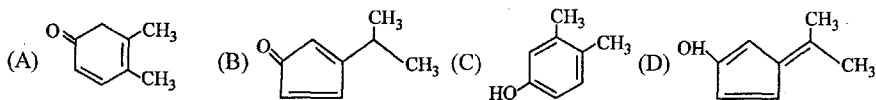
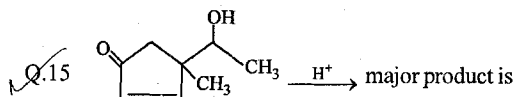
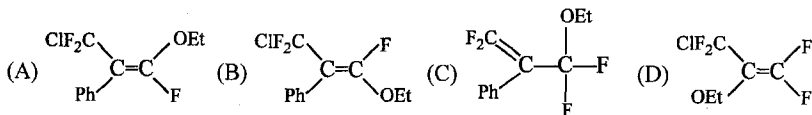
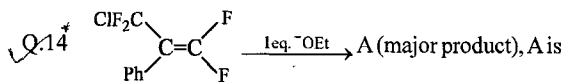
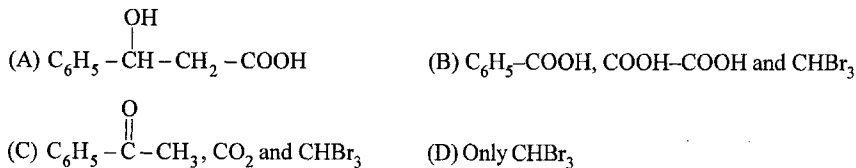
Q.12 Identify Z in the following series of reaction:



Q.13 In the reaction sequence

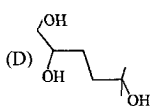
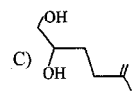
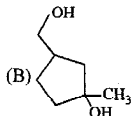
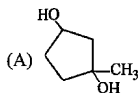
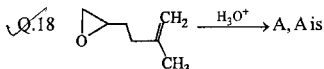
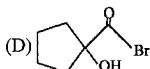
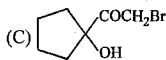
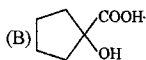
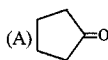
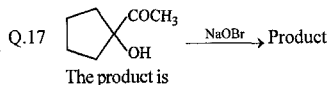


product will be

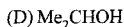
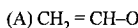


- ✓ Q.16 HBr reacts fastest with
(A) 2-methyl butane-2-ol
(C) Butane-1-ol

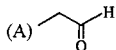
(B) 3-methyl butane-2-ol
(D) 2-methyl butane-1-ol



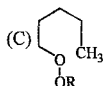
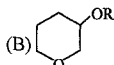
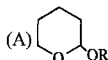
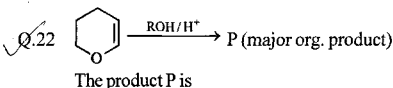
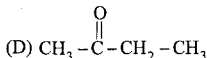
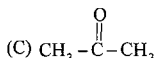
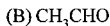
- ✓ Q.19 Which one of the following alcohol will undergo acid catalysed dehydration with greatest ease by E_1 mechanism?

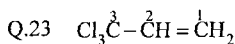


- ✓ Q.20 (A) $\text{C}_3\text{H}_6\text{O} \xrightarrow[\text{H}_2\text{SO}_4]{\text{NaCN}}$ cyanohydrin of (A). If the cyanohydrin is optically active, then the possible structure of (A) is

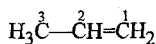


- ✓ Q.21 Which of the following has minimum reactivity towards nucleophiles?





(I)



(II)

In addition of HOBr to (I) and (II)

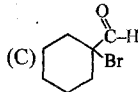
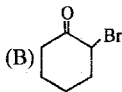
(A) Br is at C_2 in both cases

(C) Br is at C_1 in II and C_2 in I

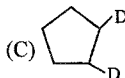
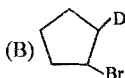
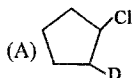
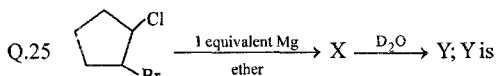
(B) Br is at C_2 in II and at C_1 in I

(D) Br is at C_1 in both cases

Q.24 Greatest amount of hydration is in



(D) All are equal



(D) none of these

Q.26 How many number of optically active isomers are possible for 2,3-dibromo butane 1,4-dioic acid?

(A) 2

(B) 3

(C) 4

(D) 1



Are which type of isomers

(A) Chain

(B) Position

(C) Metamerism

(D) functional

Q.28 Most stable conformation of ethylene glycol

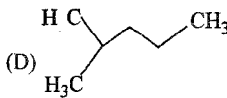
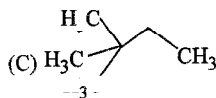
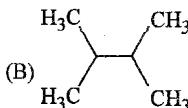
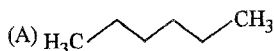
(A) Eclipsed form

(B) Staggered form

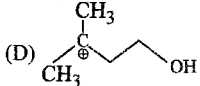
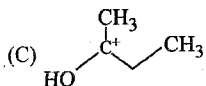
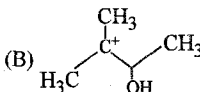
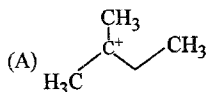
(C) Skew form

(D) None of these

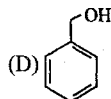
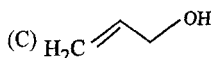
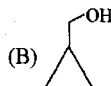
Q.29 Which has maximum B.P.?



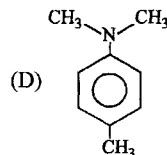
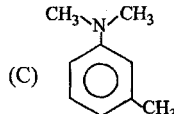
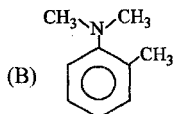
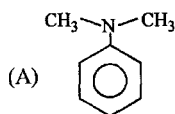
Q.30 Which of the following carbocation is most stable? (Back Bonding)



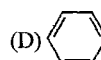
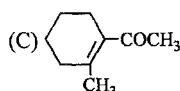
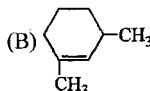
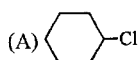
Q.31 Which of the following alcohol shows fastest reaction with HI?



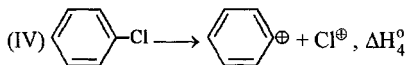
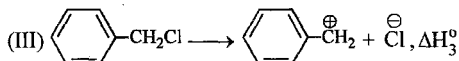
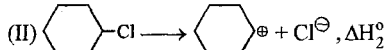
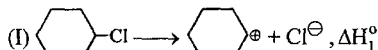
Q.32 Which of the following is strongest base. (C42)



Q.33 In which of the following molecule all the effect namely inductive, mesomeric & hyperconjugation operate



Q.34 For the reactions



The correct decreasing order of enthalpies of reaction for producing carbocation is

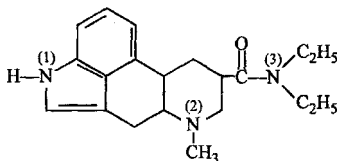
(A) $\Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ > \Delta H_4^\circ$

(B) $\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$

(C) $\Delta H_3^\circ > \Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ$

(D) $\Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ > \Delta H_3^\circ$

Q.35* Correct order of basicity of various nitrogen in LSD is



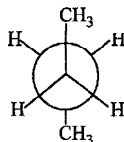
(A) $1 > 2 > 3$

(B) $2 > 1 > 3$

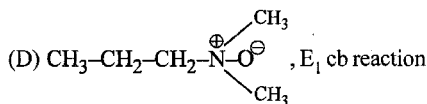
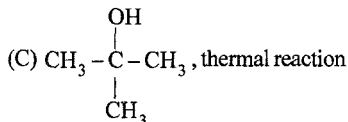
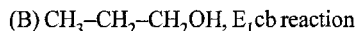
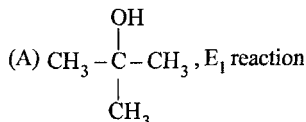
(C) $2 > 3 > 1$

(D) $3 > 2 > 1$

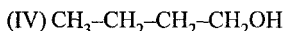
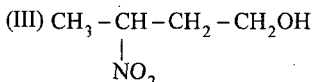
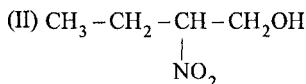
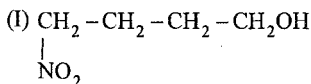
- Q.36 One of the configuration of n-butane is drawn in the given figure. Anticlockwise rotation of C_2 around C_2-C_3 bond by 120° will lead to



- (A) gauche (B) staggered (C) partially eclipsed (D) fully eclipsed
- Q.37 Which of following is correctly matched.



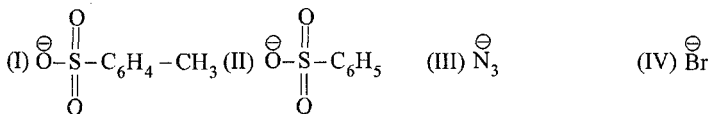
- Q.38 Arrange reactivity of given alcohol in increasing order of elimination reaction



Select answer from codes given below:

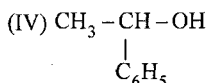
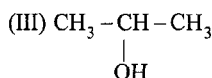
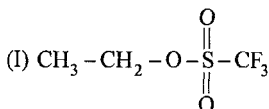
- (A) I, II, IV, III (B) I, II, III, IV (C) II, III, I, IV (D) II, III, IV, I

- Q.39* Arrange leaving power of $\text{Nu}^\ominus$ group in decreasing order

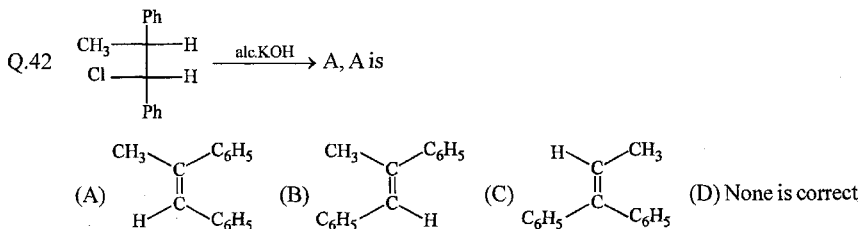
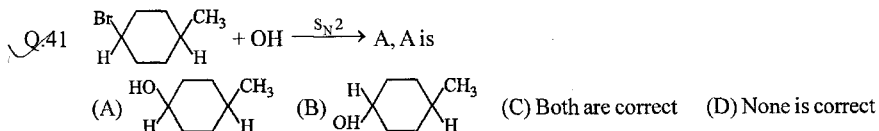


- (A) II, I, III, IV (B) II, I, IV, III (C) I, II, IV, III (D) I, II, III, IV

- Q.40* Arrange decreasing order of reactivity of these compounds for nucleophilic substitution reaction



- (A) III, IV, I, II (B) III, IV, II, I (C) I, II, III, IV (D) I, II, IV, III



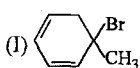
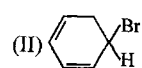
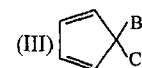
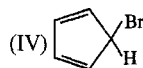
Q.43 Major product obtained in reaction of 1-phenyl-2-bromo butane with NaOMe.

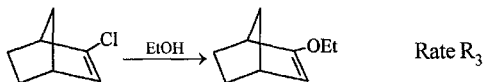
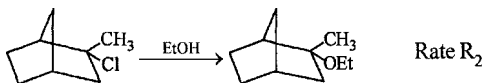
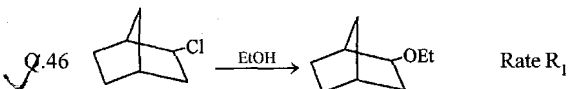
- (A) (E)-1-phenyl but-1-ene (B) (E)-1-phenyl but-2-ene
(C) 1-phenyl-1-ethoxy butane (D) (Z)-1-phenyl but-2-ene

Q.44 2-Aminocyclohexane-1-ol on treatment with HNO₂ produces

- (A) Cyclopentane carboxaldehyde (B) Cyclohexane-1,2-diol
(C) 2-amino cyclohexene (D) Cyclohex-2-enol

Q.45 Arrange the following in order of C-Br bond strength in polar solvent.

- (I)  (II)  (III)  (IV) 
- (A) I < II < III < IV (B) III < IV < I < II (C) IV < III < II < I (D) II < I < III < IV

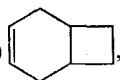
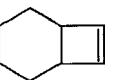


Which of following is correct :

- (A) R₁ > R₂ > R₃ (B) R₃ > R₂ > R₁ (C) R₂ > R₁ > R₃ (D) R₂ > R₃ > R₁

Q.47 Among the following pairs in which case the second compound has higher heat of hydrogenation.

- (A)  ,  (B) Cis-2-butene, trans-2-butene

- (C)  ,  (D) 1-butene, 2-butene

Q.48 (I)  Br, which is not correct about (I)

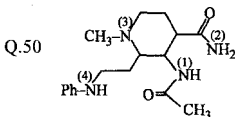
- (A) I is more soluble than bromocyclopropane
 (B) I gives pale yellow ppt. on addition with AgNO_3
 (C) I is having lower dipole moment than bromocyclopropane

(D) I have more ionic character than 

Q.49  $\text{C(CH}_3)_3 + \text{CH}_3\text{SNa} \xrightarrow{\text{C}_2\text{H}_5\text{OH}} ?$

The product formed is

- (A) $\text{CH}_3\text{SCH}_2-\underset{\text{OH}}{\text{CH}}-\text{C(CH}_3)_3$ (B) $(\text{CH}_3)_3\text{C}-\underset{\text{SCH}_3}{\text{CH}}-\text{CH}_2\text{OH}$
 (C) $\text{CH}_3\text{SCH}_2\underset{\text{OH}}{\text{CH}}-\text{C(CH}_3)_3$ (D) $(\text{CH}_3)_3\text{C}-\underset{\text{OH}}{\text{CH}_2}\text{CH}-\text{SCH}_3$



Correct order of basicity is

- (A) $3 > 1 > 2 > 4$ (B) $3 < 1 < 2 < 4$ (C) $3 < 4 < 1 < 2$ (D) $3 > 4 > 1 > 2$

Q.51 Which of the following statement about the reactivity of 1-chloro apocamphane toward AgNO_3 is true.

- (A) React by $\text{S}_{\text{N}}1$ mechanism (B) React by $\text{S}_{\text{N}}2$ mechanism
 (C) React by $\text{S}_{\text{N}}\text{i}$ mechanism (D) Does not react



Q.52 The resonance energy of following heterocycles is in the order

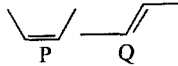
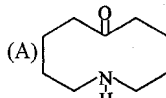
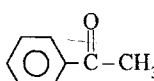
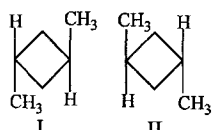
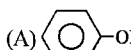
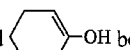
- (A) pyrrole > furan > pyridine (B) furan > pyrrole > pyridine
 (C) pyridine > pyrrole > furan (D) pyridine > furan > pyrrole

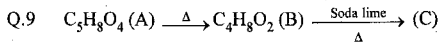
Q.53 Arrange the following halide in decreasing order of $\text{S}_{\text{N}}1$ reactivity.

- (I) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ (II) $\text{CH}_2=\underset{\text{Cl}}{\text{CH}}\text{CH}_2\text{CH}_3$ (III) $\text{CH}_3\text{CH}_2\underset{\text{Cl}}{\text{CH}}\text{CH}_3$
 (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{II} > \text{III} > \text{I}$ (D) $\text{III} > \text{II} > \text{I}$

EXERCISE-II

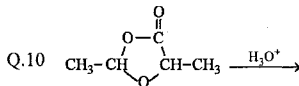
(Choose the correct option. One or more than one are correct)

- Q.1  which is/are incorrect statements?
 (A) P is cis- and Q is trans- (B) P is Z and Q is E
 (C) P is R and Q is S (D) P and Q are having same structural formula
- Q.2 $\text{CH}_3\text{CH}_2\text{NO}_2$ will not show..... isomerism:
 (A) Chain (B) functional (C) tautomerism (D) geometrical
- Q.3 $\begin{array}{c} \text{CHCOOH} \\ || \\ \text{CHCOOH} \end{array}$ (maleic acid) can form:
 (A) stereoisomer (B) geometrical isomer (C) position isomer (D) functional isomer
- Q.4 Which can show tautomerism ?
 (A)  (B) $\text{Ph}-\text{CH}_2-\text{NO}$ (C)  (D) $\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$
- Q.5 
 I and II are not
 (A) achiral-identical mirror images (B) chiral-enantiomers
 (C) geometrical isomers of each other (D) racemic mixture
- Q.6 In which of following compound chiral center is not affected on heating.
 (A) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{COOH}$ (B) 
 (C) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{COOH}$ (D) $\text{CH}_3-\underset{\text{CH}_2\text{COOH}}{\text{CH}}-\text{COOH}$
- Q.7 Which are correct statements?
 (A) methyl malonic acid is converted into propanoic acid on heating
 (B) succinic acid forms succinic anhydride on heating
 (C) 3-hydroxy propanoic acid forms Lactide on heating
 (D) $\text{CH}_3-\overset{\text{O}}{\underset{||}{\text{C}}}-\text{CH}_2\text{COOH}$ forms acetone on heating
- Q.8 Select incorrect statements:
 (A)  and  both give colour with neutral FeCl_3 solution
 (B) 2-pentanone and 3-pentanone are metamers
 (C) product formed when benzaldehyde reacts with hydroxylamine shows optical isomerism
 (D) 1,2-dibromocyclohexane shows geometrical and optical isomerism.



C is hydrocarbon occupying 0.509 litre per g approximately. Hence A and B are:

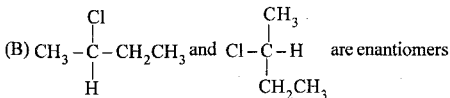
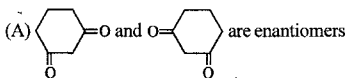
- (A) methyl malonic acid, propanoic acid (B) succinic acid, succinic anhydride
(C) Dimethylmalonic acid, 2-Methylpropanoic acid (D) Ethyl Malonic acid, Butanoic acid



products can be: (Both stable & unstable form)

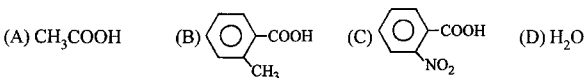
- (A) $CH_3CH(OH)_2$, $CH_3CH(OH)COOH$ (B) CH_3CHO , $CH_3CH(OH)COOH$
(C) CH_3CH_2COOH , CH_3CH_2OH (D) $CH_3CH(OH)COOH$, CH_3CH_2OH

Q.11 Which are incorrect statements:

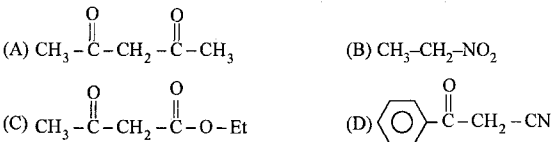


- (C) Argol (sodium potassium tartarate) will show optical isomerism.
(D) CH_3CHO on reaction with HCN gives racemic mixture.

Q.12 Which is/are less acidic than Benzoic acid?



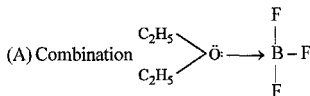
Q.13 Which have active methylene group?



Q.14 Which of the following are not resonating structures of each other:



Q.15 The unshared electron on the oxygen atom of an ether (basic centre) account for all the following except:

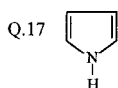
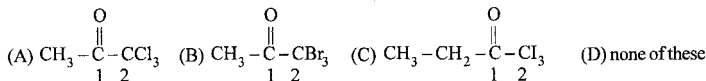


(B) Formation of oxonium salt with acids such as $\left[\text{C}_2\text{H}_5 - \text{O} - \text{C}_2\text{H}_5 \right]^+ \text{H}^- \text{Cl}^-$

(C) Formation of peroxides (explosive)

(D) Chemical inactivity of ether

Q.16 Which of the following compounds yield most stable carbanion after rupture of ($\text{C}_1 - \text{C}_2$) bond:



pyrrole (I)



pyridine (II)



aniline (III)

which is/are correct statements ?

(A) I is more basic than II

(C) III is more basic than II

(B) II is more basic than I and III

(D) all are aromatic bases

Q.18 Which is/are untrue statement(s)?

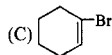
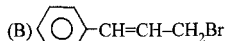
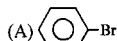
(A) Protonation increases electrophilic nature of carbonyl group

(B) CF_3SO_3^- is better leaving group than CH_3SO_3^-

(C) Benzyl carbonium ion is stabilised by resonance

(D) $\text{CCl}_3\text{CH} \begin{matrix} \text{OH} \\ \text{OH} \end{matrix}$ is unstable.

Q.19 Which reacts with AgNO_3 to give ppt.?



Q.20 Identify the correct statement(s).



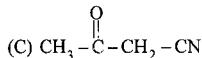
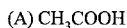
(A) I & III are antiaromatic

(C) II alone is aromatic

(B) III & IV are non-aromatic

(D) II & IV are non-aromatic

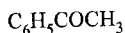
Q.21 Which have acidic hydrogen?



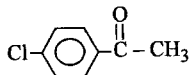
- Q.22 C-C and C=C bond lengths are unequal in:
 (A) benzene (B) 1,3 buta-di-ene
 (C) 1,3 cyclohexa-di-ene (D) None
- Q.23 Products formed when HCl adds to 2,4- hexadiene is:
 (A) 4-chloro-2-hexene (B) 2-chloro-3-hexene
 (C) 2-chloro-4-hexene (D) 1-chloro-2-hexene
- Q.24 Which of the following may be classed as an electrophilic reagent?
 (A) BCl_3 (B) NH_4^+ (C) CH_3^+ (D) AlCl_4^-
- Q.25 The acid strength of substituted carboxylic acids is known to be dependent on the nature and position of the substituent. In the following examples, an attempt has been made to arrange the acids in order of acid strength, the strongest first. One of the series is incorrect-which one?
 (A) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{COOH} > \text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH} > \text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (B) $\text{NO}_2\text{CH}_2\text{COOH} > \text{HOCH}_2\text{COOH} > \text{CH}_3\text{COOH}$
 (C) $\text{Cl}_3\text{CCOOH} > \text{BrCH}_2\text{COOH} > \text{FCH}_2\text{COOH}$
 (D) $\text{CH}_3\text{COOH} > \text{CH}_3\text{CH}_2\text{COOH} > (\text{CH}_3)_3\text{CCOOH}$
- Q.26 Which of the following would not have a dipole moment?
 (A) CCl_4 (B) CH_2Cl_2 (C) trans-1,2-dichloroethene (D) $\text{ClC}\equiv\text{CCl}$
- Q.27 Which of the following structures represent the less stable tautomeric form?
- (A) $\text{CH}_3\text{CH}_2\text{O}-\text{C}(\text{OH})=\text{CH}-\text{C}(=\text{O})\text{OCH}_2\text{CH}_3$
- (B)
- (C)
- (D)
- Q.28 Which of the following compounds cannot exhibit keto-enol tautomerism?
- (A)
- (B)
- (C)
- (D)
- Q.29 In tautomerism
 (A) a proton is moved around the molecule (B) electrons are moved around the molecule
 (C) no actual movement occurs (D) shift of double bond occurs
- Q.30 Which species is not aromatic?
- (A)
- (B)
- (C)
- (D)
- Q.31 Aromatic character is possible only when
 (A) six electrons are delocalized
 (B) the molecule is cyclic and planar having delocalized $(4n+2)\pi$ electrons where $n = 1, 2, 3, \dots$
 (C) alternate sigma (σ) & pie (π) bond are present.
 (D) None is true

EXERCISE-III

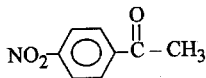
Q.1 Arrange the following compounds in decreasing order of K_{eq} for hydrate formation.



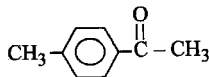
(1)



(2)



(3)



(4)

Q.2 Arrange the following compounds in order of:
Decreasing S_N1 reaction rate:



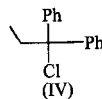
(I)



(II)

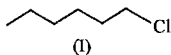


(III)



(IV)

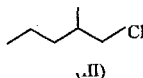
Q.3(a) Decreasing S_N2 reaction rate:



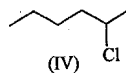
(I)



(II)

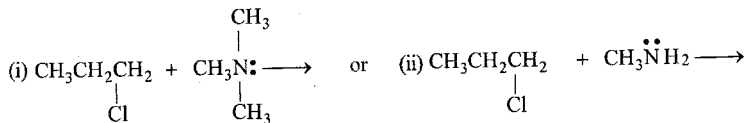


(III)

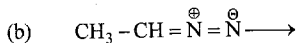
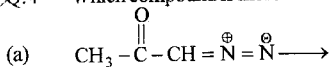


(IV)

(b) Explain which of these reactions would have the faster rate?

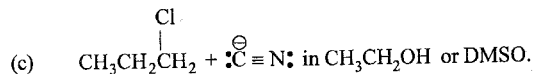
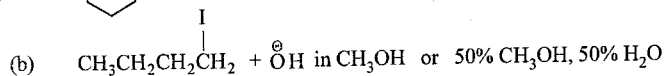
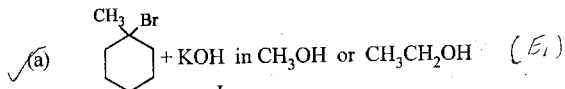


Q.4 Which compound is more stable explain

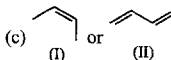
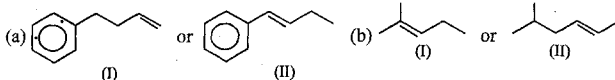


Q.5 Arrange the isomers of molecular formula C_4H_9Cl in order of decreasing rate of reaction with sodium iodide in acetone.

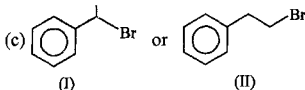
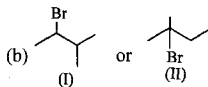
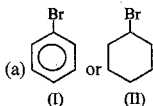
Q.6 Explain in which solvents these reactions are faster:



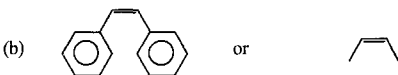
Q.7(i) Choose the member of the following pairs of unsaturated hydrocarbons that is more reactive towards acid-catalysed hydration and predict the regiochemistry of the alcohols formed from this compound.



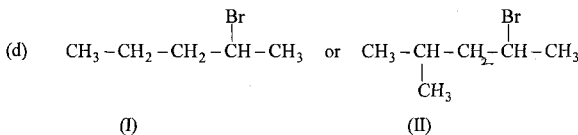
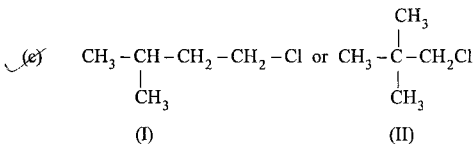
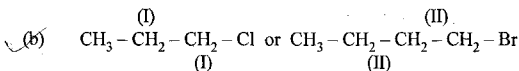
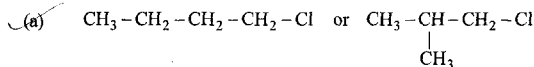
(ii) Predict on the basis of carbocation stability, which member of each of the following pairs hydrolyses at the faster rate.



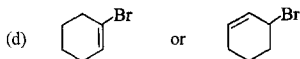
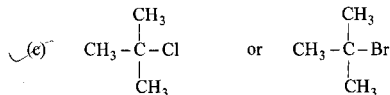
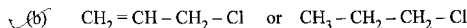
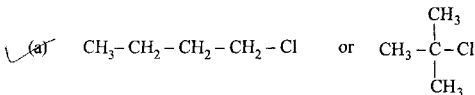
Q.8 Assuming that cation stability governs the barrier for protonation in H-X additions, predict which compound in each of the pairs in parts (a) to (b) will be more rapidly hydrochlorinated in a polar solvent.



Q.9 Select the member of each pair that shows faster rate of S_N2 reaction with KI in acetone.



Q.10 Select the member of each pair that undergoes S_N1 solvolysis in aqueous ethanol more rapidly.

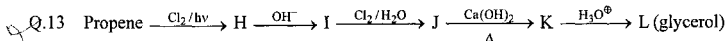


Q.11 In each of the following indicate which reaction will occur faster. Explain your reasoning.

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{I}$ with sodium cyanide in dimethyl sulfoxide.
- 1-chloro-2-methylbutane or 1-chloropentane with sodium iodide in acetone.
- Hexyl chloride or cyclohexyl chloride with sodium azide in DMF.
- Solvolysis of 1-bromo-2,2-dimethylpropane or tert-butyl bromide in ethanol.
- Reaction of 1-chlorobutane with sodium acetate in acetic acid or with sodium methoxide in methanol.
- Reaction of 1-chlorobutane with sodium azide or sodium p-toluenesulfonate in aqueous ethanol.

Q.12 Classify each of the following species as electrophile or nucleophile & explain your choice.

- CH_3CO_2^-
- CH_3O^-
- BF_3
- $\dot{\text{C}}\text{I}$
- $\bar{\text{C}}\text{Me}_3$
- $:\text{CCl}_2$
- SiF_4
- Br^-
- $:\text{P}(\text{CH}_3)_3$
- $\text{Me}_2\dot{\text{C}}\text{H}$
- $\text{CH}_2=\text{CH}_2$
- HCl



Q.14 Of the following statements, which are true for S_N2 reaction.

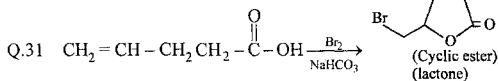
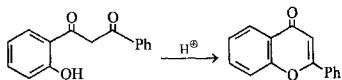
- Tertiary alkyl halides react faster than secondary.
- The absolute configuration of product is opposite to that of the reactant when an optically active substrate is used.
- The reaction shows first order kinetics.
- The rate of the reaction depends markedly on the nucleophilicity of the attacking reagent.
- The mechanism is one step.
- Carbocations are intermediate.
- $\text{Rate} \propto [\text{Alkyl halides}]$
- The rate of the reaction depends on the nature of the leaving group.

Q.15 Of the following statements which are true for S_N1 reaction.

- Tertiary alkyl halides react faster than secondary.
- The absolute configuration of the product is opposite to that of the reactant when an optical active substrate is used.
- The reaction shows first order kinetics.
- The rate of reaction depends markedly on the nucleophilicity of the nucleophile.
- The mechanism is two step.
- Carbocations are intermediate.
- $\text{Rate} \propto [\text{Alkyl halides}]$
- The rate of the reaction depends on the nature of the leaving group.

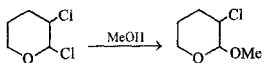
- Q.16 Explain why $\text{Cl}-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$ is hydrolysed more rapidly than $\text{CH}_3-\text{O}-\text{CH}_2-\text{CH}_2-\text{Cl}$
- Q.17 If the temperature is not kept below 25°C during the reaction of primary alcohols with p-toluenesulfonyl chloride in pyridine, it is sometimes observed that the isolated product is not the desired alkyl p-toluenesulfonate but is instead the corresponding alkyl chloride. Suggest a mechanistic explanation for this observation.
- Q.18 Because the $\text{S}_{\text{N}}1$ reaction goes through a flat carbocation, we might expect an optically active starting material to give a completely racemized product. In most cases, however, $\text{S}_{\text{N}}1$ reactions actually give more of the inversion product. In general, as the stability of the carbocation increases, the excess inversion product decreases. Extremely stable carbocations give completely racemic products. Explain these observations.
- Q.19 *cis*-1-Bromo-2-*tert*-butylcyclohexane and *trans*-1-bromo-2-*tert*-butylcyclohexane both react with sodium ethoxide in ethanol to give 1-*tert*-butylcyclohexene. The *cis* isomer reacts much more rapidly than the *trans* isomer. Explain.
- Q.20 The reaction of an amine with an alkyl halide gives an ammonium salt.
- $$\text{R}_3\text{N:} + \text{R}'\text{-X} \longrightarrow \text{R}_3\text{N}^+ - \text{R}'\text{X}^-$$
- amine alkyl halide ammonium salt
- The rate of this $\text{S}_{\text{N}}2$ reaction is sensitive to the polarity of the solvent. Draw an energy diagram for this reaction in a nonpolar solvent and another in a polar solvent. Consider the nature of the transition state, and explain why this reaction should be sensitive to the polarity of the solvent. Predict whether it will be faster or slower in a more polar solvent.
- Q.21 There is an overall 29-fold difference in reactivity of 1-chlorohexane, 2-chlorohexane towards potassium iodide in acetone.
- (a) Which one is the most reactive? Why?
- (b) Two of the isomers differ by only a factor of 2 in reactivity. Which two are these? Which one is the more reactive? Why?
- Q.22 The compound KSCN is a source of thiocyanate ion.
- Write the two most stable Lewis structures for thiocyanate ion and identify the atom in each that bears a formal charge of -1.
- Q.23 With the help of following data show HBr exhibits the peroxide effect.
- $$\text{H-X} \quad \overset{\Delta H_1^\circ}{\text{X}^\bullet + \text{CH}_2 = \text{CH}_2 \rightarrow \text{XCH}_2 - \dot{\text{C}}\text{H}_2} \quad \overset{\Delta H_2^\circ}{\text{XCH}_2 - \dot{\text{C}}\text{H}_2 + \text{H-X} \rightarrow \text{XCH}_2\text{CH}_3 + \text{X}^\bullet}$$
- | | | |
|-----|-------|--------|
| HCl | -67 | + 12.6 |
| HBr | -25.1 | - 50.2 |
| HI | +46 | -117.1 |
- All are in kJ mole^{-1} .
- Q.24 CHCl_2F is found to react more slowly than CHCl_3 in alkaline hydrolysis with plausible mechanism. Illustrate this phenomenon.
- displacement reaction.*
- Q.25 Formulate the reactions between but-1-ene in presence of small amount of benzoyl peroxide & (i) CCl_4 (ii) CBrCl_3 . Give your reasons.
- Q.26 p-Amino benzaldehyde behaves like an amide as is evident by the fact that it is not readily attacked by the nucleophile at the carbonyl carbon. Explain.
- Q.27 When CH_3MgBr react with 5-chloro-2-pentanone, cyclic ether is formed instead of alcohol. Explain.
- Q.28 The dichlorocarbene reacts with electron rich species like phenol where as it doesn't react with benzene explain.

- Q.29 Explain the following:
- (a) 3-pentanol reacts with HBr to give a mixture of 3 & 2-bromopentane. The exact composition of the mixture depends upon whether gaseous or aqueous HBr is used.
- (b) Hydration of 3-phenyl-1-butene in dil H_2SO_4 is not a satisfactory method for preparing 3-phenyl-2-butanol
- Q.30 Why is elimination preferred to hemiacetal formation in the acid catalysed cyclisation of this ketone?

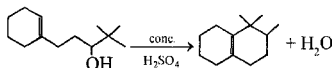


This type of reaction is called "bromolactonisation". Suggest a mechanism.

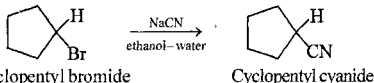
- Q.32 Draw mechanisms for these reactions, explaining why these particular products are formed.



- Q.33 Write a reasonable and detailed mechanism for the following transformation.

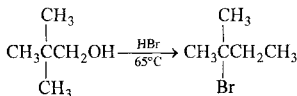


- Q.34 The reaction of cyclopentyl bromide with sodium cyanide to give cyclopentyl cyanide



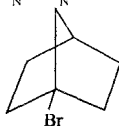
proceeds faster if a small amount of sodium iodide is added to the reaction mixture. Can you suggest a reasonable mechanism to explain the catalytic function of sodium iodide?

- Q.35 The reaction of 2,2-dimethyl-1-propanol with HBr is very slow and gives 2-bromo-2-methyl-propane as the major product.



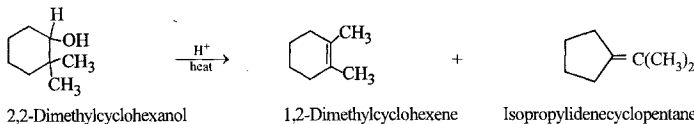
Give a mechanistic explanation for these observations.

- Q.36 1-Bromobicyclo [2.2.1] heptane (the structure of which is shown) is exceedingly unreactive towards nucleophilic substitution by either the $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ mechanism.



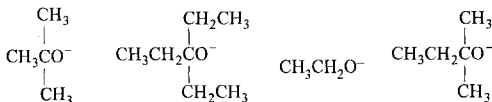
1-Bromobicyclo [2.2.1] heptane

- Q.37 We saw that acid-catalyzed dehydration of 2,2-dimethyl-cyclohexanol afforded 1,2-dimethylcyclohexene. To explain this product we must write a mechanism for the reaction in which a methyl shift transforms a secondary carbocation to a tertiary one. Another product of the dehydration of 2,2-dimethylcyclohexanol is isopropylidenecyclopentane. Write a mechanism to rationalize its formation.



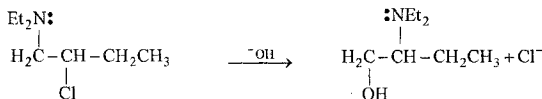
- Q.38 When 2-bromo-2,3-dimethylbutane reacts with a base under E2 conditions, two alkenes (2,3-dimethyl-1-butene and 2,3-dimethyl-2-butene) are formed.

- (a) Which of the following bases would give the highest percentage of the 1-alkene ?
 (b) Which would give the highest percentage of the 2-alkene "



- Q.39 Dr. Don T. Doit wanted to synthesize the anesthetic 2-ethoxy-2-methylpropane. He used ethoxide ion and 2-chloro-2-methylpropane for his synthesis and ended up with very little ether. What was the predominant product of his synthesis ? What reagent should he have used ?

- Q.40 The following reaction takes place under second-order conditions (strong nucleophile), yet the structure of the product shows rearrangement. Also, the rate of this reaction is several thousand times faster than the rate of substitution of hydroxide ion on 2-chlorobutane under similar conditions. Propose a mechanism to explain the enhanced rate and rearrangement observed in this unusual reaction. ("Et" is the abbreviation for ethyl.)

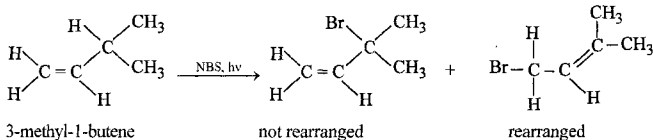


- Q.41 When ethyl bromide is added to potassium t-butoxide, the product is ethyl t-butyl ether. $\text{CH}_3\text{CH}_2-\text{Br} + (\text{CH}_3)_3\text{C}-\text{O}^-\text{K}^+ \longrightarrow (\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_2\text{CH}_3$

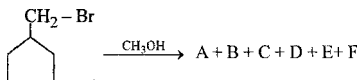
ethyl bromide potassium t-butoxide ethyl t-butyl ether

- (a) What happens to the reaction rate if the concentration of ethyl bromide is doubled ?
 (b) What happens to the rate if the concentration of potassium t-butoxide is tripled and the concentration of ethyl bromide is doubled ?
 (c) What happens to the rate if the temperature is raised ?

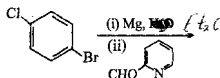
- Q.42 Give a mechanism to explain the two products formed in the following reaction.



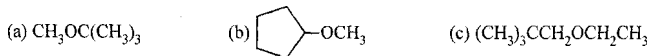
- Q.43 Give the structures of all possible products when 2-chloro-2-methylbutane reacts by the E1 mechanism.
- Q.44 Solvolysis of 1,2-dimethylpropyl p-toluenesulfonate in acetic acid (75°C) yields five different products : three are alkenes and two are substitution products. Suggest reasonable structures for these five products.
- Q.45 Solution A was prepared by dissolving potassium acetate in methanol. Solution B was prepared by adding potassium methoxide to acetic acid. Reaction of methyl iodide either with solution A or with solution B gave the same major product. Why ? What was this product?
- Q.46 Solvolysis of 2-bromo-2-methylbutane in acetic acid containing potassium acetate gave three products. Identify them.
- Q.47
- Write the structures or build molecular models of all the isomeric alkyl bromides having the molecular formula $C_5H_{11}Br$.
 - Which one undergoes E1 elimination at the fastest rate ?
 - Which one is incapable of reacting by the E2 mechanism ?
 - Which one can yield only a single alkene on E2 elimination ?
 - For which isomers does E2 elimination give two alkenes that are not constitutional isomers ?
 - Which one yields the most complex mixture of alkenes on E2 elimination ?
- Q.48 Evidence has been reported in the chemical literature that the reaction $(CH_3CH_2)_2CHCH_2Br + KNH_2 \longrightarrow (CH_3CH_2)_2C=CH_2 + NH_3 + KBr$ proceeds by the E2 mechanism. Use curved arrow notation to represent the flow of electrons for this process.
- Q.49 The given compound give mixture of six products on treatment with CH_3OH . Write structures of all five products and give mechanism of the reaction.



- Q.50 Comment on the selectivity, (that is, say what else might have happened & why it didn't) shown in this grignard addition reaction used in the manufacture of an antihistamine drug, A.



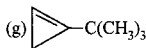
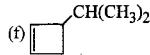
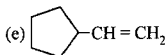
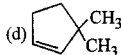
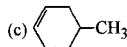
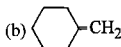
- Q.51 Outline an efficient synthesis of each of the following compounds from the indicated starting material and any necessary organic or inorganic reagents :
- Cyclopentyl cyanide from cyclopentane.
 - Cyclopentyl cyanide from cyclopentene.
 - Cyclopentyl cyanide from cyclopentanol.
 - Isobutyl iodide from isobutyl chloride.
 - Isobutyl iodide from tert-butyl chloride.
 - Isobutyl azide from isopropyl alcohol
- Q.52 Select the combination of alkyl bromide and potassium alkoxide that would be the most effective in the synthesis of the following ethers :



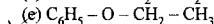
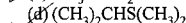
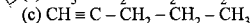
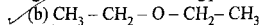
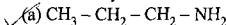
Q.53 Indicate the major alkene produced by each of the following

- (a) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\text{Br}}{\underset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}}-\text{CH}_3 \xrightarrow{\text{CH}_3-\overset{\oplus}{\text{O}}\text{K}^\oplus/\Delta}$
- (b) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\oplus}{\underset{\text{CH}_3}{\text{C}}}(\text{N}^\oplus(\text{CH}_3)_3)\overset{\oplus}{\text{O}}\text{H} \xrightarrow{\Delta}$
- (c) $(\text{CH}_3)_3\overset{\oplus}{\text{N}}-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2-\text{CH}_3 \overset{\oplus}{\text{O}}\text{H} \xrightarrow{\Delta}$
- (d) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\text{F}}{\text{CH}}-\text{CH}_3 \xrightarrow[(\Delta)]{(\text{CH}_3)_3\text{C}\overset{\oplus}{\text{O}}\text{K}^\oplus}$

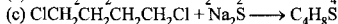
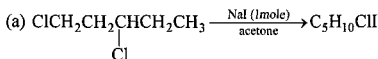
Q.54 Choose the compound of molecular formula $\text{C}_7\text{H}_{13}\text{Br}$ that gives each alkene shown as the exclusive product of E2 elimination.



Q.55 Select the alkyl halide and a nucleophile that will give each of the following products.



Q.56 Identify the product in each of the following reactions :



Q.57 Write the structure of the principal organic product to be expected from the reaction of 1-bromopropane with each of the following :

(a) Sodium iodide in acetone.

(b) Sodium acetate ($\text{CH}_3\overset{\text{O}}{\parallel}\text{CONa}$) in acetic acid.

(c) Sodium ethoxide in ethanol.

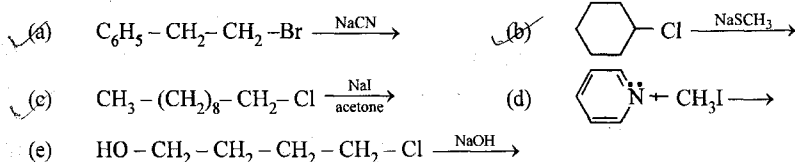
(d) Sodium cyanide in dimethyl sulfoxide.

(e) Sodium azide in aqueous ethanol

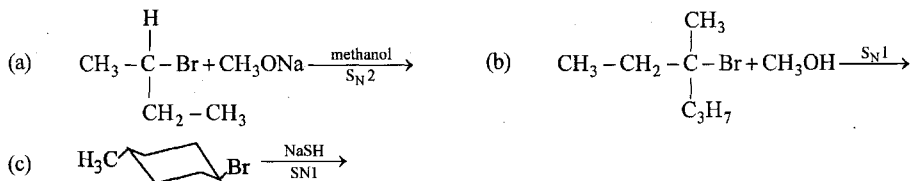
(f) Sodium hydrogen sulfide in ethanol

(g) Sodium methanethiolate (NaSCH_3) in ethanol.

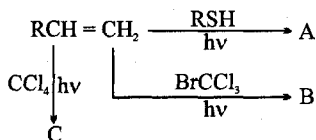
Q.58 Predict the products of the following S_N2 reaction :



Q.59 Draw each of the following reaction in a way that clearly show the stereoisomerism of the products.



Q.60 What will be the products (A, B, C) in the following reaction?

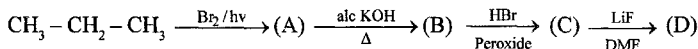


Give a mechanism for each of the reactions.

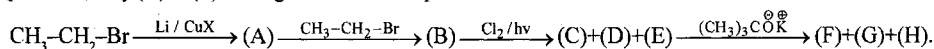
Q.61 Complete the following reactions. Give all possible products in each case. Give mechanism of the reactions.



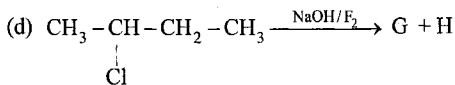
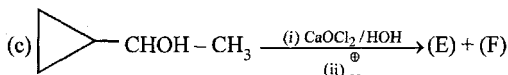
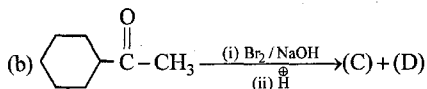
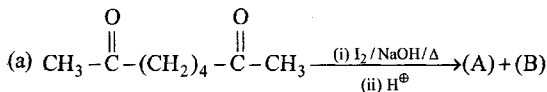
Q.62 Identify (A) to (D) in the given reaction sequence.



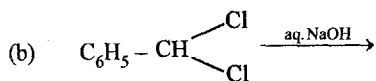
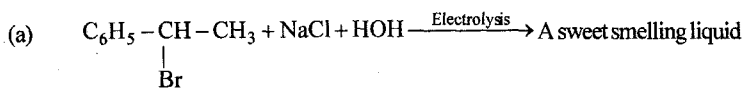
Q.63 Identify (A) to (E) in the given reaction sequence



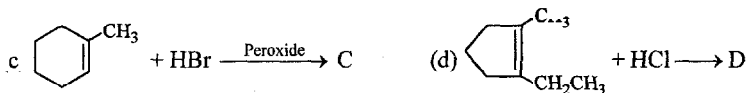
Q.64 Complete the following reactions :



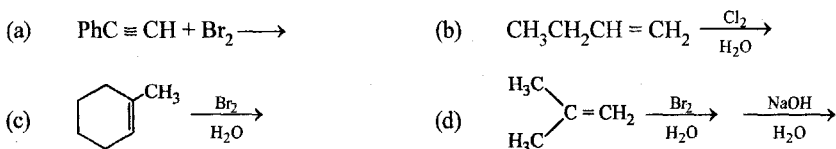
Q.65 Complete the following reactions



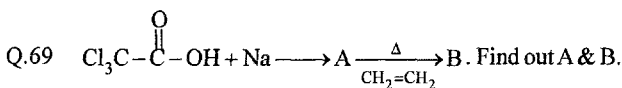
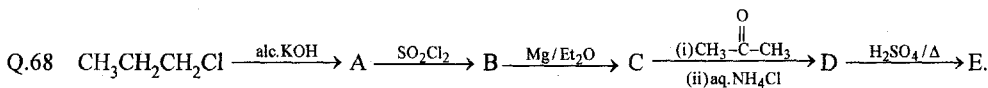
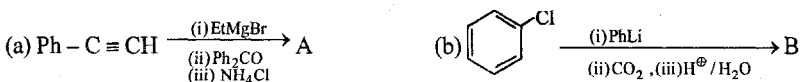
Q.66(i) What are the products of the following reactions ?



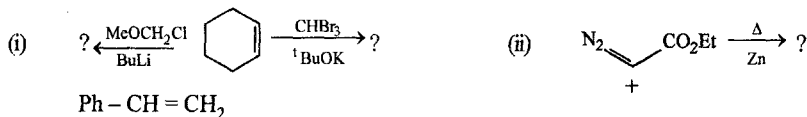
(ii) What are the products of the following reactions ?



Q.67 What products would be formed in these reactions ?



Q.70 Give product and suggest mechanism for these reactions.



EXERCISE-IV (A)

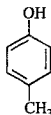
- Q.1 Reaction of $R-CO-NH_2$ with a mixture of Br_2 and KOH gives $R-NH_2$ as the main product. The intermediates involved in this reaction are: [JEE 1992]
 (A) $R-CO-NHBr$ (B) $RNHBBr$
 (C) $R-N=C=O$ (D) $R.CO.NBr_2$

- Q.2 Which one of the following has the smallest heat of hydrogenation per mole? [JEE 1993]
 (A) 1-Butene (B) trans-2-Butene
 (C) cis-2-Butene (D) 1,3-Butadiene

- Q.3 In the following compounds: [JEE 1996]



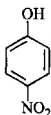
(I)



(II)



(III)

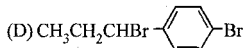
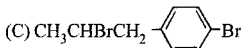
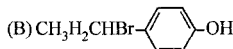
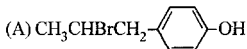


(IV)

The order of acidity is:

- (A) III > IV > I > II (B) I > IV > III > II (C) II > I > III > IV (D) IV > III > I > II

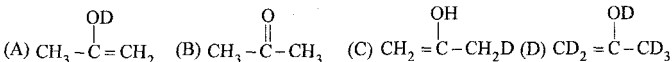
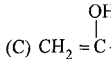
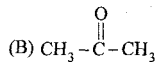
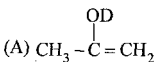
- Q.4 The reaction of $CH_3-CH=CH-$ $-OH$ with HBr gives: [JEE 1998]



- Q.5 A solution of (+) 1-chloro-1-phenylethane in toluene racemizes slowly in the presence of small amount of $SbCl_5$ due to formation of: [JEE 1999]
 (A) carbanion (B) carbene (C) free radical (D) carbocation

- Q.6 An aromatic molecule will: [JEE 1999]
 (A) have $4n$ π electrons (B) have $(4n+2)$ π electrons
 (C) be planar (D) be cyclic

- Q.7 The enol form of acetone, after Prolonged treatment with D_2O , gives: [JEE 1999]



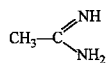
- Q.8 Amongst the following, the most basic compound is: [JEE 2000]
 (A) $C_6H_5NH_2$ (B) $p-NO_2-C_6H_4NH_2$ (C) $m-NO_2-C_6H_4NH_2$ (D) $C_6H_5CH_2NH_2$

- Q.9 An SN^2 reaction at an assymetrical carbon of a compound always gives: [JEE 2001]
 (A) enantiomer of the substrate (B) a product with opposite optical rotation
 (C) a mixture of diastereomers (D) a single stereo isomers

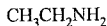
Q.10

The correct order of basicities of the following compounds is:

[JEE 2001]



(1)



(2)



(3)



(4)

(A) $2 > 1 > 3 > 4$ (B) $1 > 3 > 2 > 4$ (C) $3 > 1 > 2 > 4$

(D) 1

Q.11 Identify the correct order of reactivity in electrophilic substitution reaction of the following compounds:



(1)



(2)



(3)



(4)

(A) $1 > 2 > 3 > 4$ (B) $4 > 3 > 2 > 1$ (C) $2 > 1 > 3 > 4$ (D) $2 > 3 > 1 > 4$ [JEE 2002]Q.12 Left to right sp^2 , sp^2 , sp , sp hybridization is present in:

[JEE 2003]

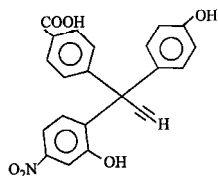
(A) $\text{H}_2\text{C}=\text{CH}-\text{C}\equiv\text{N}$ (B) $\text{H}_2\text{C}=\text{C}=\text{CH}-\text{CH}_3$ (C) $\text{HC}\equiv\text{C}-\text{C}\equiv\text{CH}$ (D) $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}_2$

Q.13 Maximum dipole moment will be of:

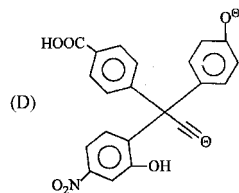
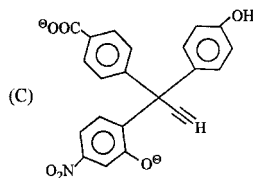
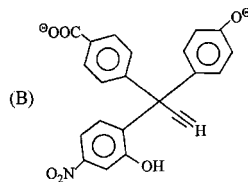
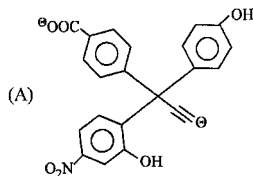
[JEE 2003]

(A) CCl_4 (B) CHCl_3 (C) CH_2Cl_2 (D) CH_3Cl

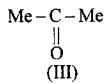
Q.14

when X is made to react with 2 eq. of NaNH_2 the product formed will be:

[JEE 2003]



Q.15 Order of rate of reaction of following compound with phenyl magnesium bromide is: [JEE 2004]



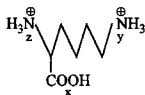
(A) I > II > III

(B) II > III > I

(C) III > I > II

(D) II > I > III

Q.16



[JEE 2004]

Correct order of acidic strength is:

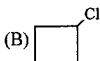
(A) $x > y > z$

(B) $z > y > x$

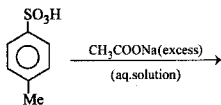
(C) $y > z > x$

(D) $x > z > y$

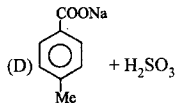
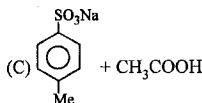
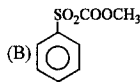
Q.17 1-Bromo-3-chloro cyclobutene on reaction with 2-equivalent of sodium in ether gives [JEE 2005]



Q.18



[JEE 2005]



Q.19 Conversion of cyclohexanol into cyclohexene is most effective in [JEE 2005]

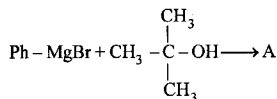
(A) concentrated H₃PO₄

(B) concentrated HCl

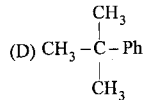
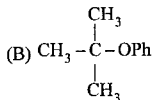
(C) concentrated HCl / ZnCl₂

(D) concentrated HBr

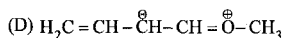
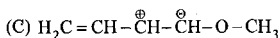
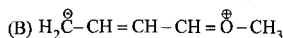
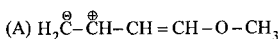
Q.20



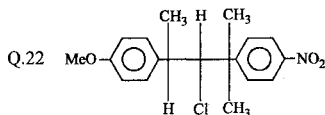
[JEE 2005]



Q.21 For 1-methoxy-1,3-butadiene, which of the following resonating structure is the least stable?

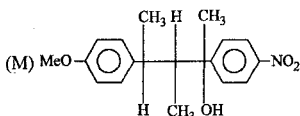
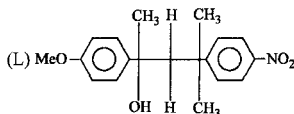
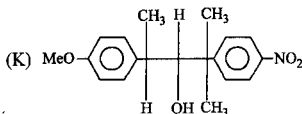


[JEE 2005]



[JEE 2005]

on hydrolysis in presence of acetone



(A) K & L

(B) only

(C) M only

(D) K & M

Q.23 When benzene sulfonic acid and p-nitrophenol are treated with NaHCO_3 , the gases released respectively are

[JEE 2006]

(A) SO_2 , NO_2

(B) SO_2 , NO

(C) SO_2 , CO_2

(D) CO_2 , CO_2

Q.24 $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + \text{KOH} \rightarrow$ Nitrogen containing compound + $\text{KCl} + \text{H}_2\text{O}$. Nitrogen containing compound is

[JEE 2006]

(A) $\text{CH}_3-\text{C}\equiv\text{N}$

(B) $\text{CH}_3-\text{NH}-\text{CH}_3$

(C) $\text{CH}_3-\text{N}^+=\text{C}^+$

(D) $\text{CH}_3-\text{N}^+\equiv\text{C}^-$

Q.25 (I) 1,2-dihydroxy benzene

(II) 1,3-dihydroxy benzene

[JEE 2006]

(III) 1,4-dihydroxy benzene

(IV) Hydroxy benzene

The increasing order of boiling points of above mentioned alcohols is

(A) $\text{I} < \text{II} < \text{III} < \text{IV}$

(B) $\text{I} < \text{II} < \text{IV} < \text{III}$

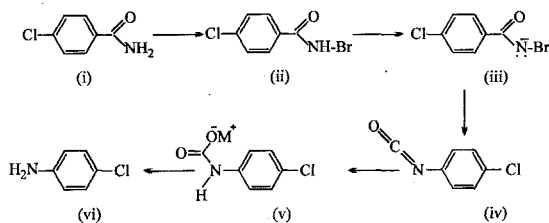
(C) $\text{IV} < \text{I} < \text{II} < \text{III}$

(D) $\text{IV} < \text{II} < \text{I} < \text{III}$

Question No. 26 to 28 (3 questions)

Comprehension I

$RCONH_2$ is converted into RNH_2 by means of Hofmann bromamide degradation.

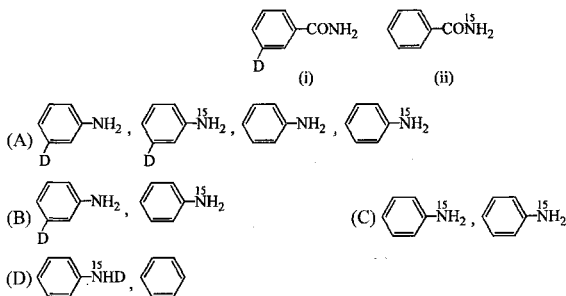


In this reaction, $RCONHBr$ is formed from which this reaction has derived its name. Electron donating group at phenyl activates the reaction. Hofmann degradation reaction is an intramolecular reaction.

- Q.26 How can the conversion of (i) to (ii) be brought about? [JEE 2006]
 (A) KBr (B) $KBr + CH_3ONa$ (C) $KBr + KOH$ (D) $Br_2 + KOH$

- Q.27 Which is the rate determining step in Hofmann bromamide degradation? [JEE 2006]
 (A) Formation of (i) (B) Formation of (ii) (C) Formation of (iii) (D) Formation of (iv)

- Q.28 What are the constituent amines formed when the mixture of (i) and (ii) undergoes Hofmann bromamide degradation?

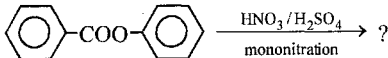


[JEE 2006]

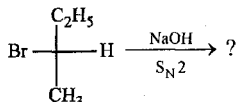
EXERCISE-IV (B)

Q.1 $\text{CH}_3\text{CH}_2\text{CHCl}_2 \xrightarrow{\text{aq.KOH}} ?$ [JEE 1991]

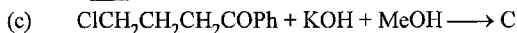
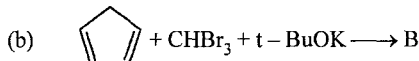
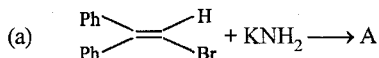
Q.2 $\text{C}_6\text{H}_6 + (\text{CH}_3)_2\text{CHCH}_2\text{Cl} \xrightarrow{\text{AlCl}_3} ?$ [JEE 1992]

Q.3  [JEE 1993]

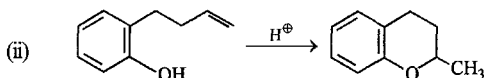
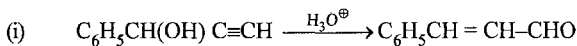
Q.4 Draw the stereochemical structure of the products in the following reaction: [JEE 1994]



Q.5 Complete the following, giving the structures of the principal organic products: [JEE 1997]



Q.6 Write the intermediate steps for each of the following reaction. [JEE 1998]

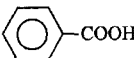
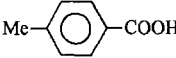
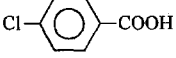
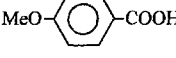
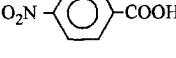


Q.7 Each of the following reaction gives two products. Write the structure of the products. [JEE 1998]



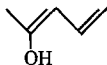
Q.8 Out of anhydrous AlCl_3 and hydrous AlCl_3 which is more soluble in diethyl ether? Explain with reason. [JEE 2003]

Q.9 Match K_a values with suitable acid: [JEE 2003]

K_a	Acid
(i) 3.3×10^{-5}	(a) 
(ii) 4.2×10^{-5}	(b) 
(iii) 6.3×10^{-5}	(c) 
(iv) 6.4×10^{-5}	(d) 
(v) 30.6×10^{-5}	(e) 

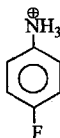
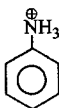
Q.10 Give resonating structures of following compound.

[JEE 2003]



Q.11 Which of the following is more acidic and why?

[JEE 2004]



Q.12 Predict whether the following molecules are iso structural or not. Justify your answer.

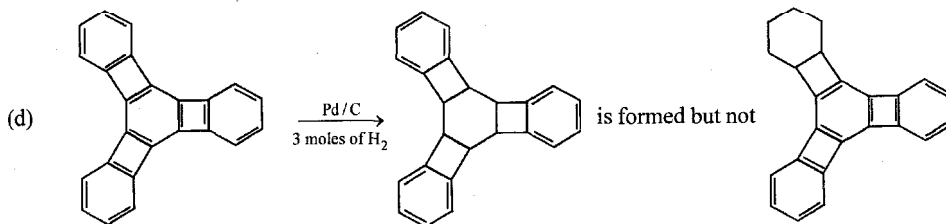
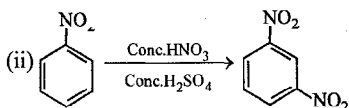
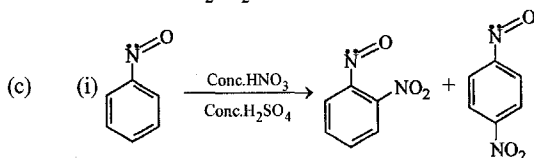
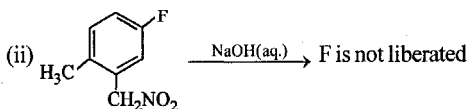
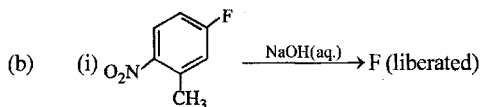
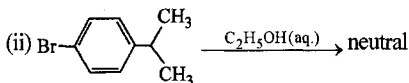
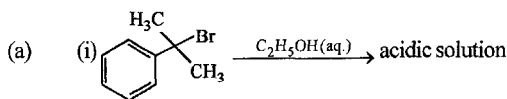
[JEE 2005]

(i) NMe_3

(ii) $\text{N}(\text{SiMe}_3)_3$

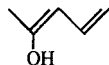
Q.13 Give reasons:

[JEE 2005]



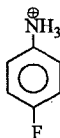
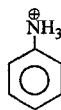
Q.10 Give resonating structures of following compound.

[JEE 2003]



Q.11 Which of the following is more acidic and why?

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Q.12 Predict whether the following molecules are iso structural or not. Justify your answer.

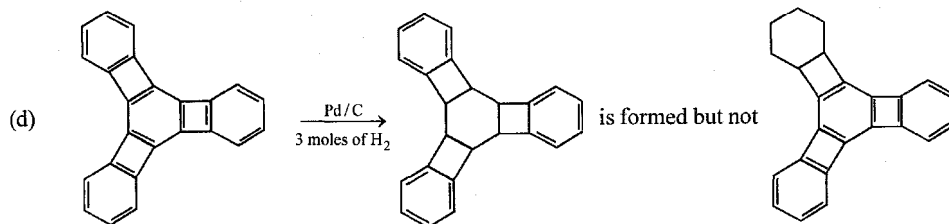
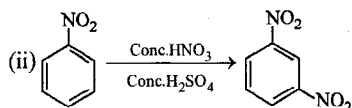
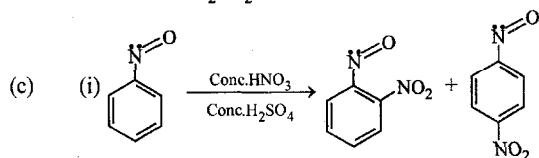
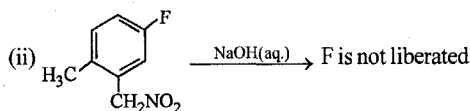
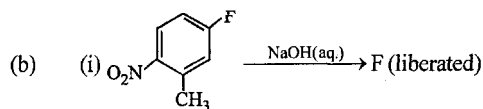
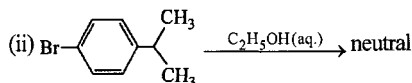
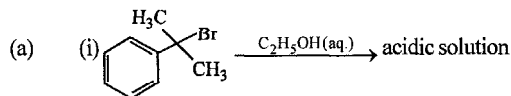
[JEE 2005]

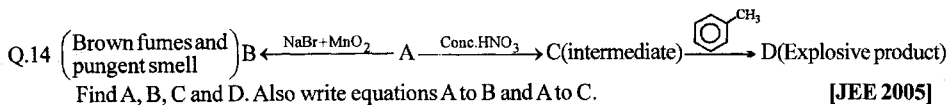
(i) NMe_3

(ii) $\text{N}(\text{SiMe}_3)_3$

Q.13 Give reasons:

[JEE 2005]





Q.15 $\mu_{\text{obs}} = \sum \mu_i x_i$ [JEE 2005]

where μ_i is the dipole moment of stable conformer and x_i is the fraction of that conformer.

- (a) Write stable conformer for $\text{Z}-\text{CH}_2-\text{CH}_2-\text{Z}$ in Newman's projection. If $\mu_{\text{solution}} = 1.0 \text{ D}$ and mole fraction of anti form = 0.82, find μ_{Gauche} .
- (b) Write most stable meso conformer of CHDY



If (i) $\text{Y} = \text{CH}_3$ about C_2-C_3 rotation and (ii) $\text{Y} = \text{OH}$ about C_1-C_2 rotation.

ANSWER KEY

EXERCISE-I

Q.1 D	Q.2 C	Q.3 B	Q.4 B	Q.5 B	Q.6 A	Q.7 A
Q.8 D	Q.9 A	Q.10 B	Q.11 A	Q.12 C	Q.13 C	Q.14 C
Q.15 C	Q.16 A	Q.17 B	Q.18 B	Q.19 C	Q.20 A	Q.21 D
Q.22 A	Q.23 C	Q.24 C	Q.25 D	Q.26 A	Q.27 D	Q.28 C
Q.29 A	Q.30 C	Q.31 B	Q.32 B	Q.33 C	Q.34 B	Q.35 B
Q.36 A	Q.37 A	Q.38 C	Q.39 B	Q.40 D	Q.41 B	Q.42 A
Q.43 A	Q.44 A	Q.45 A	Q.46 C	Q.47 C	Q.48 C	Q.49 A
Q.50 D	Q.51 D	Q.52 C	Q.53 C			

EXERCISE-II

Q.1 C	Q.2 A,D	Q.3 A,B,C,D	Q.4 A,B,C,D
Q.5 B,C,D	Q.6 A,B,D	Q.7 A,B,D	Q.8 C
Q.9 C,D	Q.10 A,B	Q.11 A,B	Q.12 A,D
Q.13 A,C,D	Q.14 C,D	Q.15 D	Q.16 A
Q.17 B,D	Q.18 D	Q.19 B,D	Q.20 A,C
Q.21 A,B,C	Q.22 B,C	Q.23 A,B	Q.24 A,C
Q.25 C	Q.26 A,C,D	Q.27 A,B,C	Q.28 D
Q.29 A,D	Q.30 B	Q.31 D	

EXERCISE-III

Q.1 $3 > 2 > 1 > 4$	Q.2 $IV > I > II > III$	Q.3 (a) $II > I > III > IV$; (b) (ii)
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Q.4 (a)	Q.5 $C-C-C-C-Cl > C-C-C-Cl > C-C-C-Cl > C-C-Cl$
	$\begin{array}{ccccccc} & & & & C & & \\ & & & & & & \\ C & - & C & - & C & - & Cl \\ & & & & & & \\ & & & & C & & \end{array}$

Q.6 (a) II, (b) I, (c) II	Q.7 (i) (a) II, (b) I, (c) II ; (ii) (a) II, (b) II, (c) I
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Q.8 (a) II, (b) I	Q.9 (a) I, (b) II, (c) I, (d) I	Q.10 (a) II, (b) I, (c) II, (d) II
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Q.11 (a) II ; (b) II ; (c) I ; (d) II ; (e) II	Q.14 b, d, e, g, h	Q.15 a, c, e, f, g, h
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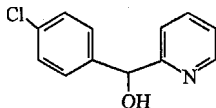
Q.27 	Q.38 (a) $Et_3CO^{\ominus}$; (b) $EtO^{\ominus}$	Q.45 $MeCOOMe$
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Q.46 $C-C-C-C, C-C=C-C, C-C-C-C$
$\begin{array}{ccccccc} & & & & OAC & & \\ & & & & & & \\ C & - & C & - & C & - & C \\ & & & & & & \\ & & & & C & & \end{array}$

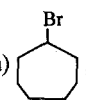
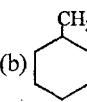
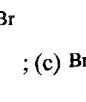
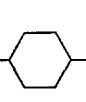
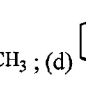
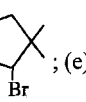
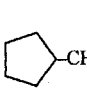
Q.47 (b) $C-C-C-Br$; (c) $Br-C-C-C$; (d) $C-C-C-C-Br$; (e) $C-C-C-C-C$
$\begin{array}{ccccccc} & & C & & C & & \\ & & & & & & \\ C & - & C & - & C & - & C \\ & & & & & & \\ & & C & & C & & \end{array}$

(f) $C-C-C-C-C$
$\begin{array}{c} Br \\ \\ C-C-C-C-C \end{array}$

Q.50



Q.53 (a) $\text{C}-\text{C}-\text{C}=\text{C}-\text{C}$; (b) $\text{C}-\text{C}-\text{C}-\underset{\text{C}}{\text{C}}=\text{C}$; (c) $\text{C}=\underset{\text{C}}{\text{C}}-\text{C}-\text{C}$; (d) $\text{C}-\text{C}-\text{C}-\text{C}=\text{C}$

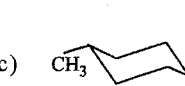
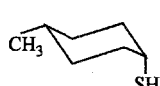
Q.54 (a)  ; (b)  ; (c)  ; (d)  ; (e)  ; (f)  ; (g) 

Q.55 (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{NH}_3$; (b) $\text{CH}_3\text{CH}_2\text{ONa} + \text{CH}_3\text{CH}_2\text{Cl}$; (c) $\text{HC}^-\text{C}^+\text{Na}^\oplus + \text{CH}_3\text{CH}_2\text{CH}_2-\text{Cl}$
(d) $\text{Me}_2\text{CH}-\text{Cl} + \text{Me}_2\text{S}$; (e) $\text{C}_6\text{H}_5\text{ONa} + \text{CH}_3\text{CH}_2-\text{Cl}$

Q.56 (a) $\text{ICH}_2\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-\text{CH}_2\text{CH}_3$; (b)  ; (c) 

Q.57 (a)  ; (b)  ; (c)  ; (d)  ; (e)  ; (f)  ; (g) 

Q.58 (a) $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_3-\text{CN}$, (b)  , (c) $\text{CH}_3-(\text{CH}_2)_9-\text{I}$, (d)  , (e) 

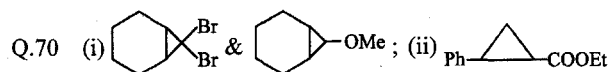
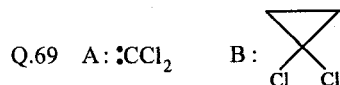
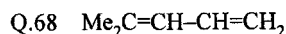
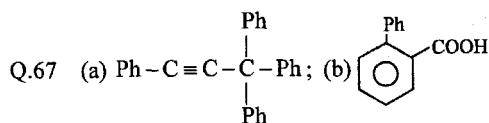
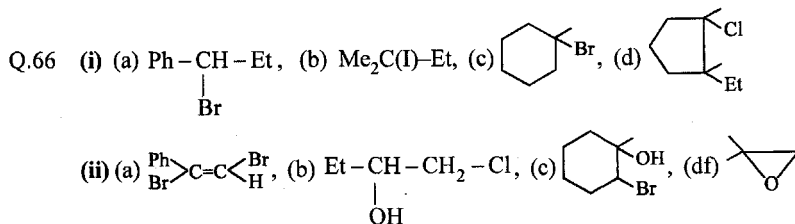
Q.59 (a) $\text{CH}_3\text{O}-\underset{\text{CH}_2\text{CH}_3}{\overset{\text{H}}{\text{C}}}-\text{CH}_3$; (b) $\text{Et}-\underset{\text{Pr}}{\overset{\text{Me}}{\text{C}}}-\text{OMe} + \text{MeO}-\underset{\text{Pr}}{\overset{\text{Me}}{\text{C}}}-\text{Et}$, (c)  + 

Q.60 (A) $\text{RCH}_2\text{CH}_2\text{SR}$ (B) $\text{RCHBrCH}_2\text{CCl}_3$ (C) $\text{RCHClCH}_2\text{CCl}_3$

Q.62 $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{F}$

Q.64 (a) $\text{CHI}_3 + (\text{CH}_2)_4(\text{COOH})_2$, (b) $\text{CHBr}_3 + \text{cyclohexyl-COOH}$, (c) $\text{CHCl}_3 + \text{cyclopropyl-COOH}$, (d) $\text{CHF}_3 + \text{EtCOONa}$

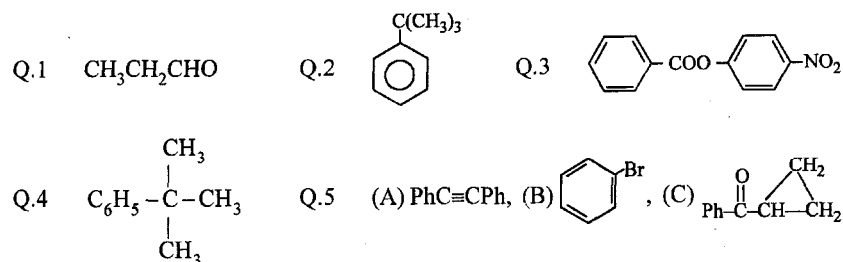
Q.65 (a) CHCl_3 , (b) PhCHO

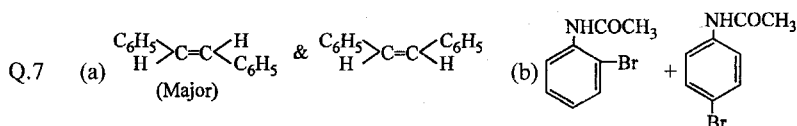
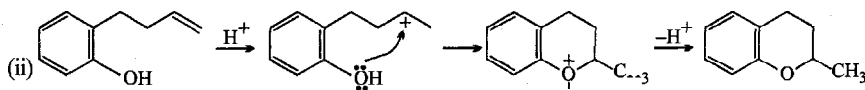
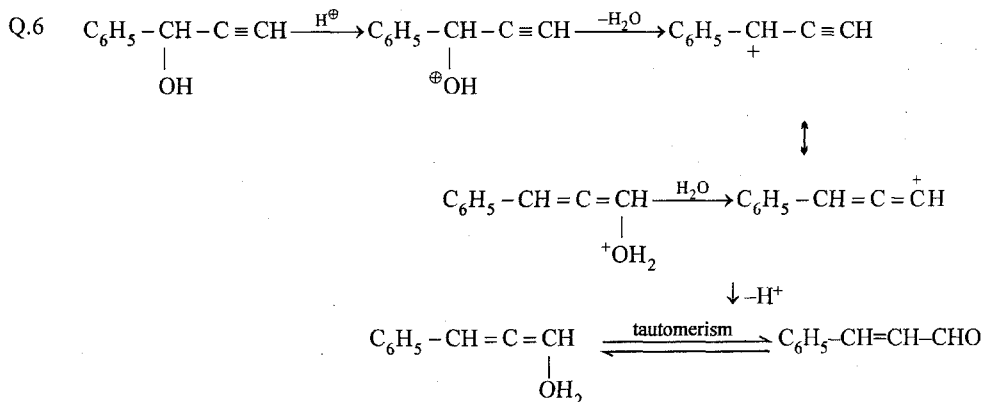


EXERCISE-IV (A)

Q.1 A, C	Q.2 D	Q.3 D	Q.4 B	Q.5 D
Q.6 B, C, D	Q.7 D	Q.8 D	Q.9 D	Q.10 B
Q.11 C	Q.12 A	Q.13 D	Q.14 C	Q.15 B
Q.16 D	Q.17 C	Q.18 C	Q.19 A	Q.20 A
Q.21 C	Q.22 A	Q.23 D	Q.24 D	Q.25 C
Q.26 D	Q.27 D	Q.28 B		

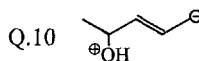
EXERCISE-IV (B)





Q.8 Anhydrous AlCl_3 is more stable than hydrous AlCl_3 because it is having vacant 3p orbital of Al which can accept lone pair of electrons from oxygen of diethylether.

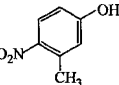
Q.9 (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c), (v)-(e)



Q.11 is more acidic as overall effect of -F is electron withdrawing, so loss of proton is easier from this compound.

Q.12 Not. NMe_3 is trigonal pyramidal while $\text{N}(\text{SiMe}_3)_3$ is trigonal planar due to back bonding.

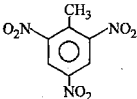
Q.13 (a) (i) $\text{H}_5\text{C}_6-\text{C}(\text{CH}_3)_2-\text{OC}_2\text{H}_5 + \text{HBr}(\text{acid})$; (ii) no reaction due to partial double bond character

(b) (i)  + F^- is liberated; (ii) Bimolecular mechanism is not possible in (ii) case

(c) (i) due to presence of lone pair of nitrogen atom NO group is electron donating and ortho, para directing

(ii) NO_2 group is electron withdrawing and meta directing

(d) Due to reduction of central ring, three four membered antiaromatic rings become stable while on reduction of terminal ring only one antiaromatic ring can be stabilized.

Q.14 (A) H_2SO_4 , (B) Br_2 , (C) NO_2^+ , (D) 

Q.15 (a) $\mu_{\text{Gauche}} = 5.55 \text{ D}$; (b) (i) , (ii) 



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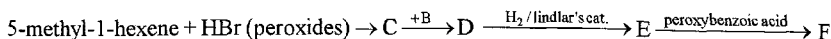
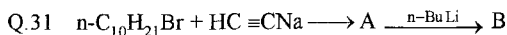
EXERCISE IV(B)

ANSWER KEY

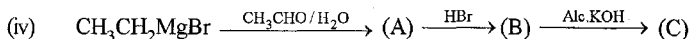
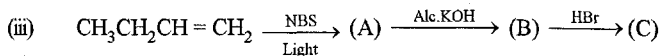
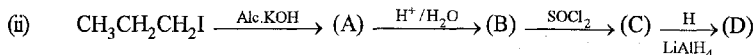
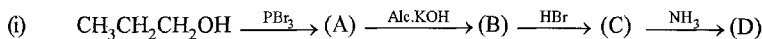
EXERCISE-I (A)

(Choose the correct option. Only one is correct)

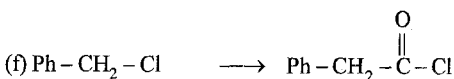
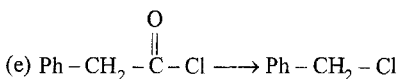
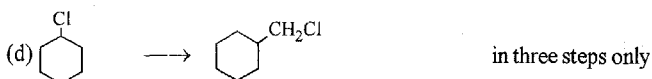
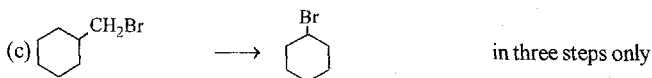
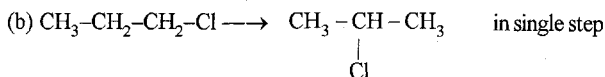
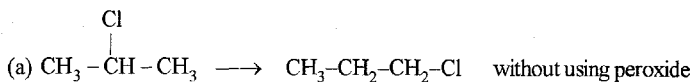
- Q.1⁺ Which one is liquid at room temperature:
 (A) CH_3Cl (B) $\text{C}_2\text{H}_5\text{Cl}$ (C) CH_3Br (D) $\text{C}_2\text{H}_5\text{Br}$
- Q.2 The correct order of melting and boiling points of the primary (1°), secondary (2°) and tertiary (3°) alkyl halides is:
 (A) $P > S > T$ (B) $T > S > P$ (C) $S > T > P$ (D) $T > P > S$
- Q.3⁺ Pick up the correct statement about alkyl halides:
 (A) They show H-bonding (B) They are soluble in water
 (C) They are soluble in organic solvent (D) They do not contain any polar bond
- Q.4 What would be the product when neopentyl chloride reacts with sodium ethoxide:
 (A) 2-Methyl-2-butanol (B) Neo pentyl alcohol
 (C) Both 1° & 2° (D) 2-Methyl-2-butene
- Q.5 Reduction of alkyl halide by LiAlH_4 is the type of reaction:
 (A) Nucleophilic substitution reaction (B) Electrophilic substitution reaction
 (C) Electrophilic Addition reaction (D) None of these
- Q.6 2-methyl butanoic acid is formed by the reaction CHCl_3 , NaOH and
 (A) Propene (B) Ethene (C) 1-Butene (D) 2-Butene
- Q.7 $\text{CCl}_4 + \text{KOH}$ (excess) $\longrightarrow$ end product of the reaction is:
 (A) K_2CO_3 (B) CO_2 (C) $\text{C}(\text{OH})_4$ (D) HCOOK
- Q.8 The yield of alkyl bromide obtained as a result of heating the dry silver salt of carboxylic acid with bromine if CCl_4 is:
 (A) $1^\circ > 3^\circ > 2^\circ$ bromides (B) $1^\circ > 2^\circ > 3^\circ$ bromides
 (C) $3^\circ > 2^\circ > 1^\circ$ bromides (D) $3^\circ > 1^\circ > 2^\circ$ bromides
- Q.9 Which of the following statements is invalid:
 (A) The more stable the carbocation the faster it is formed
 (B) Propyl cation changes to more stable isopropyl carbocation by $1,2$ shift of a hydrogen
 (C) Isopropyl chloride reacts with sodium ethoxide to form 1-ethoxypropane
 (D) Propyl halides reacts with sodium ethoxide to form 1-ethoxypropane
- Q.10 Nitrochloroform is prepared by the action of chloroform and:
 (A) Hot aqueous solution of KNO_3 (B) Hot aqueous solution of NaNO_2
 (C) Hot concentrated nitric acid (D) Hot dilute $\text{HCl} + \text{aq. NaNO}_2$ solution
- Q.11 Which of the following compounds is most rapidly hydrolysed by $\text{S}_\text{N}1$ mechanism.
 (A) $\text{C}_6\text{H}_5\text{Cl}$ (B) $\text{Cl}-\text{CH}_2-\text{CH}=\text{CH}_2$
 (C) $(\text{C}_6\text{H}_5)_3\text{CCl}$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$
- Q.12 $\text{CH}_3\text{MgBr} + \text{CH}_2 = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow{\text{H}_2\text{O}}$ product (major), product is
- (A) $\text{CH}_2 = \text{CH} - \overset{\text{OH}}{\underset{\text{CH}_3}{\text{C}}} - \text{H}$ (B) $\text{CH}_2 = \text{CH} = \text{CH} - \text{CH}_3$
 (C) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$ (D) None is correct



Q.32 Complete the following by providing (A), (B), (C) and (D):



Q.33 Carry out following conversions.



EXERCISE-III

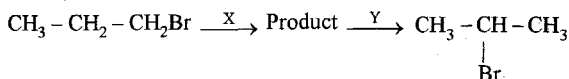
- Q.1 A primary alkyl halide (X) C_4H_9Br reacted with alcoholic KOH to give (Y). (Y) reacts with HBr to give (Z) an isomer of (X). On reacting with alkali metal such as sodium/dry ether gives (S) C_8H_{18} which was different from the compound produced when n-butylbromide was treated with Na. What (X), (Y), (Z) and (S).
- Q.2 On electrolysis an aqueous ethanoic solution of sodium chloride gives sweet smelling liquid (X). (X) gives isocyanide test and condenses with acetone to form hypnotic (Y). What are (X) and (Y)?
- Q.3 The alkyl halide C_4H_9Br (A) reacts with alcoholic KOH and gives an alkene (B), which reacts with bromine to give dibromide (C). (C) is transformed with KOH & sodamide to gas (D) which forms a precipitate when passed through an ammoniacal silver nitrate solution. Give the structure formulae of the compounds (A), (B), (C) and (D) and explain reactions involved.
- Q.4 An unsaturated hydrocarbon (A) C_6H_{10} readily gives (B) on treatment with $NaNH_2$ in liquid NH_3 . When (B) is allowed to react with 1-chloropropane, a compound (C) is obtained. On partial hydrogenation in the presence of Lindlar catalyst, (C) gives (D), C_9H_{18} . On ozonolysis (D) gives 2, 2-dimethylpropanal and 1-butanal. Give structures of (A), (B), (C) and (D) with proper reasoning.
- Q.5 One mole of each bromoderivative (A) and NH_3 react to give one mole of an organic compound (B). (B) reacts with CH_3I to give (C). Both (B) and (C) react with HNO_2 to give compounds (D) and (E), respectively. (D) on oxidation and subsequent decarboxylation gives 2-methoxy-2-methylpropane. Give structures of (A) to (E) with proper reasoning.
- Q.6 A chloro derivative (A) on treatment with zinc-copper couple gives a hydrocarbon with five carbon atoms. When 'A' is dissolved in ether and treated with sodium, 2,2,5,5-tetramethylhexane is obtained. What is the original compound 'A'?
- Q.7 A primary alkyl halide (A) C_4H_9Br reacted with alcoholic KOH to give (B). (B) reacts with HBr to give (C) an isomer of (A). (A) on reacting with sodium gives (D) C_8H_{18} which was different from the compound produced when n-butylbromide was treated with Na. What are (A), (B), (C) and (D).
- Q.8 One mole of a hydrocarbon (A) reacts with 1 mole of bromine giving a dibromo compound $C_7H_{10}Br_2$. (A) on treatment with cold dilute alkaline $KMnO_4$ solution forms a compound $C_7H_{12}O_2$. On ozonolysis (A) gives equimolar quantities of propanone and ethanal. Deduce structure of (A).
- Q.9 A hydrocarbon (A) was found to contain 85.7% carbon and 14.3% H. This compound consumes 1 molar equivalent of hydrogen to give a saturated hydrocarbon. 1 g of (A) just decolourised 38.05 g of 5% solution by weight of Br_2 in CCl_4 . Compound (A) on oxidation with conc. $KMnO_4$ gave a compound (C), C_4H_8O and (B) acetic acid. Compound (C) can easily be prepared by the action of acidic aqueous mercuric sulphate on 2-butyne. Deduce (A), (B) and (C).
- Q.10 When a hydrocarbon (A) is treated with excess of hydrogen chloride, a dihalogen derivative (B) is obtained. The compound (B) is treated separately with aqueous and alcoholic KOH to give compounds (C) and (A) respectively both having three carbon atoms. Compound (C) does not reduce Tollen's reagent and compound (A) give white precipitate with ammoniacal silver nitrate. Assign structures to compounds (A) to (C).

- Q. 11 An organic compound (A) $C_7H_{15}Cl$ on treatment with alcoholic caustic potash gives a hydrocarbon (B) C_7H_{14} . (B) on treatment with ozone and subsequent hydrolysis gives acetone and butyraldehyde. What are (A) and (B). Explain reactions.
- Q. 12 A dihalogen derivative (A) of a hydrocarbon having two carbon atoms react with alcoholic potash and forms another hydrocarbon which gives a red precipitate with ammonical cuprous chloride. Compound (A) gives an aldehyde when treated with KOH (aq). What is (A)?
- Q. 13 A white precipitate was formed slowly when $AgNO_3$ was added to a compound (A) with molecular formula $C_6H_{13}Cl$. Compound (A) on treatment with hot alcoholic KOH gave a mixture of two isomeric alkenes (B) and (C) having formula C_6H_{12} . The mixture of (B) and (C) on ozonolysis furnished four compounds (i) CH_3CHO , (ii) C_2H_5CHO , (iii) CH_3COCH_3 , (iv) $(CH_3)_2CH.CHO$. What are (A), (B) and (C).
- Q. 14 0.369 g of a bromo derivative of a hydrocarbon (A) when vaporized occupied 67.2 mL at NTP. (A) on reaction with aqueous NaOH gives (B). (B) when passed over alumina at $250^\circ C$ gives a neutral compound (C) while at $350^\circ C$ it gives a hydrocarbon (D). (D) when heated with HBr gives an isomer of (A). When (D) is treated with dil. H_2SO_4 , (E) is obtained. Identify (A) to (E) and explain the reactions.
- Q. 15 An organic compound (X) on analysis gives 24.24 % C, 4.04% H. Further sodium extract of 1.0 g of (X) gives 2.90 g of AgCl with acidified $AgNO_3$ solution. The compound (X) may be represent by two isomeric structures (Y) and (Z). (Y) on treatment with aqueous KOH solution gives a dihydroxy compound, while (Z) on similar treatment gives ethanal. Find out (X), (Y) and (Z).
- Q. 16 A chloro compound (A) showed the following properties :
- Decolourized bromine in CCl_4
 - Absorbed hydrogen catalytically.
 - Gave a precipitate with ammonical cuprous chloride
 - When vaporised 1.49 g of (A) gave 448 mL of vapours at STP.
- Identify (A) and write down the equations of reactions.

EXERCISE-IV (A)

- Q.1 Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives
 (A) o-cresol (B) p-cresol
 (C) 2,4-dihydroxytoluene (D) Benzoic acid [IIT 1990]
- Q.2 Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to
 (A) The formation of less stable carbonium ion (B) Resonance stabilization
 (C) The inductive effect (D) sp^2 hybridised carbon attached to the halogen [IIT 1990]
- Q.3 1-Chlorobutane on reaction with alcoholic potash gives:
 (A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol [IIT 1991]
- Q.4 The products of reaction of alcoholic $AgNO_2$ with ethyl bromide are [IIT 1991]
 (A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol
- Q.5 When nitrobenzene is treated with Br_2 in presence of $FeBr_3$ the major product formed is:
 m-bromonitrobenzene. Statements which are related obtain the m-isomer are
 (A) The electron density on meta carbon is more than on ortho and para position
 (B) The intermediate carbonium ion formed after initial attack of Br^+ at the meta position is least destabilized.
 (C) Loss of aromaticity when Br^+ attacks at the ortho and para position and not at meta position
 (D) Easier loss of H^+ to regain aromaticity from the meta position than from ortho and para positions. [IIT 1992]
- Q.6 Arrange the following compounds in order of increasing dipole moment
 Toluene m-dichlorobenzene o-dichlorobenzene p-dichlorobenzene
 I II III IV
 (A) $I < IV < II < III$ (B) $IV < I < II < III$ (C) $IV < I < III < II$ (D) $IV < II < I < III$ [IIT 1996]
- Q.7 In the reaction of p-chloro toluene with KNH_2 in liq. NH_3 , the major product is:
 (A) o-toluidine (B) m-toluidine (C) p-toluidine (D) p-chloraniline [IIT 1997]
- Q.8 $(CH_3)_3CMgCl$ reaction with D_2O produces:
 (A) $(CH_3)_3CD$ (B) $(CH_3)_3OD$ (C) $(CD_3)_3CD$ (D) $(CH_3)_3OD$ [IIT 1997]
- Q.9 Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with:
 (A) SO_2Cl_2 (B) $SOCl_2$ (C) Cl_2 (D) $NaOCl$ [IIT 1998]
- Q.10 Toluene, when treated with Br_2/Fe , gives p-bromotoluene as the major product, because the CH_3 group
 (A) is para directing (B) is meta directing
 (C) activates the ring by hyperconjugation (D) deactivates the ring [IIT 1999]
- Q.11 The order of reactivity of the following alkyl halides for a S_N2 reaction is: [IIT 2000]
 (A) $RF > RC > R-Br > R-I$ (B) $R-F > R-Br > R-Cl > R-I$
 (C) $R-Cl > R-Br > RF > RI$ (D) $R-I > RBr > R-Cl > R-F$

Q.12 Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation:

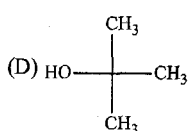
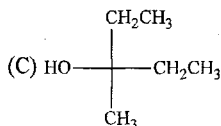
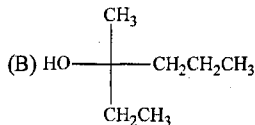
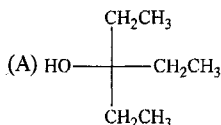


- (A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C
 (B) X = concentrated alcoholic NaOH, 80°C; Y = HBr / acetic acid 20°C
 (C) X = dilute aqueous NaOH, 20°C; Y = Br₂ / CHCl₃, 0°C
 (D) X = concentrated alcoholic NaOH, 80°C; Y = Br₂ / CHCl₃, 0°C

[IIT 2002]

Q.13 CH₃MgBr + Ethyl ester → which can be formed as product.
 (excess)

[IIT 2003]



Q.14 Match the following:

Column I

- (A) CH₃-CHBr-CD₃ on treatment with alc. KOH gives CH₂=CH-CD₃ as a major product.
 (B) Ph-CHBr-CH₃ reacts faster than Ph-CHBr-CD₃.
 (C) Ph-CH₂-CH₂Br on treatment with C₂H₅OD/C₂H₅O⁻ gives Ph-CD=CH₂ as the major product.
 (D) PhCH₂CH₂Br and PhCD₂CH₂Br react with same rate.

Column II

- (P) E1 reaction
 (Q) E2 reaction
 (R) E1cb reaction
 (S) First order reaction

EXERCISE-IV (B)

Q.1 Fill in the blanks:

- (a) Butan nitrile can be prepared by heating _____ with alcoholic KCN.

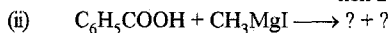
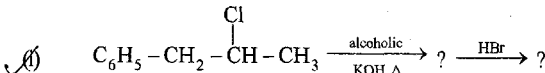
[IIT 1992]

(b) Amongst three isomers of nitrophenol, the one that is least soluble in water is _____

[IIT 1992]

Q.2 Identify the major product in the following reactions :

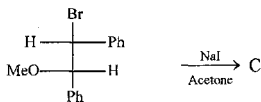
[IIT 1993]



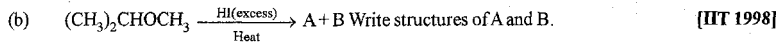
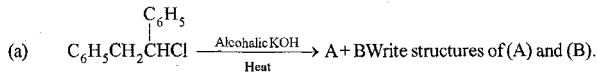
Q.3 An alkyl halide X of formula C₆H₁₃Cl on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C₆H₁₂). Both alkenes on hydrogenation give 2,3-dimethylbutane. Predict the structures of X, Y and Z.

[IIT 1996]

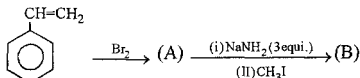
Q.4 Predict the structure of the intermediates/products in the following reaction sequence – [IIT 1996]



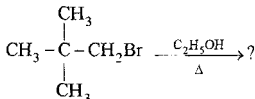
Q.5



Q.6 Complete the following reaction with appropriate structures of products/reagents. [IIT 1998]



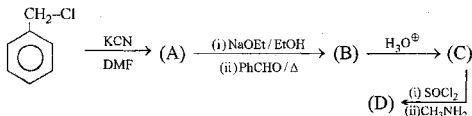
Q.7 What would be major product? [IIT 2000]



Q.8 How would you synthesis 4 methoxyphenol from bromobenzene in NOT more than five steps? State clearly the reagents used in each step and show the structures of the intermediate compounds in your synthetic scheme. [IIT 2001]

Q.9 An alkene A ($\text{C}_{16}\text{H}_{16}$) on ozonolysis give only one product B ($\text{C}_8\text{H}_8\text{O}$). Compound B on reaction with NaOH/I_2 yields sodium benzoate. Compound B reacts with $\text{KOH/NH}_2\text{NH}_2$ yielding a hydrocarbon C (C_8H_{10}). Write the structure of compound B and C. Based on this information two isomeric structures can be prepared for alkene A. Write their structures and identify the isomer which on catalytic hydrogenation ($\text{H}_2/\text{Pd-C}$) gives a racemic mixture. [IIT 2001]

Q.10 Give major products A, B, C and D in following reaction sequence. [IIT 2004]



ANSWER KEY

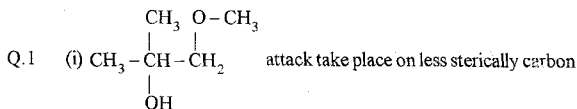
EXERCISE-I (A)

Q.1 D	Q.2 A	Q.3 C	Q.4 D	Q.5 A	Q.6 D	Q.7 A
Q.8 B	Q.9 C	Q.10 C	Q.11 C	Q.12 C	Q.13 A	Q.14 D
Q.15 C	Q.16 B	Q.17 A	Q.18 D	Q.19 C	Q.20 D	Q.21 B
Q.22 C	Q.23 D	Q.24 C	Q.25 D	Q.26 B	Q.27 B	Q.28 A
Q.29 B	Q.30 C	Q.31 B	Q.32 B	Q.33 C	Q.34 C	Q.35 A
Q.36 A	Q.37 C	Q.38 C	Q.39 A	Q.40 B	Q.41 A	Q.42 A
Q.43 A	Q.44 B	Q.45 A	Q.46 D	Q.47 B	Q.48 A	Q.49 C
Q.50 B						

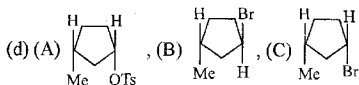
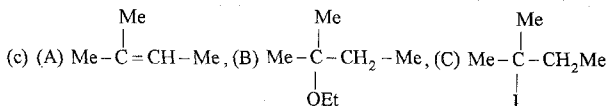
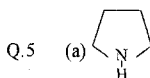
EXERCISE-I (B)

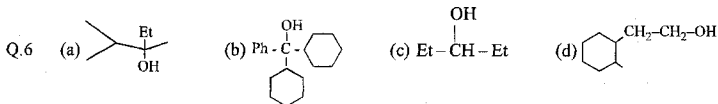
Q.1 D	Q.2 A,C	Q.3 B,C	Q.4 A,B
Q.5 A,C,D	Q.6 A,B,C, D	Q.7 B	Q.8 A,B,D
Q.9 B,C,D	Q.10 B,D	Q.11 A,D	Q.12 A,C,D
Q.13 A,C,D	Q.14 A,C,D	Q.15 B,D	Q.16 A,B,C,D
Q.17 A,B	Q.18 A,B,C	Q.19 A,B	Q.20 A,B,C
Q.21 A,C	Q.22 B,C	Q.23 A,B,C,D	Q.24 B,C,D
Q.25 A → 3, B → 2, C → 1, D → 2	Q.26 A → 4, B → 3,4, C → 3, D → 1,2		
Q.27 A → 2, B → 1, C → 4, D → 3	Q.28 (A) - S; (B) - Q; (C) - R; (D) - P		
Q.29 (A) - Q; (B) - S; (C) - R; (D) - P	Q.30 (A) - Q; (B) - R; (C) - P; (D) - Q		

EXERCISE-II



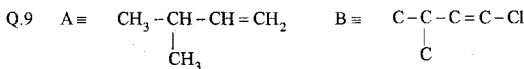
Q.4 (ii)





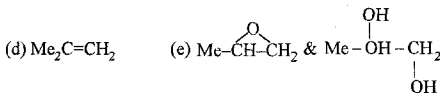
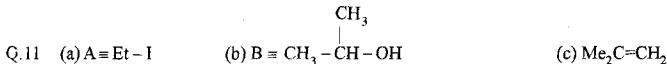
Q.7 (d)

Q.8

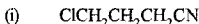


Conversion (1) NaNH_2 (2) $\text{H}_2/\text{Pd BaSO}_4$

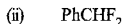
Q.10 (a) & (c)



Q.12 The mechanism of



(v) no reaction



Q.13 (i) Chloroform responds carbylamine test while ethyl alcohol does not. Carbylamine test- The compound is heated with aniline and alcoholic caustic potash-an obnoxious smell of phenyl isocyanide ($\text{C}_6\text{H}_5\text{NC}$) is produced.

(ii) Chlorine atoms of chloroform can be easily replaced by -OH groups on treatment with NaOH. Hence chlorine atoms come in the form of sodium chloride. The solution thus when treated with silver nitrate solution gives white precipitate of silver chloride. Chlorine atom in vinyl chloride is not readily replaced by NaOH, hence it does not give precipitate with AgNO_3 solution.

(iii) The two compounds are hydrolysed with aqueous NaOH when 1, 1-dibromoethane form acetaldehyde while 1, 2-dibromoethane gives glycol.

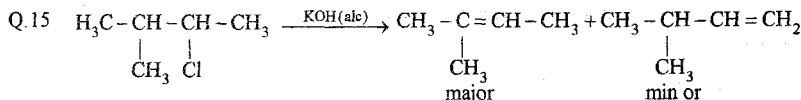
(iv) Chlorine atom in chlorobenzene is very less reactive, while chlorine in cyclohexyl chloride is very reactive. Hence the two can be differentiated as above in (ii)

(v) Apply test (ii) as above; when benzyl chloride gives white precipitate with AgNO_3 solution, while o-chlorotoluene does not give precipitate. Alternately, the compounds are oxidised when o-chlorotoluene give o-chlorobenzoic acid when benzyl chloride gives benzoic acid (having non chlorine.)

(vi) n-Hexane does not respond Cl^- ion test even after fusion with Na and treatment with acidic AgNO_3 . the remaining three compounds are distinguished by their reactivity with alcoholic AgNO_3 solution.

- (a) $\text{CH}_3 - \text{CH} = \text{CHCl}$ (a vinylic chloride) does not react with alc. AgNO_3 even on heating
 (b) $\text{CH}_2 = \text{CHCH}_2\text{Cl}$ (an allylic chloride) is the most reactive and precipitates AgCl in the cold.
 (c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$ gives precipitate of AgCl on warming with Alc. AgNO_3

Q. 14 The elimination of HI (or DI) in presence of strong base shows E_2 elimination. The rate determining step involves breaking up of $\text{C} - \text{H}$ (or $\text{C} - \text{D}$) bond. the $\text{C} - \text{D}$ bond being stronger than $\text{C} - \text{H}$ and thus elimination is faster in case of $\text{CH}_3 - \text{CH}_2\text{I}$.



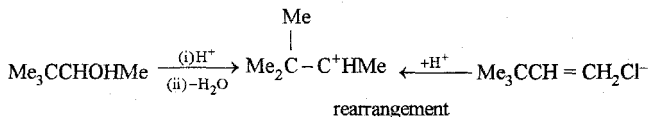
Elimination occurs according to saytzeff rule. The major product is one which involves elimination of H from less hydrogenated carbon.

Q. 16 $\text{C} - \text{I}$ bond being less stable than $\text{C} - \text{Cl}$ bond and thus on heating heterolytic cleavage of $\text{C} - \text{I}$ form I^- which gives yellow precipitate with AgNO_3

Q. 17 Due to three electronegative chlorine atoms present on carbon, the latter acquires a partial positive charge due to $-I$ effect of chlorine with the result it tends to attract electrons of the $\text{C} - \text{H}$ bond towards itself. Hence the removal of hydrogen atom as proton becomes easier.

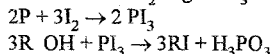
Q. 18 Alcohol acts as negative catalyst for oxidation of chloroform. Also it reacts with COCl_2 to give harmless diethyl carbonate.

Q. 19 Carbonium ions can undergo rearrangement from less stable to more stable ones. Both compounds give the same carbonium ion initially. Hence



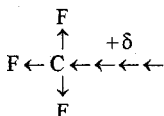
Both give the same two isomeric chloride in the same proportions, with the tertiary chloride predominating.

Q. 20 Phosphorus reacts with I_2 to give PI_3 which replaces OH group of alcohol to produce $\text{R} - \text{I}$



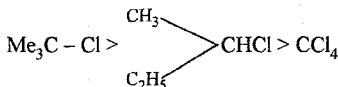
Q. 21 KCN is an ionic compound $[\text{K}^+ : \text{C} \equiv \text{N} : ^-]$ in which both C and N carry a lone pair electron. Carbon carrying lone pair of electrons is more reactive and thus alkyl attacks carbon to give alkyl cyanide AgCN being covalent has $\text{Ag} - \text{C} \equiv \text{N} :$ structure with lone pair on N thus R attacks N atom and $\text{R} - \text{C} \equiv \text{N} :$ is formed.

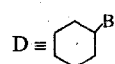
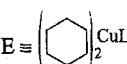
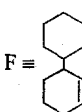
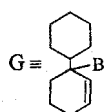
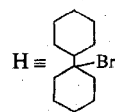
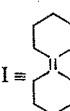
- Q. 22 The strong - I.E of fluorine atom in $F_3C-\overset{+}{C}$ produces partial +ve charge on C atom of CF_3 which ultimately increases +ve charge on 2nd carbon atom to destabilize it

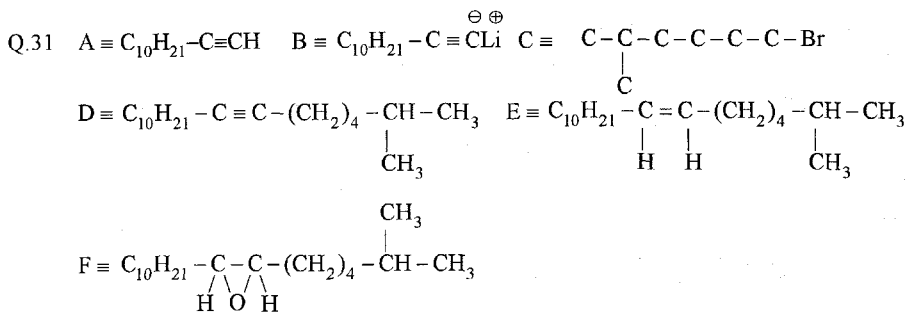
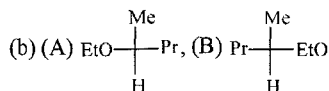


In F_3C^+ , the unshared pair of electrons in the p orbitals of each of the fluorine atom are shifted to C^+ via p-p orbital overlapping and thus stabilizing F_3C^+ ion

- Q. 23 Dry gaseous hydro halogen acids are better electrophile. Also in aqueous solution H_2O acting as nucleophile may produce alcohol.
- Q. 24 KI reacts with RCI to form RI . This RI molecule now hydrolysed easily to give ROH because alkyl iodide are more reactive than alkyl chloride.
- $$RCI + KI \xrightarrow{HOH} ROH \text{ (more fast)}$$
- Q. 25 The more pronounced the nucleophilic activity of the attacking reagent, then the more the S_N2 mechanism will be favoured. Since in S_N1 mechanism the reagent does not enter into the rate determining step of ionisation. However it can also be expected that as the nucleophilic activity is so slow that the mechanism will change from S_N2 to S_N1 .
- Q. 26 The dielectric constant of water is greater than that of aqueous acetone, and so ionisation of benzyl chloride is facilitated.
- Q. 27 On moving from 3° to 1° alcohol rate of S_N2 reaction decreases and reaches to minimum and mechanism changes from S_N2 to S_N1 . Then rate of S_N1 rate of reaction increases.
- Q. 28 Anhydrous $AlCl_3$ is used as a catalyst in Friedel Crafts alkylation.
- Q. 29 tert-Butyl chloride, sec butyl chloride and CCl_4 with alc. silver nitrate

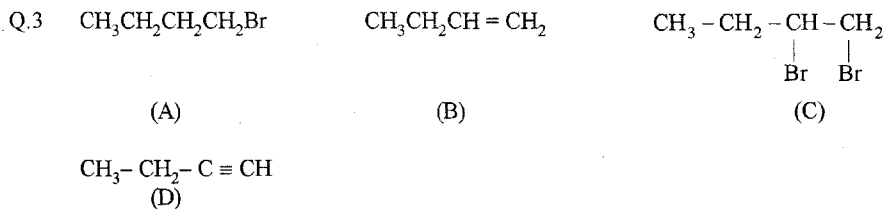
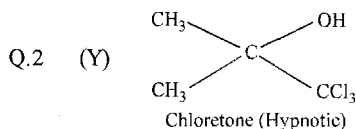
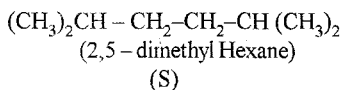
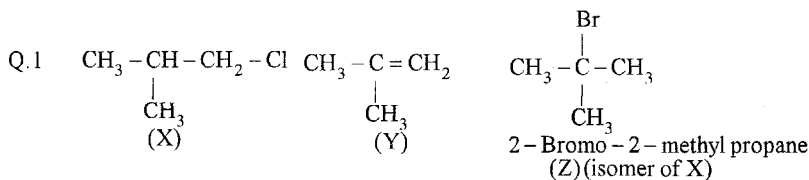


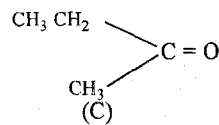
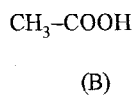
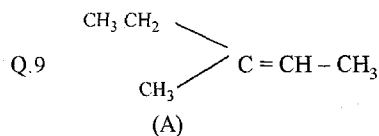
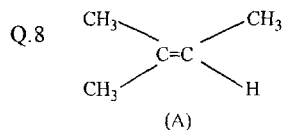
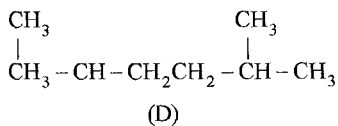
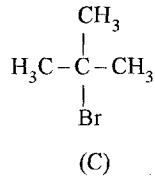
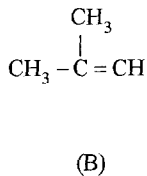
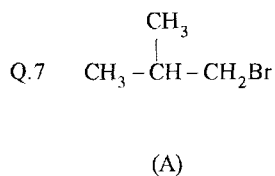
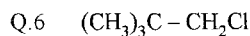
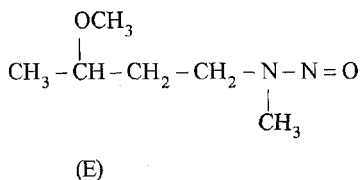
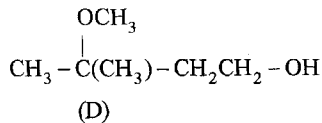
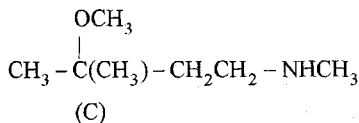
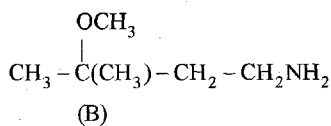
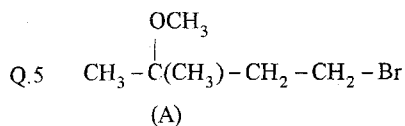
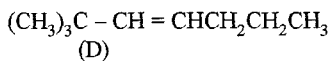
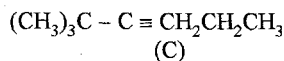
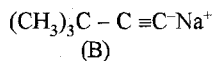
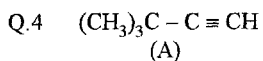
- Q. 30 (a) A $\equiv$  B $\equiv$  C $\equiv$  D $\equiv$ 
- E $\equiv$  F $\equiv$  G $\equiv$  H $\equiv$ 
- I $\equiv$ 

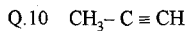


- Q.32 (i) A, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$; B, $\text{CH}_3\text{CH}=\text{CH}_2$; C, $\text{CH}_3\text{CHBrCH}_3$; D, $\text{CH}_3\text{CHNH}_2\text{CH}_3$
(ii) A, $\text{CH}_3\text{CH}=\text{CH}_2$; B, $\text{CH}_3\text{CHOHCH}_3$; C, $\text{CH}_3\text{CHClCH}_3$; D, $\text{CH}_3\text{CH}_2\text{CH}_2$
(iii) A, $\text{CH}_3\text{CHBrCH}=\text{CH}_2$; B, $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$; C, $\text{CH}_3\text{CHBrCH}=\text{CH}_2$ & $\text{CH}_3\text{CH}=\text{CH}-\text{CH}_2\text{Br}$
(iv) A, $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$; B, $\text{CH}_3\text{CH}_2\text{CHBrCH}_3$; C, $\text{CH}_3\text{CH}=\text{CH}-\text{CH}_3$

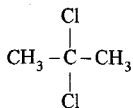
EXERCISE-III



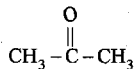




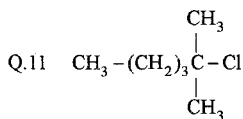
(A)



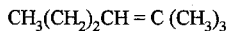
(B)



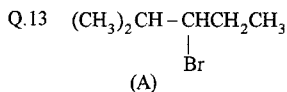
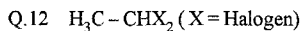
(C)



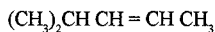
(A)



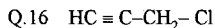
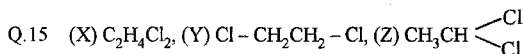
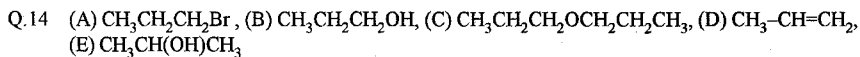
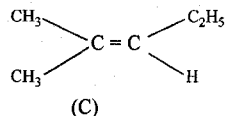
(B)



(A)



(B)

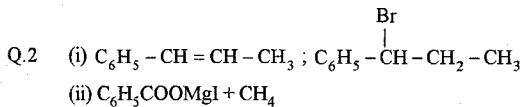


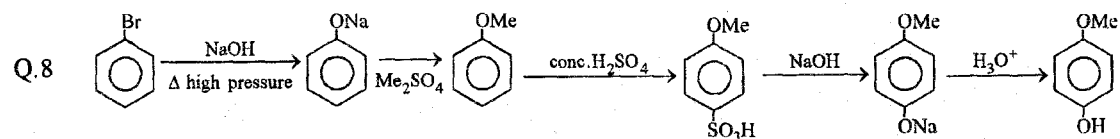
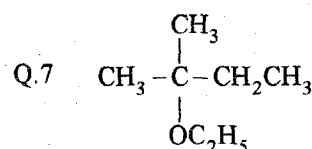
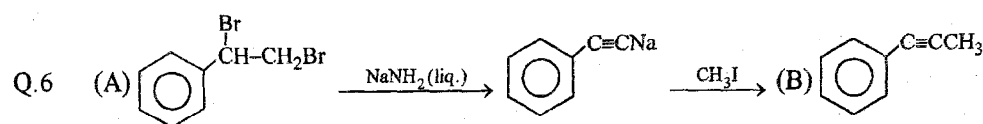
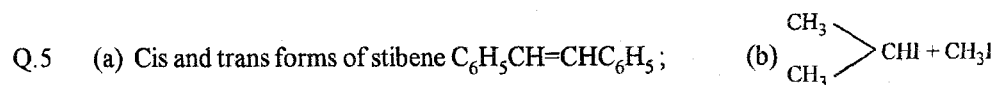
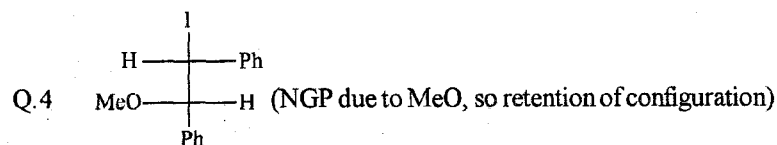
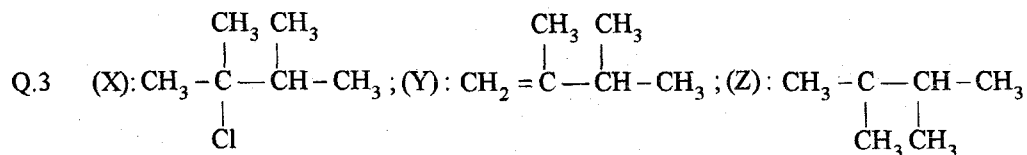
EXERCISE-IV (A)

- Q.1 D Q.2 D Q.3 A Q.4 C Q.5 A,B Q.6 B Q.7 B
Q.8 A Q.9 A,C Q.10 A,C Q.11 D Q.12 B Q.13 D
Q.14 A-Q; B-Q; C-R,S; D-P,S

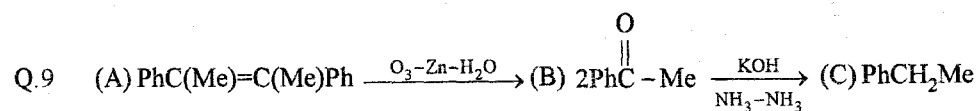
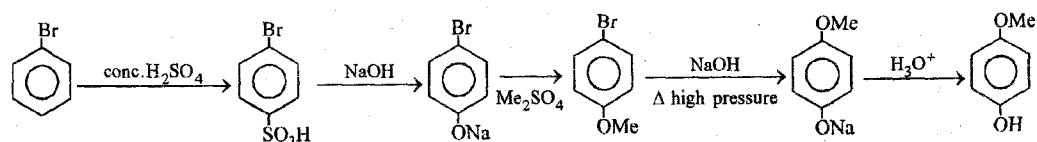
EXERCISE-IV (B)

- Q.1 (a) propyl chloride, (b) ortho

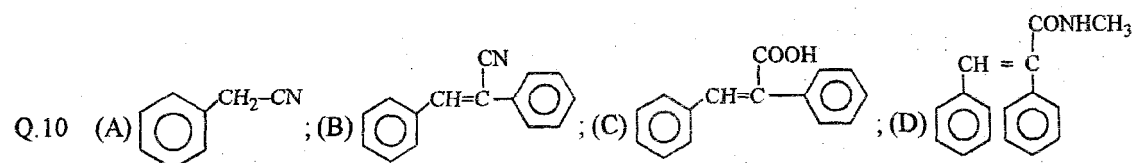




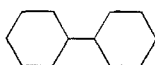
Alternative route

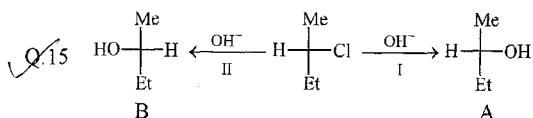
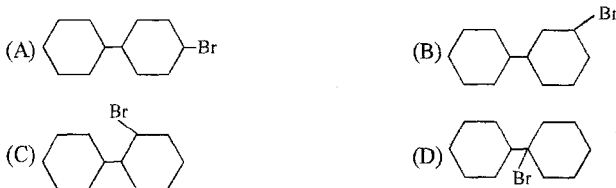


Two isomers of A are cis and trans



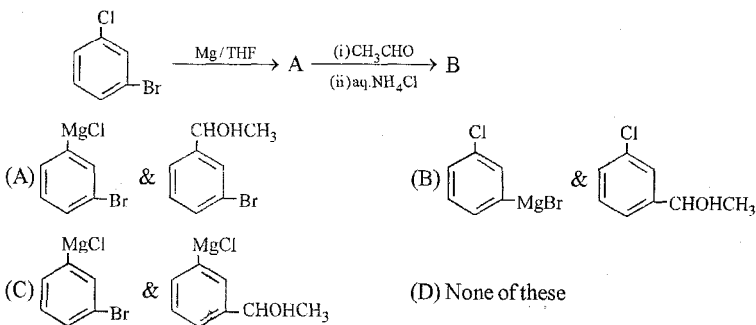
- Q. 13 Which is incorrect about Hunsdicker reaction?
 (A) Only Cl_2 can give alkyl halide
 (B) I_2 will give ester when treated with RCOOAg
 (C) The reaction proceeds through free radical
 (D) F_2 cannot give alkyl halide

- Q. 14 Major product for the reaction  is:



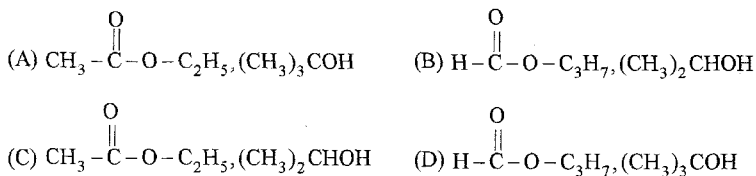
Steps I and II are

- (A) both $\text{S}_{\text{N}}1$ (B) both $\text{S}_{\text{N}}2$ (C) $\text{IS}_{\text{N}}1$, II $\text{S}_{\text{N}}2$ (D) I $\text{S}_{\text{N}}2$, II $\text{S}_{\text{N}}1$
 Q. 16 What are A & B in the following reaction?



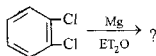
- Q. 17 Ester A ($\text{C}_4\text{H}_8\text{O}_2$) + $\text{CH}_3\text{MgBr} \xrightarrow{\text{H}_3\text{O}^+} \text{C}_4\text{H}_{10}\text{O}$
 (2 pair) (alcohol)
 (B)

Alcohol B reacts fastest with Lucas reagent. Hence A and B are



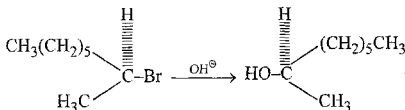
- Q.18 The hydrolysis of alkyl isocyanides ($R-CH_2-N \equiv C$) gives
 (A) $RNHCOOH$ (B) $R-CH-NH_2-COOH$ (C) $RNH_2 + HCOOH$ (D) $RCH_2NH_2 + HCOOH$

Q.19 What is product of the following reaction?

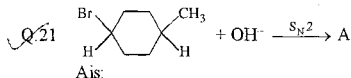


- (A) (B) (C) (D) None of these

Q.20 Following reaction is:



- (A) $\text{S}_\text{E}2$ (B) $\text{S}_\text{N}1$ (C) $\text{S}_\text{N}0$ (D) $\text{S}_\text{N}2$



- (A) (B) (C) Both (D) None

Q.22 $\text{CHCl}_3 + \text{OH}^- \longrightarrow \text{HCOO}^-$
 Intermediates of this reaction are

- (A) :CCl_3^- (B) :CCl_2 (C) both (D) None

Q.23 Racemic mixture is obtained due to halogenation of:

- (A) n-pentane (B) Isopentane (C) neopentane (D) Both A & B

Q.24 For $\text{CH}_3\text{Br} + \text{OH}^- \longrightarrow \text{CH}_3\text{OH} + \text{Br}^-$
 the rate of reaction is given by the expression:

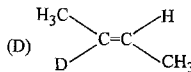
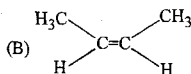
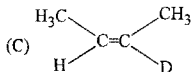
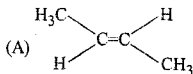
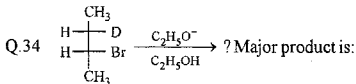
- (A) $\text{rate} = k[\text{CH}_3\text{Br}]$ (B) $\text{rate} = k[\text{OH}^-]$
 (C) $\text{rate} = k[\text{CH}_3\text{Br}][\text{OH}^-]$ (D) $\text{rate} = k[\text{CH}_3\text{Br}]^0[\text{OH}^-]^0$

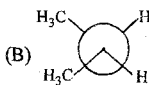
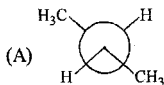
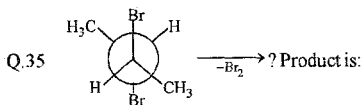
Q.25 Which will give white ppt. with AgNO_3 ?

- (A) (B) (C) (D) Both A & C

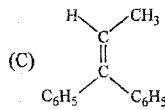
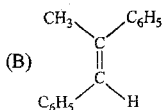
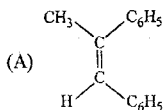
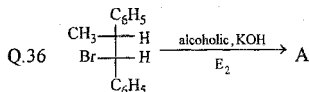
Q.26 The reaction of SOCl_2 on alkanols to form alkyl chlorides gives good yields because
 (A) Alkyl chlorides are immiscible with SOCl_2
 (B) The other products of the reaction are gaseous and escape out
 (C) Alcohol and SOCl_2 are soluble in water
 (D) The reaction does not occur via intermediate formation of an alkyl chloro sulphite

- Q. 27 True about alkyl halides is / are:
 (A) Tertiary alkyl halides undergo S_N2 substitutions
 (B) Alkyl iodides on exposure to sunlight gradually darken
 (C) Alkyl chlorides do not give beilstein test
 (D) A nucleophilic substitution is most difficult in alkyl iodides
- Q. 28 The product formed when glycerol reacts with PCl_5 is—
 (A) 1,2,3-trichloropropane (B) Glycerol monochlorohydrin
 (C) Glycerol dichlorohydrin (D) All of these
- Q. 29 On heating glycerol with excess amount of HI, the product formed is—
 (A) Allyl iodide (B) Isopropyl iodide (C) Propylene (D) 1,2,3-tri-iodopropane
- Q. 30 To prepare 3-ethyl-3-pentanol the reagents needed are:
 (A) $CH_3CH_2MgBr + CH_3 - \overset{\overset{O}{||}}{C} - CH_2CH_3$
 (B) $CH_3MgBr + CH_3CH_2 - \overset{\overset{O}{||}}{C} - CH_2CH_3$
 (C) $CH_3CH_2MgBr + CH_3CH_2 - \overset{\overset{O}{||}}{C} - CH_2 - CH_3$
 (D) $CH_3CH_2CH_2MgBr + CH_3 - \overset{\overset{O}{||}}{C} - CH_2CH_3$
- Q. 31 Non-occurrence of the following reaction
 $Br^- + CH_3OH \longrightarrow BrCH_3 + OH^-$, is due to
 (A) Attacking nucleophile is stronger one (B) Leaving group is a strong base
 (C) Alcohols are not good substrate (D) Hydroxide ions are weak bases
- Q. 32 When ethyl bromide is treated with moist Ag_2O , main product is:
 (A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of the above
- Q. 33 When ethyl bromide is treated with dry Ag_2O , main product is:
 (A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of the above





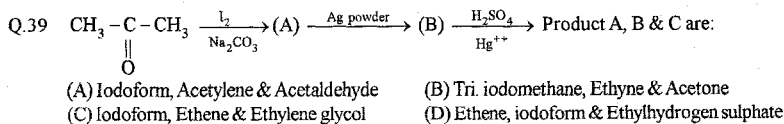
(C) both (A) & (B) (D) none of these



(D) None is correct

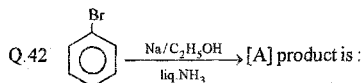
Q.37 To remove 'phosgene' from chloroform, _____ is added:
(A) Ethyl amine (B) Ethyl chloride (C) Ethyl alcohol (D) Ethane

Q.38 A gem dichloride is formed in the reaction except:
(A) CH_3CHO and PCl_5 (B) CH_3COCH_3 and PCl_5
(C) $\text{CH}_2 = \text{CH}_2$ and Cl_2 (D) $\text{CH}_2 = \text{CHCl}$ and HCl

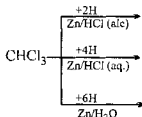


Q.40 Which of the following statement is wrong:
(A) All carbonyl compounds of the general structure $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{R}$ give a positive iodoform test.
(B) All secondary alcohols give iodoform reaction
(C) Alkanols of the structure $\text{CH}_3\text{CH}(\text{OH}) - \text{R}$ (where $\text{R} = \text{H}$, alkyl or aryl) give iodoform reaction.
(D) The only aldehyde giving iodoform reaction is acetaldehyde.

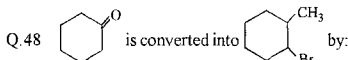
Q.41 $\text{A} \xrightarrow{\text{O}^-\text{Br}} \text{CHBr}_3$
Here A is nothing but:
(A) Isopropyl alcohol (B) Methanol (C) Ethanoic acid (D) n-Butyl alcohol



- Q.43 The appropriate naming for the second step in the haloform reaction is:
 (A) Halogenation (B) Dehydrohalogenation
 (C) Reduction (D) Dehydrogenation
- Q.44 Which product is obtained when bleaching powder is distilled with acetone:
 (A) CCl_4 (B) CHCl_3 (C) CH_2Cl_2 (D) CH_3Cl
- Q.45 What will be the reduction product of following reactions:



- (A) CH_2Cl_2 , CH_3Cl , CH_4 (B) CH_4 , CH_3Cl , CH_2Cl_2
 (C) CH_3Cl , CH_2Cl_2 , CH_4 (D) CH_3Cl , CH_4 , CH_2Cl_2
- Q.46 Fe/HCl reduction of 'pyrene' (CCl_4) gives:
 (A) CH_4 (B) CH_2Cl_2 (C) CH_3Cl (D) CHCl_3
- Q.47 Addition of bromine on propene in the presence of brine yields a mixture of:
 (A) $\text{CH}_3\text{CHClCH}_2\text{Br}$ and $\text{CH}_3\text{CHBrCH}_2\text{Cl}$ (B) $\text{CH}_3\text{CHClCH}_2\text{Br}$ and $\text{CH}_3\text{CHBrCH}_2\text{Br}$
 (C) $\text{CH}_3\text{CHClCH}_2\text{Cl}$ and $\text{CH}_3\text{CHBrCH}_2\text{Br}$ (D) $\text{CH}_2\text{CHClCH}_2\text{Cl}$ and $\text{CH}_3\text{CHBrCH}_2\text{Cl}$

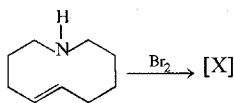


- (A) (i) $\text{CH}_3\text{-MgI}$, H_3O^+ (ii) H_2SO_4 , Δ , (iii) HBr , R_2O_2
 (B) (i) $\text{CH}_3\text{-MgI}$, H_3O^+ (ii) H_2SO_4 , Δ , (iii) HBr
 (C) (i) $\text{CH}_3\text{-MgI}$, H_3O^+ (ii) HBr
 (D) (i) HBr , R_2O_2 (ii) $\text{CH}_3\text{-MgI}$, H_3O^+
- Q.49 To form Malonic acid, by shortest possible route, we have to start a reaction from:
 (A) Ethyldiene chloride (B) Methyl chloride (C) Methylene chloride (D) Chloro ethan
- Q.50 Frankland reagent is:
 (A) Dialkyl lithium (B) Dialkyl Zinc (C) Dialkyl magnesium (D) Dialkyl aluminium

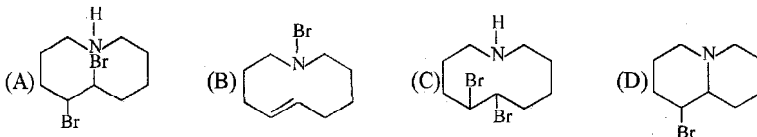
EXERCISE-I (B)

(Choose the correct option. One or more than one are correct)

Q.1 In the given reaction:



[X] is:



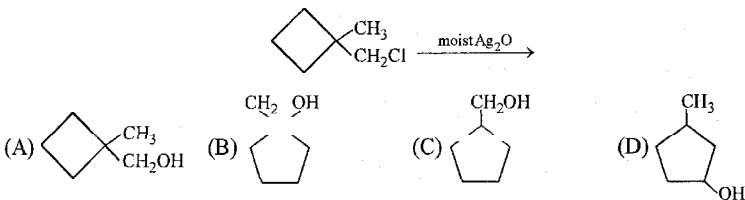
Q.2 Consider the following statements?

- (A) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-I}$ will react more readily than $(\text{CH}_3)_2\text{CHI}$ for $\text{S}_\text{N}2$ reactions.
 (B) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-Cl}$ will react more readily than $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-Br}$ for $\text{S}_\text{N}2$ reaction.
 (C) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-Br}$ will react more readily than $(\text{CH}_3)_3\text{C-CH}_2\text{-Br}$ for $\text{S}_\text{N}2$ reactions
 (D) $\text{CH}_3\text{-O-C}_6\text{H}_5\text{-Br}$ will react more readily than $\text{NO}_2\text{-C}_6\text{H}_5\text{-CH}_2\text{Br}$ for $\text{S}_\text{N}2$ reaction

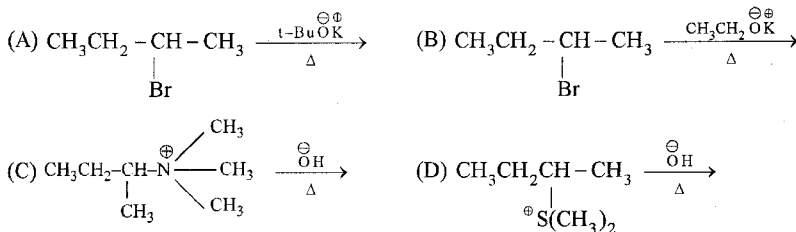
Q.3 A solution of (R)-2-iodobutane, $[\alpha] = -15.9^\circ$ in acetone is treated with radioactive iodine, until 1.0% of iodobutane contain radioactive iodine, the specific rotation of recovered iodobutane is found to be 15.58° , which statement is correct about above information.

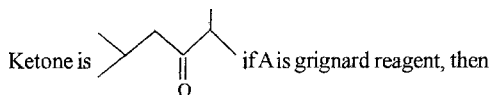
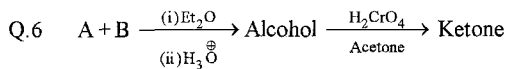
- (A) It has optical purity 96% (B) % of R in solution is 99%
 (C) Racemic mixture is 2% (D) Final solution is dextrorotatory

Q.4 Which are possible products in following



Q.5 In which product formation takes place according to Hoffmann's rule



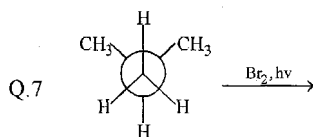


(A) A is isobutyl magnesium bromide

(B) B is $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - CH = O$

(C) B is $CH_3 - \underset{\substack{| \\ CH_3}}{CH} COCl$

(D) A is isopropyl magnesium iodide



(A) $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - CH_2Br$

(B) $CH_3 - \underset{\substack{| \\ CH_3}}{C}Br - CH_3$

(C) $CH_3CH_2CH_2Br$

(D) $CH_3 - \underset{\substack{| \\ Br}}{CH} - CH_3$

Q.8 Select the correct statements from following

(A) $CH_3CH_2CH_2I$ will react more readily than $(CH_3)_2CHI$ for S_N2

(B) $CH_3CH_2CH_2Cl$ will react more readily than $CH_3CH_2CH_2Br$ for S_N2

(C) $CH_3CH_2CH_2CH_2Br$ will react more readily than $(CH_3)_2CH-CH_2Br$ for S_N2

(D) $CH_3-O-C_6H_4-CH_2Br$ will react more readily than $NO_2-C_6H_4-CH_2Br$ for S_N2

Q.9 Which of the following will react with $AcOAg$

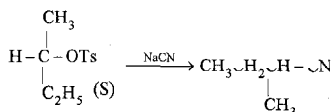
(A) $R-NH_2$

(B) $R-OTs$

(C) $R-N_3$

(D) $R-Br$

Q.10 Consider the given reaction



which of following statements are correct for above reaction.

(A) Product formation takes place due to the breaking of O-Ts

(B) The reaction is $\text{S}_{\text{N}}2$

(C) The reaction is $\text{S}_{\text{N}}1$

(D) Configuration of product is (R)

Q.11 Benzoyl chloride is less reactive than acetyl chloride for nucleophilic acyl substitution reaction because

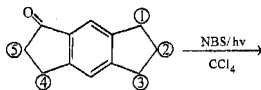
(A) $-\text{C}-\text{O}-$ group of benzoyl chloride is in conjugation with benzene ring. This makes benzoyl chloride more stable than acyl chloride.

(B) CH_3- group of acetyl chloride make C-Cl bond stronger due to the +I effect.

(C) C_6H_5- group of benzoyl chloride make C-Cl bond weaker due to +R effect.

(D) Carbonyl carbon in benzoyl chloride is less electron deficient than acetyl chloride

Q.12 Bromination can take place at



(A) 1

(B) 5

(C) 3

(D) 4

Q.13 Which of following will give syn-elimination

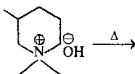
(A) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{OCOCH}_3$

(B) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})-\text{CH}_3$

(C) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{O}-\text{C}(=\text{S})-\text{S}-\text{CH}_3$

(D) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{N}^+(\text{CH}_3)_2-\text{O}^-$

Q.14 Which of following are correct for given reaction



(A) Major product of reaction is



(B) Major product is



(C) The reaction is thermal elimination reaction

(D) The reaction is cyclic E_2 reaction

Q.15 Haloform reaction is given by

(A) All 2° alcohol

(B) All methyl ketones

(C) All the compounds having $\text{CH}_3 - \text{C} -$ group

(D) Chloral



Q.16 Identify the compounds that will undergo NGP reaction

(A) $\text{CH}_3\text{CH}_2 - \text{S} - \text{CH}_2\text{CH}_2\text{Br}$

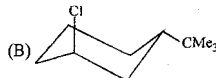
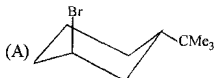
(B) $\text{CH}_3 - \text{NHCH}_2\text{CH}_2\text{CH}_2\text{Br}$

(C) $\text{CH}_3 - \text{CH} - \text{C} = \text{O}$

(D) $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$



Q.17 Which alkyl halide undergo E_2 elimination



Q.18 3-Phenyl-3-pentanol can be prepared from grignard reagent & other component which can be

(A) 3-pentanone

(B) Ethylbenzoate

(C) Ethyl phenyl ketone

(D) Propanoyl chloride

Q.19 In which of following reaction inverted product will be obtained.

(A) S_N1

(B) S_N2

(C) S_{N1GP}

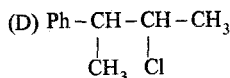
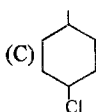
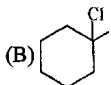
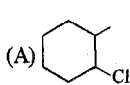
(D) S_Ni

Q.20 Rate of S_N2 will be negligible in



Q.21 S_N1 & S_N2 is not favourable in
 (A) $H_2C=CH-Cl$ (B) $Ph-CH_2-Cl$ (C) $Ph-Cl$ (D) $H_2C=CH-CH_2-Cl$

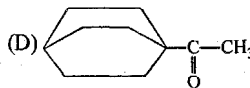
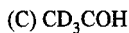
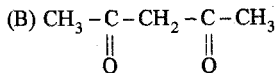
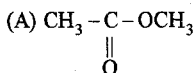
Q.22 S_N1 & S_N2 product are same in (excluding stereoisomer)



Q.23 Rate of S_N2 depends on
 (A) Conc of Nucleophile
 (C) Nature of leaving group

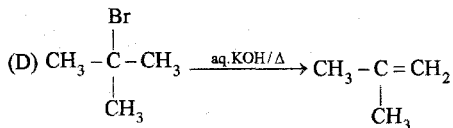
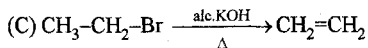
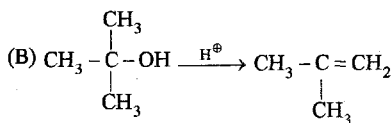
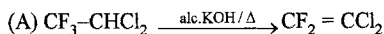
(B) Conc of substrate
 (D) Nature of solvent

Q.24 $G \xrightarrow{NaOI} \text{yellow solid}$
 G can be



Q.25 Match the List I (reaction) with List II (reaction intermediate) and select the correct answer using the codes given below the Lists.

List I



List II

(1) Transition state

(2) Carbocation

(3) Carbanion

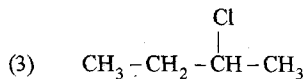
(4) Free radical

Q. 26 Match the List I with List II and select the correct answer using the codes given below the Lists.

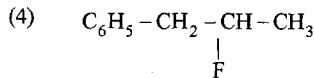
- List I**
- (A) E1CB
(B) Saytzeff alkene as major product

- List II**
- (1) Quaternary Amine oxide
(2) Xanthate

(C) E2



(D) E_C

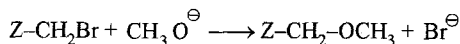


Q. 27 Match List I with List II and select the correct answer from the codes given below:

- List I**
- (A) $\text{CH}_3\text{O}-\text{SO}_2\text{CH}_3 + \text{C}_2\text{H}_5\text{O}^-$
(B) $\text{CH}_3-\text{CH}_2-\text{I} + \text{PH}_3$
(C) $\text{HC} \equiv \overset{\ominus}{\text{C}} \overset{\oplus}{\text{Na}} + \text{CH}_3-\text{CH}_2-\text{Br}$
(D) $\text{CH}_3-\text{Cl} + \text{CH}_3-\text{O}^-$

- List II**
- (1) $\text{CH}_3-\text{CH}_2-\text{PH}_2$
(2) $\text{CH}_3-\text{O}-\text{C}_2\text{H}_5$
(3) $\text{CH}_3-\text{O}-\text{CH}_3$
(4) $\text{CH} \equiv \text{C}-\text{CH}_2-\text{CH}_3$

Q. 28 Match List-I with List-II for given S_N2 reaction & select the correct answer from the codes given below



- List-I**
- (A) H-
(B) CH₃-
(C) C₂H₅-
(D) $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH} - \\ \diagdown \\ \text{CH}_3 \end{array}$
- List-II (Rates)**
- (P) 0.1
(Q) 3
(R) 1
(S) 100

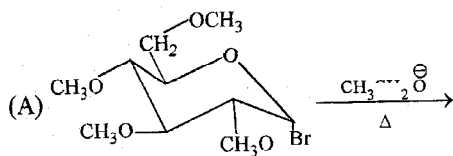
Q. 29 Match List-I & List-II and select correct answer

- | List-I | List-II |
|--|-------------------------------|
| (A) $\text{R}-\text{MgX} + \text{HCHO} \longrightarrow \text{Adduct} \xrightarrow{\text{H}_3\text{O}^+}$
n-carbon | (P) Ketone |
| (B) $\text{R}-\text{MgX} + (\text{CH}_2)_2\text{O} \longrightarrow \text{Adduct} \xrightarrow{\text{H}_3\text{O}^+}$
n-carbon | (Q) 1° Alcohol (n + 1) carbon |
| (C) $\text{R}-\text{MgX} + \text{CO}_2 \longrightarrow \text{Adduct} \xrightarrow{\text{H}_3\text{O}^+}$
n-carbon | (R) Acid (n + 1) carbon |
| (D) $\text{R}-\text{MgX} + \text{Ph}-\text{C} \equiv \text{N} \longrightarrow \text{Adduct} \xrightarrow{\text{H}_3\text{O}^+}$ | (S) 1° Alcohol (n + 2) carbon |

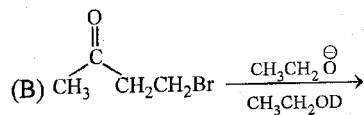
Q.30 Match the following

List-I

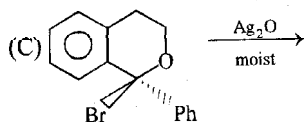
List-II



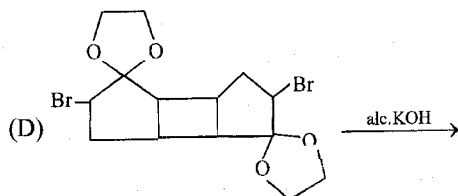
(P) E1



(Q) E2



(R) E1cb



(S) Ec

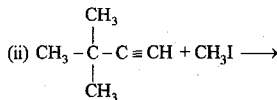
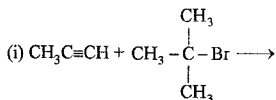
EXERCISE-II

- Q.1 Explain the following giving proper reasoning :**
- Treatment of 1,1-dimethyloxirane with sodium methoxide in methanol gives primarily one product. Also identify the product giving its IUPAC name
 - Trimethyl amine reacts with alkyl iodide (RI). As the reaction proceeds the reaction rate abnormally increases, inspite of depleting concentration of reactants.
 - Optically active 2-iodo butane on treatment with NaI*/Acetone gives a product which do not show optical activity.
 - In the solvent DMSO the order of reactivity of halide ions with methyl bromide is $F^- > Cl^- > Br^- > I^-$ opposite to that observed in methanol solution.
 - $(n-C_3H_7)_3N + CH_3I \rightarrow (n-C_3H_7)_3NCH_3^+ I^-$
Relative rate : in hexane, 1 ; in chloroform, 13000
 - $Br^- + CH_3OTs \rightarrow CH_3Br + TsO^-$
Relative rates : in methanol, 1 ; in HMPT, 10^5 .
 - Although ethers are weakly polar, are used as solvent of choice for Grignard reagents.
 - Treatment of either $CH_3-CH(O)-CH_2-CH_2Br$ or $CH_3-CH(Br)-CH(O)-CH_2$ with aqueous HO^- gives the same product. Also write down mechanism involved, & IUPAC name of product.

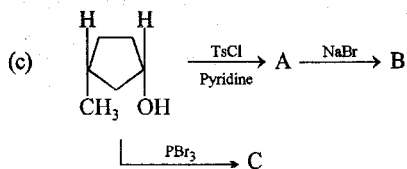
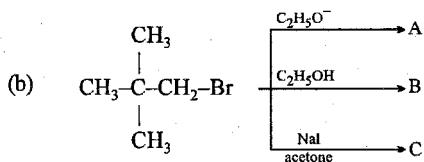
- Q.2** On reaction between an alkyl halide and KOH in a mixture of water and ethanol, compare the two mechanisms SN_1 and SN_2 :
- relative rates of CH_3X , C_2H_5X , iso- C_3H_7X & tert- C_4H_9X
 - relative rates for RCI , RBr and RI
 - effect on rate of a rise in temperature
 - effect on rate of increasing water content of the solvent.
 - effect on rate increasing alcohol content of the solvent.

- Q.3** In the following reaction is carried out in the weakly ionizing solvent, acetone $(CH_3)_2C=O$. (Bs is brosyl, p-bromo benzene sulfonyl)
- $$n-C_4H_9OBs + X^- \longrightarrow n-C_4H_9X + BsO^-$$
- The order of reactivity of halide ions depends on the salt used on their source :
- if Li^+X^- is used, $I^- > Br^- > Cl^-$
- if $(n-C_4H_9)_4N^+X^-$ is used $Cl^- > Br^- > I^-$
- How do you account for this contrast in behavior ?

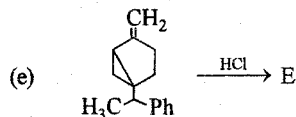
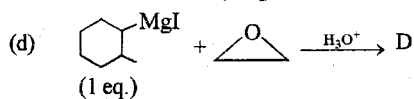
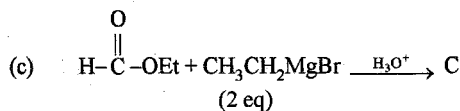
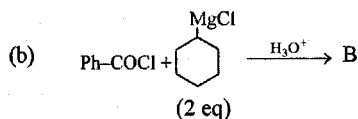
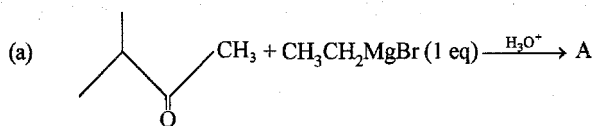
- Q.4** Which of the following is the correct method for synthesizing 4,4-dimethyl-pent-2-yne, using sodamide in liquid NH_3 ? Given reasons.



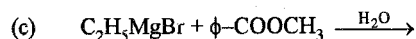
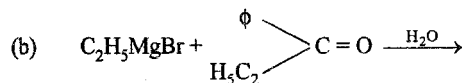
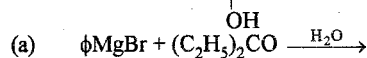
- Q.5** Give the products and find all unknowns.
- $H_2NCH_2CH_2CH_2CH_2Br \xrightarrow{\Delta}$

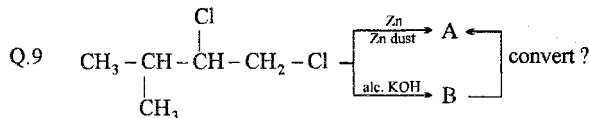
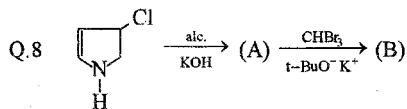
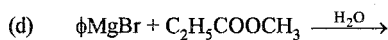


Q.6 Find out the unknown alphabets.



Q.7 To prepare $\begin{array}{c} \text{C}_2\text{H}_5 \\ | \\ \phi-\text{C}-\text{C}_2\text{H}_5 \\ | \\ \text{OH} \end{array}$ by RMgX which is the incorrect pair



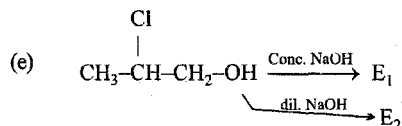


Q. 10 What is/are the limitation in the formation of Grignard reagent by $\text{RX} + \text{Mg} \xrightarrow{\text{dry ether}}$

- R should not contain any acidic group
- R should not contain any electronegative group
- Alkyl halide should not be vicinal dihalide
- Alkyl halide can't be Tertiary.

Q. 11 Identify the unknowns

- $\text{CH}_3\text{CH}_2\text{Cl} + \text{I}^- \rightarrow \text{A}$
- $(\text{CH}_3)_2\text{CH}-\text{Br} + \text{KOH (aq.)} \rightarrow \text{B}$
- $(\text{CH}_3)_3\text{C}-\text{Br} + \text{KCN (alc.)} \rightarrow \text{C}$
- $(\text{CH}_3)_3\text{C}-\text{Br} + \text{KOH (aq.)} \rightarrow \text{D}$



Q. 12 Write major product of the the following reactions:

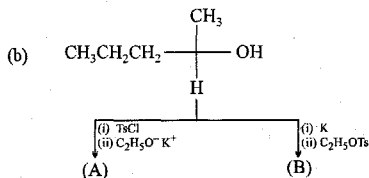
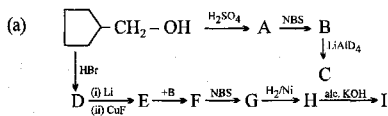
- $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{KCN} \xrightarrow[\text{heat}]{\text{EtOH}} \text{A}$
- $\text{PhCHO} \xrightarrow{\text{SF}_4} \text{B}$
- $\text{BrCH}_2\text{CH}=\text{CHCO}_2\text{Me} \xrightarrow[\text{DMF}]{\text{AgF}} \text{C}$
- $\text{EtOH} + \text{HI} \longrightarrow \text{D}$
- $\text{EtOH} + \text{HCN} \longrightarrow \text{E}$

Q. 13 State at least one characteristic test which can distinguish :

- $\text{C}_2\text{H}_5\text{OH}$ from CHCl_3
- Halogen atoms present in CHCl_3 and $\text{CH}_2=\text{CHCl}$
- 1, 1- dibromoethane from 1, 2-Dibromomethane
- Chlorobenzene from chlorocyclohexane
- o-Chlorotoluene from benzyl chloride
- n-Hexane, $\text{CH}_3\text{CH}=\text{CHCl}$, $\text{CH}_2=\text{CHCH}_2\text{Cl}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

Q. 14 $\text{CH}_3-\text{CH}_2\text{I}$ reacts more rapidly with strong base in comparison to $\text{CD}_3\text{CH}_2\text{I}$

- Q.15 2-chloro-3-methylbutane on treatment with alcoholic potash gives 2-methylbutene-2 as major product.
- Q.16 Iodoform gives precipitate with AgNO_3 on heating while CHCl_3 does not.
- Q.17 Hydrogen atom of chloroform is definitely acidic, but that of methane is not.
- Q.18 A small amount of alcohol is added to chloroform bottles.
- Q.19 Treatment of $\text{Me}_3\text{CCH}=\text{CH}_2$ and $\text{Me}_3\text{CCHOHMe}$ with concentrated hydrochloric acid gives the same two isomeric alkyl chlorides what are these two products.
- Q.20 Iodine reacts with alcohols to give alkyl iodide only in presence of phosphorous.
- Q.21 KCN reacts with $\text{R}-\text{I}$ to give alkyl cyanide, while AgCN results in isocyanide as major product.
- Q.22 The carbocation $\text{F}_3\text{C}-\dot{\text{C}}<$ is unstable whereas carbocation F_3C^+ is more stable.
- Q.23 Dry gaseous hydro halogen acid and not their aqueous solutions are used to prepare alkyl halides from alkenes.
- Q.24 RCl is hydrolysed to ROH slowly but reaction fastens on addition of KI .
- Q.25 The mechanism of decomposition of $\text{Me}_3\text{S}^+\text{OH}^-$ is $\text{S}_{\text{N}}2$ whereas of $\text{Me}_3\text{S}^+\text{I}^-$ is $\text{S}_{\text{N}}1$.
- Q.26 Alkaline hydrolysis of benzyl chloride in 50% aqueous acetone proceeds by both $\text{S}_{\text{N}}2$ and $\text{S}_{\text{N}}1$ mechanism, when water is used as solvent, mechanism was now mainly $\text{S}_{\text{N}}1$.
- Q.27 Ethanol is less reactive than methanol as well as isopropyl alcohols when treated with dry HCl gas.
- Q.28 Anhydrous AlCl_3 is used as a catalyst in Friedel Crafts alkylation.
- Q.29 Arrange following compounds according to their reactivity with alc silver nitrate.
t-Butyl chloride, sec butyl chloride and CCl_4 .
- Q.30



What are (A) and (B) explain giving their stereochemistry.



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REACTIONS CHART

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EXERCISE IV (A)

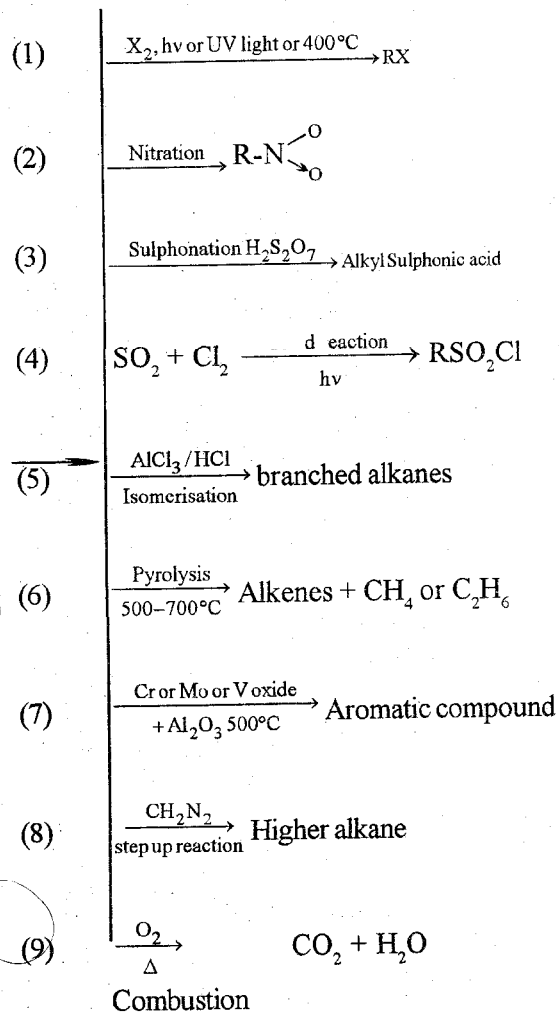
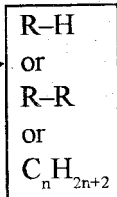
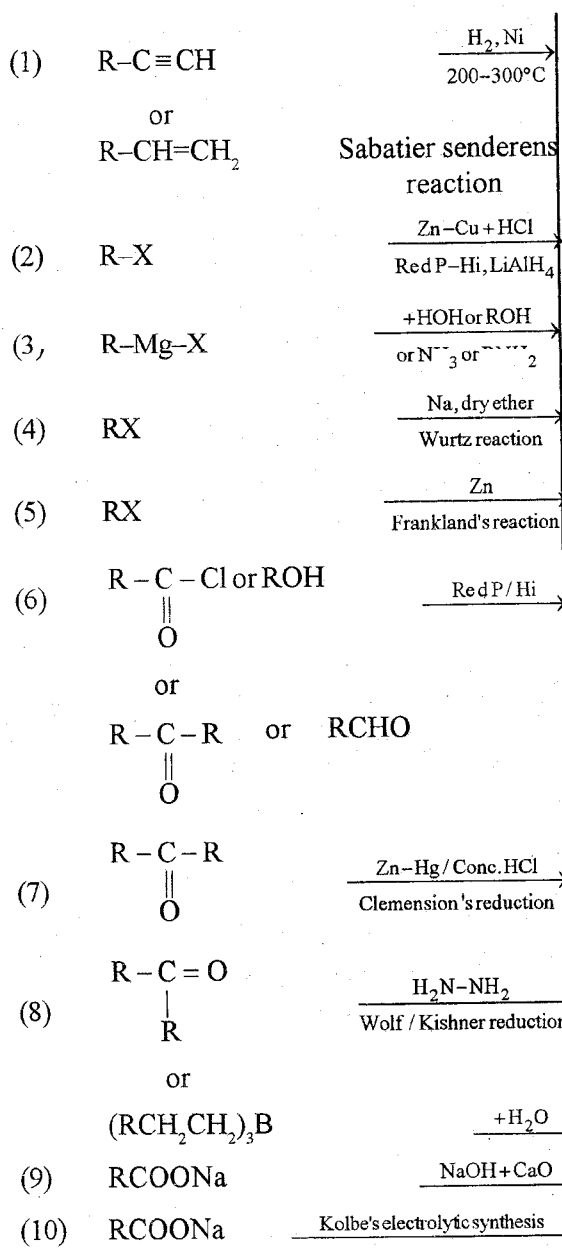
EXERCISE IV (B)

ANSWER KEY

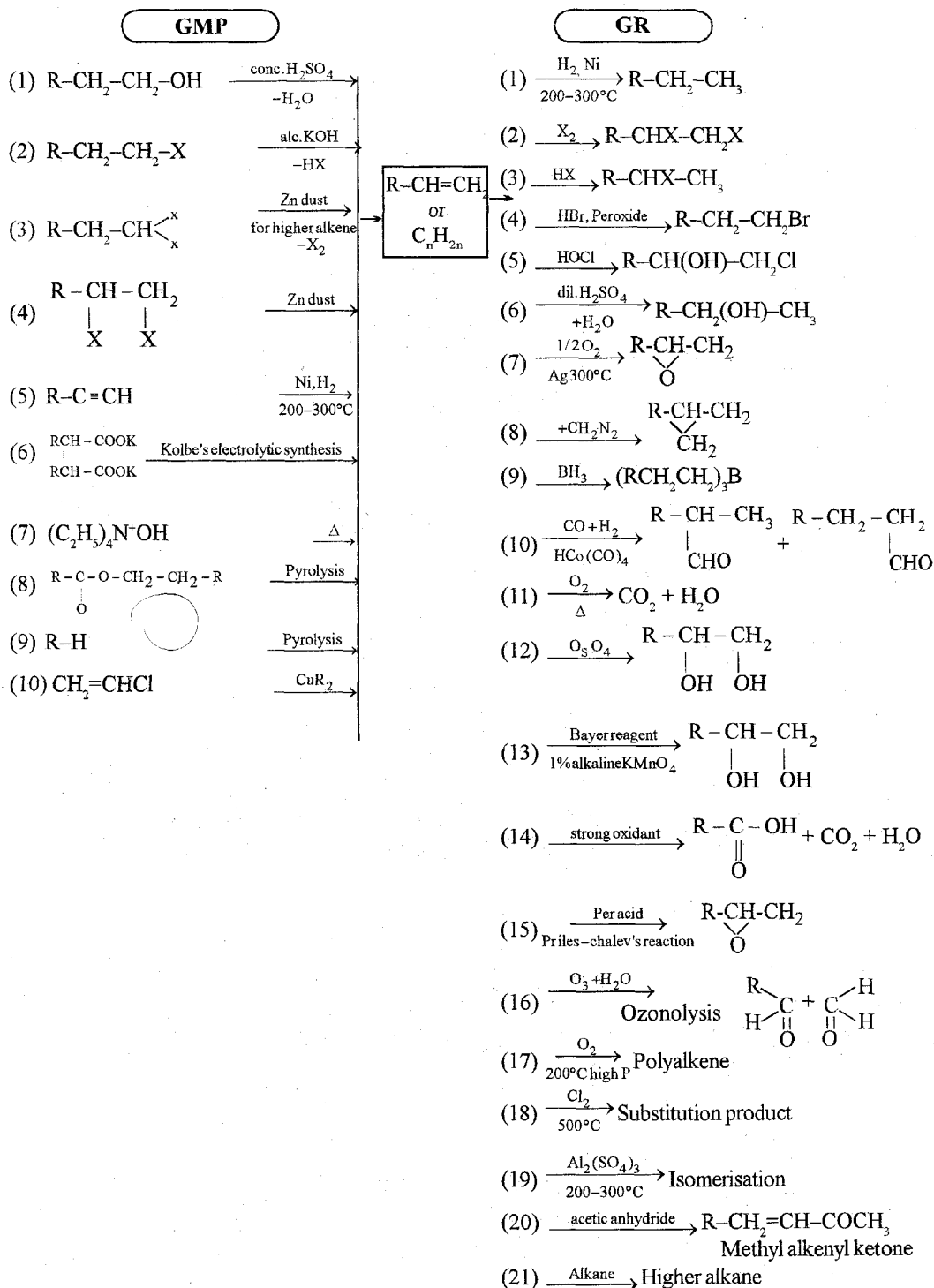
REACTION CHART FOR ALKANES

GMP

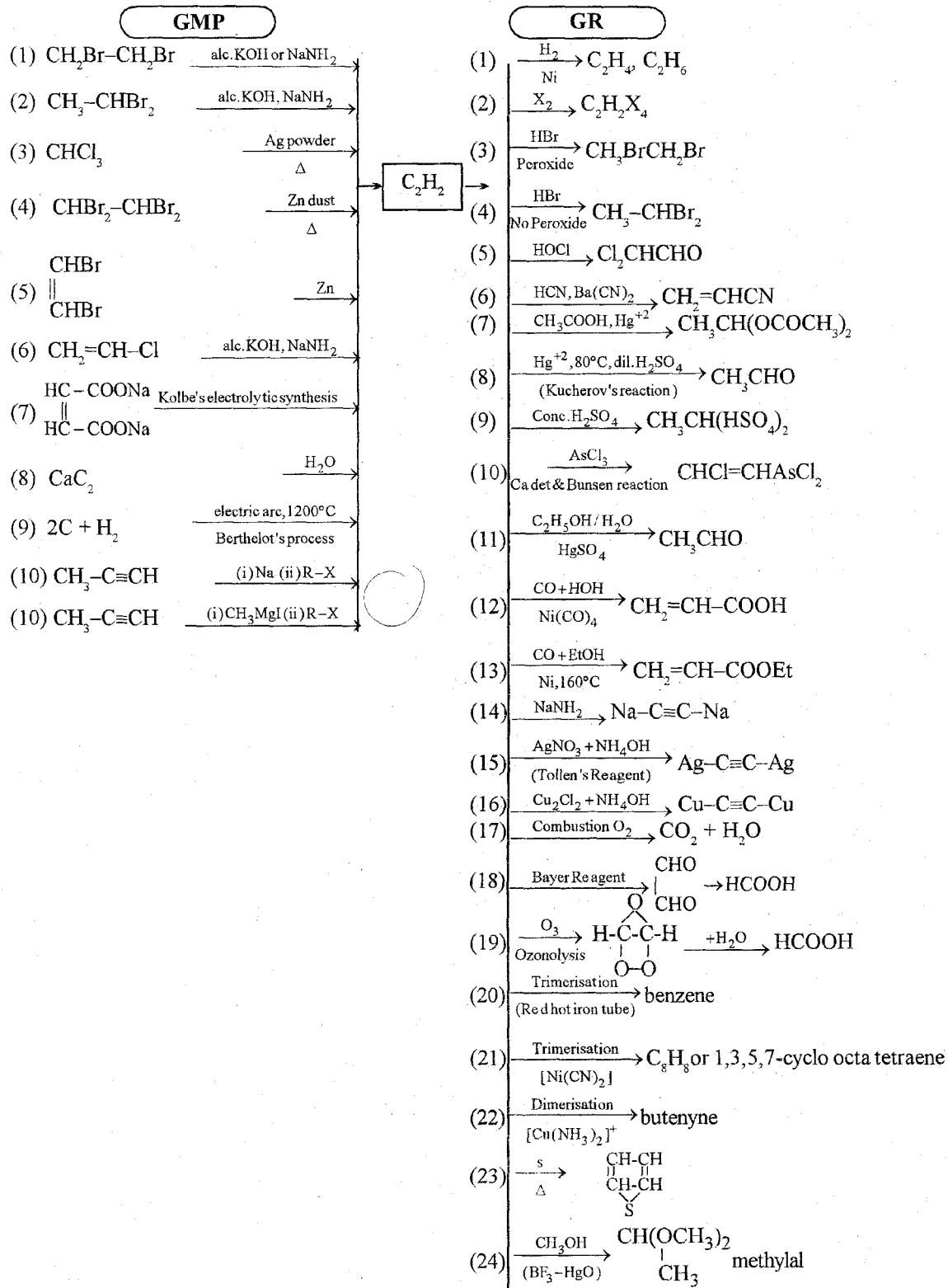
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REACTION CHART FOR ALKENES

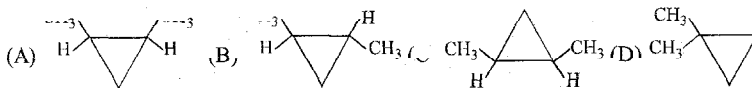


REACTION CHART FOR ALKYNES

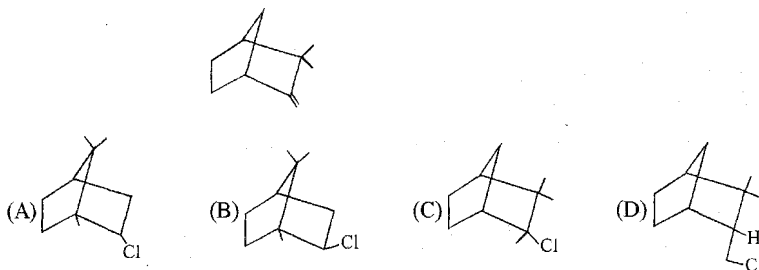


EXERCISE-I (A)

Q.1 The reaction of E-2-butene with CH_2I_2 and Zn-Cu Couple in either medium leads to formation of

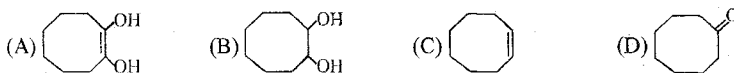


Q.2 The product formed upon heating camphene with HCl is

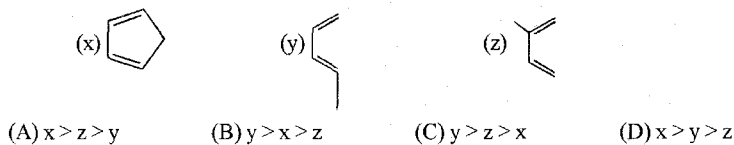


Q.3 (E)-3-bromo-3-hexene when treated with $\text{CH}_3\text{O}^\ominus$ in CH_3OH gives
 (A) 3-hexyne (B) 2-hexyne (C) 2,3-hexadiene (D) 2,4-hexadiene

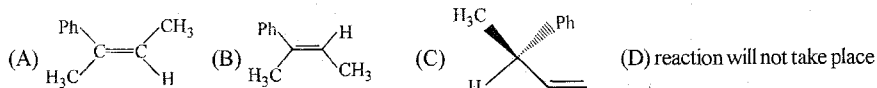
Q.4 The reaction of cyclooctyne with HgSO_4 in the presence of aq. H_2SO_4 gives



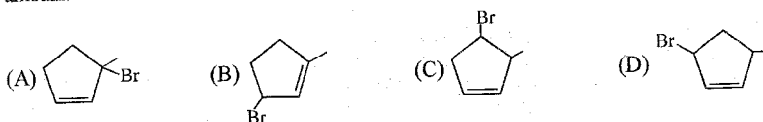
Q.5 Order of reactivity of the following diene x, y and z in Diels-Alder reaction is



Q.6 The major product P formed in given  $\xrightarrow{\text{H}_2\text{SO}_4}$ P reaction is

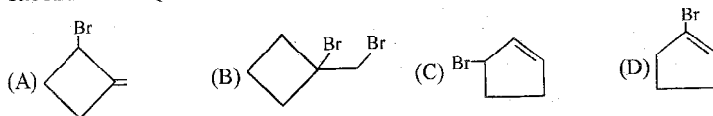


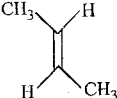
Q.7  + Br₂ $\xrightarrow{h\nu}$ mixture of product. Among the following which product will formed minimum amount.



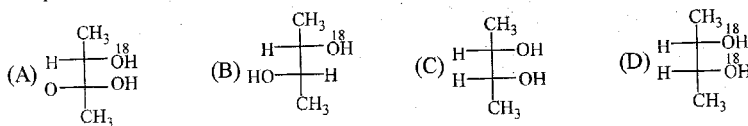
Q.8  $\xrightarrow[\Delta]{H_2SO_4}$ P (Major) $\xrightarrow{NBS}$ Q (Major)

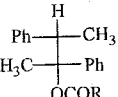
The structure of Q is



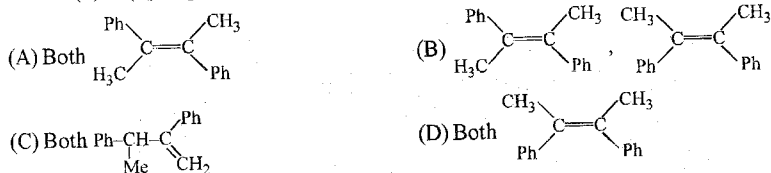
Q.9  $\xrightarrow[(ii) H_3O^+]{(i) CH_3COOOH}$ X

The probable structure of 'X' is



Q.10  $\xrightarrow{\Delta}$ P (Alkene) $\xrightarrow[CCl_4]{Br_2}$ Q (Alkene) $\xrightarrow[Acetone]{NaI}$

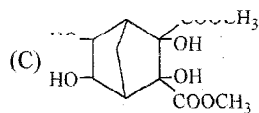
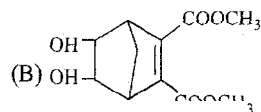
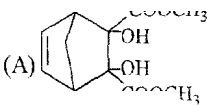
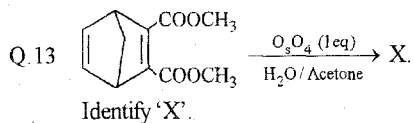
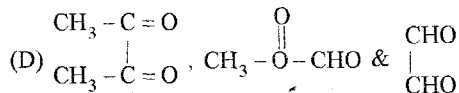
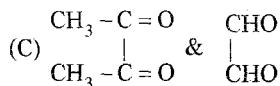
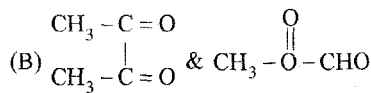
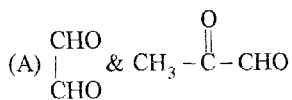
Alkene (P) & (Q) respectively are



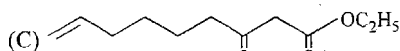
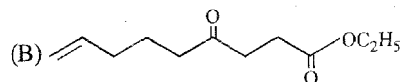
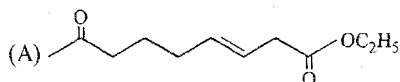
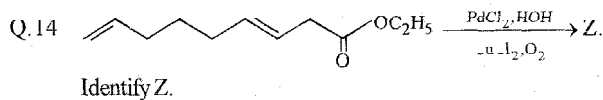
Q.11 Ozonolysis of CH₃-CH=C=CH₂ will give
(A) Only CH₃CHO
(C) Only CO₂

(B) Only HCHO
(D) Mixture of CH₃CHO, HCHO & CO₂

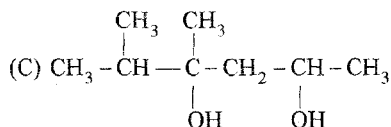
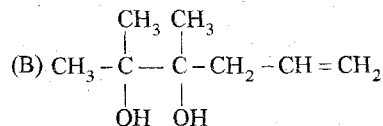
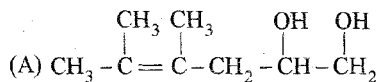
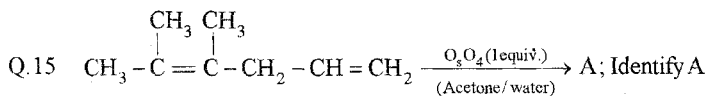
Q.12 O-xylene on ozonolysis will give



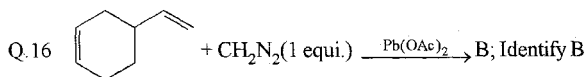
(D) Reaction will not occur



(D) All are correct



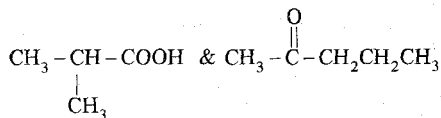
(D) Reaction will not occur



Q. 17 1-Penten-4-yne reacts with bromine at -80°C to produce:

- (A) 4,4,5,5-Tetrabromopentene (B) 1,2-Dibromo-1,4-pentadiene
(C) 1,1,2,2,4,5-hexabromopentane (D) 4,5-dibromopentyne

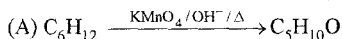
Q. 18 Compound (A) on oxidation with hot $\text{KMnO}_4/\text{OH}^\ominus$ gives two compound



compound A will have structure.

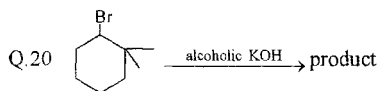
- (A) $\text{CH}_3\text{CH}_2 - \underset{\text{CH}_3}{\text{C}} = \underset{\text{CH}_3}{\text{C}} - \text{CH}_2\text{CH}_3$ (B) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CH}_2\text{CH}_2\text{CH}_3$
(C) $\text{CH}_3\underset{\text{CH}_3}{\text{CH}} - \text{C} \equiv \text{C} - \text{CH}_3$ (D) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{C} \equiv \text{C} - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$

Q. 19 Consider the following reaction



In the above reaction (A) will be

- (A) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH} = \text{CH}_2$ (B) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{CH} = \text{CH}_2$
(C) $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} - \text{CH} = \text{CH}_2$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2 - \underset{\text{CH}_3}{\text{C}} = \text{CH}_2$



Major product is:

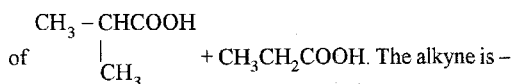


Q. 21 Number of required O_2 mole for complete combustion of one mole of propane –

- (A) 7 (B) 5 (C) 16 (D) 10

- Q.22 How much volume of air will be needed for complete combustion of 10 lit. of ethane –
 (A) 135 lit. (B) 35 lit. (C) 175 lit. (D) 205 lit.
- Q.23 When n-butane is heated in the presence of AlCl_3/HCl it will be converted into –
 (A) Ethane (B) Propane (C) Butene (D) Isobutane
- Q.24 The reacting species of alc. KOH is –
 (A) OH^- (B) OR^+ (C) OK^- (D) RO^-
- Q.25 The product of reaction between one mole of acetylene and two mole of HCHO in the presence of Cu_2Cl_2 –
 (A) $\text{HOCH}_2 - \text{C} \equiv \text{C} - \text{CH}_2\text{OH}$ (B) $\text{H}_2\text{C} = \text{CH} - \text{C} \equiv \text{C} - \text{CH}_2\text{OH}$
 (C) $\text{HC} \equiv \text{C} - \text{CH}_2\text{OH}$ (D) None of these
- Q.26 PMA polymer is formed by methyl acrylate, which is prepared as follows –
 (A) $\text{R} - \text{C} \equiv \text{CH} \xrightarrow{\text{CO} + \text{ROH}}$ (B) $\text{HC} \equiv \text{CH} \xrightarrow[\text{Ni}(\text{CO})_4]{\text{CO} + \text{CH}_3\text{OH}}$
 (C) $\text{HC} \equiv \text{CH} \xrightarrow[\text{Ni}(\text{CO})_4]{\text{CO} + \text{H}_2\text{O}}$ (D) None of these
- Q.27 During the preparation of ethane by Kolbe's electrolytic method using inert electrodes the pH of the electrolyte –
 (A) Increases progressively as the reaction proceeds
 (B) Decreases progressively as the reaction proceeds
 (C) Remains constant throughout the reaction
 (D) May decrease if the concentration of the electrolyte is not very high
- Q.28 Ethylene forms ethylene chlorohydrin by the action of –
 (A) Dry HCl gas (B) Dry chlorine gas
 (C) Solution of chlorine gas in water (D) Dilute hydrochloric acid
- Q.29 Anti-Markownikoff's addition of HBr is not observed in –
 (A) Propene (B) But-2-ene (C) Butene (D) Pent-2-ene
- Q.30 Which alkene on heating with alkaline KMnO_4 solution gives acetone and a gas, which turns lime water milky –
 (A) 2-Methyl-2-butene (B) Isobutylene
 (C) 1-Butene (D) 2-Butene
- Q.31 Acetylene may be prepared using Kolbe's electrolytic method employing –
 (A) Pot. acetate (B) Pot. succinate (C) Pot. fumarate (D) None of these
- Q.32 $\text{B} \xleftarrow{\text{I indlar}} \text{R} - \text{C} \equiv \text{C} - \text{R} \xrightarrow{\text{Na}/\text{NH}_3} \text{A}$
 A and B are geometrical isomers ($\text{R} - \text{CH} = \text{CH} - \text{R}$) –
 (A) A is trans, B is cis (B) A and B both are cis
 (C) A and B both are trans (D) A is cis, B is trans
- Q.33 Which is expected to react most readily with bromine –
 (A) $\text{CH}_3\text{CH}_2\text{CH}_3$ (B) $\text{CH}_2 = \text{CH}_2$ (C) $\text{CH} \equiv \text{CH}$ (D) $\text{CH}_3 - \text{CH} = \text{CH}_2$
- Q.34 By the addition of CO and H_2O on ethene, the following is obtained –
 (A) Propanoic acid (B) Propanal (C) 2-Propanoic acid (D) None of the above

Q.35 An alkyne C_7H_{12} on reaction with alk. $KMnO_4$ and subsequent acidification with HCl yields a mixture



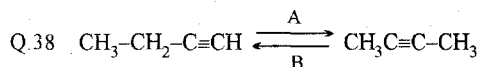
- (A) 3-Hexyne (B) 2-Methyl-3-hexyne
(C) 2-Methyl-2-hexyne (D) 2-Methyl-2-hexene

Q.36 A compound (C_5H_8) reacts with ammonical $AgNO_3$ to give a white precipitate and reacts with excess of $KMnO_4$ solution to give $(CH_3)_2CH-COOH$. The compound is -

- (A) $CH_2=CH-CH=CH-CH_3$ (B) $(CH_3)_2CH-C\equiv CH$
(C) $CH_3(CH_2)_2C\equiv CH$ (D) $(CH_3)_2C=C=CH_2$

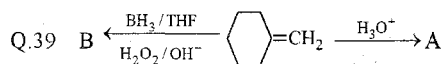
Q.37 Which of the following reagents cannot be used to locate the position of triple bond in $CH_3-C\equiv C-CH_3$

- (A) Br_2 (B) O_3 (C) Cu_2^{2+} (D) $KMnO_4$



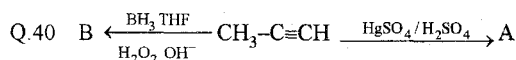
A and B are -

- (A) alcoholic KOH and $NaNH_2$ (B) $NaNH_2$ and alcoholic KOH
(C) $NaNH_2$ and Lindlar (D) Lindlar and $NaNH_2$



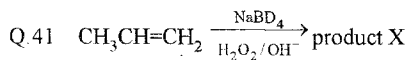
A and B are -

- (A) Both Cyclohexyl-CH_2OH (B) Both Cyclohexyl-CH_2OH
(C) Cyclohexyl-CH_2OH , Cyclohexyl-CH_2OH (D) Cyclohexyl-CH_2OH , Cyclohexyl-CH_2OH

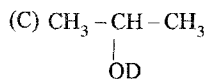
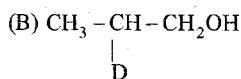
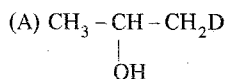


A and B are -

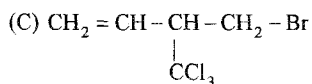
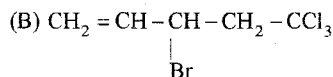
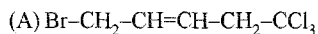
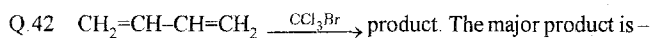
- (A) CH_3CH_2CHO , $CH_3-C(=O)-CH_3$ (B) $CH_3-C(=O)-CH_3$, CH_3CH_2CHO
(C) CH_3CH_2CHO (both) (D) $CH_3-C(=O)-CH_3$ (both)



X is -



(D) none is correct



(D) None is correct

Q 43 Mixture of one mole each of ethene and propyne on reaction with Na will form H_2 gas at S.T.P. -

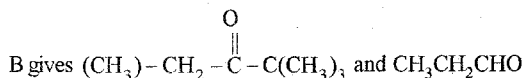
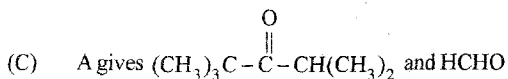
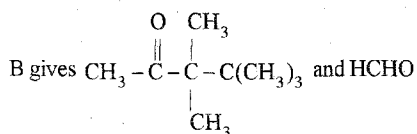
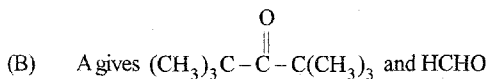
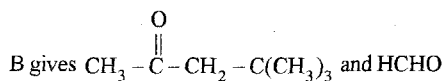
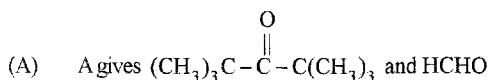
(A) 22.4 L

(B) 11.2 L

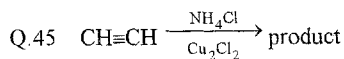
(C) 33.6 L

(D) 44.8 L

Q 44 Dehydration of 2, 2, 3, 4, 4-pentamethyl-3-pentanol gave two alkenes A and B. The ozonolysis products of A and B are -

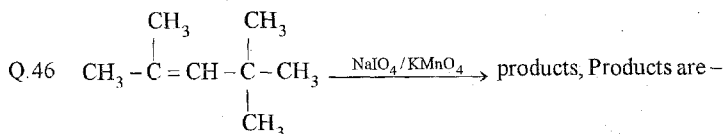


(D) None of these

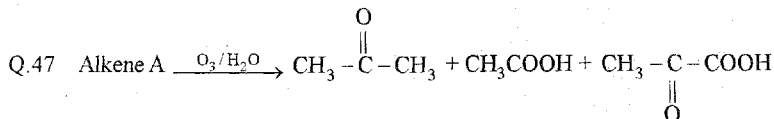


Product is –

- (A) $\text{Cu}-\text{C}\equiv\text{C}-\text{Cu}$ (B) $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$ (C) $\text{CH}\equiv\text{C}-\text{Cu}$ (D) $\text{Cu}-\text{C}\equiv\text{C}-\text{NH}_4$

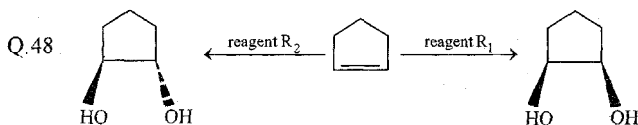


- (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$, $(\text{CH}_3)_3\text{C}-\text{COOH}$ (B) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$, $(\text{CH}_3)_3\text{C}-\text{CHO}$
 (C) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$, $(\text{CH}_3)_3\text{C}-\text{COOH}$ (D) None is correct



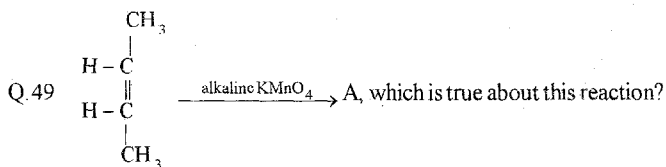
A can be –

- (A) $\text{CH}_3-\overset{\text{CHCH}_3}{\parallel}{\text{C}}-\text{CH}=\text{C}-\text{CH}_3$ (B) $\text{CH}_3-\overset{\text{C}(\text{CH}_3)_2}{\parallel}{\text{C}}-\text{CH}=\text{HC}-\text{CH}_3$
 (C) Both correct (D) None is correct

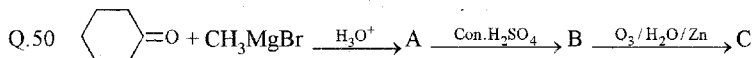


R_1 and R_2 are –

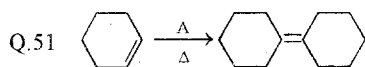
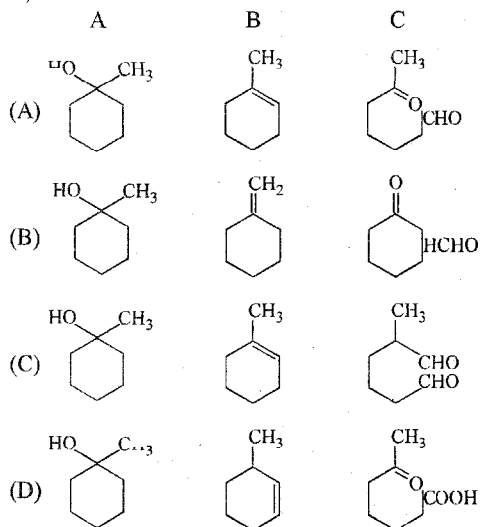
- (A) Cold alkaline KMnO_4 , $\text{OsO}_4/\text{H}_2\text{O}_2$ (B) Cold alkaline KMnO_4 , HCO_3H
 (C) Cold alkaline KMnO_4 , $\text{CH}_3-\text{O}-\text{O}-\text{CH}_3$ (D) $\text{C}_6\text{H}_5\text{CO}_3\text{H}$, HCO_3H



- (A) A is meso 1, 2-butan-di-ol formed by *syn* addition
 (B) A is meso 1, 2-butan-di-ol formed by *anti* addition
 (C) A is a racemic mixture of d and l, 1, 2-butan-di-ol formed by anti addition
 (D) A is a racemic mixture of d and l, 1, 2-butan-di-ol formed by *syn* addition



A, B and C are –



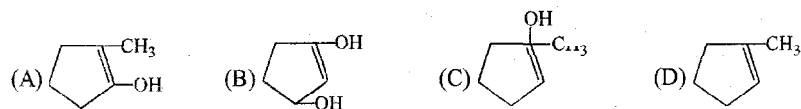
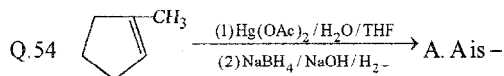
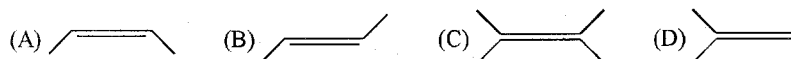
A can be –

- (A) Conc. H_2SO_4 (B) alcoholic KOH (C) Et_3N (D) t-BuOK

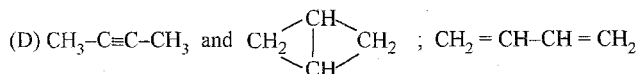
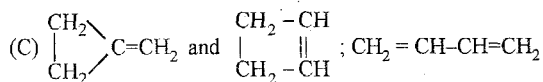
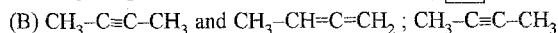
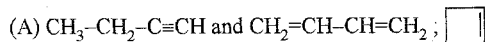
Q.52 $BrCH_2-CH_2-CH_2Br$ reacts with Na in the presence of ether at $100^\circ C$ to produce –

- (A) $BrCH_2-CH=CH_2$ (B) $CH_2=C=CH_2$ (C)
(D) All of these

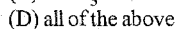
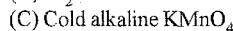
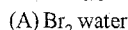
Q.53 Which has least heat of hydrogenation –



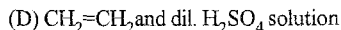
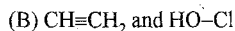
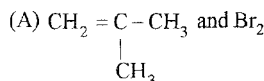
Q. 55 An organic compound of molecular formula C_4H_6 , (A), forms precipitates with ammoniacal silver nitrate and ammoniacal cuprous chloride. 'A' has an isomer 'B', one mol of which reacts with one mol of Br_2 to form 1, 4-dibromo-2-butene. Another isomer of A is 'C', one mole of C reacts with only 1 mol. of Br_2 to give vicinal dibromide. A, B & C are



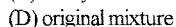
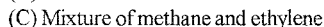
Q. 56 $CH_3-CH=CH-CH_3 \xrightarrow[\text{cis}]{X}$ product is Y (non-resolvable) then X can be –



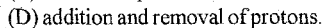
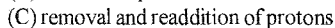
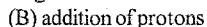
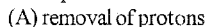
Q. 57 Electrophilic addition reaction is not shown by



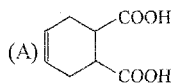
Q. 58 A mixture of CH_4 , C_2H_4 and C_2H_2 gaseous are passed through a Wolf bottle containing ammonical cuprous chloride. The gas coming out is



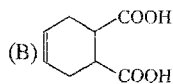
Q. 59 In the presence of strong bases, triple bonds will migrate within carbon skeletons by the



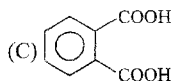
Q. 60 $CH_2=CH-CH=CH_2 + \begin{array}{c} CHCOOH \\ || \\ CHCOOH \end{array} \xrightarrow{\Delta}$ product X by reaction R. X and R are



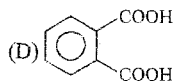
Diels Alder



Friedel-Crafts



Diels Alder



Friedel-Crafts

Q. 61 For the ionic reaction of hydrochloric acid with the following alkenes, predict the correct sequence of reactivity as measured by reaction rates:

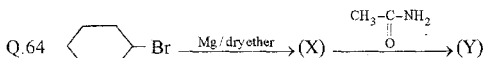
- (I) $\text{ClCH}=\text{CH}_2$ (II) $(\text{CH}_3)_2\text{C}=\text{CH}_2$ (III) $\text{OHC}.\text{CH}=\text{CH}_2$ (IV) $(\text{NC})_2\text{C}=\text{C}(\text{CN})_2$
 (A) $\text{IV} > \text{I} > \text{III} > \text{II}$ (B) $\text{I} > \text{IV} > \text{II} > \text{III}$ (C) $\text{III} > \text{II} > \text{IV} > \text{I}$ (D) $\text{II} > \text{I} > \text{III} > \text{IV}$

Q. 62 The addition of bromine to 2-cyclohexenyl benzoate in 1,2-dichloroethane produces ____ dibromo derivatives:

- (A) 2 (B) 3 (C) 4 (D) 6

Q. 63 How many products will be formed when methylenecyclohexane reacts with NBS?

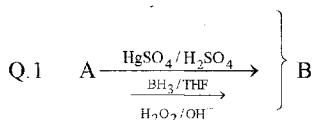
- (A) 3 (B) 1 (C) 2 (D) 4



The structures of (X) and (Y) respectively are

- (A) $\text{X} = \text{Cyclohexyl-MgBr}$; $\text{Y} = \text{Cyclohexyl-OH}$
 (B) $\text{X} = \text{Cyclohexyl-Mg-Br}$; $\text{Y} = \text{Cyclohexyl-OH}$
 (C) $\text{X} = \text{Cyclohexyl-MgBr}$; $\text{Y} = \text{Cyclohexyl}$
 (D) $\text{X} = \text{BrMg-Cyclohexyl-MgBr}$; $\text{Y} = \text{HO-Cyclohexyl-OH}$

EXERCISE-I (B)

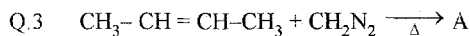


B is identical when A is -

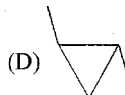
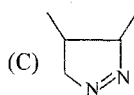
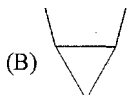
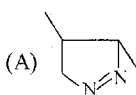
- (A) $\text{H}-\text{C}\equiv\text{C}-\text{H}$ (B) $\text{CH}_3-\text{C}\equiv\text{C}-\text{H}$ (C) $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3$ (D) $\text{CH}_3-\text{CH}=\text{CH}_2$

Q.2 An alkene on ozonolysis yields only ethanal. There is an isomer of this which on ozonolysis yields:

- (A) propanone (B) ethanal
(C) methanal (D) only propanal

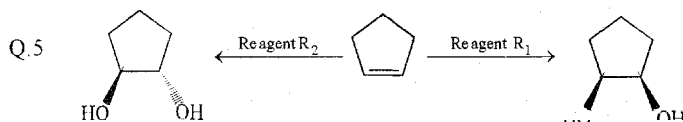


A can be



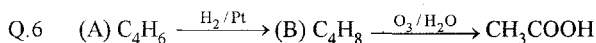
Q.4 Aqueous solution of potassium acetate is electrolysed. Possible organic products are:

- (A) n-Butane (B) $\text{C}_2\text{H}_5\text{COOC}_2\text{H}_5$ (C) CH_3-CH_3 (D) $\text{CH}_2=\text{CH}_2$



R_1 and R_2 are:

- (A) cold alkaline KMnO_4 , $\text{OsO}_4/\text{H}_2\text{O}_2$ (B) $\text{OsO}_4/\text{NaHSO}_3$; AgO_2 , H_3O^+
(C) cold alkaline KMnO_4 , $\text{C}_6\text{H}_5\text{CO}_3\text{H}/\text{H}_3\text{O}^+$ (D) $\text{C}_6\text{H}_5\text{CO}_3\text{H}$, $\text{OsO}_4/\text{NaHSO}_3$

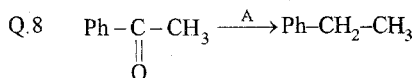


Hence A and B are

- (A) $\text{CH}_3\text{C}\equiv\text{CCH}_3$, $\text{CH}_3\text{CH}=\text{CHCH}_3$ (B) $\text{CH}_2=\text{CHCH}_3$, $\text{CH}_3\text{CH}=\text{CHCH}_3$
(C) , $\text{CH}_3\text{CH}=\text{CHCH}_3$ (D) None

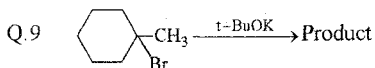
Q.7 Which is/are true statements/ reactions?

- (A) $\text{Al}_2\text{C}_3 + \text{H}_2\text{O} \longrightarrow \text{CH}_4$ (B) $\text{CaC}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2$
(C) $\text{Mg}_2\text{C}_3 + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{C}\equiv\text{CH}$ (D) $\text{Me}_3\text{C}-\text{H} + \text{aq. KOH} \longrightarrow \text{Me}_3\text{C}-\text{OH}$

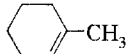
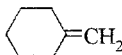
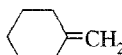
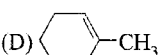


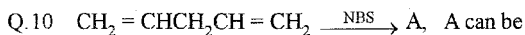
A could be:

- (A) NH_2NH_2 , glycol/ OH^- (B) $\text{Sn}(\text{Hg})/\text{conc. HCl}$
(C) P/HI (D) $\text{CH}_2=\text{CH}_2$; Raney Ni - H_2



which is / are correct statements about the product:

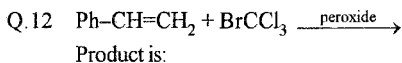
- (A)  is an endocyclic Saytzeff product
- (B)  is an exocyclic Saytzeff product
- (C)  is an exocyclic Hoffmann product
- (D)  is an endocyclic Hoffmann product

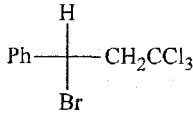
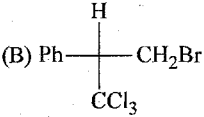
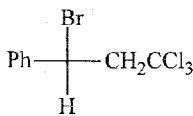
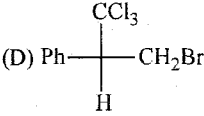


- (A) $\text{CH}_2 = \underset{\text{Br}}{\text{CH}}\text{CHCH} = \text{CH}_2$ (B) $\text{CH}_2 = \text{CHCH} = \text{CH}-\text{CH}_2\text{Br}$
- (C) $\text{CH}_2 = \text{CHCH}_2\text{CH} = \text{CHBr}$ (D) $\text{CH}_2 = \text{CHCH}_2\underset{\text{Br}}{\text{C}} = \text{CH}_2$

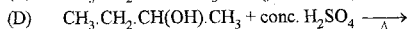
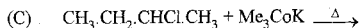
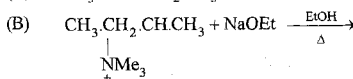
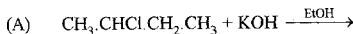
Q.11 Which are correct statements?

- (A) meso-1, 2-dibromo-butane on reaction with NaI / acetone gives trans-2-butene
- (B) d-or l- 1, 2-dibromobutane on reaction with NaI/acetone gives cis-2-butene
- (C) meso-1, 2-dibromo-butane on reaction with NaI / acetone gives cis-2-butene
- (D) d-or l- 1, 2-dibromobutane on reaction with NaI/acetone gives trans-2-butene

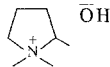


- (A)  (B) 
- (C)  (D) 

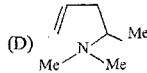
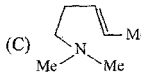
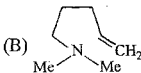
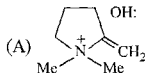
Q. 13 Which of the following elimination reactions will occur to give but-1-ene as the major product?



Q. 14



The above compound undergoes ready elimination on heating to yield which of the following products?



Q. 15 Which of the following will give same product with HBr in presence or absence of peroxide.

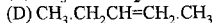
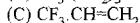
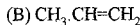
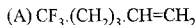
(A) Cyclohexene

(B) 1-methylcyclohexene

(C) 1,2-dimethylcyclohexene

(D) 1-butene

Q. 16 The ionic addition of HCl to which of the following compounds will produce a compound having Cl on carbon next to terminal.



Q. 17 Select true statement(s):

(A) I_2 does not react with ethane at room temperature even though I_2 is more easily cleaved homolytically than the other halogens.

(B) Stereochemical outcome of a radical substitution and a radical addition reaction is identical.

(C) The rate of bromination of methane is decreased if HBr is added to the reaction mixture.

(D) Allylic chloride adds halogens faster than the corresponding vinylic chloride.

Q. 18 Select true statement(s):

(A) Instead of radical substitution, cyclopropane undergoes electrophilic addition reactions in sun light.

(B) In general, bromination is more selective than chlorination.

(C) The 2,4,6-tri-tert, butylphenoxy radical is resistant to dimerization.

(D) The radical-catalysed chlorination, $\text{ArCH}_3 \rightarrow \text{ArCH}_2\text{Cl}$, occurs faster when Ar = phenyl than when Ar = p-nitrophenyl.

Q. 19 Nitrene is an intermediate in one of the following reactions:

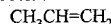
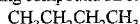
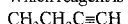
(A) Schmidt rearrangement

(B) Beckmann rearrangement

(C) Baeyer-Villiger oxidation

(D) Curtius reaction

Q. 20 Which reagent is the most useful for distinguishing compound I from the rest of the compounds



I

II

III

IV

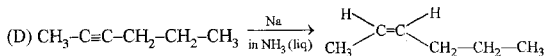
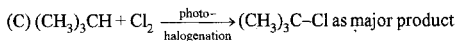
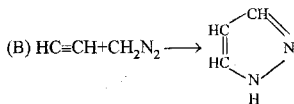
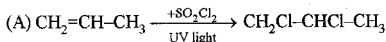
(A) alk. KMnO_4

(B) Br_2/CCl_4

(C) $\text{Br}_2/\text{CH}_3\text{COOH}$

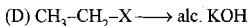
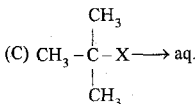
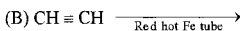
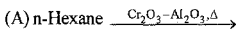
(D) Ammonical AgNO_3

- Q.21 In most of the rearrangements, the migrating group migrates with
 (A) inversion of its configuration (B) retention of its configuration
 (C) racemization of its configuration (D) retention of its structure
- Q.22 Indicate among the following the reaction **not correctly** formulated.



Q.23

List I



List II

(1) Substitution reaction

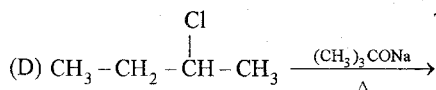
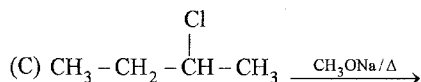
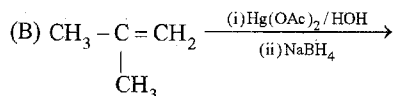
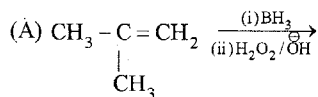
(2) Elimination reaction

(3) Aromatisation

(4) Cyclization

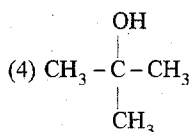
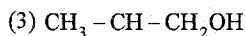
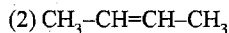
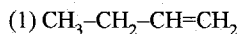
Q.24

List I



Codes:	A	B	C	D
(a)	4	3	1	2
(b)	4	3	2	1
(c)	3	4	2	1
(d)	3	4	1	2

List II

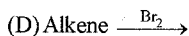
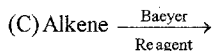


Q.25

List I

(A) Walden Inversion

(B) Racemic mixture



Codes:	A	B	C	D
(a)	3	4	2	1
(b)	3	4	1	2
(c)	4	3	1	2
(d)	4	3	2	1

List II

(1) Cis addition

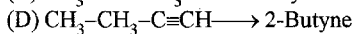
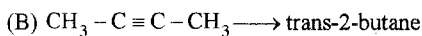
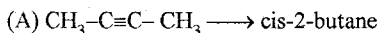
(2) Trans addition

(3) SN_1 reaction

(4) SN_2 reaction

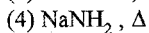
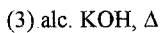
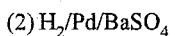
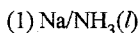
Q.26

List I



Codes:	A	B	C	D
(a)	2	1	3	4
(b)	1	2	4	3
(c)	1	2	3	4
(d)	2	1	4	3

List II



Q. 27

List I

- (A) $\text{RCOONa} \xrightarrow{\text{electrolysis}} \text{R-R}$
 (B) $\text{R-CH}_2\text{-COOH} \xrightarrow[\Delta]{\text{Soda lime}} \text{R-CH}_3$
 (C) $\text{RCOOH} \xrightarrow[\text{(ii) Cl}_2/\Delta]{\text{(i) AgNO}_3} \text{R-Cl}$
 (D) $\text{R'-X} + \text{R}_2\text{CuLi} \longrightarrow \text{R-R'}$

Codes:

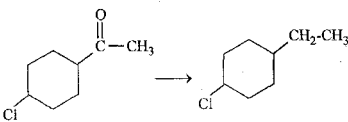
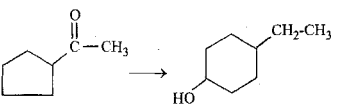
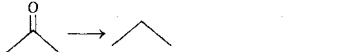
	A	B	C	D
(a)	2	3	4	1
(b)	1	3	4	2
(c)	2	4	3	2
(d)	2	4	1	3

List II

- (1) Corey-Housh reaction
 (2) Kolbe electrolysis
 (3) Oakwood degradation
 (4) Hunsdiecker reaction

Q. 28

List I

- (A) 
 (B) 
 (C) 
 (D) 

List II

- (1) Birch reduction
 (2) Stephen's reduction
 (3) Wolf-Kishner reduction
 (4) Clemmensen reduction

Q. 29

List I

- (A) n-Hexane $\longrightarrow$ Benzene
 (B) $\text{CH}\equiv\text{CH} \longrightarrow$ Benzene
 (C) $\text{CH}_3\text{-(CH}_2\text{)}_6\text{CH}_3 \longrightarrow$ 2,2,3,3 tetramethyl butane
 (D) $\text{CH}_3\text{-CH}_2\text{-X} \longrightarrow$ n-Butane

List II

- (1) Wurtz reaction
 (2) Coupling of reactants is taking place
 (3) $\text{AlCl}_3 + \text{HCl}$ at 300°C
 (4) Polymerisation
 (5) Aromatic products is formed
 (6) $\text{Zn} + \Delta$ used as reagent
 (7) Al_2O_3 at high temperature

List I(A) Alkene $\xrightarrow[\text{Reagent}]{\text{Baeyer}}$?(B) Alkene $\xrightarrow[\text{in CCl}_4]{\text{Br}_2}$?(C) $\text{—}\equiv\text{—}$ $\xrightarrow[\text{BaSO}_4]{\text{H}_2 \text{ Pd.}}$?(D) Alkene $\xrightarrow[\text{by H}_3\text{O}^+]{\text{PhCO}_3\text{H}}$?(E) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{CH}_2 \xrightarrow[\text{(ii) NaBH}_4]{\text{(i) Hg(OAc)}_2 / \text{HOH}}$ **List II**

(1) Neighbouring group participation seen

(2) Markovnikov's rule is followed.

(3) Cis isomer formed

(4) Syn addition

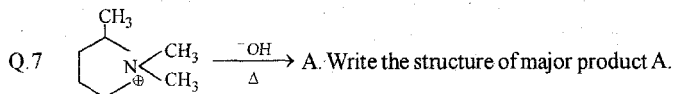
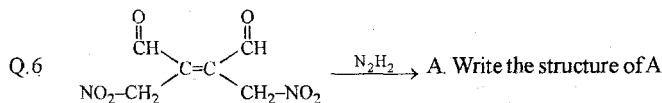
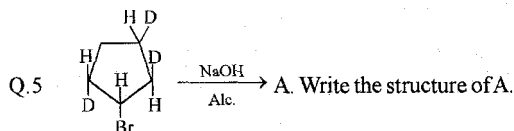
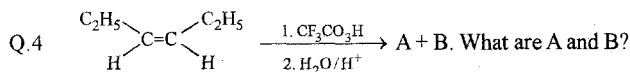
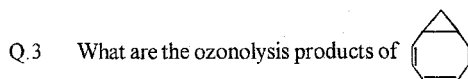
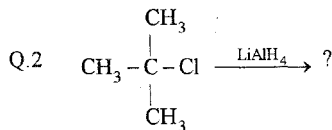
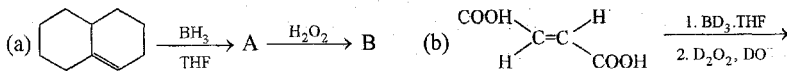
(5) Anti addition

(6) Racemic mixture in case of But-2-ene (cis)

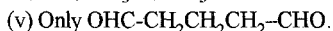
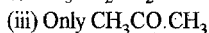
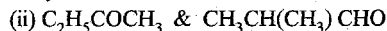
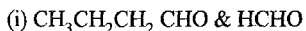
(7) Meso compound in case of But-2-ene (cis)

EXERCISE-II

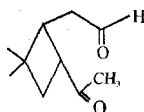
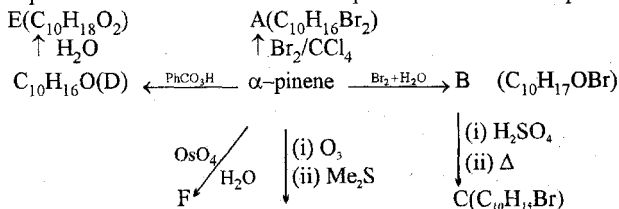
Q.1 Give the product of



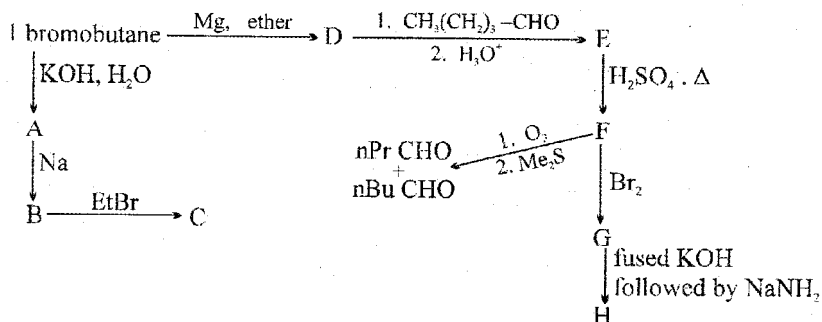
Q.8 Give the structure of the alkene that yields on ozonolysis



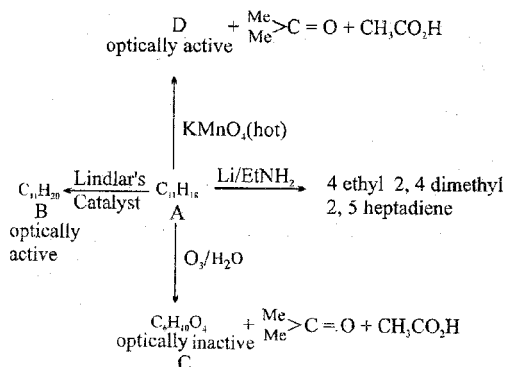
Q.9 One of the constituent of turpentine is α -pinene having molecular formula $\text{C}_{10}\text{H}_{16}$. The following scheme give reaction of α -pinene. Determine the structure of α -pinene & of the reaction products A through E.



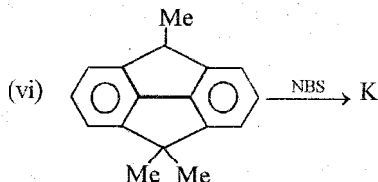
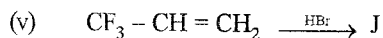
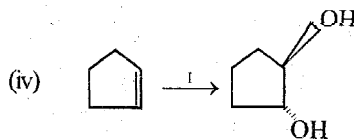
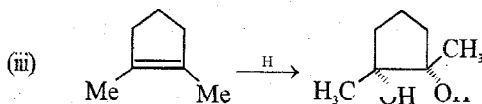
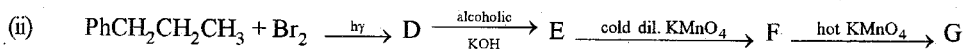
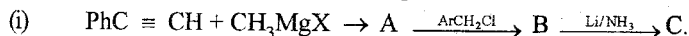
Q.10 Propose structures for intermediates & products A to K



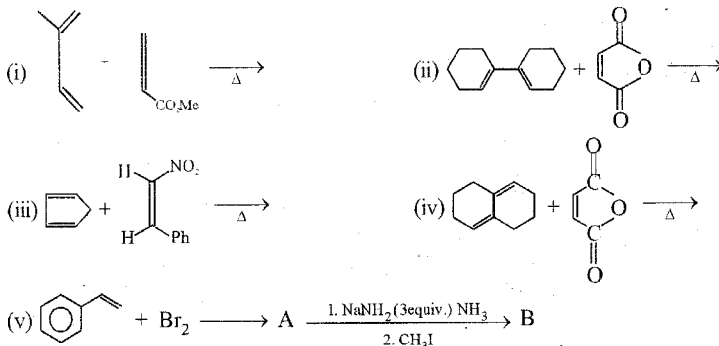
Q.11 Identify the following (A to D).



Q.12 What are A to K for the following reactions



Q. 13 What will be the product in the following reaction



Q. 14 (i) Compare the reaction of $\text{CH}_2=\text{CH}_2$ & $\text{CF}_2=\text{CF}_2$ with NaOEt in EtOH
 (ii) $\text{CCl}_2=\text{CCl}_2$ does not decolourise Br_2 solution - explain.

Q. 15 Account for the following facts

- Ozonolysis if carried out in MeOH solvent a hydroxy peroxy ether is formed as unexpected product.
- When 2, 3 dimethyl 2 butene is treated with O_3 in presence of HCHO in CH_2Cl_2 medium, an ozonide other than that expected of the starting alkene is formed. Identify the unexpected ozonide.

Q. 16 Explain the following:

- 1, 2 shift does not take place during oxymercuration demercuration. Why?
- Halogenation of alkene is anti addition but not syn addition. Why?
- Anti markovnikov addition is not applicable for HCl . Why?
- 1,4-addition takes place in butadi-ene. Why?
- C-H bond is stronger than C-C bond but in chlorination C-H bonds get cleaved but not C-C bond. Why?

Q. 17 Conversion:

- (i) $\text{C}_2\text{H}_2 \longrightarrow$ racemic 2, 3 dibromobutane (ii) 2 butyne $\longrightarrow$ 2 pentyne
 (iii) Ethyne $\longrightarrow$ Acetone (iv) Methane $\longrightarrow$ n Butane (v) Ethene $\longrightarrow$ Propionic Acid

Q. 18 Conversion:

- (i) $\text{C}_2\text{H}_2 \longrightarrow$ ethylidene diacetate (ii) $\text{C}_2\text{H}_2 \longrightarrow$ Butyne diol
 (iii) $\text{C}_2\text{H}_2 \longrightarrow$ m nitroaniline (iv) cis but 2 ene $\longrightarrow$ Trans but 2 ene

Q. 19 Outline a stereospecific synthesis of meso 3, 4 dibromohexane from ethyne.

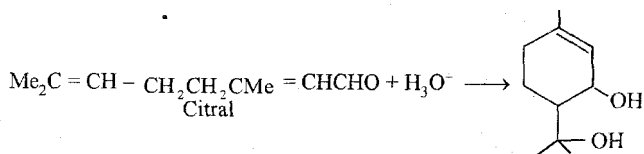
Q. 20 How can you convert

- Ethane into meso 2, 3 dimethyl oxiran
- CaC_2 into 1, 3, 5 hexatriene
- Trimethylsecbutyl ammonium hydroxide into 1,4-butan-dial
- Cyclo hexanol into trans cyclo hexane-1, 2-diol

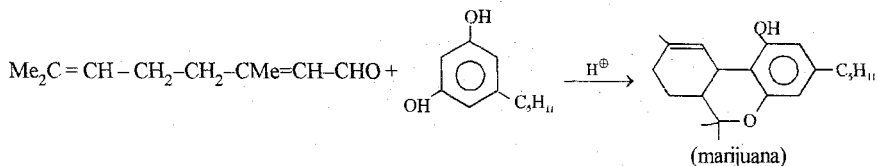
Q. 21 How will you convert

- Hexane dial into 1,3,5 hexatriene
- 1-methyl propyl ethanoate into 1,4-dichloro-2-butene

Q.22 Explain the mechanism of following conversion.

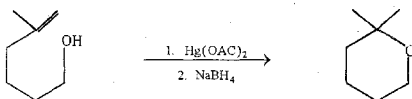


Q.23 When citral is allowed to react in presence of dilute acid with olivetol, there is obtained a mixture of products, one of which is drug marijuana. Reaction is as follows.



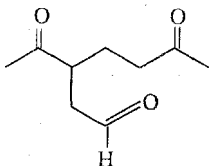
Explain the mechanism.

Q.24 The following cyclisation has been observed in the oxymercuration & demercuration of this unsaturated alcohol. Propose a mechanism for this reaction.



Q.25 Write the structural formula of limonene from the following observation:

- Limonene when treated with excess H_2 & Pt catalyst, the product formed is 1 isopropyl - 4 methyl cyclohexane
- When it is treated with O_3 & then $\text{Zn}/\text{H}_2\text{O}$ the products of the reaction are HCHO & following compound



Q.26 (a) $\text{MeCH}_2-\text{C}\equiv\text{CBr} + \text{CH}\equiv\text{CMe} \xrightarrow{\text{Cu}^+} \text{A}$

(b) $\text{CH}_2(\text{Cl})-\text{CHCl}_2 \xrightarrow{\text{OH}^-} \text{B}$

(c) $\text{CH}_2=\text{CH}-\underset{\text{OH}}{\text{CH}}-\text{CH}=\text{CH}-\text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{MeOH}} \text{C}$

(d) $\text{Cyclohexyl}-\text{C}\equiv\text{CH} \xrightarrow[\text{H}_2\text{SO}_4/\text{H}_2\text{O}]{\text{Hg}^{2+}} \text{D}$

(e) $\text{Cl}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow{\text{HOBr}} \text{E}$

(f) $\text{Cyclohexyl}-\text{C}(\text{OH})(\text{Me})_2 \xrightarrow{\text{H}^+} \text{F} \xrightarrow{\text{O}_3/\text{Zn}/\text{H}_2\text{O}} \text{G}$

- Q.27 Acetylene is acidic but it does not react with NaOH or KOH. Why?
- Q.28 $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}_2$, adds up HBr to give $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CHBr}-\text{CH}_3$ while $\text{CH}\equiv\text{C}-\text{CH}=\text{CH}_2$ adds up HBr to give $\text{CH}_2=\text{C}(\text{Br})-\text{CH}=\text{CH}_2$.
- Q.29 Chlorination of ethane to ethyl chloride is more practicable than the chlorination of n-pentane to 1-chloropentane.
- Q.30 Why n-pentane has higher boiling point than neopentane?

EXERCISE-III

- Q.1 0.37 gm of ROH was added to CH_3MgI and the gas evolved measured 112 cc at STP. What is the molecular wt. of ROH? On dehydration ROH gives an alkene which on ozonolysis gives acetone as one of the products. ROH on oxidation easily gives an acid containing the same number of carbon atoms. Give the structures of ROH and the acid with proper reasoning.
- Q.2 An alkane A (C_5H_{12}) on chlorination at 300° gives a mixture four different mono chlorinated derivatives B, C, D and E. Two of these derivatives give the same stable alkene F on dehydrohalogenation. On oxidation with hot alkaline KMnO_4 followed by acidification of F gives two products G and H. Give structures of A to H with proper reasoning.
- Q.3 There are six different alkene A, B, C, D, E and F. Each on addition of one mole of hydrogen gives G which has the lowest molecular wt hydrocarbon containing only one asymmetric carbon atom. None of the above alkene give acetone as a product on ozonolysis. Give the structures of A to F. Identify the alkenes that is likely to give a ketone containing more than five carbon atoms on treatment with a warm conc. solution of alkaline KMnO_4 .
- Q.4 3,3-dimethyl-1-butene and HI react to give two products, $\text{C}_6\text{H}_{13}\text{I}$. On reaction with alc. KOH one isomer, (I) gives back 3,3-dimethyl-1-butene the other (J) gives an alkene that is reductively ozonized to $\text{Me}_2\text{C}=\text{O}$. Give the structures of (I) and (J) and explain the formation of the later.
- Q.5 Three isomeric alkenes A, B and C, C_5H_{10} are hydrogenated to yield 2-methylbutane A and B gave the same 3° ROH on oxymercuration – demercuration. B and C give different 1° ROH's on hydroboration-oxidation. Supply the structures of A, B & C.
- Q.6 Two isomeric alkyl bromides A and B ($\text{C}_5\text{H}_{11}\text{Br}$) yield the following results in the laboratory. A on treatment with alcoholic KOH gives C and D (C_5H_{10}). C on ozonolysis gives formaldehyde and 2 methyl propanal. B on treatment with alcoholic KOH gives only C (C_5H_{10}). Deduce the structures of A, B, C and D. Ignore the possibility of geometrical and optical isomerism.
- Q.7 Give the structure of A, B and C.
- A (C_4H_8) which adds on HBr in the presence and in the absence of peroxide to give the same product $\text{C}_4\text{H}_9\text{Br}$.
 - B (C_4H_8) which when treated with $\text{H}_2\text{SO}_4 / \text{H}_2\text{O}$ give ($\text{C}_4\text{H}_{10}\text{O}$) which cannot be resolved into optical isomers.
 - C (C_6H_{12}), an optically active hydrocarbon on catalytic hydrogenation gives an optically inactive compound C_6H_{14} .
- Q.8 An alkylhalide, X, of formula $\text{C}_6\text{H}_{13}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C_6H_{12}). Both alkenes on hydrogenation gives 2,3-dimethylbutane predict the structures of X, Y and Z.
- Q.9 Identify a chiral compound C, $\text{C}_{10}\text{H}_{14}$, that is oxidized with hot KMnO_4 to Ph COOH, and an achiral compound D, $\text{C}_{10}\text{H}_{14}$, inert to oxidation under the same conditions.

- Q. 10 C_7H_{14} (A) decolorises Br_2 in CCl_4 and reacts with $Hg(OAc)_2$ in THF. H_2O followed by reduction with $NaBH_4$ to produced a resolvable compound B. (A) undergoes reductive ozonolysis to give the same compound (C) obtained by oxidation of 3-hexanol with $KMnO_4$. Identity A, B and C compound D, and isomers of A reacts with BH_3 . THF and then H_2O_2/OH to give chiral E. Oxidation of E with $KMnO_4$ or acid dichromate affords a chiral carboxylic acid F. Reductive Ozonolysis of D, gives the same compound G which is obtained by oxidation of 2-methyl-3-pentanol with $KMnO_4$ identify D, E, F and G.
- Q. 11 Three compounds A, B and C are isomers of the formula C_5H_8 . All of them decolorises bromine in CCl_4 and gives a positive test with Baeyer's reagent. All the three compounds dissolve in conc. H_2SO_4 . Compound A gives a white ppt. with ammonical silver nitrate whereas B and C do not. On hydrogenation in presence of Pt catalyst, A and B both yield n-pentane whereas C gives a product of formula C_5H_{10} . On oxidation with hot alkaline $KMnO_4$ (B) gives acetic acid and CH_3CH_2COOH . Identify A, B & C.
- Q. 12 An unsaturated hydrocarbon (A) C_6H_{10} readily gives (B) on treatment with $NaNH_2$ in liquid NH_3 . When (B) is allowed to react with 1-chloropropane a compound (C) is obtained. On partial hydrogenation in the presence of Lindlar's catalyst, (C) gives (D), C_9H_{18} . On ozonolysis, (D) gives 2, 2-dimethylpropanal and 1-butanal with proper reasoning give the structures of (A) (B), (C) and (D).
- Q. 13 A hydrocarbon A, of the formula C_8H_{10} , on ozonolysis gives compound B ($C_4H_6O_2$) only. The compound B can also be obtained from the alkylbromide (C_3H_5Br) upon treatment with magnesium in dry ether, followed by carbon dioxide and acidification. Identify A, B and C and also give equations for the reactions.
- Q. 14 An organic compound (A), C_6H_{10} on reduction first gives (B), C_6H_{12} and finally (C), C_6H_{14} . (A) on ozonolysis followed by hydrolysis gives two aldehydes (D), C_2H_4O and (E) $C_2H_2O_2$. Oxidation of (B) with acidified $KMnO_4$ gives the acid (F), $C_4H_8O_2$. Determine the structures of the compounds (A) to (F) with proper reasoning.
- Q. 15 Compound A (C_6H_{12}) is treated with Br_2 to form compound B ($C_6H_{12}Br_2$). On treating B with alcoholic KOH followed by $NaNH_2$ the compound C (C_6H_{10}) is formed. C on treatment with H_2/Pt forms 2-methylpentane. The compound 'C' does not react with ammonical Cu_2Cl_2 or $AgNO_3$. When A is treated with cold $KMnO_4$ solution, a diol D is formed which gives two acids E and F when heated with $KMnO_4$ solution. Compound E is found to be ethanoic acid. Deduce the structures from A to F.
- Q. 16 An optically active hydrocarbon (A), C_8H_{12} gives an optically inactive compound (B) after hydrogenation. (A) gives no ppt. with $Ag(NH_3)_2^+$ and gives optically inactive (C), C_8H_{14} with H_2 in presence of $Pd/BaSO_4$. Determine the structures, give suitable names for A, B, C & give your reasoning.
- Q. 17 A organic compound A having carbon and hydrogen, adds one mole of H_2 in presence of Pt catalyst to form normal hexane. On vigorous oxidation with $KMnO_4$, it gives a simple carboxylic acid containing 3 carbon atoms. Assign the structure to A.
- Q. 18 An organic compound A, C_6H_{10} , on catalytic reduction first gives B, C_6H_{12} , and finally C, C_6H_{14} . A on ozonolysis followed by hydrolysis gives two aldehydes D, C_2H_4O and E, $C_2H_2O_2$. Oxidation of B with acidified $KMnO_4$ gives acid F.
- Q. 19 A hydrocarbon has 88.89% carbon and 11.11% hydrogen. 0.405 g sample of the hydrocarbon occupies 229.54 ml at $100^\circ C$ and 1 atm pressure. It decolorises potassium permanganate solution and bromine water without evolving hydrobromic acid. It gave no precipitate with either ammoniacal silver nitrate or cuprous chloride solution. When it reacts with dilute H_2SO_4 in presence of mercuric sulphate, under appropriate conditions, methyl ethyl ketone is formed. What is the hydrocarbon. Write the structural formulae of the eight possible isomer of this compound.

- Q. 20 6g sample of a natural gas consisting of methane (CH_4) and ethylene (C_2H_4) was burned with excess of oxygen and 17.2g of carbon dioxide and some water was obtained as products. What percent by weight of the sample was ethylene.

EXERCISE-IV (A)

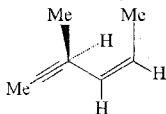
- Q. 1 Alcoholic solution of KOH is a specific reagent for – [IIT '90]
 (A) Dehydration (B) Dehydrogenation
 (C) Dehydro halogenation (D) Dehalogenation
- Q. 2 Of the following, unsaturated hydrocarbons are – [IIT '90]
 (A) ethyne (B) cyclohexane (C) n-propane (D) ethene
- Q. 3 1-chlorobutane on reaction with alcoholic potash gives – [IIT '91]
 (A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol
- Q. 4 The hybridisation of carbon atoms in C–C single bond of $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}_2$ is – [IIT '91]
 (A) sp^3-sp^3 (B) sp^2-sp^3 (C) $\text{sp}-\text{sp}^2$ (D) sp^2-sp^2
- Q. 5 The product(s) obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be –
 (A) $\text{CH}_3-\text{CH}_2-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$ (B) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CHO}$
 (C) $\text{CH}_3-\text{CH}_2-\text{CHO} + \text{HCHO}$ (D) $\text{CH}_3-\text{CH}_2-\text{COOH} + \text{HCOOH}$
- Q. 6 When cyclohexane is poured on water, it floats, because – [IIT '97]
 (A) Cyclohexane is in 'boat' form (B) Cyclohexane is in 'chair' form
 (C) Cyclohexane is in 'crown' form (D) Cyclohexane is less dense than water
- Q. 7 Which of the following compounds will show geometrical isomerism? [IIT '98]
 (A) 2-butene (B) Propene (C) 1-phenylpropene (D) 2-methyl-2-butene
- Q. 8 In the compound $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH}$, the C_2-C_3 bond is of the type – [IIT '99]
 (A) $\text{sp}-\text{sp}^2$ (B) sp^3-sp^3 (C) $\text{sp}-\text{sp}^3$ (D) sp^2-sp^3
- Q. 9 Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition – [IIT '2000]
- (A)

(B)

(C)

(D)
- Q. 10 Propyne and propene can be distinguished by – [IIT '2000]
 (A) conc. H_2SO_4 (B) Br_2 in CCl_4 (C) dil. KMnO_4 (D) AgNO_3 in ammonia
- Q. 11 In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkene because – [IIT S '2001]
 (A) both are highly ionic
 (B) one is oxidising and the other is reducing
 (C) one of the step is endothermic in both the cases
 (D) All the steps are exothermic in both cases

Q.12



Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives –

- (A) An optically active compound (B) An optically inactive compound [IIT '2001]
(C) A racemic mixture (D) A diastereomeric mixture

Q.13

The reaction of propene with HOCl proceeds via the addition of –

[IIT '2001]

- (A) H^+ in first step (B) Cl^+ in first step
(C) OH^- in first step (D) Cl^+ and OH^- in single step

Q.14

The nodal plane in the π -bond of ethene is located in –

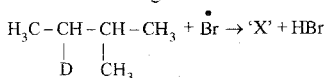
[IIT '2002]

- (A) the molecular plane
(B) a plane parallel to the molecular plane
(C) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond at right angle
(D) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond

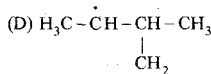
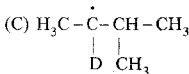
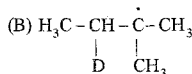
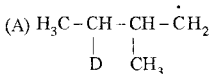
Q.15

Consider the following reactions –

[IIT '2002]



Identify the structure of the major product 'X'



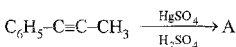
Q.16

Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne-

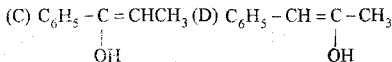
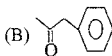
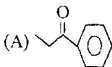
[IIT '2002]

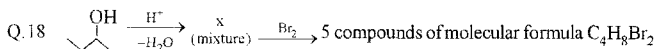
- (A) bromine, CCl_4 (B) H_2 , Lindlar catalyst
(C) dilute H_2SO_4 , HgSO_4 (D) ammoniacal Cu_2Cl_2 solution

Q.17



[IIT '2003]





Number of compounds in X will be:

- (A) 2 (B) 3 (C) 4 (D) 5

[IIT '2003]

Q.19 2-hexyne can be converted into trans-2-hexene by the action of:

- (A) $H_2-Pd-BaSO_4$ (B) Li in liq. NH_3 (C) H_2-PtO_2 (D) $NaBH_4$

[IIT '2004]

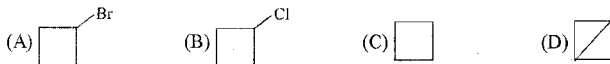
Q.20 When Phenyl Magnesium Bromide reacts with tert. butanol, which of the following is formed?

- (A) Tert. butyl methyl ether (B) Benzene
(C) Tert. butyl benzene (D) Phenol

[IIT '2005]

Q.21 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed?

[IIT '2005]



Q.22 Cyclohexene is best prepared from cyclohexanol by which of the following:

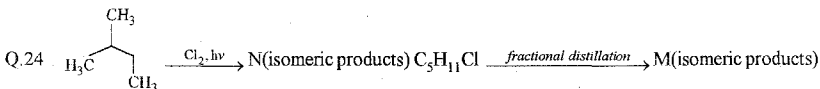
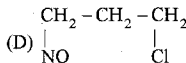
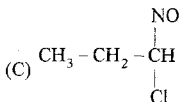
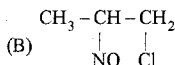
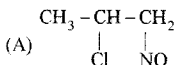
- (A) conc. H_3PO_4 (B) conc. $HCl/ZnCl_2$ (C) conc. HCl (D) conc. HBr

[IIT '2005]

Q.23 $CH_3-CH=CH_2 + NOCl \rightarrow P$

[IIT 2006]

Identify the adduct.



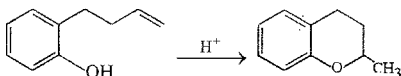
What are N and M?

- (A) 6, 6 (B) 6, 4 (C) 4, 4 (D) 3, 3

[IIT 2006]

EXERCISE-IV (B)

- Q.1 $(\text{CH}_3)_2\overset{\text{Cl}}{\underset{|}{\text{C}}}-\text{CH}_2\text{CH}_3 \xrightarrow{\text{alc. KOH}} ?$ [IIT 1992]
- Q.2 $\text{C}_6\text{H}_5\text{CH}_2\underset{\text{Br}}{\underset{|}{\text{CH}}}\text{CH}_3 \xrightarrow[\text{heat}]{\text{alc. KOH}} ? \xrightarrow{\text{HBr}} ?$ [IIT 1993]
- Q.3 $\text{C}(\text{C}_6\text{H}_{12})_4$, an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound, C_6H_{14} . [IIT 1993]
- Q.4 Draw the stereochemical structure of the product in the following reactions. [IIT 1994]
- $\text{R}-\text{C}\equiv\text{C}-\text{R} \xrightarrow[\text{Lindlar catalyst}]{\text{H}_2}$
- Q.5 Write down the structures of the stereoisomers formed when cis-2-butene is reacted with bromine. [IIT 1995]
- Q.6 An organic compound E (C_5H_8) on hydrogenation gives compound F (C_5H_{12}). Compound E on ozonolysis gives formaldehyde and 2-ketopropanal. Deduce the structure of compound E. [IIT 1995]
- Q.7 Give the structures of the major organic products from 3-ethyl-2-pentene under each of the following reaction conditions. [IIT 1996]
- (a) HBr in the presence of peroxide (b) $\text{Br}_2/\text{H}_2\text{O}$
- (c) $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}$; NaBH_4
- Q.8 An alkyl halide, (X) of formula $\text{C}_6\text{H}_{13}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes (Y) and (Z) (C_6H_{12}). Both alkenes on hydrogenation give 2, 3-dimethylbutane. Predict the structures of (X), (Y) and (Z). [IIT 1996]
- Q.9 3,3-Dimethylbutan-2-ol loses a molecule of water in the presence of concentrated sulphuric acid to give tetramethylethylene as a major product. Suggest a suitable mechanism. [IIT 1996]
- Q.10 One mole of the compound A (molecular formula C_8H_{12}), incapable of showing stereoisomerism, reacts with only one mole of H_2 on hydrogenation over Pd. A undergoes ozonolysis to give a symmetrical diketone B ($\text{C}_8\text{H}_{12}\text{O}_2$). What are the structure of A and B? [IIT 1997]
- Q.11 Compound (A) C_6H_{12} gives a positive test with bromine in carbon tetrachloride. Reaction of (A) with alkaline KMnO_4 yields only (B) which is the potassium salt of an acid. Write structure formulae and IUPAC name of (A) and (B). [IIT 1997]
- Q.12 The central carbon-carbon bond in 1,3-butadiene is shorter than that of n-butane. Why? [IIT 1998]
- Q.13 Write the intermediate steps for each of the following reaction [IIT 1998]
- $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{C}\equiv\text{CH} \rightarrow \text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$
- Q.14 Write the intermediate steps for each of the following reaction. [IIT 1998]

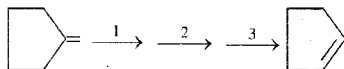


Q. 15 Discuss the hybridisation of carbon atoms in allene (C_3H_4) and show the π -orbital overlaps.

[IIT 1999]

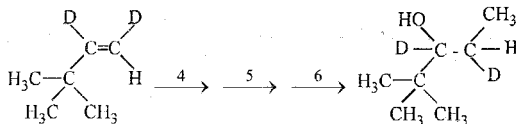
Q. 16 Complete the following –

[IIT 1999]

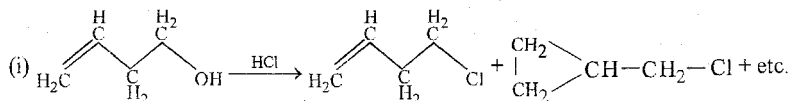


Q. 17 Complete the following –

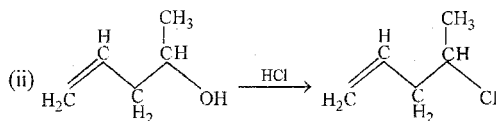
[IIT 1999]



Q. 18 Explain briefly the formation of the products giving the structures of the intermediates. [IIT 1999]



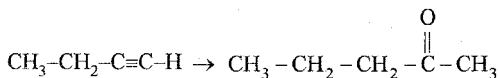
But



Explain the non formation of cyclic product in (ii)

Q. 19 Carry out the following transformation in not more than three steps.

[IIT 1999]

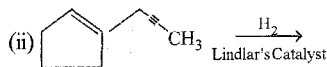
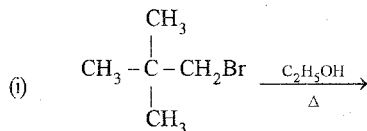


Q. 20 $CH_2=CH^-$ is more basic than $HC\equiv C^-$

[IIT 2000]

Q. 21 What would be the major product in each of the following reactions?

[IIT 2000]

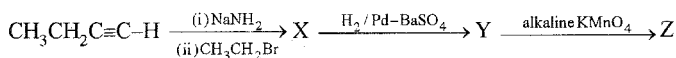


Q. 22 On reaction with 4N alcoholic KOH at 175 °C 1-pentyne is slowly converted into equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% 1,2-pentadiene (C). Give the suitable mechanism of formation of A, B and C with all intermediates.

[IIT 2001]

Q. 23 Identify X, Y and Z in the following synthetic scheme and write their structures. Is the compound Z optically active? Justify your answer.

[IIT 2002]



- Q.24 A biologically active compound, Bombykol ($C_{16}H_{30}O$) is obtained from a natural source. The structure of the compound is determined by the following reactions.
- On hydrogenation, Bombykol gives a compound A, $C_{16}H_{34}O$, which reacts with acetic anhydride to give an ester.
 - Bombykol also reacts with acetic anhydride to give another ester, which on oxidative ozonolysis (O_3/H_2O_2) gives a mixture of butanoic acid, oxalic acid and 10-acetoxy decanoic acid. Determine the number of double bonds in Bombykol. Write the structures of compound A and Bombykol. How many geometrical isomers are possible for Bombykol? [IIT 2002]
- Q.25 If after complete ozonolysis of one mole of monomer of natural polymer gives two moles of CH_2O

$$\begin{array}{c} CH_3 \\ | \\ O = C - CH = O \end{array}$$

and one mole of $O = C - CH = O$. Identify the monomer and draw the all-cis structure of natural polymer. [IIT 2005]

- Q.26  $\xrightarrow{H^+, \Delta}$ X $\xrightarrow[(ii) Zn / Cl_3COOH]{(i) O_3}$ Y.
- Identify X and Y. [IIT 2005]

ANSWER KEY

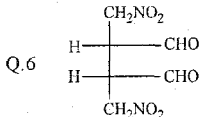
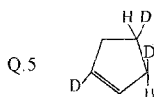
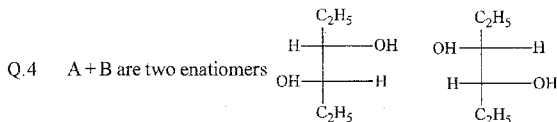
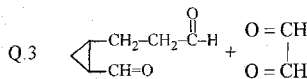
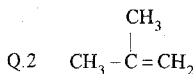
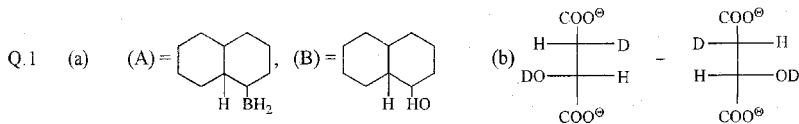
EXERCISE-I (A)

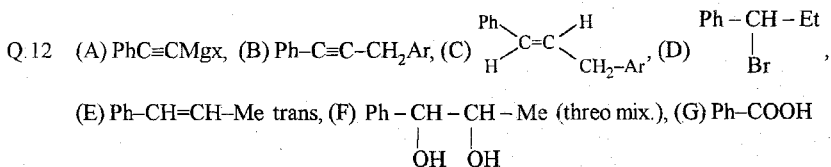
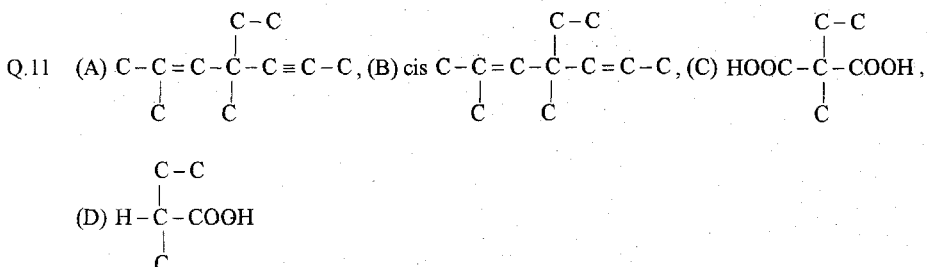
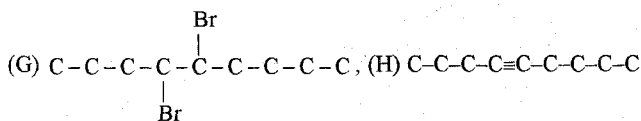
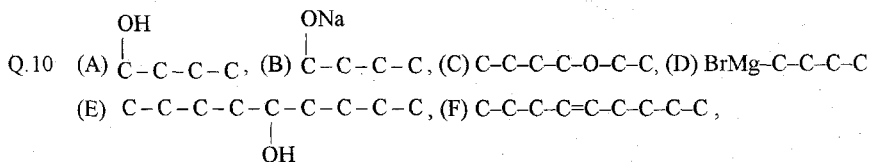
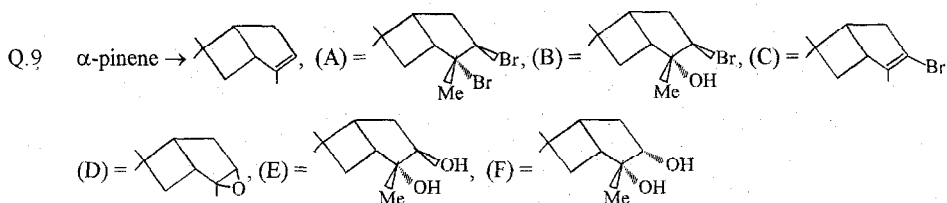
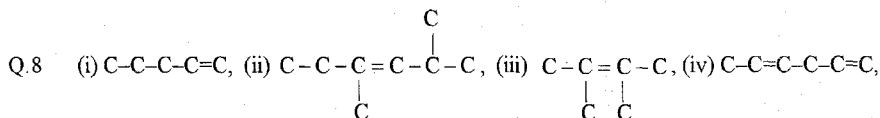
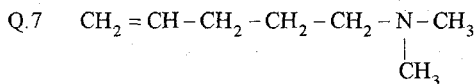
Q.1	B	Q.2	A	Q.3	A	Q.4	D	Q.5	A	Q.6	A	Q.7	C
Q.8	C	Q.9	C	Q.10	C	Q.11	D	Q.12	A	Q.13	B	Q.14	A
Q.15	B	Q.16	B	Q.17	D	Q.18	B	Q.19	D	Q.20	B	Q.21	B
Q.22	C	Q.23	D	Q.24	A	Q.25	A	Q.26	B	Q.27	A	Q.28	C
Q.29	B	Q.30	B	Q.31	C	Q.32	A	Q.33	D	Q.34	A	Q.35	B
Q.36	B	Q.37	A	Q.38	A	Q.39	D	Q.40	B	Q.41	B	Q.42	A
Q.43	B	Q.44	B	Q.45	B	Q.46	A	Q.47	C	Q.48	B	Q.49	A
Q.50	A	Q.51	A	Q.52	C	Q.53	C	Q.54	C	Q.55	A	Q.56	C
Q.57	C	Q.58	C	Q.59	C	Q.60	A	Q.61	D	Q.62	A	Q.63	A
Q.64	C												

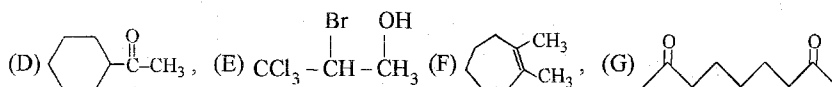
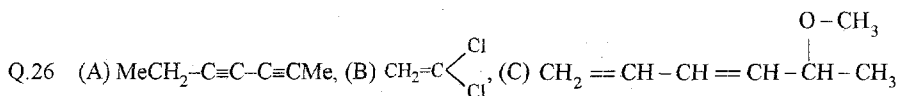
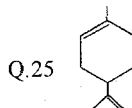
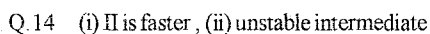
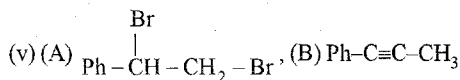
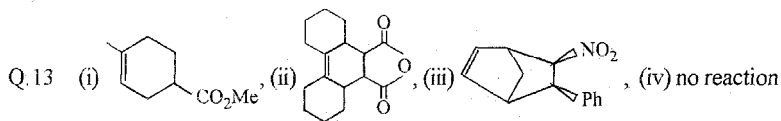
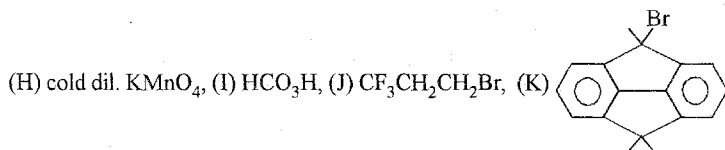
EXERCISE-I (B)

Q.1	A,C	Q.2	A,C	Q.3	A,B,C,D	Q.4	A,B,C,D
Q.5	B,C	Q.6	A,B	Q.7	A,B,C	Q.8	A,B,C,D
Q.9	A,C	Q.10	A,B	Q.11	A,B	Q.12	A,C
Q.13	B,C	Q.14	B,D	Q.15	A,C	Q.16	A,B,D
Q.17	A,C,D	Q.18	B,C,D	Q.19	A,D	Q.20	D
Q.21	B,D	Q.22	A,C,D				
Q.23	(A) 3,4; (B) 3,4; (C) 1,2; (D) 2	Q.24	(c)	Q.25	(c)		
Q.26	(d)	Q.27	(a)	Q.28	(A) 4; (B) 3; (C) 1; (D) 3,4		
Q.29	(A) 5,7; (B) 4,5; (C) 3; (D) 1,2,6	Q.30	(A) 4,7; (B) 5,6; (C) 3,4; (D) 5,6; (E) 2				

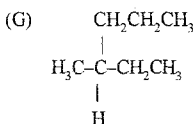
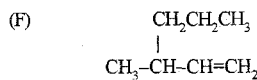
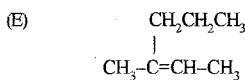
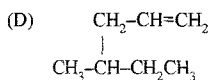
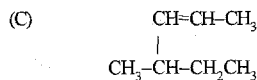
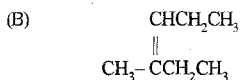
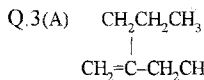
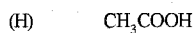
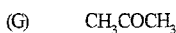
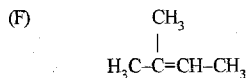
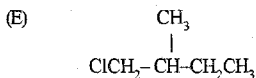
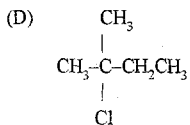
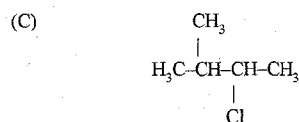
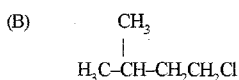
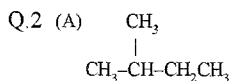
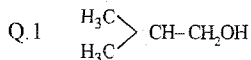
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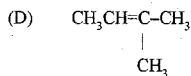
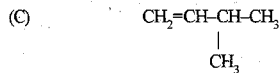
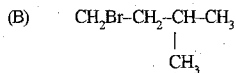
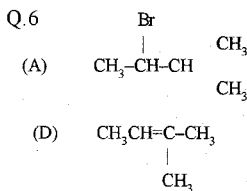
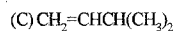
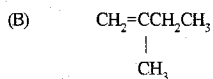
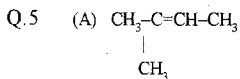
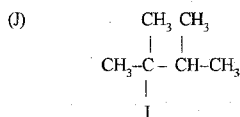
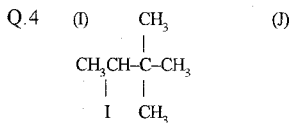




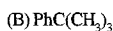
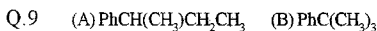
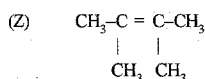
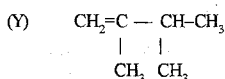
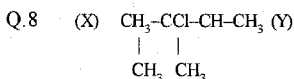
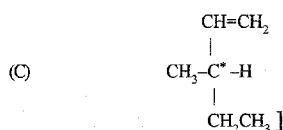
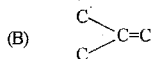
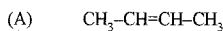


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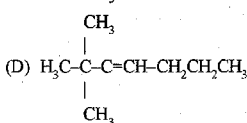
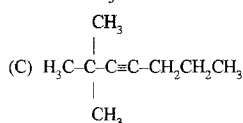
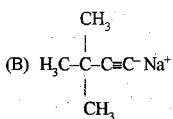
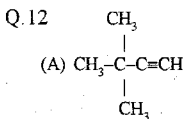
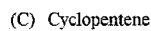
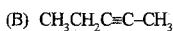
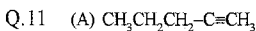
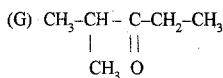
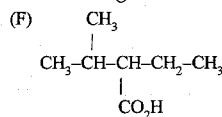
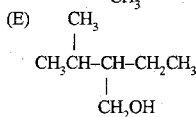
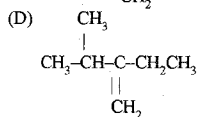
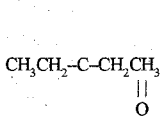
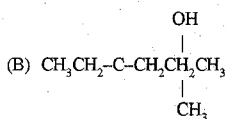
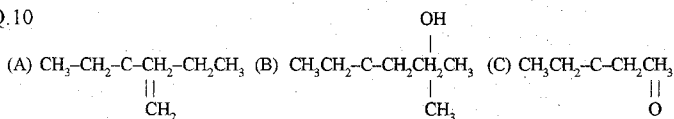


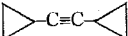
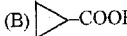


Q.7



Q.10



- Q.13 (A)  (B)  (C) 
- Q.14 (A) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}_3$ (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{CH}_2\text{CH}_3$
 (D) CH_3CHO (E) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$ (F) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- Q.15 (A) $\begin{array}{c} \text{CH}_3-\text{CH}=\text{CH}-\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ (B) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}-\text{CH}-\text{CH}_3 \\ | \quad | \quad | \\ \text{Br} \quad \text{Br} \quad \text{CH}_3 \end{array}$ (C) $\begin{array}{c} \text{CH}_3-\text{C}=\text{C}-\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
 (D) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}-\text{CH}-\text{CH}_3 \\ | \quad | \quad | \\ \text{OH} \quad \text{OH} \quad \text{CH}_3 \end{array}$ (E) CH_3COOH (F) $\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{COOH} \\ | \\ \text{CH}_3 \end{array}$
- Q.16 $\begin{array}{c} \text{H} \\ | \\ \text{CH}_3-\text{C}=\text{C}-\text{C}-\text{C}\equiv\text{C}-\text{CH}_3 \\ | \quad | \quad | \\ \text{H} \quad \text{H} \quad \text{CH}_3 \end{array}$

- Q.17 $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_3$
- Q.18 (A) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$, (B) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$,
 (C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$, (D) CH_3CHO , (E) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$, (F) $\text{CH}_3\text{CH}_2\text{COOH}$

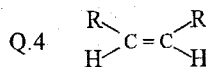
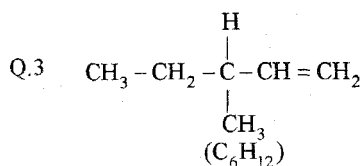
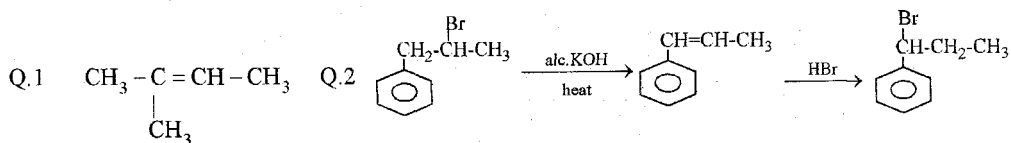
- Q.19 Isomer are : $\text{C}\equiv\text{C}-\text{C}-\text{C}$, $\text{C}=\text{C}-\text{C}=\text{C}$, $\text{C}=\text{C}=\text{C}-\text{C}$, , , , , 

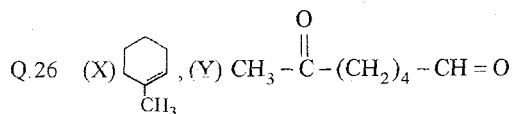
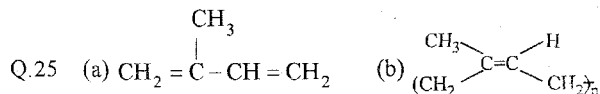
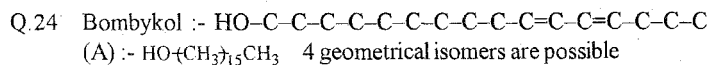
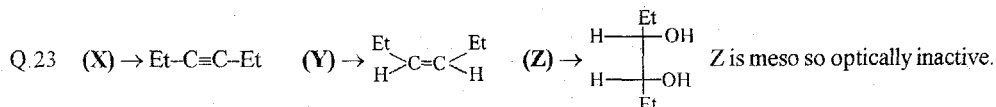
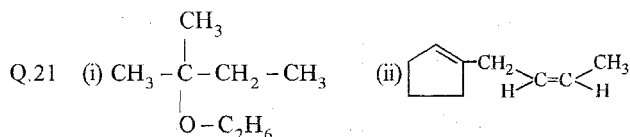
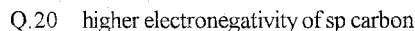
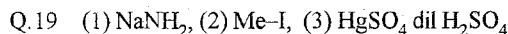
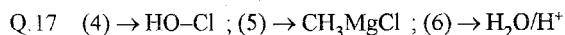
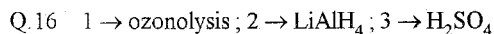
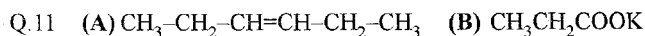
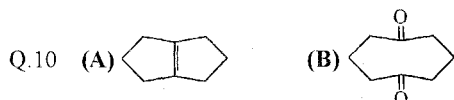
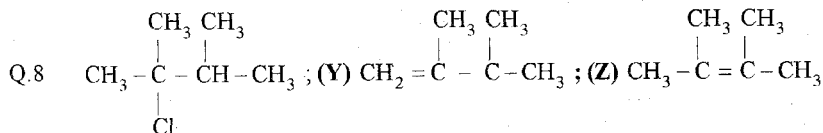
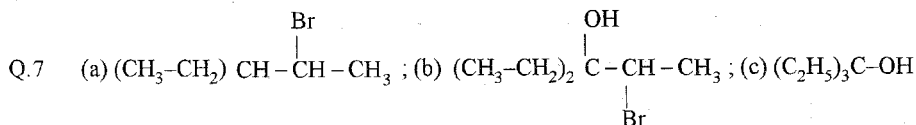
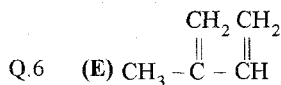
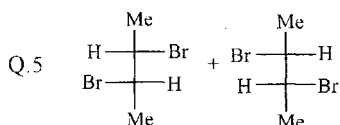
Q.20 23.7

EXERCISE-IV (A)

- | | | | | |
|--------|---------|--------|--------|--------|
| Q.1 C | Q.2 A,D | Q.3 A | Q.4 C | Q.5 A |
| Q.6 D | Q.7 A,C | Q.8 D | Q.9 A | Q.10 D |
| Q.11 C | Q.12 B | Q.13 B | Q.14 A | Q.15 B |
| Q.16 D | Q.17 A | Q.18 B | Q.19 B | Q.20 B |
| Q.21 D | Q.22 A | Q.23 A | Q.24 B | |

EXERCISE-IV (B)







BANSAL CLASSES

CHEMISTRY

TARGET IIT JEE 2007

XI (PQRS)

***NOMENCLATURE
OF
ORGANIC COMPOUNDS***

CONTENTS

EXERCISE - I

EXERCISE - II

ANSWER KEY

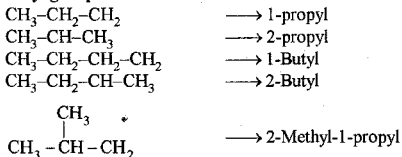
IUPAC

Nomenclature according to IUPAC system involves use of following terms:

- (i) **Word root :** The word root represents the number of C atoms in parent chain.
- | No. of C atom | W.R. | |
|---------------|-------|--------------|
| 1 | Meth | Octane 8 |
| 2 | Eth | Nonane 9 |
| 3 | Prop. | Decane 10 |
| 4 | But | Undecane 11 |
| 5 | Pent | Dodecane 12 |
| 6 | Hex | Tridecane 13 |
| 7 | Hep | |
- (ii) **Primary suffix :** Primary suffix is used to indicate saturation or unsaturation in carbon chain. While writing name, primary suffix is added to word root.
- | Nature of C chain | Primary suffix |
|-------------------------|----------------|
| (1) Saturated C chain | ane |
| (2) Unsaturated C chain | |
| C=C | ene |
| C≡C | yne |
| 2 C≡C | diyne |
- (iii) **Secondary suffix :** Secondary suffix is used to indicate functional group in organic comp. It is added primary suffix by dropping its terminal "e".

Prefix : The part of the name C appears before the word root is called prefix. Different prefixes are used for dif categories of group as:

- (a) **Alkyl groups:**



- (b) In IUPAC system, for nomenclature some groups are not considered as functional group but treated as substituent. These functional group are always indicated by prefixes instead of secondary suffixes.

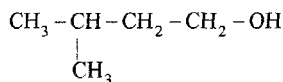
-NO ₂	Nitro
-OR	Alkoxy
-Cl	Chloro
-Br	Bromo
-I	Iodo
-F	Fluoro
-N=O	Nitroso
-NO ₂	Nitro
=N	Diazo

- (c) In polyfunctional compound, one of the functional group is treated as principal functional group & indicated by secondary suffix while other functional groups are treated as substituents & indicated by prefixes.

GF	O.Co.	Suffix	IUPAC Name	Prefix
R-OH	Alcohols	ol	alkan ol	Hydroxy
R-SH	Thioalcohols	thiol	alkanethiol	Mercapto
R-NH ₂	Amines	amine	alkanamin o	Amino
R-CHO	Aldehyde	al	alkanal	formyl
RCOR	Ketone	one	alkanone	Keto or oxo
RCOOH	carboxylic acid	-oic acid	alkanoic acid	Carboxy
Amides	RCONH ₂	amide	alkanamide	Carbamoyl
Acid chloride	RCOCl	oyl halide	alkanoyl chloride	Chloroformyl
Ester	RCOOR	oate	alkylalkanoate	Carbalkoxy
Nitriles	R-C≡N	nitrile	alkanenitrile	Cyano

Arrangement :

Prefix(es) + word root + p. suffix + sec. suffix



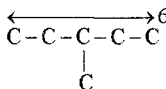
Methyl but an ol

Rules:

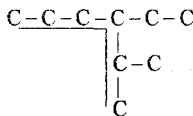
For saturated compounds:

(i) Selection of longest chain :

- (a) Longest continuous chain of carbon atoms is selected. This is called parent chain while all other C atoms C are not included in parent chain is called side chain.

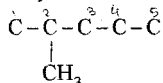


- (b) If more than one set of longest chains are possible, the chain C max. no. of substituent acts as parent chain.

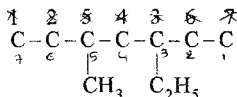


(ii) Numbering of selected chain :

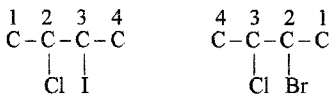
- (a) The selected chain is numbered from one end to other. The number are called locants. Numbering is done in such a way that lowest no. is assigned to side chain or substituent get



- (b) If two different alkyl groups or substituents are at same position from opposite ends, lowest no. is given in alphabetical order.

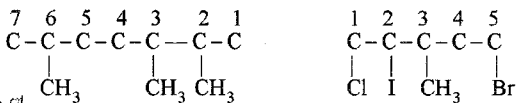


- (c) If two different substituents are at same position from opp. ends, lowest no. is assigned in order of their alphabets.

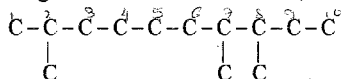


The numbering is done in such a way that the substituted carbon atoms have the lowest possible numbers. When series of locants containing the same no. of terms are compared, term by term, the chosen series should contain the lowest no. on the occasion of 1st diff.

- (d) If more than two substituents & side chains are present, the sum of their numbers should be lowest at the first preference irrespective of the nature of substituent or side chain. This is K/a lowest sum rule.

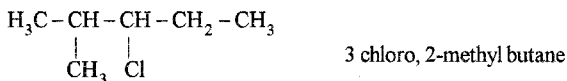


- In case when starting locant is lower from one side, then lowest sum rule is not obeyed.

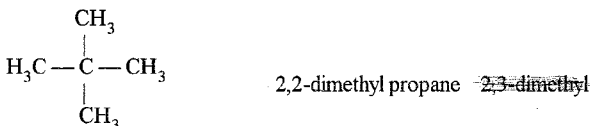


(iii) Arrangement of prefixes :

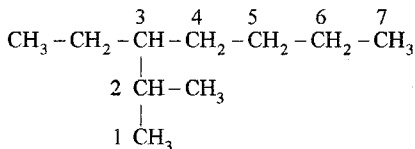
- (a) Side chain or substituent group are added as prefix with its locant in alphabetical order.

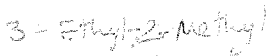


- (b) If more than one similar alkyl group or substituents are present terms, tri like are used.



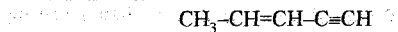
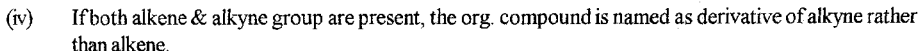
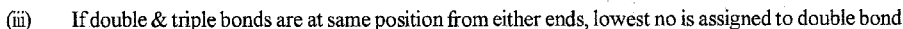
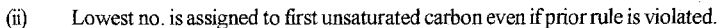
- (c) In case side chain is also branched, it is also numbered from carbon atom attached to main chain & is generally written in brackets.





4-(1-methylpropyl)benzene

(i) Select the longest possible carbon chain having maximum no. of unsaturated carbon atoms or max. no. of double or triple bonds, even if puor rule is violated.

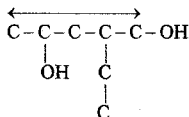


Pent-3-en-1-yne

$$\text{CH}_2 =$$


For functional groups :

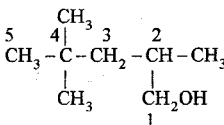
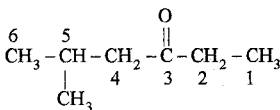
- (i) Select the longest possible carbon chain having max. no. of functional even if prior rules are violated.



- (ii) The carbon atom of functional group is to be included in deciding the longest carbon chain.

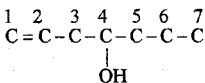
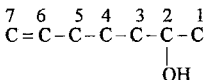


- (iii) The lowest no is assigned to functional group even if prior rules are violated.

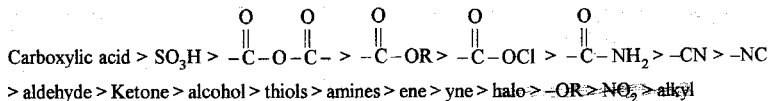


- (iv) The order of numbering a carbon chain, thus follows the order:

- (a) Functional group
- (b) Unsaturation
- (c) Substituents & side chains



- (v) If more than 1 functional group; then choice of principal functional group is made on the basis following order of preference:



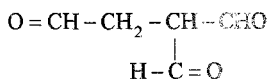
EXERCISE - I

Q.1 How many 1° carbon atom will be present in a simplest hydrocarbon having two 3° and one 2° carbon atom?
(A) 3 (B) 4 (C) 5 (D) 6

Q.2 C₃H₆Br₂ can show:
(A) Two gem dibromide (B) Two vic dibromide
(C) Two tert. dibromo alkane (D) Two sec. dibromo alkane

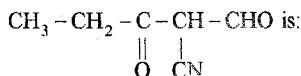
Q.3 The IUPAC name of the compound CH₃CH=CHCH=CHC≡CCH₃ is:
(A) 4,6-octadiene-2-yne (B) 2,4-octadiene-6-yne
(C) 2-octyn-4,6-diene (D) 6-octyn-2,4-diene

Q.4 The correct IUPAC name of the following compound is:



(A) 1,1-diformyl propanal (B) 3-formyl butanedial
(C) 2-formyl butanedial (D) 1,1,3-ethane tricarbaldehyde

Q.5 The correct IUPAC name of compound:



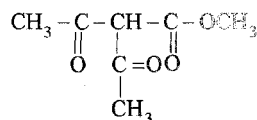
(A) 2-cyano-3-oxopentanal (B) 2-formyl-3-oxopentanenitrile
(C) 2-cyano-1,3-pentanedione (D) 1,3-dioxo-2-cyanopentane

Q.6 All the following IUPAC names are correct except:

(A) 1-chloro-1-ethoxy propane (B) 1-amino-1-ethoxypropane
(C) 1-ethoxy-2-propanol (D) 1-ethoxy-1-propanamine

(amino is funct. group, not a substituent)

Q.7 IUPAC name of:



(A) Methyl-2,2 acetyl ethanoate (B) 2,2 acetyl-1-methoxy ethanone
(C) Methyl-2-acetyl-3-oxobutanate (D) None

Q.8 The IUPAC name of β-ethoxy-α-hydroxy propionic acid (trivial name) is:

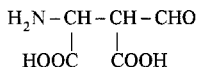
(A) 1,2-dihydroxy-1-oxo-3-ethoxy propane (B) 1-carboxy-2-ethoxy ethanol
(C) 3-Ethoxy-2-hydroxy propionic acid (D) All above

- Q.9 The IUPAC name of compound $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{CHO}}{\text{CH}} - \text{CH}_3$ is:
- (A) 3,5-Dimethyl-4-Formyl pentanone (B) 1-Isopropyl-2-methyl-4-oxo butanal
(C) 2-Isopropyl-3-methyl-4-oxo pentanal (D) None of the above

- Q.10 The IUPAC name of compound $\text{CH}_3 - \overset{\text{HO}-\text{C}=\text{O}}{\parallel}{\text{C}} - \underset{\text{NH}_2}{\text{C}} = \underset{\text{Cl}}{\text{C}} - \text{H}$ is:

- (A) 2-amino-3-chloro-2-methyl-2-pentenoic acid
(B) 3-amino-4-chloro-2-methyl-2-pentenoic acid
(C) 4-amino-3-chloro-2-methyl-2-pentenoic acid
(D) All of the above

- Q.11 The IUPAC name of the structure is:



- (A) 3-amino-2-formyl butane-1, 4-dioic acid (B) 3-amino-2, 3-dicarboxy propanal
(C) 2-amino-3-formyl butane-1, 4-dioic acid (D) 1-amino-2-formyl succinic acid
- Q.12 $\text{C}_4\text{H}_6\text{O}_2$ does not represent:
(A) A diketone (B) A compound with two aldehyde
(C) An alkenoic acid (D) An alkanolic acid
- Q.13 Esters are functional isomers of:
(A) Hydroxy aldehyde (B) Ketone (C) Diketone (D) Diols
- Q.14 How many carbons are in simplest alkyne having two side chains?
(A) 5 (B) 6 (C) 7 (D) 8
- Q.15 Which of the following is not correctly matched:

(A) Lactic acid $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{COOH}$

(B) Tartaric acid $\text{HO} - \text{CH} - \text{COOH}$

(C) Pyvaldehyde $\text{HO} - \underset{\text{CH}_3}{\text{CH}} - \text{COOH}$
 $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CHO}$

(D) Iso-octane $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{CH} - \text{CH}_3$
 $\quad \quad \quad |$
 $\quad \quad \quad \text{CH}_3$

- Q.16 Which of the following pairs have absence of carbocyclic ring in both compounds?
 (A) Pyridine, Benzene (B) Benzene, Cyclohexane
 (C) Cyclohexane, Furane (D) Furane, Pyridine
- Q.17 The commercial name of trichloroethene is:
 (A) Westron (B) Perclene (C) Westrosol (D) Orlone
- Q.18 The compound which has one isopropyl group is:
 (A) 2,2,3,3-tetramethyl pentane (B) 2,2-dimethyl pentane
 (C) 2,2,3-trimethyl pentane (D) 2-methyl pentane
- Q.19 How many secondary carbon atoms does methyl cyclopropane have?
 (A) None (B) One (C) Two (D) Three
- Q.20 The IUPAC name of the compound $\text{CH}_2 - \text{CH} - \text{CH}_2$ is:
 $\begin{array}{ccc} \text{OH} & \text{OH} & \text{OH} \\ | & | & | \end{array}$
 (A) 1,2,3-tri hydroxy propane (B) 3-hydroxy pentane-1,5-diol
 (C) 1,2,3-hydroxy propane (D) Propane-1,2,3-triol
- Q.21 As per IUPAC rules, which one of the following groups, will be regarded as the principal functional group?
 (A) $-\text{C}\equiv\text{C}-$ (B) $-\text{OH}$ (C) $-\text{C}-$
 $\text{O}=\text{C}$ (D) $-\text{C}-\text{H}$
 $\text{O}=\text{C}$
- Q.22 Which of the following is the first member of ester homologous series?
 (A) Ethyl ethanoate (B) Methyl ethanoate (C) Methyl methanoate (D) Ethyl methanoate
- Q.23 The correct IUPAC name of 2-ethyl-3-pentyne is:
 (A) 3-methyl hexyne-4 (B) 4-ethyl pentyne-2 (C) 4-methyl hexyne-2 (D) None of these

- Q.24 IUPAC name for the compound $\begin{array}{c} \text{Cl} \\ \diagup \\ \text{C}=\text{C} \\ \diagdown \end{array}$ is
 $\begin{array}{c} \text{CH}_2\text{CH}_3 \\ \diagup \\ \text{C} \\ \diagdown \\ \text{I} \end{array}$
 $\begin{array}{c} \text{H}_3\text{C} \\ \diagup \end{array}$
 (A) E-3-iodo-4-chloro-3-pentene (B) E-2-chloro-3-iodo-2-pentene
 (C) Z-2-chloro-3-iodo-2-pentene (D) Z-3-iodo-4-chloro-3-pentene

- Q.25 The IUPAC name of the compound is $\text{CH}_3 - \text{CH} - \text{CH} - \text{NH}_2$
 $\begin{array}{c} \text{Ph} \\ | \\ \text{CH}_3 \end{array}$
 (A) 1-amino-1-phenyl-2-methyl propane (B) 2-methyl-1-phenyl propane-1-amine
 (C) 2-methyl-1-amino-1-phenyl propane (D) 1-isopropyl-1-phenyl methyl amine

Q. 26 Which of the following compound is wrongly named?

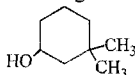
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$; 2-Chloro pentanoic acid

(B) $\text{CH}_3\text{C}(\text{CH}_3)=\text{CHCOOH}$; 2-Methyl hex-3-enoic acid

(C) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCOCH}_3$; Hex-3-en-2-one

(D) $\text{CH}_3-\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CHO}$; 4-Methyl pentanal

Q. 27 The IUPAC name of the given compound is:



- (A) 1,1-dimethyl-3-hydroxy cyclohexane (B) 3,3-dimethyl-1-hydroxy cyclohexane
(C) 3,3-dimethyl-1-cyclohexanol (D) 1,1-dimethyl-3-cyclohexanol

Q. 28 The IUPAC name of $(\text{C}_2\text{H}_5)_2\text{NCH}_2\text{CH}(\text{Cl})\text{COOH}$ is:

- (A) 2-chloro-4-N-ethylpentanoic acid (B) 2-chloro-3-(N,N-diethyl amino)-propanoic acid
(C) 2-chloro-2-oxo diethylamine (D) 2-chloro-2-carboxy-N-ethyl ethane

Q. 29 The IUPAC name of the compound $\text{Br}(\text{Cl})\text{CH}_2\text{CF}_3$ is:

- (A) haloethane (B) 1,1,1-trifluoro-2-bromo-2-chloroethane
(C) 2-bromo-2-chloro-1,1,1-trifluoroethane (D) 1-bromo-1-chloro-2,2,2-trifluoro ethane

Q. 30 The group of hetrocyclic compounds is:

- (A) Phenol, Furane (B) Furane, Thiophene (C) Thiophene, Phenol (D) Furane, Aniline

Q. 31 The correct IUPAC name of $\text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)=\text{CH}-\text{COOH}$ is:

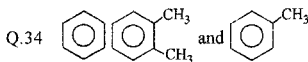
- (A) 2-methyl butanoic acid (B) 2-ethyl-2-propenoic acid
(C) 2-carboxy-1-butene (D) None of the above

Q. 32 The IUPAC name of the following structure $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)\text{C}\equiv\text{C}(\text{CH}_3)\text{CH}(\text{CH}_3)_2$ is:

- (A) 3-methyl-4-hexynene-2 (B) 3-methyl-2-hexenyne-4
(C) 4-methyl-4-hexenyne-4 (D) all are correct

Q. 33 The IUPAC name of the following structure is $[\text{CH}_3\text{CH}(\text{CH}_3)]_2\text{C}(\text{CH}_2\text{CH}_3)\text{C}(\text{CH}_3)\text{C}(\text{CH}_2\text{CH}_3)_2$

- (A) 3,5-diethyl-4,6-dimethyl-5-[1-methylethyl]-3-heptene
(B) 3,5-diethyl-5-isopropyl-4,6-dimethyl-2-heptene
(C) 3,5-diethyl-5-propyl-4,6-dimethyl-3-heptene
(D) None of these



Number of secondary carbon atoms present in the above compounds are respectively:

- (A) 6,4,5 (B) 4,5,6 (C) 5,4,6 (D) 6,2,1

Q.35 The IUPAC name of acetyl acetone is:

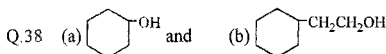
- (A) 2,5-Pentane dione (B) 2,4-Pentane dione (C) 2,4-Hexane dione (D) 2,4-butane dione

Q.36 When vinyl & allyl are joined each other, we get

- (A) Conjugated alkadiene (B) cumulative alkadiene
(C) Isolated alkadiene (D) Allenes

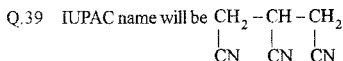
Q.37 Glycerine is:

- (A) Propane triol-1,2,3 (B) Propylene trialcohol
(C) Propyl glycol (D) Hydroxy methyl glycol



True statement for the above compounds is:

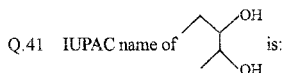
- (A) (a) is phenol while (b) is alcohol (B) Both (a) and (b) are primary alcohol
(C) (a) is primary and (b) is secondary alcohol (D) (a) is secondary and (b) is primary alcohol



- (A) 1,2,3-Tricyano propane (B) Propane trinitrile-1,2,3
(C) 1,2,3-cyano propane (D) 3-cyano pentane-1,3-dinitrile

Q.40 A substance containing an equal number of primary, secondary and tertiary carbon atoms is:

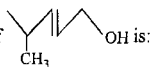
- (A) Mesityl Oxide (B) Mesitylene (C) Maleic acid (D) Malonic acid



- (A) But-2-ene-2,3-diol (B) Pent-2-ene-2,3-diol
(C) 2-methylbut-2-ene-2,3-diol (D) Pent-3-ene-3,4-diol

Q.42 The IUPAC name of $\text{BrCH}_2 - \text{CH} - \text{CO} - \text{CH}_2 - \text{CH}_2\text{CH}_3$ is:

- |
 CONH₂
(A) 2-bromo methyl-3-oxo hexanamide (B) 1-bromo-2-amino-3-oxo hexane
(C) 1-bromo-2-amino-n-propyl ketone (D) 3-bromo-2-propyl propanamide

Q.43 IUPAC name of  is:

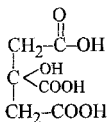
- (A) 5-methyl hexanol (B) 2-methyl hexanol
(C) 2-methyl hex-3-enol (D) 4-methyl pent-2-en-1-ol

Q.44 The IUPAC name of $\text{CH}_3\text{CH}_2-\text{N}(\text{CH}_3)\text{CH}_2\text{CH}_3$ is:

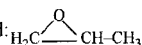
- (A) N-methyl-N-ethylethyl amine (B) diethyl methyl amine
(C) N-ethyl-N-methyl ethyl amine (D) methyl diethyl amine

Q.45 The molecular formula of the first member of the family of alkenynes and its name is given by the set

- (A) C_3H_2 , alkene (B) C_3H_6 , 1-penten-3-yne
(C) C_6H_8 , 1-hexen-5-yne (D) C_4H_4 , butenyne

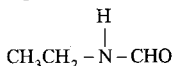
Q.46 The IUPAC name of compound 

- (A) 1,2,3-tricarboxy-2,1-propane (B) 3-Carbox-3-hydroxy-1,5-pentane dioic acid
(C) 3-hydroxy-3-Carboxy-1,5-pentane dioic acid (D) None

Q.47 The IUPAC name of the compound: 

- (A) Propylene Oxide (B) 1,2-Oxo propane
(C) 1,2-Epoxy propane (D) 1,2-Propoxide

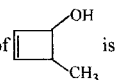
Q.48 One among the following is the correct IUPAC name of the compound



- (A) N-Formyl aminoethane (B) N-Ethyl formyl amine
(C) N-Ethyl methanamide (D) Ethylamino methanal

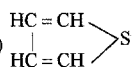
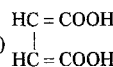
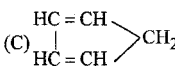
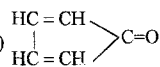
Q.49 Which among the following is the correct IUPAC name of isoamylene:

- (A) 1-Pentene (B) 2-Methyl-2-butene (C) 3-Methyl-1-butene (D) 2-Methyl-1-butene

Q.50 The IUPAC name of  is

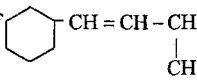
- (A) 3-Methyl cyclo-1-butene-2-ol (B) 4-Methyl cyclo-2-butene-1-ol
(C) 4-Methyl cyclo-1-butene-3-ol (D) 2-Methyl cyclo-3-butene-1-ol

Q.51 Which of the following is a heterocyclic compound

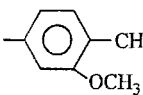
- (A)  (B)  (C)  (D) 

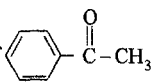
Q. 52 The number of primary, secondary and tertiary amines possible with the molecular formula C_3H_9N is:
 (A) 1,2,2 (B) 1,2,1 (C) 2,1,1 (D) 3,0,1

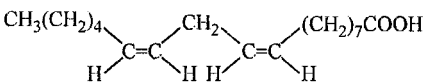
Q. 53 The IUPAC name of $C_6H_5CH=CH-COOH$ is:
 (A) cinnamic acid (B) 1-phenyl-2-carboxy ethane
 (C) 3-phenyl prop-2-enoic acid (D) dihydroxy-3-phenyl propionic acid

Q. 54 The IUPAC name of  is:
 (A) 1-cyclohexyl-3-methyl-1-pentene (B) 3-methyl-5-cyclohexyl-pent-ene
 (C) 1-cyclohexyl-3-ethyl-but-1-ene (D) 1-cyclohexyl-3,4-dimethyl-but-1-ene

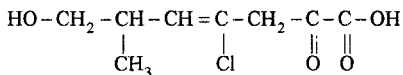
Q. 55 The IUPAC name of $CH-C(=O)-O-CH_2-C(=O)-OH$ is:
 (A) 1-acetoxy acetic acid (B) 2-acetoxy ethanoic acid
 (C) 2-ethanoyl oxyacetic acid (D) 2-ethanoyl oxyethanoic acid

Q. 56 The IUPAC name of  is:
 (A) 2-methoxy-4-nitro benzaldehyde (B) 4-nitro anisaldehyde
 (C) 3-methoxy-4-formyl nitro benzene (D) 2-formyl-4-nitro anisole

Q. 57 The IUPAC name of  is:
 (A) phenyl ethanone (B) methyl phenyl ketone
 (C) acetophenone (D) phenyl emethyl ketone

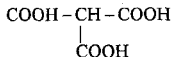
Q. 58 The IUPAC name of  is:
 (A) cis-cis-9, 12-octadecan dienoic acid (B) cis-trans-9,12-octadecan dienoic acid
 (C) 9,10-octa decadienoic acid (D) 9,14-octa decadienoic acid

Q. 59 The suffix of the principal group, the prefixes for the other groups and the name of the parent in the structure



(A) -oic acid, chloro, hydroxy, oxo, methyl, 4-heptene
 (B) -oic acid, chloro, hydroxy, methyl, oxo, 4-heptene
 (C) -one, carboxy, chloro, methyl, hydroxy, 4-heptene
 (D) -one, carboxy, chloro, methyl, hydroxy, 4-heptene

Q.60 The IUPAC name of compound



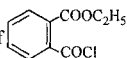
- (A) Tricarboxy methane
(C) Tributanoic acid

- (B) Propane triotic acid
(D) 2-carboxy propanedioic acid

Q.61 The IUPAC name of $\text{OHC} - \text{CH}_2 - \text{CH}_2 - \overset{\text{CH}_2 - \text{CHO}}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CHO}$ is:

- (A) 4,4-di(formylmethyl) butanal
(C) hexane-3-acetal-1, 6-dial

- (B) 2-(formylmethyl) butane-1, 4-dicarbaldehyde
(D) 3-(formylmethyl) hexane-1, 6-dial

Q.62 The IUPAC name of  is:

- (A) 2-chlorocarbonyl ethylbenzoate
(C) ethyl-2-(chlorocarbonyl) benzoate

- (B) 2-carboxyethyl benzoyl chloride
(D) ethyl-1-(chlorocarbonyl) benzoate

Q.63 Which of the following is crotonic acid:

- (A) $\text{CH}_2 = \text{CH} - \text{COOH}$
(C) $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$

- (B) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \text{COOH}$
(D) $\text{CH} - \text{COOH}$
 $\quad \quad \quad \parallel$
 $\quad \quad \quad \text{CH} - \text{COOH}$

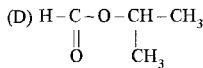
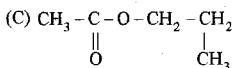
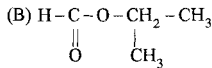
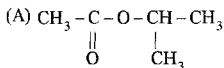
Q.64 $\text{CH}_3 - \text{O} - \underset{\parallel}{\underset{\text{O}}{\text{C}}} - \text{CH}_2 - \text{COOH}$

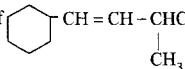
The correct systematic name of the above compound is:

- (A) 2-acetoxy ethanoic acid
(C) 3-methoxy formyl ethanoic acid

- (B) 2-methoxy carbonyl ethanoic acid
(D) 2-methoxy formyl acetic acid

Q.65 Structural formula of isopropyl methanoate is:



Q.66 The IUPAC name of  is:

- (A) 1-cyclohexyl-3-methyl-1-pentene
(C) 1-cyclohexyl-3-ethyl-but-1-ene

- (B) 3-methyl-5-cyclohexyl-pent-1-ene
(D) 1-cyclohexyl-3,4-dimethyl-but-1-ene

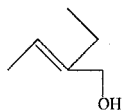
Q.67 The correct IUPAC name of the compound:

- (A) 5-ethyl-3, 6-dimethyl non-3-ene
(C) 4-methyl-5, 7-diethyl oct-2-ene

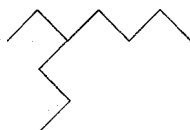
- (B) 5-ethyl-4, 7-dimethyl non-3-ene
(D) 2,4-ethyl-5-methyl oct-2-ene

EXERCISE - II

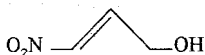
Q.1



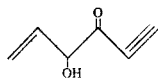
Q.2



Q.3



Q.4



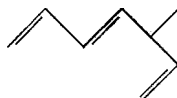
Q.5



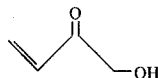
Q.6



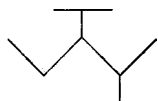
Q.7



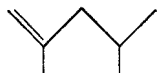
Q.8



Q.9



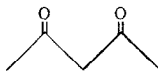
Q.10



Q.11



Q.12



Q.13



Q.14



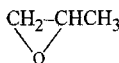
Q.15



Q.16

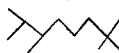


Q.17

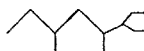


Give the IUPAC names for each of the following :

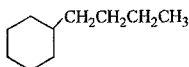
Q.18



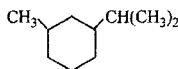
Q.19



Q.20

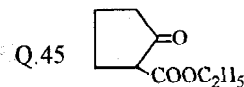
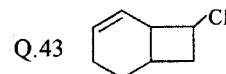
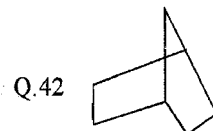
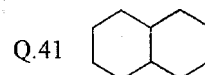
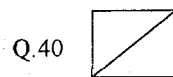
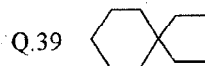
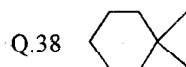
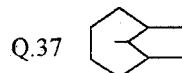
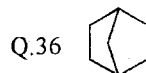
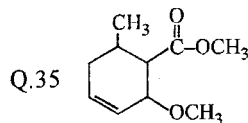
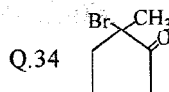
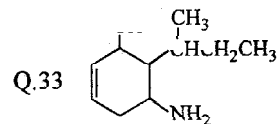
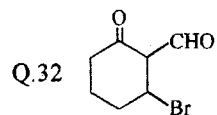
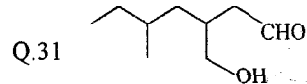
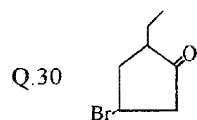
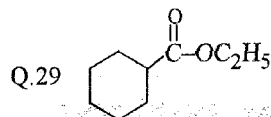
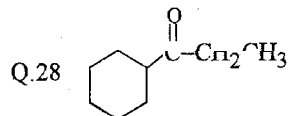
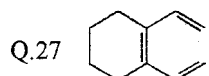
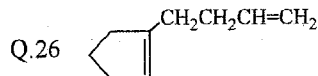
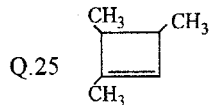
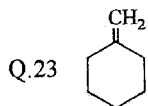


Q.21



Q.22

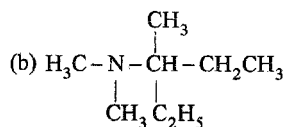




O=C1C=CC(=O)C=C1OC1CCCCC1C#CCC(Cl)CC(=O)CC1CCCC1CC1(CCCN)CCCCC1OCC(C)C1=CCCCC1CC(C(=O)O)C1CCCCC1=OCC(=C)C(OC)C(=O)C1(CCC1)[N+](=O)[O-]C1=CC=CC=C1C=CC(C2=CC=CC=C2)C=CC=C

Q.55 Write IUPAC Name of following :

(a)  Me = methyl group



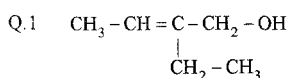
Q.56 Write IUPAC name of succinic acid

ANSWER KEY

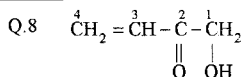
EXERCISE - I

Q.1	B	Q.2	A	Q.3	B	Q.4	C	Q.5	B	Q.6	B	Q.7	C
Q.8	C	Q.9	C	Q.10	B	Q.11	C	Q.12	D	Q.13	A	Q.14	B
Q.15	D	Q.16	D	Q.17	C	Q.18	D	Q.19	C	Q.20	D	Q.21	D
Q.22	C	Q.23	C	Q.24	C	Q.25	B	Q.26	B	Q.27	C	Q.28	B
Q.29	D	Q.30	B	Q.31	B	Q.32	B	Q.33	A	Q.34	A	Q.35	B
Q.36	C	Q.37	A	Q.38	D	Q.39	A	Q.40	B	Q.41	B	Q.42	A
Q.43	D	Q.44	C	Q.45	D	Q.46	B	Q.47	C	Q.48	C	Q.49	C
Q.50	B	Q.51	A	Q.52	C	Q.53	C	Q.54	A	Q.55	D	Q.56	A
Q.57	A	Q.58	A	Q.59	B	Q.60	D	Q.61	D	Q.62	C	Q.63	C
Q.64	B	Q.65	D	Q.66	A	Q.67	A						

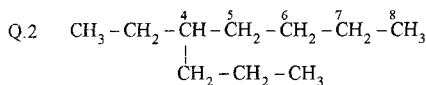
EXERCISE - II



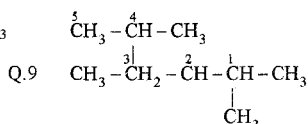
2-ethyl-2-butene-1-ol



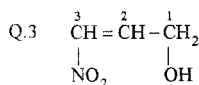
1-Hydroxy-3-Butene-2-one



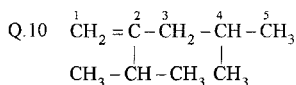
4-Ethyl octane



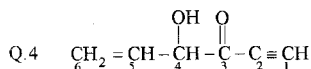
3-Ethyl-2,4-dimethyl pentane



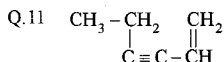
3-nitro-2-propene-1-ol



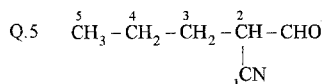
2-isopropyl-4-methyl-1-pentene



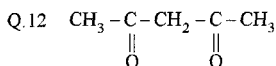
4-hydroxy-5-hexene-1-yne-3-one



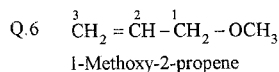
1-Hexene-3-yne



2-Formyl pentane nitrile



2,4, pentane dione



1-Methoxy-2-propene

Q.13 Cyclopropanecarboxylic acid

Q.7 3-methyl-1,4,6-Heptatriene

- | | |
|--|--|
| Q.14 Cyclopropane carboxylic acid | Q.39 spiro(4.5) decane |
| Q.15 1,3,5-cyclohexatriene | Q.40 Bicyclo(2.2.1) heptane |
| Q.16 1,3-cyclobutadiene | Q.41 Bicyclo(4.4.0) decane |
| Q.17 1,2-epoxy propane | Q.42 Bicyclo(2.2.1) heptane |
| Q.18 2,2,6,7-tetramethylcatane | Q.43 8-chloro bicyclo(4.2.0) oct-2-ene |
| Q.19 3-ethyl-4,6-dimethyloctane | Q.44 2-cyclopenten-1-ol |
| Q.20 Butyl cyclohexane | Q.45 2-carbethoxy cyclopentanone |
| Q.21 3-isopropyl-1-methylcyclohexane | Q.46 Bicyclo(3.1.0) hexane |
| Q.22 2-ethyl-1-methylcyclopentane | Q.47 Cyclohex-2-en-1,4-dione |
| Q.23 Methyl enyl cyclohexane | Q.48 2-ethynyl cyclohexanol |
| Q.24 Isopylidenecyclopentane | Q.49 4-chloro-1-cyclopentyl pentane-2-one |
| Q.25 1,3,4-dimethyl-1-cyclobutene | Q.50 1-Amino methyl-2-ethyl cyclohexanol |
| Q.26 1-(3-butenyl) cyclopentene | Q.51 1-propyl-4-isopropyl-1-cyclohexene |
| Q.27 1,2-diethenyl cyclohexene | Q.52 2-(β -keto cyclohexyl) propanoic acid |
| Q.28 1-cyclohexyl-1-propanone | Q.53 3-ethoxyl-1(1-nitrocyclohexyl)-hexe-4-one-1 |
| Q.29 Ethyl cyclohexanecarboxylate | Q.54 1,3-diphenyl-1,4-pentadiene |
| Q.30 4-Bromo-2-ethyl cyclopentaneone | Q.55 (a) 5,6-diethyl-3-methyl-dec-4-ene
(b) N,N,3-trimethyl-3-pentanamine |
| Q.31 3-(1-hydroxyethyl)-5-methylheptanal | Q.56 Butane-1,4-dioic acid |
| Q.32 6-Bromo-2-oxocyclohexanecarbaldehyde | |
| Q.33 3-amino-2-sec-butyl-5-cyclohexen-1-ol | |
| Q.34 2-bromo-2-methyl cyclopentanone | |
| Q.35 Methyl-2-methoxy-6-methyl-3- cyclohexene
carboxylate | |
| Q.36 Bicyclo(2.2.1)heptane | |
| Q.37 9-methyl bicyclo(4.2.1) nonane | |
| Q.38 spiro(2.5) octane | |